

Exact Derandomization of Logarithmic Space: $L = RL = BPL$

Lecture notes on the September 23, 2026 construction

Original mathematics: OpenAI

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Abstract

We present the September 23, 2026 construction for deterministic simulation of polynomial-time randomized logarithmic-space computation. Its probability-approximation theorem gives any fixed inverse-polynomial accuracy in logarithmic space and polynomial time, yielding $L = RL = BPL$. The construction repeatedly resolves part of a system of continuation probabilities, using partial propagation and copying to reduce the remaining transition weights. It evaluates the resulting success contributions through sparse tables, conditional averaging, and recursive estimates at different accuracy levels. We explain how these ingredients work together and why local sparsity alone does not prove the space bound: the implementation must also control every suspended call and halt on every seed. The complete structural, probabilistic, and implementation arguments are included, together with variable-precision approximation, accepting-computation extraction, and the effective compiler.

Provenance. Original mathematics: OpenAI, *Exact Derandomization of Logarithmic Space: $L = RL = BPL$* , September 23, 2026, at commit `adc7f1241b42e`. This exposition also adapts the pedagogical source in `sragavan99/deslop/bpl_example`, at commit `2021122121dc`. The presentation, notation, and reading order have been revised; the mathematical claims and proof route are those of the original. Both source versions are identified in the accompanying README. Distributed under the Apache License, Version 2.0.

Contents

1	The problem and the results	4
1.1	Where the difficulty lies	5
1.2	How to read the proof	5
2	Technical overview: resolve probabilities without storing them	5
2.1	The exact target and the stopping condition	6
2.2	One update: partial propagation and the two extremes	6
2.3	Choosing the correction: where to propagate, how much, and what to copy	7
2.4	Sparse tables: sampling, support bounds, and bias correction	10
2.5	What expansion does, and why an arbitrary expander is insufficient	10
2.6	What an accuracy level asks the estimator to do	11
2.7	From local queries to a deterministic answer	12
2.8	Undirected connectivity and Laplacian solvers	13
3	The transition model and resource parameters	13
4	Sampling vertex subsets with a short description	16
4.1	The sampling interface	16
4.2	Polynomial channels with independent nonconstant columns	16
5	The exact reduction: propagation with controlled incoming support	17
5.1	Correction with a fixed column multiplier	18
5.2	Choosing which columns permit correction	19
5.3	The finite residual	24
5.4	Divide the remaining entries between copies	25
5.5	Iterate the construction and bound the remaining error	27
6	The estimator: what is queried and what is returned	28
6.1	One mathematical table, several ways to query it	29
6.2	The order in which quantities are defined	29
6.3	A row query, step by step	29
6.4	The guarantees proved in the next sections	30
7	Sparse random tables for the exact construction	31
7.1	Conditioning and normalization	32
7.2	Separate entries by size and normalize each part	33
7.3	Corrections with an incoming gate	35
7.4	Detector scores	37
7.5	Accessing entries when the input estimates are inaccurate	37
7.6	How errors propagate through table assembly	41
8	Conditional averaging in the shared environment	42
8.1	Describe a walk before the queried vertex is available	45
9	Estimating by successive accuracy improvements	47
9.1	Tasks and dependency order	47
9.2	Which expressions are averaged	48
9.3	Mean comparisons and the cost of releasing a source	49
9.4	Choosing precision to pay for scale and domain changes	52
9.5	Average differences and group them by accuracy level decrease	53
9.6	Absolute difference bounds and averaging noise	55

10 Representing a row by a default and exceptions	57
10.1 The row properties that compression uses	57
10.2 The compression update	59
10.3 Compare with a row whose relevant outputs stay fixed	60
10.4 Distinguish truly crowded rows from estimation errors	63
10.5 Bound numerical ranges without changing the needed outputs	65
10.6 The accuracy available before fingerprints	66
11 Comparing vertices by short fingerprints	67
11.1 Choose one short hash seed for each accuracy level block	68
11.2 Compare keys while restoring a shared bit vector	70
11.3 The actual update and the exact-equality comparison	71
11.4 Find a candidate set independent of the current hash seed	73
11.5 Bound the additional error from collisions	75
11.6 Close the accuracy and comparison bounds together	77
12 Computing numerical values one digit at a time	79
12.1 Exact values from bounded signed digits	80
12.2 The numerical interface	82
13 Evaluate the estimator with logarithmic total memory	83
13.1 Shared memory, call records, and query types	84
13.2 What belongs to a local program state	86
13.3 Find incoming entries by traversing a program tree	91
13.4 Verify the storage bounds for the actual estimator operations	94
13.5 Add the memory retained by all active calls	101
14 From the estimator to deterministic simulation	104
14.1 Choose constants before choosing the input-dependent ranges	105
14.2 Pass from average error to the specified start	106
14.3 The class equality and promise separation	109
14.4 Handle an accuracy request supplied as input	109
14.5 Construct an accepting computation from continuation values	110
A Detailed arithmetic constructions	111
A.1 Compare finite truncations before taking a digit	112
A.2 Multiply without retaining a long partial product	113
A.3 Take a reciprocal using an early prefix and a small remainder	115
A.4 Order statistics, sums, and finite-accuracy access tests	117
B Compile a supplied machine into a deterministic simulation	119
B.1 An interpreter that enforces its own resource limits	120
B.2 Fix the deterministic approximator once	122
B.3 Use padding to fit the promised source computation	123
B.4 Read the padded input without storing it	124
B.5 Compute explicit bounds for the generated machine	125
C Historical context and related work	127
D Notation and a guide to the source symbols	129
References	130

1 The problem and the results

A logarithmic-space machine can remember a bounded number of positions in its input, but cannot keep a polynomial-length computation or random tape. Can fresh random bits increase what such a machine can decide? We answer this question by giving a deterministic logarithmic-space simulation of every polynomial-time randomized logarithmic-space computation with bounded error.

The difficulty is simultaneous control of time and memory. A deterministic simulation may revisit earlier work, but it cannot save a long random tape or a table containing one probability for every configuration. The construction in this paper instead estimates acceptance probabilities by repeatedly recomputing small pieces of information. Its analysis must establish both that these estimates are accurate and that all information retained during recomputation fits in logarithmic space.

Throughout, n is the input length and logarithms have base two. Randomized running-time bounds hold on every random tape. The complete machine conventions are given in Section 3.

The class L consists of languages decidable by deterministic machines in $O(\log(n+2))$ work space. Such a halting deterministic machine has only polynomially many configurations and hence runs in polynomial time. The class RL permits polynomial-time randomized machines in this space, with acceptance probability zero on no inputs and at least $1/2$ on yes inputs. For BPL , the corresponding probabilities are at most $1/3$ and at least $2/3$.

Notation and reading conventions. Bold lowercase symbols denote vectors, and bold uppercase symbols denote matrices, whether random or deterministic. Ordinary uppercase letters denote scalar random variables; ordinary lowercase letters denote deterministic scalars. Sets, groups, operators, and named functions are identified separately by their definitions and, where helpful, calligraphic or sans-serif type. Entries inherit the typeface of their parent object: $\mathbf{C}_{xy}$ and $\mathbf{w}_x$. A stage index is separated by a comma, as in $\mathbf{C}_{l,xy}$. Parentheses are reserved for an entry of a composite expression, such as $(\mathbf{EC})_{xy}$. Fixing an environment does not rename the objects. Appendix D collects the recurring symbols and their relation to the source notation.

Theorem 1.1 (Exact logarithmic-space derandomization).

$$L = RL = BPL.$$

The statement follows from a more informative approximation result. We estimate the acceptance probability even when it lies between the usual acceptance and rejection thresholds. This separates the numerical task from the final decision: after obtaining an approximation, a bounded-error computation is decided by comparison with $1/2$. The next theorem also makes explicit how a request for greater accuracy changes the resources.

Theorem 1.2 (Acceptance-probability approximation). *Fix a randomized machine $\mathcal{M}$ with polynomial worst-case running time and $O(\log(n+2))$ work space. Let $p_{\mathcal{M}}(x)$ be its acceptance probability on an input x of length n . There is a uniform deterministic algorithm which, on input $(x, 1^q)$ with $q \geq 1$, outputs an integer $0 \leq a \leq 2^{q+2}$ satisfying*

$$\left| a2^{-(q+2)} - p_{\mathcal{M}}(x) \right| \leq 2^{-q}.$$

It uses $O(\log(n+2) + q)$ work space and $(n+2)^{O(1)}2^{O(q)}$ time. The constants depend only on $\mathcal{M}$.

In particular, every fixed inverse-polynomial accuracy is attained in logarithmic space and polynomial time. The precision parameter is unary; the running-time bound is exponential in general

q . We first prove the fixed inverse-polynomial result and then obtain the single variable-precision algorithm by padding. Section 14 also derives total deterministic separators for bounded-error promise problems and constructs accepting runs when the machine’s acceptance probability has a specified inverse-polynomial lower bound.

The simulation is also effective at the level of machine descriptions. Given a source program and supplied polynomial-time and logarithmic-space bounds, one terminating compiler produces a deterministic decider and explicit numerical bounds for its space and time (Theorem B.1). The supplied bounds are hypotheses on the source, rather than properties the compiler must decide. Section B constructs this compiler by specializing one fixed deterministic library.

1.1 Where the difficulty lies

Two familiar approaches explain why the claimed bound requires more than a routine simulation. Enumerating all random tapes uses too many choices. Replacing those tapes by a shorter seed helps, but Nisan’s generator still uses $O(\log^2 n)$ seed bits for a polynomial-time logarithmic-space computation [23]; direct enumeration of those seeds takes quasipolynomial time. Alternatively, one can work directly with the configuration graph and recursively evaluate transition probabilities. That approach has to account for the information retained at every recursive level, not merely the space used by one local calculation.

Earlier work connects acceptance-probability estimation to stochastic matrix products and inverses [1], and uses different accuracies at different recursion levels [17]. These are useful points of comparison for the construction below. Our full discussion of prior work, including the distinction between a space bound and simultaneous polynomial-time and space bounds, appears at the end of these notes.

1.2 How to read the proof

Section 2 explains the construction before introducing its implementation details. The formal proof then has three parts. First, Sections 3–5 replace the original transition probabilities by a sequence of exact success contribution calculations whose remaining error decreases geometrically. Second, Sections 7–11 define and analyze a recursive estimator for these success contributions. Third, Sections 12–13 show how to evaluate that estimator while retaining only logarithmically many bits. Section 14 combines the three parts and derives the consequences stated above.

The longer arithmetic constructions and the effective compiler are included after the main argument. Their statements and proofs are retained: moving them later changes the order of reading, not the hypotheses on which the simulation rests.

2 Technical overview: resolve probabilities without storing them

The algorithm has two layers. The first is an exact algebraic construction: repeatedly replace a system of continuation probabilities by an equivalent one in which less probability remains unresolved. The second is a small-space evaluation procedure: estimate a requested entry of that construction without ever storing its matrices. We explain the exact construction first, including how it chooses the correction matrix. We then explain what sampling, conditional expansion, and recursive accuracy contribute to its evaluation.

For this overview we suppress vertex encodings, inverse ports, short fingerprints, and numerical digit procedures. Sections 11–13 restore these details; in particular, the space bound must count information retained by *all* suspended calls. We also describe thresholds by their purpose before

giving their smooth implementations. Neither simplification removes the incoming-support restriction or the dependence among recursive estimates.

2.1 The exact target and the stopping condition

Start with a polynomial-time randomized logarithmic-space machine. Its configurations form a polynomial-size directed graph. Add elapsed time to each configuration, and keep halted computations in distinct successive layers until a common final time. Let $\mathbf{S}$ be the transition matrix and $\mathbf{e}$ the indicator of acceptance in the final layer. Every edge goes forward in time, so

$$\mathbf{p}_0 = (\mathbf{I} - \mathbf{S})^{-1} \mathbf{e} = \sum_{j \geq 0} \mathbf{S}^j \mathbf{e}$$

is a finite sum. Its entry at a vertex is the probability of eventual acceptance from that vertex. The answer is its entry at the designated start. Every entry lies in $[0, 1]$, including at configurations the original start never reaches.

During one stage, write the system as

$$\mathbf{p} = \mathbf{w} + \mathbf{C}\mathbf{p}. \quad (2.1)$$

The matrix $\mathbf{C}$ contains *residual transition weights*, and the vector $\mathbf{w}$ contains *accumulated success contributions*. At a starting vertex x ,

$$\mathbf{p}_x = \mathbf{w}_x + \sum_y \mathbf{C}_{xy} \mathbf{p}_y.$$

The value $\mathbf{w}_x$ has already been resolved. Each remaining weight $\mathbf{C}_{xy}$ still multiplies a continuation probability at y . These weights represent the answer; they are not a probability distribution over stored algorithmic states.

Since $0 \leq \mathbf{p}_y \leq 1$, it suffices to make the residual *row sum* small:

$$0 \leq \mathbf{p}_x - \mathbf{w}_x \leq \sum_y \mathbf{C}_{xy}. \quad (2.2)$$

Small individual entries alone would not suffice: a million entries of size 10^{-6} can still sum to one. The construction proves two bounds together. At stage l , every residual entry is at most 2^{-hl} , while at most $2^l |V_0|$ vertices are incident to positive entries, for a fixed $h \geq 4$. Every residual row sum is therefore at most

$$|V_0| 2^{-(h-1)l}.$$

After $O(\log n)$ stages, the accumulated success contribution at the primary copy of the start gives any prescribed fixed inverse-polynomial accuracy. We explain the copies below.

2.2 One update: partial propagation and the two extremes

For a nonnegative correction matrix $\mathbf{E}$, define

$$\mathbf{w}' = \mathbf{w} + \mathbf{E}\mathbf{w}, \quad \mathbf{C}' = \mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}. \quad (2.3)$$

The equation for $\mathbf{p}$ is preserved:

$$\mathbf{w}' + \mathbf{C}'\mathbf{p} = \mathbf{w} + \mathbf{C}\mathbf{p} + \mathbf{E}(\mathbf{w} - \mathbf{p} + \mathbf{C}\mathbf{p}) = \mathbf{p}.$$

This is an algebraic identity for any $\mathbf{E}$. Choosing $\mathbf{E}$ well means arranging progress while keeping the new weights nonnegative, their row sums at most one, and their computation accessible in small space.

Suppose only $\mathbf{E}_{xy} = \delta$ is nonzero. In row x , remove weight δ from the pending continuation at y , add $\delta \mathbf{w}_y$ to the accumulated success contribution, and send weight $\delta \mathbf{C}_{yz}$ to each successor z . We have substituted the known equation for $\mathbf{p}_y$ into part of the equation for $\mathbf{p}_x$. Multiple substitutions create shortcuts along directed paths.

Two extreme choices show what remains to be designed.

- If $\mathbf{E} = 0$, nothing changes. Access stays easy, but no additional probability is resolved.
- If $\mathbf{E} = \mathbf{C}$, then $\mathbf{C}' = \mathbf{C}^2$ and $\mathbf{w}' = \mathbf{w} + \mathbf{C}\mathbf{w}$. Every pending continuation is expanded through another step. This resolves longer walks at once, but does not ensure a decrease in the largest entry or control incoming support. Squaring a deterministic forward chain, for example, leaves weight-one edges to vertices two layers later.

Figure 1 illustrates the two extremes and a partial update. The actual construction selects which continuations to expand and how far to expand them. This is the sense in which it resembles *truncated squaring*. It does not compute $\mathbf{C}^2$ and discard small entries. Moreover, $\mathbf{E}$ records cumulative propagation through several sweeps and can exceed $\mathbf{C}$ entrywise. The choices 0 and $\mathbf{C}$ are motivating examples, not endpoints of an entrywise interpolation.

For the linear-system interpretation, put $\mathbf{A} = \mathbf{I} - \mathbf{C}$. Then

$$\mathbf{I} - \mathbf{C}' = (\mathbf{I} + \mathbf{E})\mathbf{A}, \quad \mathbf{w}' = (\mathbf{I} + \mathbf{E})\mathbf{w}.$$

The step left-preconditions the system by $\mathbf{I} + \mathbf{E}$, preserving its solution. A Richardson step for this preconditioned system, starting at zero, returns $\mathbf{w}'$, with error $\mathbf{C}'\mathbf{p}$. The outer construction repeatedly improves the representation of the system; it is not simply Richardson iteration with one fixed preconditioner.

2.3 Choosing the correction: where to propagate, how much, and what to copy

At one stage, let f bound the entries of $\mathbf{C}$. A row has at most $1/(\delta f)$ entries of size at least δf , because its total weight is at most one. A column has no such automatic bound: many different rows can send substantial weight to the same destination.

For a raw propagation matrix $\mathbf{G}$, the relevant *incoming load* at y is the count

$$\#\{x : \mathbf{G}_{xy} \geq \delta f\},$$

at the fixed support threshold δ under discussion. A column is crowded when this exceeds a fixed constant times $1/f$. This is a scale-dependent count of substantial predecessors, not the column sum or a per-vertex memory allocation.

Why control this count? Later we retain each source with probability comparable to f . A bound of order $1/f$ gives a constant expected number of retained predecessors. A further gate imposes a hard constant bound on the actual correction table in every environment. This supports forward and backward movement through a sampled correction and bounds how many old sources can charge an error at one midpoint. Both uses matter. Low incoming load alone does not prove that the whole recursive state fits in logarithmic space.

Suppose a fixed multiplier vector $\mathbf{g}$, with entries in $[0, 1]$, says where propagation is allowed. One enables it fully; zero disables it. Beginning with $\mathbf{E}_0 = 0$, perform a constant number of sweeps:

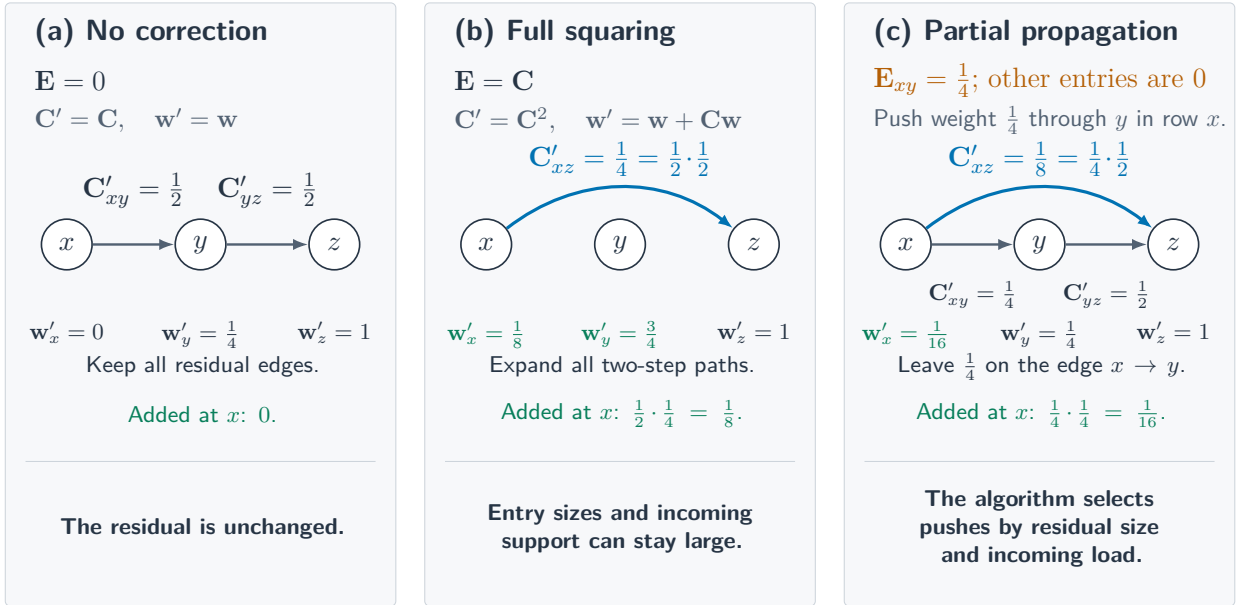
$$\mathbf{G}'_i = \mathbf{C} + \mathbf{E}_{i-1}\mathbf{C}, \quad \mathbf{E}_{i,xy} = f \mathbf{Q}(\mathbf{G}'_{i,xy}/f)\mathbf{g}_y. \quad (2.4)$$

One probability-preserving update, three choices of $\mathbf{E}$

$$\mathbf{w}' = \mathbf{w} + \mathbf{E}\mathbf{w}, \quad \mathbf{C}' = \mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}$$

Same input in all panels: $\mathbf{C}_{xy} = \mathbf{C}_{yz} = \frac{1}{2}$, all other entries zero, and $\mathbf{w} = (0, \frac{1}{4}, 1)^\top$.

Each panel depicts the directed graph with edge weights $\mathbf{C}'$ and success contributions $\mathbf{w}'$ below its vertices.



All three preserve $\mathbf{p} = \mathbf{w}' + \mathbf{C}'\mathbf{p}$, with $\mathbf{p} = (\frac{3}{8}, \frac{3}{4}, 1)^\top$.

The arrow labels are entries of $\mathbf{C}'$; below x, y, z are $\mathbf{w}'_x, \mathbf{w}'_y, \mathbf{w}'_z$, respectively. Panel (a) also shows the original system because $\mathbf{C}' = \mathbf{C}$ and $\mathbf{w}' = \mathbf{w}$. Panel (c) illustrates one partial push. The full construction uses pilot detection, thresholded propagation, and copies of exceptional destinations. Its cumulative $\mathbf{E}$ can exceed $\mathbf{C}$ entrywise; it is not obtained by simply truncating $\mathbf{C}^2$.

Figure 1: The two extreme choices and one partial propagation. Edge labels give residual transition weights; node values give accumulated success contributions. The full construction adds pilot selection, thresholds, and copying.

The function Q smoothly removes the excess above a small threshold:

$$(t - \theta_2)_+ \leq Q(t) \leq (t - \theta_1)_+, \quad 0 < \theta_1 < \theta_2.$$

The raw matrix includes newly propagated weight. The assignment to $\mathbf{E}_i$ replaces the cumulative correction; it does not add that correction again to $\mathbf{E}_{i-1}$.

The positive remainder serves more than nonnegativity. At a fixed point, every corrected position retains at least $\theta_1 f$, so a row can correct only $O(1/f)$ positions. Paths move forward through this bounded set. Counting those ordered paths bounds the tail after k sweeps by a constant multiple of $c^k/k!$. A sufficiently large *constant* number of sweeps therefore suffices, independently of the graph size. The remainder also makes subtraction stable: the residual dominates a fixed multiple of the sum of the unsigned terms in $\mathbf{C} - \mathbf{E} + \mathbf{EC}$.

We still need $\mathbf{g}$. Testing the incoming load of $\mathbf{C}$ once would miss crowding created by propagation. The construction uses a pilot calculation to detect crowding as it develops, freezes its decisions, and restarts the actual correction.

Algorithm 2.1 One exact outer stage

Input. A forward substochastic matrix $\mathbf{C}$ with entries at most f , and accumulated success vector $\mathbf{w}$. This describes exact targets; the expectations are not assumed available as small-space subroutines.

1. **Pilot.** From zero pilot correction, form $\mathbf{G}_i = \mathbf{C} + \hat{\mathbf{E}}_{i-1}\mathbf{C}$. Sample sources and destinations at rate comparable to f , suppressing sampled rows with too many substantial entries. At each column, take the conditional expected score for having many substantial retained predecessors, conditioned on retaining that column.
2. **Freeze.** A smooth decreasing function of the largest score seen so far gives a multiplier for each column. After the fixed number of pilot sweeps, freeze these multipliers as $\mathbf{g}$.
3. **Restart and propagate.** Set the actual correction to zero and run (2.4) with this fixed $\mathbf{g}$. Let $\mathbf{E}$ be the last correction.
4. **Update.** Form $\mathbf{D} = \mathbf{C} - \mathbf{E} + \mathbf{EC}$ and $\tilde{\mathbf{w}} = \mathbf{w} + \mathbf{Ew}$.
5. **Copy.** Divide remaining large entries among a fixed number of destination copies using prescribed routing fractions and row factors. Transport $\tilde{\mathbf{w}}$ by the same weighted projection. The next entry bound is $f/2^h$.

Output. The next exact residual matrix and accumulated success vector.

Pilot scores are deterministic expectations: later noisy estimates do not redefine these targets. The score is an order statistic that becomes large when sufficiently many sampled predecessors contribute. The row gate lets the scores themselves have bounded-access representations. The running maximum remembers crowding in earlier sweeps. Freezing before restarting gives a monotone actual recurrence to which the fixed-point analysis applies.

Fully enabled columns acquire small residual entries. Large remaining entries occur only at exceptional columns. Each exceptional column accounts for many substantial pilot incidences; row-sum bounds limit the total number of incidences. There are therefore few exceptional columns, and their extra copies increase the active-vertex count by at most a factor of two.

Copying changes the domain. A weighted projection $\mathbf{P}$ from new vertices to old ones satisfies

$$\mathbf{C}^+\mathbf{P} = \mathbf{PD}, \quad \mathbf{w}^+ = \mathbf{P}\tilde{\mathbf{w}}.$$

The new solution vector is $\mathbf{Pp}$. A primary copy has projection weight one and preserves its parent's value. Following primary copies preserves the original start value. This qualifies the simpler fixed-domain picture (2.1).

2.4 Sparse tables: sampling, support bounds, and bias correction

We do not build the large matrices and conditional expectations of the exact construction. We represent their contributions as conditional means of random tables that can be queried locally.

Entries of size about a occur only $O(1/a)$ times in a row of bounded total weight. Retaining destinations with probability comparable to a leaves a constant expected number. This explains the division into magnitude scales: one sampling rate would oversample some rows or miss their fine contributions.

An expected bound is insufficient for an algorithm that must fit in space on every seed. A row gate suppresses a table row whenever too many retained entries are substantial; correction tables also have an incoming gate. These gates introduce bias. A preparatory conditional mean corrects it. For a fixed scalar contribution k_{xy} , a random gate R , and the relevant retention event $\mathcal{C}_{xy}$, the basic calculation is

$$q_{xy} = \mathbb{E}[R \mid \mathcal{C}_{xy}], \quad B_{xy} = k_{xy}R/q_{xy}, \quad \mathbb{E}[B_{xy} \mid \mathcal{C}_{xy}] = k_{xy}.$$

The construction bounds q_{xy} away from zero wherever the contribution is needed. The actual formula includes support cutoffs and a clipped denominator, with the preparatory task defined before the normalized task. Exact identities apply to true inputs; estimated inputs introduce errors that require analysis.

The connection with Benczúr–Karger sampling [7] is sampling followed by compensating weights. Their cut sparsifier preserves all cuts of an undirected graph simultaneously. Here a sparse table has the required conditional mean, bounded access, and controlled error after averaging. A single environment need not approximate the whole residual matrix well, preserve all cuts, or preserve all quadratic forms. Later row compression is a different operation: it represents a raw row by a default and exceptions.

Polynomial evaluation supplies a short sampling description: keep a bounded number of coefficients and test membership at the queried identifier. We require comparable marginals, a conditional pair bound, a lower-tail count bound, and invariance under the common affine change induced by appending copy symbols. Full independence of all retained vertices is unnecessary.

2.5 What expansion does, and why an arbitrary expander is insufficient

The expander operates on *sampling environments*, not on the configuration graph. For an entry (x, y) , the environment is conditioned to retain x on one channel and y on another. Averaging must preserve these events and approach the corresponding conditional mean uniformly over identifiers and rates.

The interface has two parts. Analytically, each conditional slice has a stationary averaging operator with a uniform spectral gap, giving exponentially decreasing L_8 error with walk length. Computationally, the walk has a short description that survives recursive calls without retaining all the identifiers on which it is conditioned. The construction provides boundedly many invertible variants whose validity is tested when an identifier becomes available. Exactly one variant represents each genuine conditional step.

A sampling channel uses a polynomial coefficient matrix $\mathbf{A} = (\mathbf{a}_0, \mathbf{F})$ over $\mathbb{F}_{p_{\text{hd}}}$, conditioned on the nonconstant columns $\mathbf{F}$ being linearly independent. Conditional on retaining an identifier y , the

environment can be parametrized by an independent uniform frame $\mathbf{F}$ and a uniform vector $\mathbf{z}$ in a box of side l_{box} . These choices and y determine the constant coefficient.

The infinite group is

$$\mathcal{G}_h = \text{SL}_h(\mathbb{Z}) \ltimes \mathbb{Z}^h, \quad h \geq 3.$$

It acts by

$$(\mathbf{F}, \mathbf{z}) \mapsto (\mathbf{M}\mathbf{F} \bmod p_{\text{fld}}, \mathbf{M}\mathbf{z} + \mathbf{c} \bmod l_{\text{box}}).$$

The moduli need not agree. The linear part moves any frame to any other, and translations move the box coordinate, so the finite action is transitive. Its image is a finite quotient of $\mathcal{G}_h$: the action is a homomorphism into a finite permutation group, whose kernel can be quotiented out. The state space itself is a homogeneous space, and the averaging graph is a Schreier graph; it need not be a Cayley graph of the whole quotient.

Property (T) of $\text{SL}_h(\mathbb{Z})$, together with relative property (T) for the two-dimensional affine group, supplies a uniform gap. Section 8 derives it from Shalom’s quantitative results. This is not merely expansion of $\text{SL}_2(\mathbb{F}_p)$.

A replacement based on zig-zag expanders [32] would be useful if it supplied both conditional mixing and the implementation properties above, along with sampling identities and copy invariance. A strongly explicit graph with a spectral gap alone does not provide this interface. Transitivity alone does not specify how retention events, actions, and short validated descriptions fit together. This list explains the concrete construction’s role; we do not assert a general substitution theorem for arbitrary expanders or transitive codes.

2.6 What an accuracy level asks the estimator to do

There is no probability budget assigned to a vertex. The integer m is an *accuracy level of a query*. With the task’s stage, scale, start-retention rate, and dependency rank, it defines an estimator and its access relation. A larger level changes the calculation, not the true target.

The level has three operational effects. Small levels make a task inactive: return zero and empty access. Active tasks call earlier tasks at smaller levels. The parent–child decrease determines how much conditional averaging to perform and how much information a suspended call can retain.

The assigned effective accuracy exponent is

$$p_{\text{acc}} = \frac{m}{k_{\text{prec}}} - \frac{1}{2} \log_2 \frac{1}{a} - \frac{1}{2} \log_2 \frac{1}{u} - a_* i_*. \quad (2.5)$$

Here a is entry scale, u is start-retention rate, and i_* is dependency rank. These deductions pay for scale changes, changes of conditioning, and propagation through earlier tasks. A nonpositive exponent triggers the zero answer. Otherwise the intended error is a fixed constant times $2^{-p_{\text{acc}}}$ in specified moment norms. Thus m is not itself a number of correct output bits.

Let T_r denote one scalar sample expression after replacing all earlier inputs by child-level- r estimates. The level-zero term is zero; exact base contributions are inserted separately. Write successive improvements as

$$T_{r_{\text{max}}} = \sum_{r=1}^{r_{\text{max}}} (T_r - T_{r-1}).$$

The algorithm averages $T_r - T_{r-1}$ using a conditional walk of length proportional to $m - r$, with scale-dependent cutoffs on r . Both terms use the *same* walk. Early terms can be large but receive longer averaging. Later differences are smaller because both estimates are already accurate. Mean error and remaining averaging noise are bounded separately.

This relates to varying-accuracy evaluation of stochastic-matrix products [17]. Here the schedule must also pay for conditioning, sparse access, and suspended recursion. If $d \leq m - r < 2d$, a candidate contribution has an $O(d)$ -bit description and calls a child of level at most $m - d$. These decreases telescope along a chain to $O(m)$, allowing the accuracy level to serve both error analysis and space accounting.

Algorithm 2.2 Estimate one entry at accuracy level m

Input. A task specifying its exact target and common parameters, a queried entry, and one shared sampling environment.

1. If the effective accuracy exponent is nonpositive, return zero with empty access.
2. Express the raw signal as direct, product, projection, or vector sample terms. Average consecutive-level differences at each included scale.
3. Process contributions in groups of comparable level decrease. Combine contributions at the same destination, then compress the row to a default and a controlled number of exceptions. Small-decrease groups use short fingerprints; adaptive collisions are included in the analysis.
4. Apply row functions, preparatory normalization, incoming gates, and frozen multipliers in dependency order. Return a requested value or access information by recomputation.

Convention. A numerical digit request evaluates this fixed estimator more precisely; it does not redefine the task or its access graph.

2.7 From local queries to a deterministic answer

The approximate values must be accurate often enough; their implicit tables must always have bounded access; and their evaluation must always fit in space and terminate. These are separate obligations.

Compression controls row descriptions, but later destinations can depend on earlier fingerprint collisions. A fixed-pair hashing bound is insufficient. The proof compares the actual computation with one using exact equality throughout a block of accuracy levels. This yields a candidate set independent of that block's hash seed; departures from the set are charged to lower-level errors. Estimator accuracy and comparison error are proved together.

For space, bounding one local call by $O(\log n)$ would still permit an $O(\log^2 n)$ stack. The implementation shares long-lived workspace and regenerates local data. Each suspended record is charged to a decrease in call allowance and, for arithmetic, a change in requested digit place. These charges telescope. The proof counts shared workspace, all suspended records, traversal data, and arithmetic scratch. It proves termination even on seeds giving poor estimates.

The environment has $O(\log n)$ bits. After paying for the conversion from average error to the specified start, the moment bound shows that a fixed majority of appropriate environments give accurate answers. Enumerate them deterministically and take a median by repeated counting passes. Recompute estimates as needed instead of storing their list. Preserve the multiplicity of original seed encodings so enumeration matches the analyzed probability law.

Choose hierarchy depth, query level, and output digit precision so their three errors sum to the requested tolerance. For bounded-error decision, compare the answer with $1/2$; for arbitrary acceptance probabilities, return the approximation.

2.8 Undirected connectivity and Laplacian solvers

Undirected connectivity is a special case. A sufficiently long polynomial-time random walk finds a reachable target with constant probability by the usual hitting bound [3]. Apply this algorithm to the time-layered walk and threshold its acceptance probability. Even for an undirected input, the construction works with a directed forward configuration graph and residual continuation values.

Reingold’s algorithm [30] instead improves expansion within connected components, using powering and degree reduction, then enumerates short walks. The zig-zag product of Reingold, Vadhan, and Wigderson [32] controls degree while retaining expansion; Rozenman–Vadhan derandomized squaring [34] is another related route. Here partial propagation resolves a linear system, while the auxiliary expander mixes conditional sampling environments. These are distinct uses of expansion.

The connection with Spielman–Teng [38] is preconditioning and recursive approximate computation. Their nearly-linear-time solver uses symmetric diagonally dominant structure and energy-norm approximation. Here the directed, time-layered system $\mathbf{I} - \mathbf{C}$ is analyzed using nonnegativity, forward order, conditional moments, and total live memory. Its sparse tables are not spectral preconditioners. A closer space-complexity precedent is the Laplacian solver of Murtagh, Reingold, Sidford, and Vadhan [22], which combines fast-solver ideas with small-space graph operations.

The formal proof now defines the model and sampling interface, constructs the exact reduction, builds and averages sparse tables, analyzes recursive accuracy and compression, realizes the queries within bounded total memory, and takes the final median.

3 The transition model and resource parameters

We now make the configuration-graph description precise. Two choices in this reduction will be used throughout the proof. Time is part of the vertex, so the transition matrix has no directed cycles. Also, a vertex retains the complete configuration, even after the machine halts. This gives a bounded list of predecessors as well as successors. Merging all accepting configurations into one sink would lose the predecessor bound.

The reduction gives continuation probabilities at every vertex, not only at the designated start. This stronger conclusion is needed because the later construction takes weighted copies of these probabilities.

We use machines with finite control, an endmarked read-only input, finitely many work tapes, and fresh independent fair bits. Input heads are confined between the endmarkers. All randomized running-time bounds hold on every random tape. Space counts traversed writable cells in bits, including every simultaneously live counter, numerical register and suspended call; visiting a blank work cell counts toward space. A transducer has a one-way write-only output tape. Throughout, n is the input length and logarithms have base two.

A matrix indexed by a finite set V is *strictly forward* if there is an integer time attached to every vertex and every nonzero entry goes to a strictly later time. It is *substochastic* if its entries are nonnegative and each row sums to at most one. An edge is specified by a vertex and a port from a fixed finite set. A *paired* edge operation returns the other endpoint together with its inverse port; applying the inverse operation restores the first endpoint. Absent ports are explicitly recognized.

Lemma 3.1 (Configuration reduction). *Fix a polynomial-time randomized machine using $O(\log(n + 2))$ work space. On an input of length n one can specify, uniformly in logarithmic space, a polynomial-size vertex set V_0 , a strictly forward substochastic matrix $\mathbf{S}$, a vector $\mathbf{e} \in \{0, 1\}^{V_0}$, and a start vertex x_0 with the following properties.*

1. The incoming and outgoing port degrees of $\mathbf{S}$ are bounded by constants depending only on the machine. Port validity, edge weights, and pairing are computable in logarithmic space.
2. The vector $\mathbf{p}_0 = (\mathbf{I} - \mathbf{S})^{-1}\mathbf{e}$ is well defined and belongs to $[0, 1]^{V_0}$.
3. The coordinate $\mathbf{p}_{0,x_0}$ is the acceptance probability of the machine.

In particular, an RL computation gives the promise $\mathbf{p}_{0,x_0} = 0$ or $\mathbf{p}_{0,x_0} \geq 1/2$.

Proof. Let $t(n)$ be a fixed polynomial upper bound on the running time. A vertex records a time in $\{0, \dots, t(n)\}$, the finite control, all input and work head positions, and the contents of a sufficiently large box of $O(\log(n+2))$ work cells. Take elementary machine steps with fixed dyadic coin probabilities. A transition advances the time by one. Invalid configurations or moves leaving the box have no corresponding transition. The box is chosen large enough that this never affects a computation from the specified start.

We need local access in both directions. Successor access is immediate: each elementary step has constantly many successor choices. To find predecessors, guess the old finite control, locally overwritten symbols, head movements, and coin choice, then check that the forward step gives the specified configuration. There are constantly many such guesses. Combine transitions with the same ordered pair of endpoints by adding their probabilities. Order the distinct successors and predecessors by their configuration encodings; their positions give paired ports for the resulting matrix entries. These lists have constant length, so duplicate removal, summed weights, and the inverse port are computed in logarithmic space by local enumeration. This base computation makes no recursive graph queries. All records use $O(\log(n+2))$ bits, and the read-only input can be consulted during these checks.

The simulated input head is an address stored in a vertex. It must not become an uncounted storage device for the algorithm carrying out the simulation. To ensure this, all input-head coordinates in the vertex universe are restricted to a fixed polynomial interval containing the positions reachable within $t(n)$ local steps from their specified initial positions. Configurations and predecessor candidates outside that interval are invalid. To obtain a simulated head's scanned symbol, compare its stored coordinate with the input boundaries, supply any required endmarker or off-data blank according to the source tape convention, and otherwise scan the real endmarked input to the designated position using a logarithmic-space counter. A position forbidden by the source tape convention remains invalid. The physical scan stays between the real endmarkers, even when a valid simulated coordinate is outside the data. This routine makes no recursive graph query.

To collect acceptance at one common time, let an already halted computation keep its complete tape and head data while its time advances. These are distinct vertices in successive layers, not self-loops. Keeping the full configuration prevents many unrelated halted computations from merging at one vertex. Let $\mathbf{e}$ be one precisely at accepting configurations at time $t(n)$, and zero elsewhere. Final-time rows have no outgoing transitions. Thus $\mathbf{S}$ is strictly forward and substochastic, and it is nilpotent. Its inverse expression is the finite sum

$$(\mathbf{I} - \mathbf{S})^{-1}\mathbf{e} = \sum_{j=0}^{t(n)} \mathbf{S}^j \mathbf{e}.$$

Starting from any valid configuration, at most one success contribution can be collected, at final time. The sum is therefore a probability in $[0, 1]$. At x_0 it is exactly the machine's acceptance probability. $\square$

The coordinatewise bound on $\mathbf{p}_0$ will be used after weighted projections to larger vertex sets. It holds at every configuration, independently of the acceptance probability at the designated

start. Thus the matrix construction and the probability approximation below apply to arbitrary randomized computations satisfying the resource bounds.

The parameters to remember. There are three size quantities with different roles. The input has length n ; a vertex address and the shared randomness will have length $O(b_{\text{mem}})$; and n_v is an upper bound on the number of vertices at any one stage. The stage index counts copy steps, whereas a sampling-rate index measures how many vertices a random subset is expected to retain. We specify these quantities below before introducing any of the local thresholds.

Size convention. Fix an integer parameter b_{mem} , a power of two, comparable with $\log_2(n+2)$ and at least a sufficiently large fixed constant. For example, take the least power of two at least the maximum of that constant and $\lceil \log_2(n+2) \rceil$. This choice is computable in logarithmic space and includes all bounded input lengths in the same construction. All constants throughout the construction are independent of the input; they may depend on the fixed machine and, for fixed inverse-polynomial accuracy, its fixed exponent. The uniform precision parameter of Theorem 1.2 is handled separately by padding. The number of stages is $l_{\text{max}} = O(b_{\text{mem}})$. At stage l , a vertex has a base identifier followed by l symbols from a fixed radix d_0 . The radix will be selected by the hierarchy. Encode each base vertex by a fixed-width radix- d_0 word and prepend the digit 1; unused base words are invalid. Appending one copy digit at each stage gives injective numerical identifiers even across different stage lengths. Indeed, a base width w places stage- l identifiers in $[d_0^{w+l}, 2d_0^{w+l})$, and these intervals are disjoint. Appending a fixed suffix is an affine map on identifiers.

Every stage has size at most a common bound

$$n_v = 2^{c_{\text{bd}, n_v} b_{\text{mem}}}.$$

We may increase the fixed constant c_{bd, n_v} as needed. This is a bound on one stage, not a claim that there are only n_v pairs of vertices. Integrated entry errors below carry a factor $1/n_v$ while summing over both endpoints. This is a normalization per source vertex, not per ordered pair. Later degree bounds control the additional sum over destinations; inserting a second factor of $1/n_v$ would change the error statement.

Fix integers $h \geq 4$ and $2 \leq h_0 < h$, and put

$$b_k = 2^{-hk} \quad (k \geq 0).$$

These are the nominal sampling rates. Stage l will have density $f = b_l$. The stochastic tables will use rates from this same grid for their source and terminal retention tests. The error power in the later analysis is always $s = 8$.

Uniform constants and finite ranges. All thresholds needed by the algorithm can be chosen rational, with fixed gaps, and all discrete constants can be chosen integer. We first prove estimates with constants uniform in the ambient finite universe. The sampling construction applies to every finite range of identifiers and rates once its prime field is sufficiently large. The hierarchy uses $O(b_{\text{mem}})$ copy digits and rates with $k = O(b_{\text{mem}})$; Section 14.1 will verify this rate bound also for the recursive estimator and give a compatible order of constant choices.

We can then take the prime field to have size $2^{O(b_{\text{mem}})}$, with its exponent chosen after these constants. Its elements and all identifiers occupy $O(b_{\text{mem}})$ bits. Increasing that exponent does not change the probabilistic constants proved in the next section.

4 Sampling vertex subsets with a short description

The detector needs to sample rows and columns without writing down a random bit for every vertex. We therefore describe an entire random subset by a small collection of polynomial coefficients. Given the coefficients and one vertex identifier, membership can be tested on demand.

4.1 The sampling interface

Before choosing coordinates, specify what the rest of the proof needs. A channel is a short random description of nested subsets of the vertex universe, one subset for each nominal rate. A membership query supplies a vertex and a rate, and returns one bit. The required guarantees are: correct marginal probabilities up to fixed factors, a conditional pair bound after one specified membership is imposed, a lower-tail estimate for the number of hits in a deterministic set, and invariance when a common copy suffix is appended to every identifier. Lemma 4.1 gives their exact quantifiers and constants.

Conditional averaging imposes a further requirement on this same family: it must admit rapidly mixing actions that preserve the required memberships and have short, validated descriptions. Section 8 proves this additional property for the construction below. Separating these requirements explains both why pairwise-independent hashing is a useful starting point and why an arbitrary such hash family is not yet sufficient.

4.2 Polynomial channels with independent nonconstant columns

The environment consists of the independent rank channels below together with a finite string of independent uniform auxiliary bits. The length and uses of that string will be specified when needed.

One independently chosen collection of coefficients is called a channel. We use a fixed number of independent rank channels, including channels denoted X, Y, Z . Extra channels, when needed, are also a fixed finite number. A channel is a uniformly random matrix

$$\mathbf{A} = (\mathbf{a}_0, \mathbf{F}) \in \mathbb{F}_{p_{\text{fld}}}^{h \times h_0}$$

conditioned on the last $r = h_0 - 1$ columns $\mathbf{F}$ being linearly independent. We call such an ordered collection of columns a *frame*. The integer p_{fld} is prime. It is chosen larger than every identifier and sufficiently large for all rate comparisons below. For a numerical identifier y , set

$$\mathbf{v}(y) = (1, y, \dots, y^{h_0-1})^\top, \quad \mathbf{a}(y) = \mathbf{A}\mathbf{v}(y) \in \mathbb{F}_{p_{\text{fld}}}^h.$$

Field elements in a retention test use their representatives in $\{0, \dots, p_{\text{fld}} - 1\}$. At nominal rate b_k , let

$$l_{\text{box},k} = \lceil p_{\text{fld}} 2^{-k} \rceil, \quad \pi_{b_k} = (l_{\text{box},k} / p_{\text{fld}})^h.$$

Evaluate the same polynomial array at each identifier. The identifier is retained if all coordinates of $\mathbf{a}(y)$ lie in $\{0, \dots, l_{\text{box},k} - 1\}$. Write $X_a(y)$ for the indicator on channel X at rate a , and similarly for other channels. The same notation permits a specified fixed suffix in the channel argument. Retention boxes are nested. All rates in this section belong to the grid $b_k = 2^{-hk}$, and the admissible range satisfies $p_{\text{fld}} 2^{-k} \geq 1$. The choices of h, h_0 and the identifier range are those of Section 3.

Lemma 4.1 (Rank-channel estimates). *There are positive constants $c_\pi, c_\pi^+, c_{\text{bd,pair}}, c_{\text{bd,var}}$, depending only on h, h_0 , with the following properties in every admissible finite range.*

1. *A specified identifier is retained at rate a with probability π_a , where $c_\pi a \leq \pi_a \leq c_\pi^+ a$.*
2. *For distinct identifiers x, y and any admissible rates u, a ,*

$$\Pr\{X_a(y) = 1 \mid X_u(x) = 1\} \leq c_{\text{bd,pair}} a.$$

3. *If $\mathcal{A}_0$ is a deterministic set of q distinct identifiers and $T_a = \sum_{y \in \mathcal{A}_0} X_a(y)$, then*

$$\Pr\{T_a < q\pi_a/2\} \leq \frac{c_{\text{bd,var}}}{q\pi_a} \quad (q > 0).$$

4. *Replacing every channel argument y by the same injective affine map $\alpha y + \beta$ leaves the joint law of all evaluations unchanged, provided $\alpha \neq 0$ in $\mathbb{F}_{p_{\text{fld}}}$. In particular, a common fixed padding suffix leaves their law unchanged.*

Proof. We first compare the conditioned sampling rule with unrestricted polynomial coefficients. The rank condition has probability bounded below by an absolute positive constant:

The probability that an unrestricted h -by- r matrix is a frame is

$$q_{\text{fr}} = \prod_{j=0}^{r-1} (1 - p_{\text{fld}}^{j-h}) \geq c_{\text{fr}} > 0.$$

For fixed positive-degree coefficients, $\mathbf{a}_0$ is uniform, so any one evaluation is uniform in $\mathbb{F}_{p_{\text{fld}}}^h$ and independent of frame success. This proves the exact marginal retention probability. In the ranges used, take $p_{\text{fld}}2^{-k} \geq 1$. Then $p_{\text{fld}}2^{-k} \leq l_{\text{box},k} \leq 2p_{\text{fld}}2^{-k}$, proving comparability. The prime exponent can in fact make all these sides much larger than one.

For the conditional-pair bound, full independence after rank conditioning is unnecessary. We can first use independence in the unrestricted distribution and then pay for conditioning by the reciprocal of its probability. Without conditioning on a frame, two evaluations at distinct field points are independent uniform vectors: the two evaluation functionals have rank two since $h_0 \geq 2$. Consequently

$$\Pr\{X_a(y) = 1, X_u(x) = 1 \mid \mathbf{F} \text{ is a frame}\} \leq \frac{\pi_a \pi_u}{q_{\text{fr}}}.$$

The first marginal after conditioning is still π_u , which proves the second assertion.

The count estimate uses the same comparison. For the third assertion, in the unrestricted distribution the hit count has mean $q\pi_a$ and variance at most $q\pi_a$, by pairwise independence. Chebyshev's inequality bounds the stated lower-tail event by $4/(q\pi_a)$. Conditioning on frame success increases this probability by at most $1/q_{\text{fr}}$.

For reuse after copying vertices, substituting $\alpha y + \beta$ in a polynomial acts on its positive-degree coefficient columns by an invertible triangular matrix, with diagonal $\alpha, \dots, \alpha^{h_0-1}$. It therefore preserves the frame condition. Its action on the full coefficient array is an invertible linear change of variables, so it preserves uniform measure conditional on that condition. Appending digits in radix d_0 has α a power of d_0 ; taking $p_{\text{fld}} > d_0$ ensures the required invertibility. $\square$

5 The exact reduction: propagation with controlled incoming support

This section constructs the exact target of the estimator. It is useful to keep its final guarantee in view: at each stage, the largest transition entry decreases by a factor 2^h , while the number of

vertices incident to positive entries increases by at most a factor of two. The residual transition mass then tends to zero geometrically. Propositions 5.6 and 5.8 state the one-stage guarantee and its consequence for acceptance probabilities.

There are three operations to justify. A correction transfers transition contributions into the success contribution. A detector restricts that correction in columns with too many substantial incoming entries. A copy step divides the remaining large entries while controlling how many copies become active. We begin with the identity that makes the first operation exact.

Given a transition matrix $\mathbf{C}$ and a nonnegative correction $\mathbf{E}$, put

$$\mathbf{D} = \mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}.$$

Then $\mathbf{I} - \mathbf{D} = (\mathbf{I} + \mathbf{E})(\mathbf{I} - \mathbf{C})$. Thus a relation $\mathbf{w} = (\mathbf{I} - \mathbf{C})\mathbf{p}$ becomes $(\mathbf{I} + \mathbf{E})\mathbf{w} = (\mathbf{I} - \mathbf{D})\mathbf{p}$: the same vector $\mathbf{p}$ is represented by the new transition and the updated success contribution. The correction must make $\mathbf{D}$ nonnegative and keep its row sums at most one. It must also reduce entries sufficiently fast that, after a controlled increase in the number of active vertices, the remaining transition contribution becomes small. The construction uses only the model reduction, with no one-sided acceptance promise.

At one stage the input is a finite ordered set V , a rate $0 < f \leq 1$ on the grid of Section 4, and a nonnegative matrix $\mathbf{C}$ with

$$\mathbf{C}_{xy} = 0 \quad \text{unless } x < y, \quad \mathbf{C}\mathbf{1} \leq \mathbf{1}, \quad \mathbf{C}_{xy} \leq f. \quad (5.1)$$

All matrix and vector inequalities are entrywise. A scalar cutoff applied to a matrix acts entrywise. Multiplying such an array by a column multiplier means scaling column y by that multiplier at y ; all transition products, such as $\mathbf{E}\mathbf{C}$, are ordinary matrix products. Write

$$\mathcal{R}(\mathbf{C}) = \{x : \text{some } \mathbf{C}_{xy} > 0\}, \quad \mathcal{T}(\mathbf{C}) = \{y : \text{some } \mathbf{C}_{xy} > 0\}, \quad \mathcal{A}(\mathbf{C}) = \mathcal{R}(\mathbf{C}) \cup \mathcal{T}(\mathbf{C}).$$

The stage will reduce the entry bound from f to $f/2^h$ and increase $|\mathcal{A}(\mathbf{C})|$ by at most a factor of two. We first analyze the correction with an arbitrary fixed column multiplier. We then choose that multiplier by a finite detector calculation, and use copies to divide the entries in columns where correction is not fully enabled.

5.1 Correction with a fixed column multiplier

The correction will remove most of a sufficiently large entry, but it must leave a small positive remainder. That remainder has two uses. It ensures nonnegativity of the new transition matrix, and its positive size bounds the number of corrected positions in a row. The cutoff $\mathbf{Q}$ expresses this rule continuously, so that replacing an input by a nearby estimate has a controlled effect. The gap between its two thresholds leaves room for later errors.

Put $q = 2^h$. Fix rational numbers $0 < \theta_1 < \theta_2 < q^{-1}/32$ and a nondecreasing Lipschitz function $\mathbf{Q} : [0, \infty) \rightarrow [0, \infty)$ satisfying

$$(t - \theta_2)_+ \leq \mathbf{Q}(t) \leq (t - \theta_1)_+. \quad (5.2)$$

Its eighth root is required to be Lipschitz on bounded intervals. This extra regularity is used when errors in support decisions are bounded by eighth moments. It is a choice we can build into the cutoff, rather than an additional hypothesis on the input matrix. Here and below the scalar cutoffs can be chosen piecewise polynomial with rational coefficients. To see this explicitly, let β be the normalized integral of $t^7(1-t)^7$ on $[0, 1]$, extended by zero to the left and one to the right. Both

$\beta^{1/8}$ and $(1 - \beta)^{1/8}$ are Lipschitz: near their vanishing endpoints the corresponding functions vanish to order eight, and away from those endpoints the assertion follows by differentiation. For example,

$$Q(t) = (t - \theta_1)_+^\beta \left(\frac{t - \theta_1}{\theta_2 - \theta_1} \right)$$

has all the stated properties. Near θ_1 its eighth root is of order $(t - \theta_1)_+^{9/8}$; elsewhere its root has bounded derivative on every bounded interval. Let l_Q be a fixed Lipschitz constant for Q . For real arguments, extend Q by $Q(t) = 0$ when $t < 0$. This extension preserves its Lipschitz and root-Lipschitz properties and permits evaluation on real approximations. The exact hierarchy itself uses only nonnegative arguments.

For the moment, suppose we have already decided how much correction each column permits. Write this decision as a fixed multiplier $\mathbf{g} : V \rightarrow [0, 1]$. A value of one enables the full correction, and a value of zero disables it. Starting from zero, define

$$\mathbf{E}_0 = 0, \quad \mathbf{G}'_i = \mathbf{C} + \mathbf{E}_{i-1}\mathbf{C}, \quad \mathbf{E}_{i,xy} = fQ(\mathbf{G}'_{i,xy}/f)\mathbf{g}_y \quad (i \geq 1). \quad (5.3)$$

The cutoff leaves a positive residual at each corrected entry. Keeping $\mathbf{g}$ fixed makes the update map monotone. We next show that a fixed number of iterations suffices for every such multiplier, independently of the size of V . At a nonnegative fixed point $\mathbf{E}_*$ of this recurrence, put $\mathbf{T} = \mathbf{C} - \mathbf{E}_* + \mathbf{E}_*\mathbf{C}$. In a column with $\mathbf{g}_y = 1$, the lower bound on Q gives $\mathbf{T}_{xy} \leq \theta_2 f$. The truncation estimate below will transfer this entry bound to a finite correction, with a small additional error. The fixed point is only a comparison object; the hierarchy will use a finite iteration.

What the fixed-multiplier calculation will establish. The remainder left by Q limits how many positions a single row can correct. Forward order then turns iteration error into an ordered-path count, giving a factorial tail rather than a contraction assumption. We postpone that proof until after the pilot and actual constructions are fully specified.

5.2 Choosing which columns permit correction

The fixed-multiplier argument did not explain how to choose $\mathbf{g}$. We need correction to be enabled only in columns with a controlled number of substantial entries. A preliminary sequence of corrections, called the pilot, estimates how crowded each column is. After that sequence, its final choice of $\mathbf{g}$ is frozen and the actual correction starts again from zero. The two sequences serve different purposes: the pilot chooses the multiplier; the actual sequence is the monotone recurrence analyzed above.

At this stage these are exact mathematical definitions. Every detector is a true expectation of an already defined input. Computing an approximation to the detector comes later, so an approximate value never defines its own exact target.

For the detectors we use independent channels X, Y from Section 4. Denote their common retention probability at rate f by π_f . The consequences of Lemma 4.1 needed here can be written, with fixed positive constants, as

$$c_\pi f \leq \pi_f \leq c_\pi^+ f, \quad \Pr\{Y_f(w) = 1 \mid Y_f(y) = 1\} \leq c_{\text{pair}} f \quad (w \neq y), \quad (5.4)$$

$$\Pr\left\{\sum_{x \in \mathcal{A}} X_f(x) < \frac{1}{2}|\mathcal{A}|\pi_f\right\} \leq \frac{c_{\text{var}}}{|\mathcal{A}|\pi_f} \quad (\emptyset \neq \mathcal{A} \subseteq V \text{ deterministic}). \quad (5.5)$$

These constants are uniform in the stage, the admissible moduli, and common fixed padding suffixes. No independence between different rows of a detector trial will be assumed.

All threshold choices and ranks below are fixed constants. The pilot uses the iteration count from Lemma 5.2. Choose

$$0 < \delta_0 < \delta_1 < \theta_1/100, \quad 0 < \eta_- < \eta_+ < \delta_0/3.$$

If other row scores require still smaller detector thresholds, make those choices before fixing the ranks. Let ϕ be a nondecreasing cutoff which is zero through δ_0 and one from δ_1 on. Let ρ be a nonincreasing cutoff which is one through η_- and zero from η_+ on. Finally, let $\gamma : [0, 1] \rightarrow [0, 1]$ be nonincreasing, one on $[0, 1/2]$, and zero on $[4/5, 1]$. These functions and their complements, when used, have the root regularity just described.

A detector looks for many retained rows that contribute substantially to the same column. It must first exclude rows with too many sampled entries, because those rows will not support a bounded-access representation. The row gate $\mathbf{r}_i^{\text{gate}}$ performs this exclusion; the j_{det} -th-largest score then tests whether enough eligible rows remain.

For a finite multiset of nonnegative numbers, rank_j means its j -th largest member, with enough additional zeros that this is always defined. Starting from $\hat{\mathbf{E}}_0 = 0$, perform k_* pilot steps. At step i , first set

$$\mathbf{G}_i = \mathbf{C} + \hat{\mathbf{E}}_{i-1} \mathbf{C}. \quad (5.6)$$

For a draw of the terminal channel, set

$$\mathbf{r}_{i,x}^{\text{gate}} = \rho(\text{rank}_{k_{\text{row}}} \{\mathbf{G}_{i,xw}/f : Y_f(w) = 1\}). \quad (5.7)$$

Define the deterministic column number

$$\mathbf{u}_{i,y} = \mathbb{E} \left[\text{rank}_{j_{\text{det}}} \{\phi(\mathbf{G}_{i,xy}/f) \mathbf{r}_{i,x}^{\text{gate}} : X_f(x) = 1\} \mid Y_f(y) = 1 \right]. \quad (5.8)$$

Then put

$$\hat{\mathbf{g}}_{i,y} = \gamma \left(\max_{1 \leq r \leq i} \mathbf{u}_{r,y} \right), \quad \hat{\mathbf{E}}_{i,xy} = fQ(\mathbf{G}_{i,xy}/f) \hat{\mathbf{g}}_{i,y}. \quad (5.9)$$

The order of these definitions is essential: $\mathbf{G}_i$ is already deterministic when its detector is formed, and $\mathbf{u}_i$ is defined before $\hat{\mathbf{E}}_i$. The running maximum makes $\hat{\mathbf{g}}_i$ nonincreasing. Its final value is therefore no larger than any earlier pilot multiplier; monotonicity of Q will let us bound the actual corrections by the pilot corrections.

After the pilot, set $\mathbf{g} = \hat{\mathbf{g}}_{k_*}$ and specialize the fixed-multiplier recurrence of Equation (5.3) to the actual correction

$$\mathbf{E}_0 = 0, \quad \mathbf{G}'_i = \mathbf{C} + \mathbf{E}_{i-1} \mathbf{C}, \quad \mathbf{E}_{i,xy} = fQ(\mathbf{G}'_{i,xy}/f) \mathbf{g}_y \quad (1 \leq i \leq k_*). \quad (5.10)$$

Write

$$\mathbf{E} = \mathbf{E}_{k_*}, \quad \mathbf{D} = \mathbf{C} - \mathbf{E} + \mathbf{E} \mathbf{C}. \quad (5.11)$$

Algorithm 5.1 Pilot, frozen correction, and copying

Input. The stage data in (5.1), its success vector $\mathbf{w}$, and the fixed cutoffs and ranks specified here.

1. Set $\hat{\mathbf{E}}_0 = 0$. For $i = 1, \dots, k_*$, compute $\mathbf{G}_i$, the row gate $\mathbf{r}_i^{\text{gate}}$, and the exact detector vector $\mathbf{u}_i$ by (5.6)–(5.8). Update $\hat{\mathbf{g}}_i, \hat{\mathbf{E}}_i$ by (5.9).
2. Freeze $\mathbf{g} = \hat{\mathbf{g}}_{k_*}$. Restart $\mathbf{E}_0 = 0$, then perform the k_* updates in (5.10). Set $\mathbf{E} = \mathbf{E}_{k_*}$.

Algorithm 5.1 Pilot, frozen correction, and copying (continued)

3. Form $\mathbf{D} = \mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}$ and $\tilde{\mathbf{w}} = \mathbf{w} + \mathbf{E}\mathbf{w}$.
4. Use the routing fractions and row factors in (5.21)–(5.22). Return

$$\mathbf{C}_{x_i, y_j}^+ = \mathbf{r}_{i,x} \mathbf{D}_{xy} s_j(\mathbf{D}_{xy}/f), \quad \mathbf{w}_{x_i}^+ = \mathbf{r}_{i,x} \tilde{\mathbf{w}}_x.$$

Access convention. This algorithm defines exact mathematical targets. The later estimator evaluates its entries and expectations recursively; it does not store these arrays.

We now analyze that construction. The fixed-multiplier lemma applies to every possible final pilot choice, so its constants can be fixed before the detector ranks. The pilot estimates then justify which columns retain positive multipliers, and the copying proof converts that information into an active-vertex bound.

Lemma 5.2 (Fixed point and factorial truncation). *For every fixed column multiplier $0 \leq \mathbf{g} \leq 1$, there is a unique strictly forward nonnegative solution of*

$$\mathbf{E}_{*,xy} = f\mathbf{Q}((\mathbf{C} + \mathbf{E}_*\mathbf{C})_{xy}/f)\mathbf{g}_y. \quad (5.12)$$

For the recurrence in Equation (5.3), $0 \leq \mathbf{E}_i \leq \mathbf{E}_*$, and for every $k \geq 0$,

$$\max_{x,y} \frac{\mathbf{E}_{*,xy} - \mathbf{E}_{k,xy}}{f} \leq e^{1/\theta_1} \frac{(l_{\mathbf{Q}}/\theta_1)^k}{k!}. \quad (5.13)$$

In particular k_* can be chosen as a fixed integer so that the right side of Equation (5.13) at $k = k_*$ is at most $q^{-1}/32$, uniformly in $\mathbf{C}, f, V$, and $\mathbf{g}$.

Proof. We prove the lemma in three steps: construct the comparison fixed point, bound the number of corrected positions in one row, and count the increasing paths along which a truncation error can propagate.

Order the vertices compatibly with the strictly forward relation. For a fixed row x , the entry $(\mathbf{E}_*\mathbf{C})_{xy}$ involves only $\mathbf{E}_{*,xz}$ with $z < y$. Equation (5.12) therefore defines the row successively, in increasing column order. Columns at or before x receive zero. This proves existence and uniqueness in the stated class, without a contraction hypothesis. Monotonicity of the update map proves $\mathbf{E}_i \leq \mathbf{E}_*$ and the nonnegativity of the differences.

To bound the support of a fixed row, use the positive remainder deliberately left by $\mathbf{Q}$. Set $\mathbf{T} = \mathbf{C} - \mathbf{E}_* + \mathbf{E}_*\mathbf{C}$. If $a = (\mathbf{C} + \mathbf{E}_*\mathbf{C})_{xy}$, then Equation (5.2) implies

$$0 \leq \mathbf{E}_{*,xy} \leq (a - \theta_1 f)_+.$$

Thus $\mathbf{T} \geq 0$, and at every positive position of $\mathbf{E}_*$ one has $\mathbf{T}_{xy} \geq \theta_1 f$. Also

$$\mathbf{T}\mathbf{1} = \mathbf{C}\mathbf{1} - \mathbf{E}_*(\mathbf{1} - \mathbf{C}\mathbf{1}) \leq \mathbf{C}\mathbf{1} \leq \mathbf{1}. \quad (5.14)$$

Consequently, for each fixed row x , its set $\mathcal{A}_x = \{y : \mathbf{E}_{*,xy} > 0\}$ has cardinality $m_x \leq 1/(\theta_1 f)$.

We next bound an entry of the fixed point by expanding its recurrence. The useful count is the size of the corrected support of this one row, not the total size of the graph. Unroll the inequality $\mathbf{E}_* \leq \mathbf{C} + \mathbf{E}_*\mathbf{C}$ along this row's support. Each resulting term for $\mathbf{E}_{*,xy}$ is a product of entries of $\mathbf{C}$ along a sequence

$$x < z_1 < \cdots < z_j < y, \quad z_1, \dots, z_j \in \mathcal{A}_x.$$

Every substitution keeps the source coordinate equal to x : the recursive factor is $\mathbf{E}_{*,xz}$, while only its terminal coordinate changes. All intermediate vertices therefore belong to the same set $\mathcal{A}_x$. Strict ordering makes a choice of intermediate vertices specify at most one such sequence. There are at most $\binom{m_x}{j}$ such sequences, and each product is at most f^{j+1} . The expansion terminates because every sequence is strictly increasing. Therefore

$$\mathbf{E}_{*,xy} \leq f \sum_{j=0}^{m_x} \binom{m_x}{j} f^j = f(1+f)^{m_x} \leq f e^{1/\theta_1}. \quad (5.15)$$

Only the support of the single fixed row x was used here; the number of vertices in the whole graph does not enter the exponent.

The same path count now controls the error after finitely many iterations. Each iteration forces an error contribution to traverse one more strictly increasing step; choosing those steps produces the factorial in the denominator. Let $\Delta_i = \mathbf{E}_* - \mathbf{E}_i$. These matrices are nonnegative, and their row- x supports are contained in $\mathcal{A}_x$. The Lipschitz bound gives

$$\Delta_{i,xy} \leq l_Q \sum_{z < y} \Delta_{i-1,xz} \mathbf{C}_{zy}.$$

Outside $\mathcal{A}_x$ the left side is zero, so when this inequality is unrolled it suffices to use intermediate vertices from $\mathcal{A}_x$ at every step. After k steps there are at most $\binom{m_x}{k}$ strictly increasing sequences of intermediate vertices. Their initial differences are at most $f e^{1/\theta_1}$ by Equation (5.15), and their k transition factors are at most f^k . Thus

$$\Delta_{k,xy} \leq f e^{1/\theta_1} l_Q^k f^k \binom{m_x}{k} \leq f e^{1/\theta_1} \frac{(l_Q/\theta_1)^k}{k!}.$$

For $k > m_x$ the first upper bound is zero. This proves Equation (5.13). The factorial eventually dominates the fixed exponential factor, giving the stated fixed choice of k_* . $\square$

The iteration count k_* is now fixed independently of the detectors. We next choose $\mathbf{g}$ so that correction is supported on columns with only boundedly many substantial entries. A pilot selects this multiplier; the actual recurrence then uses that one multiplier for all k_* steps.

Lemma 5.3 (Elementary pilot bounds). *All pilot and actual inputs and corrections are nonnegative and strictly forward. For $1 \leq i \leq k_*$,*

$$\begin{aligned} \mathbf{G}_i \mathbf{1}, \hat{\mathbf{E}}_i \mathbf{1}, \mathbf{G}'_i \mathbf{1}, \mathbf{E}_i \mathbf{1} &\leq i \mathbf{1}, \\ \mathbf{G}_{i,xy}, \hat{\mathbf{E}}_{i,xy}, \mathbf{G}'_{i,xy}, \mathbf{E}_{i,xy} &\leq i f. \end{aligned}$$

Moreover,

$$\mathbf{E}_{i-1} \leq \mathbf{E}_i \leq \hat{\mathbf{E}}_i, \quad \mathbf{G}'_i \leq \mathbf{G}_i.$$

The row and column supports of every matrix in this statement are contained in $\mathcal{R}(\mathbf{C})$ and $\mathcal{T}(\mathbf{C})$, respectively.

Proof. Equation (5.2) gives $\mathbf{Q}(0) = 0$ and $0 \leq \mathbf{Q}(t) \leq t$ for $t \geq 0$. Consequently $\hat{\mathbf{E}}_i \leq \mathbf{G}_i$. If $\hat{\mathbf{E}}_{i-1} \mathbf{1} \leq (i-1) \mathbf{1}$, then

$$\mathbf{G}_i \mathbf{1} = \mathbf{C} \mathbf{1} + \hat{\mathbf{E}}_{i-1} \mathbf{C} \mathbf{1} \leq \mathbf{1} + \hat{\mathbf{E}}_{i-1} \mathbf{1} \leq i \mathbf{1}.$$

For an individual entry,

$$\mathbf{G}_{i,xy} \leq f + f \sum_z \hat{\mathbf{E}}_{i-1,xz} \leq i f.$$

This proves the pilot bounds by induction.

The sequence $\widehat{\mathbf{g}}_i$ is nonincreasing, so $\mathbf{g} \leq \widehat{\mathbf{g}}_i$ for every i . Induction in Equations (5.6)–(5.10), using monotonicity of $\mathbf{Q}$, proves $\mathbf{E}_i \leq \widehat{\mathbf{E}}_i$ and $\mathbf{G}'_i \leq \mathbf{G}_i$. The map $\mathbf{F} \mapsto f\mathbf{Q}((\mathbf{C} + \mathbf{F}\mathbf{C})/f)\mathbf{g}$, applied entrywise after forming the matrix product, is monotone. Its iteration from zero is therefore nondecreasing. The actual bounds follow from the pilot bounds.

A product of strictly forward matrices is strictly forward. A zero row of $\mathbf{C}$ remains a zero row in all these formulas, by induction from the zero correction. A zero column of $\mathbf{C}$ is a zero column of every product of the form $\mathbf{F}\mathbf{C}$ and hence of every input and correction. This proves the remaining assertions. $\square$

The next lemma supplies both directions of the count comparison. A sufficiently crowded column is detected with large score. Conversely, a positive score above the stated threshold certifies many substantial entries. The first implication limits correction support; the second later limits the number of extra copies.

Lemma 5.4 (The detector bounds overloaded columns). *Let*

$$d_{\text{row}} = 1 + c_{\text{pair}}k_*/\eta_-.$$

Choose the integer $k_{\text{row}} \geq 160d_{\text{row}}$, increasing it if necessary for the other finitely many row-gate estimates. For any fixed positive integer j_{det} , choose

$$r_* \geq c_\pi^{-1} \max(4j_{\text{det}}, 40c_{\text{var}}). \quad (5.16)$$

Then the pilot has the following properties.

1. *If $\#\{x : \mathbf{G}_{i,xy} \geq \delta_1 f\} > r_*/f$, then $\mathbf{u}_{i,y} \geq 19/20 > 9/10$.*
2. *If $\mathbf{u}_{i,y} > 1/4$, then*

$$\#\{x : \mathbf{G}_{i,xy} > \delta_0 f\} > \frac{j_{\text{det}}}{4c_\pi^+ f}.$$

3. *For any pilot correction with multiplier $\widehat{\mathbf{g}}_{i,y} > 0$,*

$$\#\{x : \mathbf{G}_{i,xy} \geq \delta_1 f\} \leq r_*/f.$$

For every actual correction, $\mathbf{g}_y > 0$ implies

$$\#\{x : \mathbf{G}'_{i,xy} \geq \delta_1 f\} \leq r_*/f.$$

Proof. First bound how often the row gate removes a particular row. This estimate is conditional on retaining the queried column, as the detector requires. Fix i, y and condition on $Y_f(y) = 1$. In a fixed row, the set $\{w : \mathbf{G}_{i,xw} > \eta_- f\}$ has cardinality at most $k_*/(\eta_- f)$ by Lemma 5.3. Its expected number of retained members is at most d_{row} : the specified terminal y , if in this set, contributes at most one, and Equation (5.4) applies to every other member. If the row gate is not full, at least k_{row} such members have been retained. Markov's inequality therefore gives

$$\Pr\{\mathbf{r}_{i,x}^{\text{gate}} < 1 \mid Y_f(y) = 1\} \leq d_{\text{row}}/k_{\text{row}}. \quad (5.17)$$

For a crowded column, compare the number of sampled substantial rows with the number lost to row gates. It is enough to bound the expected number lost; the losses need not be independent. Let $\mathcal{A} = \{x : \mathbf{G}_{i,xy} \geq \delta_1 f\}$ and $n = |\mathcal{A}|$. Let $N_{\mathcal{A}}$ count its retained starts, and let $L_{\mathcal{A}}$ count those

retained starts whose row gate is not full. Independence of the two channels and Equation (5.17) imply

$$\mathbb{E}[L_{\mathcal{A}} \mid Y_f(y) = 1] \leq n\pi_f d_{\text{row}}/k_{\text{row}}.$$

Thus

$$\Pr\{L_{\mathcal{A}} > n\pi_f/4 \mid Y_f(y) = 1\} \leq 1/40.$$

The law of the start channel is unchanged by the conditioning, so Equation (5.5) gives

$$\Pr\{N_{\mathcal{A}} < n\pi_f/2 \mid Y_f(y) = 1\} \leq c_{\text{var}}/(n\pi_f).$$

If $n > r_*/f$, Equation (5.16) ensures $n\pi_f \geq \max(4j_{\text{det}}, 40c_{\text{var}})$. Outside an event of probability at most $1/20$, there are at least j_{det} retained high rows with full gates. Each has score exactly one, so the trial's j_{det} -th largest score is one. All trial scores lie in $[0, 1]$, proving the first assertion. This argument used no independence among the row-gate failures.

For the converse, a positive j_{det} -th-largest score requires at least j_{det} sampled eligible starts. Counting those starts bounds the mean score from above. For the second assertion let $\mathcal{A}_0 = \{x : \mathbf{G}_{i,xy} > \delta_0 f\}$, and let N_0 be its number of retained starts. A positive score must come from $\mathcal{A}_0$. Pointwise the trial is at most $\mathbf{1}_{\{N_0 \geq j_{\text{det}}\}}$, and hence at most N_0/j_{det} . Its conditional expectation is at most $|\mathcal{A}_0|\pi_f/j_{\text{det}} \leq |\mathcal{A}_0|c_{\pi}^+ f/j_{\text{det}}$. The asserted lower bound on $|\mathcal{A}_0|$ follows.

Finally, a positive $\widehat{\mathbf{g}}_{i,y}$ implies $\mathbf{u}_{i,y} < 4/5$, so the first assertion excludes a high-entry count greater than r_*/f . If $\mathbf{g}_y > 0$, all pilot multipliers at y are positive. Apply the same bound to every $\mathbf{G}_i$ and then use $\mathbf{G}'_i \leq \mathbf{G}_i$ from Lemma 5.3. $\square$

5.3 The finite residual

The actual construction uses the finite correction $\mathbf{E} = \mathbf{E}_{k_*}$, not the comparison fixed point. We therefore check that stopping the iteration preserves nonnegativity and bounded row sums, and that the error paid for stopping is small enough in fully corrected columns.

There is a further requirement for estimation. The residual is the signed expression $\mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}$, so its constituent terms can be larger than the final answer. We need a bound on each constituent relative to the nonnegative residual. Without it, a small relative error in a constituent could become a large error after cancellation. The domination estimate below supplies exactly this bound.

Lemma 5.5 (Residual bounds and unsigned-term domination). *For the k_* chosen in Lemma 5.2, put*

$$k_{\text{entry}} = k_* + 1, \quad \varepsilon_{\text{good}} = \theta_2 + q^{-1}/32 < q^{-1}/16, \quad c_{\text{dom}} = \theta_1/(2k_*).$$

The matrix $\mathbf{D}$ in Equation (5.11) is nonnegative, strictly forward, and substochastic. Its entries satisfy

$$\mathbf{D}_{xy} \leq k_{\text{entry}} f, \quad \mathbf{g}_y = 1 \implies \mathbf{D}_{xy} \leq \varepsilon_{\text{good}} f. \quad (5.18)$$

Furthermore,

$$\mathbf{D} \geq c_{\text{dom}}(\mathbf{C} + \mathbf{E} + \mathbf{E}\mathbf{C}). \quad (5.19)$$

Thus every unsigned term in $\mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}$ is bounded by a fixed multiple of $\mathbf{D}$. In the formulas $\mathbf{G}_i = \mathbf{C} + \widehat{\mathbf{E}}_{i-1}\mathbf{C}$ and $\mathbf{G}'_i = \mathbf{C} + \mathbf{E}_{i-1}\mathbf{C}$, each unsigned term is bounded by the corresponding input itself. Moreover, $\mathcal{R}(\mathbf{D}) \subseteq \mathcal{R}(\mathbf{C})$ and $\mathcal{T}(\mathbf{D}) \subseteq \mathcal{T}(\mathbf{C})$.

Proof. The last correction was computed from $\mathbf{G}'_{k_*}$, whereas the residual uses $\mathbf{E}_{k_*}$ in its product with $\mathbf{C}$. Separate their difference first. Set $\mathbf{A} = \mathbf{G}'_{k_*}$ and $\mathbf{\Delta}_{\text{inc}} = (\mathbf{E} - \mathbf{E}_{k_*-1})\mathbf{C} \geq 0$. Then

$$\mathbf{D} = \mathbf{A} - \mathbf{E} + \mathbf{\Delta}_{\text{inc}}, \quad \mathbf{A} - \mathbf{E} \geq \min(\mathbf{A}, \theta_1 f). \quad (5.20)$$

Here and in the minimum the scalar $\theta_1 f$ denotes the constant entrywise bound. The second inequality follows from $\mathbf{E} \leq (\mathbf{A} - \theta_1 f)_+$. It proves $\mathbf{D} \geq 0$. Since $\mathbf{C}\mathbf{1} \leq \mathbf{1}$ and $\mathbf{E} \geq 0$,

$$\mathbf{D}\mathbf{1} = \mathbf{C}\mathbf{1} - \mathbf{E}(\mathbf{1} - \mathbf{C}\mathbf{1}) \leq \mathbf{C}\mathbf{1} \leq \mathbf{1}.$$

Also $\mathbf{D} \leq \mathbf{C} + \mathbf{E}\mathbf{C}$, whose entries are at most $f + f \sum_z \mathbf{E}_{xz} \leq (k_* + 1)f$ by Lemma 5.3. Strict forwardness and the support claims follow from the same formulas.

Because $\mathbf{A}_{xy} \leq k_* f$, Equation (5.20) gives $\mathbf{A} - \mathbf{E} \geq (\theta_1/k_*)\mathbf{A}$. On the other hand,

$$\mathbf{C} + \mathbf{E} + \mathbf{E}\mathbf{C} = \mathbf{A} + \mathbf{E} + \mathbf{\Delta}_{\text{inc}} \leq 2\mathbf{A} + \mathbf{\Delta}_{\text{inc}}.$$

Since $\theta_1/k_* < 1$, these two inequalities prove Equation (5.19). The remaining unsigned-term assertions are immediate from nonnegativity of the summands in the two input formulas.

For the good-column bound use the fixed point with this same multiplier $\mathbf{g}$, and set $\mathbf{\Delta} = \mathbf{E}_* - \mathbf{E}$. With $\mathbf{T} = \mathbf{C} - \mathbf{E}_* + \mathbf{E}_*\mathbf{C}$,

$$\mathbf{D} = \mathbf{T} + \mathbf{\Delta} - \mathbf{\Delta}\mathbf{C} \leq \mathbf{T} + \mathbf{\Delta}.$$

If $\mathbf{g}_y = 1$, the lower bound in Equation (5.2) gives $\mathbf{T}_{xy} \leq \theta_2 f$. Equation (5.13) gives $\mathbf{\Delta}_{xy} \leq q^{-1}f/32$. This proves Equation (5.18). $\square$

The domination in Equation (5.19) is also an estimation bound. It lets us compare the three unsigned contributions $\mathbf{C}$, $\mathbf{E}$, and $\mathbf{E}\mathbf{C}$ separately against their estimates, measuring each error relative to the same nonnegative target $\mathbf{D}$. Their signs are then combined by a triangle inequality in Lemma 9.3. The estimated raw arrays may have negative entries; positivity here is a property of the exact target.

5.4 Divide the remaining entries between copies

The residual already has small entries in columns with $\mathbf{g}_y = 1$. We distribute the remaining large entries among copies of their destinations. The detector incidence bound will limit the number of parents with active extra copies, and a weighted projection will relate the new transition to the residual.

A primary copy is always present. An extra copy receives transition mass only when an entry is still too large, and its outgoing row is enabled only when the detector permits it. The routing fractions below specify how the incoming mass is divided; the row factors specify which copies can continue the computation. Their compatibility is what makes the projection identity exact.

Choose an integer $d_0 > k_{\text{entry}}q$, and let $\tau_0 = q^{-1}/8$, $\tau_1 = q^{-1}/4$. Choose a rising cutoff χ which is zero through τ_0 and one from τ_1 on. For $0 \leq j < d_0$ set

$$s_0(t) = 1 - (1 - d_0^{-1})\chi(t), \quad s_j(t) = d_0^{-1}\chi(t) \quad (j > 0). \quad (5.21)$$

They are nonnegative, sum to one, and have Lipschitz eighth roots on bounded ranges. Define $\mathbf{u}_x^{\max} = \max_{1 \leq i \leq k_*} \mathbf{u}_{i,x}$. Let κ be a rising cutoff, zero through $1/4$ and one from $1/2$ on, and put

$$\mathbf{r}_{0,x} = 1, \quad \mathbf{r}_{j,x} = \kappa(\mathbf{u}_x^{\max}) \quad (j > 0). \quad (5.22)$$

On $V^+ = V \times \{0, \dots, d_0 - 1\}$ write $x_i = (x, i)$ and set

$$\mathbf{C}_{x_i, y_j}^+ = \mathbf{r}_{i,x} \mathbf{D}_{xy} s_j(\mathbf{D}_{xy}/f). \quad (5.23)$$

Order the copies first by their parent and then by their copy symbol.

Proposition 5.6 (One-stage density and active-count reduction). *Choose the detector rank j_{det} large enough that*

$$j_{\text{det}} \geq \frac{2c_{\pi}^+(d_0 - 1)k_*(k_* + 1)}{\delta_0}, \quad (5.24)$$

and then choose r_ as in Equation (5.16). The matrix $\mathbf{C}^+$ is strictly forward and substochastic, and*

$$\mathbf{C}_{x_i, y_j}^+ \leq f/q, \quad |\mathcal{A}(\mathbf{C}^+)| \leq 2|\mathcal{A}(\mathbf{C})|.$$

If $\mathbf{P}^r$ is the matrix from V^+ to V defined by

$$\mathbf{P}_{x_i x}^r = \mathbf{r}_{i, x}, \quad \mathbf{P}_{x_i z}^r = 0 \quad (z \neq x), \quad (5.25)$$

then

$$\mathbf{C}^+ \mathbf{P}^r = \mathbf{P}^r \mathbf{D}. \quad (5.26)$$

Proof. We check the entry and row-sum bounds first, then prove that projection recovers the old transition, and finally count the active copies.

The row sum at x_i is $\mathbf{r}_{i, x} \sum_y \mathbf{D}_{xy} \leq 1$. Strict forwardness follows from that of $\mathbf{D}$. Put $t = \mathbf{D}_{xy}/f$. If $t < \tau_1$, then every routed entry is at most $ft < f/q$. If $t \geq \tau_1$, all fractions equal $1/d_0$, so every entry is at most $k_{\text{entry}}f/d_0 < f/q$. This proves the density bound.

If a nonprimary fraction $s_j(\mathbf{D}_{xy}/f)$ is positive on a positive entry, then $\mathbf{D}_{xy}/f > \tau_0 > \varepsilon_{\text{good}}$. Lemma 5.5 therefore implies $\mathbf{g}_y < 1$. Since γ equals one through $1/2$, this gives $\mathbf{u}_y^{\max} > 1/2$, and hence $\mathbf{r}_{j, y} = 1$. The primary factor is always one. Thus for every positive entry of $\mathbf{D}$,

$$\sum_{j=0}^{d_0-1} s_j(\mathbf{D}_{xy}/f) \mathbf{r}_{j, y} = 1. \quad (5.27)$$

For a zero entry no identity of its fractions is needed. Multiplying Equation (5.23) by $\mathbf{P}^r$ and using Equation (5.27) proves Equation (5.26) entrywise.

The formal vertex set grew by the factor d_0 , but most extra vertices can be inactive. We count parents of active extra copies by charging each such parent to many substantial pilot entries. Row-mass bounds then limit the total number of those incidences. Put

$$\mathcal{O} = \{y : \mathbf{u}_y^{\max} > 1/4\}, \quad n' = |\mathcal{A}(\mathbf{C})|.$$

Every $y \in \mathcal{O}$ has, for some pilot step i , more than $j_{\text{det}}/(4c_{\pi}^+f)$ entries with $\mathbf{G}_{i, xy} > \delta_0 f$, by Lemma 5.4. Counting these incidences over all steps and all active rows, and using Lemma 5.3, gives

$$\begin{aligned} |\mathcal{O}| \frac{j_{\text{det}}}{4c_{\pi}^+f} &\leq \sum_{i=1}^{k_*} \#\{(x, y) : \mathbf{G}_{i, xy} > \delta_0 f\} \\ &\leq \frac{n'}{\delta_0 f} \sum_{i=1}^{k_*} i = \frac{n'k_*(k_* + 1)}{2\delta_0 f}. \end{aligned}$$

The assertion is also valid when $n' = 0$, since then all matrices and detectors vanish. Thus

$$|\mathcal{O}| \leq \frac{2c_{\pi}^+ n' k_*(k_* + 1)}{\delta_0 j_{\text{det}}}.$$

Every active child has an active parent, by the support conclusion of Lemma 5.5. A nonprimary child active as a row has $\mathbf{r}_{i, x} > 0$, so its parent belongs to $\mathcal{O}$. A nonprimary child active as a column

has a positive extra routing fraction and, as shown above, its parent has $\mathbf{u}^{\max} > 1/2$, so again belongs to $\mathcal{O}$. Therefore

$$|\mathcal{A}(\mathbf{C}^+)| \leq n' + (d_0 - 1)|\mathcal{O}| \leq 2n',$$

where the last inequality is Equation (5.24). $\square$

Remark 5.7 (Order of the one-stage constants). The bounds used to select k_* in Lemma 5.2 hold for every multiplier in $[0, 1]$. Thus $\theta_1, \theta_2, \mathbf{Q}, k_*, k_{\text{entry}}$ and d_0 can be fixed before any detector rank. Next fix all finitely many correction, detector, and access-score threshold ladders, taking δ_0, δ_1 below the correction support thresholds as required. Choose k_{row} large enough for the corresponding row estimates. Choose j_{det} by Equation (5.24), and then r_* by Equation (5.16). An incoming rank used later can be chosen after r_* : its full-gate probability is controlled by the column bound in Lemma 5.4, independently of the auxiliary row gates. There is therefore no requirement to enlarge k_{row} again in response to that incoming rank. Subsequent estimator constants may depend on these already fixed ranks and thresholds.

5.5 Iterate the construction and bound the remaining error

The copy step controls transition entries and active vertices. We now iterate it and transport the terminal success contribution through the same weighted projections. The resulting success contribution will approximate the original acceptance probability, including probabilities strictly between zero and one.

The remaining calculation connects the local construction to the original acceptance probability. Each copy step has a projection back to its parent. Composing these projections tells us which weighted base value a current vertex represents; composing the success contribution updates tells us how much of that value has already been collected.

Starting with $\mathbf{C}_0 = \mathbf{S}$ on V_0 , repeat the one-stage construction with

$$f_l = q^{-l}, \quad V_{l+1} = V_l \times \{0, \dots, d_0 - 1\}, \quad \mathbf{C}_{l+1} = \mathbf{C}_l^+.$$

Let $\mathbf{E}_l, \mathbf{D}_l, \mathbf{P}_l^r$ denote the correction, residual, and projection at stage l . Define matrices from V_l to V_0 by

$$\mathbf{P}_0^c = \mathbf{I}, \quad \mathbf{P}_{l+1}^c = \mathbf{P}_l^r \mathbf{P}_l^c, \quad \mathbf{B}_0 = \mathbf{I}, \quad \mathbf{B}_{l+1} = \mathbf{P}_l^r (\mathbf{I} + \mathbf{E}_l) \mathbf{B}_l. \quad (5.28)$$

The superscript c here denotes the cumulative projection, not a transpose. Let $\mathbf{e}$ be the base final-success success contribution and $\mathbf{p}_0 = (\mathbf{I} - \mathbf{S})^{-1} \mathbf{e}$, so $0 \leq \mathbf{p}_0 \leq 1$ by the model reduction.

Proposition 5.8 (Success contribution identity and a shallow target). *For every stage,*

$$\mathbf{C}_l \mathbf{P}_l^c = \mathbf{P}_l^c - \mathbf{B}_l (\mathbf{I} - \mathbf{S}). \quad (5.29)$$

The vectors $\mathbf{w}_l = \mathbf{B}_l \mathbf{e}$ satisfy

$$0 \leq \mathbf{w}_l \leq 1, \quad \mathbf{w}_l = \mathbf{P}_l^c \mathbf{p}_0 - \mathbf{C}_l \mathbf{P}_l^c \mathbf{p}_0, \quad (5.30)$$

and their updates are

$$\mathbf{w}_{l+1, x_i} = \mathbf{r}_{i, x} (\mathbf{w}_{l, x} + (\mathbf{E}_l \mathbf{w}_l)_x). \quad (5.31)$$

If $\bar{x}_l$ is the all-primary copy of a base vertex x , then

$$0 \leq \mathbf{p}_{0, x} - \mathbf{w}_{l, \bar{x}_l} \leq |V_0| 2^{-(h-1)l}. \quad (5.32)$$

Consequently there is $l_{\max} = O(b_{\text{mem}})$ for which the last upper bound is strictly less than $1/16$ for every base vertex x . In particular, $\mathbf{p}_{0, x} = 0$ implies $\mathbf{w}_{l_{\max}, \bar{x}_{l_{\max}}} = 0$, and $\mathbf{p}_{0, x} \geq 1/2$ implies $\mathbf{w}_{l_{\max}, \bar{x}_{l_{\max}}} > 7/16$. Every full stage domain has size $2^{O(b_{\text{mem}})}$ and admits $O(b_{\text{mem}})$ -bit addresses.

Proof. First combine the correction identity and the copy projection into one invariant. The assertion at $l = 0$ is $\mathbf{S} = \mathbf{I} - (\mathbf{I} - \mathbf{S})$. The residual identity can be written as

$$\mathbf{D}_l = \mathbf{I} - (\mathbf{I} + \mathbf{E}_l)(\mathbf{I} - \mathbf{C}_l).$$

Using Equation (5.26) and then the inductive hypothesis,

$$\begin{aligned} \mathbf{C}_{l+1}\mathbf{P}_{l+1}^c &= \mathbf{P}_l^r \mathbf{D}_l \mathbf{P}_l^c \\ &= \mathbf{P}_l^r (\mathbf{P}_l^c - (\mathbf{I} + \mathbf{E}_l)(\mathbf{P}_l^c - \mathbf{C}_l \mathbf{P}_l^c)) \\ &= \mathbf{P}_{l+1}^c - \mathbf{P}_l^r (\mathbf{I} + \mathbf{E}_l) \mathbf{B}_l (\mathbf{I} - \mathbf{S}) \\ &= \mathbf{P}_{l+1}^c - \mathbf{B}_{l+1} (\mathbf{I} - \mathbf{S}). \end{aligned}$$

This proves Equation (5.29).

Every $\mathbf{B}_l$ is nonnegative. Every $\mathbf{P}_l^r$ has at most one nonzero entry per row, of value in $[0, 1]$, and thus $\mathbf{P}_l^c$ is substochastic. Apply Equation (5.29) to $\mathbf{p}_0$, using $(\mathbf{I} - \mathbf{S})\mathbf{p}_0 = \mathbf{e}$. This gives the identity in Equation (5.30). Its lower bound follows from $\mathbf{B}_l \mathbf{e} \geq 0$; its upper bound follows from $\mathbf{w}_l \leq \mathbf{P}_l^c \mathbf{p}_0 \leq 1$. Equation (5.31) is Equation (5.28) applied to $\mathbf{e}$.

By Proposition 5.6, induction gives

$$|\mathcal{A}(\mathbf{C}_l)| \leq 2^l |\mathcal{A}(\mathbf{S})| \leq 2^l |V_0|, \quad \mathbf{C}_{l,xy} \leq q^{-l}.$$

There are at most $|\mathcal{A}(\mathbf{C}_l)|$ columns containing any positive entry, so each row sum of $\mathbf{C}_l$ is at most $|V_0|2^l q^{-l} = |V_0|2^{-(h-1)l}$. The all-primary projection has every row factor equal to one and hence $(\mathbf{P}_l^c \mathbf{p}_0)_{\bar{x}_l} = \mathbf{p}_{0,x}$. The identity in Equation (5.30), together with $0 \leq \mathbf{P}_l^c \mathbf{p}_0 \leq 1$, now proves Equation (5.32).

Choose an integer $l_{\max}$ such that $(h-1)l_{\max} > \log_2(16|V_0|)$. Since $\log_2 |V_0| = O(b_{\text{mem}})$, this is $O(b_{\text{mem}})$. If $\mathbf{p}_{0,x} = 0$, nonnegativity forces $\mathbf{w}_{l_{\max}, \bar{x}_{l_{\max}}} = 0$. If $\mathbf{p}_{0,x} \geq 1/2$, the strict error bound gives $\mathbf{w}_{l_{\max}, \bar{x}_{l_{\max}}} > 7/16$. Finally,

$$|V_l| = |V_0|d_0^l = 2^{O(b_{\text{mem}})} \quad (0 \leq l \leq l_{\max}),$$

because d_0 is fixed. Parent projection deletes one radix symbol, and appending a specified symbol is affine in the numerical ID. Thus these domain changes preserve the address convention required by the sampler and by the later implementation. $\square$

The remainder of the proof will estimate the finite targets just constructed. In particular, it need only implement the k_* pilot and k_* actual updates at a stage and the success contribution update in Equation (5.31); it does not evaluate $\mathbf{E}_*$. The same error bound in Equation (5.32) gives any fixed inverse-polynomial accuracy by increasing the constant in $l_{\max} = O(b_{\text{mem}})$. Section 14 makes that choice together with the statistical and numerical precisions.

6 The estimator: what is queried and what is returned

The hierarchy gives an exact success contribution, but its formulas involve matrices and expectations too large to materialize. The next several sections define a recursive estimator for individual entries and give its error analysis. Before proving the components separately, we describe how a query uses them. This description assembles the construction proved below; the formulas and their parameter choices remain in the referenced sections.

6.1 One mathematical table, several ways to query it

A task identifies the quantity being estimated: a bin of a transition matrix, a correction, a detector score, a normalization denominator, or a success contribution. Its parameters specify the stage, iteration, sampling rates, channel roles, inherited copy suffixes, and accuracy level m . Those parameters and the shared environment define one table or vector. The source and destination vertices are arguments to a table query. They are not extra hidden parameters that change when the same entry is found by another path.

There are two kinds of output. A value query asks for a numerical entry. An access query asks which entries can be visited, or for a paired port that moves to an adjacent vertex and permits return. The access relation can include zero-weight entries, but it must contain every nonzero contribution used in the calculation and satisfy the proved degree bounds. Its finite-accuracy selection rules are fixed as part of the task, independently of how accurately a later value query is answered.

This distinction prevents a possible circularity. Increasing numerical precision must not change the graph on which the algorithm is already navigating. A task's exact estimator value is defined first; numerical digit queries then evaluate that fixed value. The implementation of these queries is postponed to Sections 12–13.

6.2 The order in which quantities are defined

At a fixed stage, first define the bins representing $\mathbf{C}_l$. At each pilot iteration, define its raw scores, the detector vector $\mathbf{u}_i$, and then its correction. After the pilot has fixed the multiplier, define the actual corrections in iteration order. The residual bins and success contribution update follow. Place the unnormalized preparatory task immediately before each normalized task that uses its mean as a denominator.

There is also an order within one row task. Its earlier dependencies determine its raw arrays. Complete those arrays before forming the row functions and auxiliary scores. Form an incoming gate from those completed scores. Only then multiply by a column multiplier or impose the incoming access cap. Thus a gate does not call itself to decide which entries exist.

The accuracy level and this dependency order play different roles. The dependency order says which mathematical objects have already been defined. The accuracy level specifies how accurate their estimates are and limits how much work can be passed to a child. The precision assignment in Equation (9.8) combines the accuracy level, task rank, and sampling rates. If the assigned precision is nonpositive, the task is inactive and returns zero values with empty access classes. Boundedness of the true outputs makes this an admissible low-accuracy answer.

6.3 A row query, step by step

For an active row task, the following is the order of evaluation. All lists and arrays in this description are defined implicitly; the algorithm accesses their entries by recomputation.

Algorithm 6.1 A row query at a specified accuracy level

Input. An active row task, the queried source, a shared environment, and the desired value or access query. All lists below are implicit.

1. *Write the required raw signal as sample terms.* The formulas in Equations (9.1)–(9.5) give direct entries, correction products, copy projections, and vector updates. The first row signal is measured in units of its scale a . A normalized task also has a second signal obtained from

Algorithm 6.1 A row query at a specified accuracy level (continued)

its preparatory conditional mean. The factors involving actual retention probabilities correct the relevant changes of conditioning.

2. *Use successive child accuracy levels.* Replace earlier inputs by their estimates at accuracy level r , and subtract the same sample expression at accuracy level $r - 1$. For a term at scale b , include only $1 \leq r \leq m - c_b$. Average the difference by a conditional walk of length $\ell(m - r)$. Its two parts use the same generator word and guessed wraps. This pairing is necessary for the small-difference bound.
3. *Process comparable accuracy level differences together.* Group terms with $d \leq m - r < 2d$, in decreasing dyadic values of d . Include exact base contributions in the first group. A candidate in such a group has an $O(d)$ -bit description and calls only children of accuracy level at most $m - d$. Its description identifies the formula, scale, accuracy level, averaging path, sign, and constituent ports.
4. *Combine new increments at the appropriate destinations.* Validate each candidate separately, including its wrap tests. The accuracy-level block width h_* is specified in Equation (11.3). For $d > 2h_*$ use exact vertex equality. For $d \leq 2h_*$ use the current block's fingerprint key. Sum the new increments with a common key. Add their sum to every old representative with that key; if there is no old representative, introduce the first valid new candidate. Old stored values are never added together merely because their keys agree. Section 11 specifies this rule and analyzes its collisions.
5. *Compress and bound the row after each group.* Apply the threshold update of Section 10, followed by its range clamps and shortlist rule. The represented row consists of a default and finitely many exceptions. The capacity depends on d , and the update is continuous in the numerical input values even though different calculations can select different presentations of their exceptions.
6. *Assemble the output from the completed row.* After the last group, apply the row functions, preparatory normalizations, auxiliary scores, incoming gates, and column multipliers in the order above. The separate column-vector calls have the specified smaller accuracy level. This produces the row values and their access relation.

Output. The prescribed entry or access information of the task's fixed estimator.

A vector task uses the same successive-level averaging schedule for scalar contributions, with the prescribed range bounds and final clipping. It does not require destination grouping or row-list compression. A success contribution query at the final primary start is one such vector task.

6.4 The guarantees proved in the next sections

The construction is established in a definite sequence. Section 7 proves the sparse-table identities and access bounds, including bounds valid on inaccurate computed inputs. Section 8 supplies the conditional walk and its short path descriptions. Section 9 bounds the mean error and averaging noise before compression. Section 10 shows that compression preserves enough accuracy when equal destinations are grouped exactly. Section 11 then proves the coupled error bound for the actual fingerprint-based calculation.

The exact-grouping calculation used in the last two steps is an analytical comparison, not a claim that exact identifiers can be stored at every small-accuracy level call. It helps isolate the

extra error caused by fingerprints. Both calculations still use the same compression and numerical range conventions.

The final accuracy theorem is Theorem 11.10. The separate implementation theorem is Theorem 13.1. The former says how often the estimator is accurate; the latter bounds memory and proves termination for every environment. The median construction needs both guarantees, so the proof never substitutes one for the other.

7 Sparse random tables for the exact construction

The hierarchy defines large matrices that we cannot store. The next step is to represent a requested entry by the conditional expectation of a sparse random table. In one environment, a row of such a table has only a bounded number of nonzero entries. Averaging over environments recovers the exact contribution we need.

There are three kinds of contributions: transition and residual entries, products involving corrections, and column detector scores. The tables called bins below separate entries by magnitude; correction tables permit local movement in both directions; detector tables supply the scores used to decide which columns permit correction. A normalized table also needs an earlier preparatory table to compute the probability by which it divides. These are roles within one construction, not separate approximation algorithms.

We first prove the exact conditional-mean identities. We then extend the same row formulas to computed raw arrays and prove the support and access bounds that hold even when those arrays are inaccurate. Finally we bound the effect of errors in the incoming gates and column multipliers. The later implementation supplies enumeration and pairing of the resulting ports.

Exact conditional-mean identities refer only to the true tables defined here. Applying the same formulas to approximate raw signals gives approximate tables; their normalization need not be unbiased.

Weighted pseudorandomness provides a related way to represent expectations with few random bits; see Braverman, Cohen and Garg [9] and the error-reduction work of Cohen, Doron, Renard, Sberlo and Ta-Shma [16]. The tables below have exact conditional-mean identities. Their estimates use a common environment, so products and subsequent normalization are analyzed through joint error bounds rather than independent sampling.

Algorithm 7.1 From a raw row to a sparse table

Input. A task’s raw array, its retained source and destination sets, and its row functions. True inputs give the exact identities below; computed inputs give the estimator tables.

1. Select the magnitude bin or correction scale, and compute the raw entry and row rank.
2. Apply the row gate and support cutoff. For a correction, complete the auxiliary scores before forming its incoming gate.
3. In a preparatory task, output the unnormalized gated quantity. In a normalized task, obtain the already-defined preparatory mean, clip its denominator to the prescribed positive interval, and divide.
4. Apply the prescribed membership mask and, when present, the frozen column multiplier. Use the support margins and capped access classes to expose ports.

Output. A sparse value table with the specified access relation. The access bounds hold for every computed input, including inaccurate ones.

The normalization corrects bias from the gates as well as from sampling. Its exact mean is a separate target; it is not estimated from the same row by declaring a ratio unbiased. The analysis first proves the true mean identities, then the hard support bounds, and finally stability under replacing true inputs by estimates.

7.1 Conditioning and normalization

Throughout this section $s = 8$. Fix a stage, write $f = b_l$, and let $\mathbf{G}$ be one of its nonnegative raw matrix inputs. The hierarchy gives absolute constants $m_{\mathbf{G}}, k_{\mathbf{G}}$ such that

$$\sum_y \mathbf{G}_{xy} \leq m_{\mathbf{G}}, \quad \mathbf{G}_{xy} \leq k_{\mathbf{G}} f. \quad (7.1)$$

These bounds include pilot and actual inputs and the residual $\mathbf{D}$; see Lemmas 5.3 and 5.5. For a correction $\mathbf{E} = f\mathbf{Q}(\mathbf{G}/f)\mathbf{g}_0$, the additional input is

$$\mathbf{g}_{0,y} > 0 \implies \#\{x : \mathbf{G}_{xy} \geq \delta_1 f\} \leq r_*/f, \quad (7.2)$$

from Lemma 5.4.

A row task has start rate $u \leq f$, end scale a , and, when needed, a usage rate $t \leq a$, all on the rate grid. Its domain is $X_u(x) = Y_a(y) = 1$, with distinct start and terminal channels. We use $a \leq k_f f$ for an absolute k_f , and use $a = f$ for corrections and detector scores. Here and in the estimation and compression sections, t denotes a sampling rate; the numerical clock t is introduced separately in Section 12. The true first raw signal is

$$\mathbf{d}_y = \mathbf{G}_{xy}/a. \quad (7.3)$$

An optional second raw signal is $\zeta \mathbf{1}_{\{Y_t(y)=1\}} \mathbf{q}_y$, where $\mathbf{q}_y = q(x, y) \in [0, 1]$ is a deterministic conditional mean and $\zeta > 0$ is a fixed sufficiently small constant. That mean will be specified separately for each normalized task.

A query must refer to the same table regardless of the path by which the queried entry was found. Otherwise repeated contributions to one entry could not be combined consistently. For this reason, all parameters of a table are common to the whole table: its stage and iteration, task type, channel roles, channel padding suffixes, rates u, a, t , and subsequently its accuracy level and numerical convention. They do not depend on the row or column at which the table is queried.

Write rk_k for the k th largest value, with unlimited zero padding. For two arrays on the same padded index set,

$$|\text{rk}_k(\mathbf{v}) - \text{rk}_k(\mathbf{w})| \leq \sup_z |\mathbf{v}_z - \mathbf{w}_z|. \quad (7.4)$$

Indeed, if every coordinate changes by at most ϵ , at least k coordinates at or above the old k th value remain at or above that value minus ϵ ; apply this in both directions.

Here is the basic tradeoff. A gate can force every sampled row to be sparse by setting crowded rows to zero. That deletion changes the expectation. Dividing by the gate's true conditional mean corrects the expectation, provided the mean is positive and not too small. The normalization can be seen before choosing the row functions. Fix a pair (x, y) on its conditioning slice, a deterministic nonnegative value $k(\mathbf{d}_y)$, and a random row gate $R \in [0, 1]$. Choose a preparatory factor h that is one wherever k is positive. If

$$q(x, y) = \mathbb{E}[h(\mathbf{d}_y)R \mid X_u(x) = Y_t(y) = 1] \geq \frac{1}{2}$$

on that support, then $k(\mathbf{d}_y)R/q(x, y)$ has conditional mean $k(\mathbf{d}_y)$ there. A clipped denominator keeps the formula bounded at every other position and for computed inputs. The constructions

below arrange a uniform positive lower bound on this mean while the row gate limits the number of nonzero retained entries. Corrections use an additional incoming gate, with the same normalization principle.

We will use several continuous switches to distinguish small from large entries. The numerical values of their thresholds are less important than the gaps between them: an entry can be safely included in an access list before it becomes nonzero, and small numerical perturbations cannot destroy that inclusion. The next definitions fix one concrete set of switches and margins so that these requirements can be checked exactly.

For real $\alpha < \beta$, let $\chi_{\alpha,\beta}$ be a fixed increasing switch, zero on $(-\infty, \alpha]$ and one on $[\beta, \infty)$. Choose it by rescaling the normalized integral of $z^{s-1}(1-z)^{s-1}$ on $[0, 1]$. Both $\chi_{\alpha,\beta}^{1/s}$ and $(1 - \chi_{\alpha,\beta})^{1/s}$ are Lipschitz: at either endpoint the vanishing function has a zero of order s , and on the interior ordinary differentiation applies. Products of bounded functions with Lipschitz s th roots again have Lipschitz s th roots.

Every scalar saturation below uses fixed min/max clipping,

$$\text{clip}_{[\alpha,\beta]}(v) = \min\{\beta, \max\{\alpha, v\}\},$$

which is 1-Lipschitz. We specify its bounds when defining the function. Functions initially defined for nonnegative first entries are extended by zero below zero; each already vanishes below a positive support threshold. These conventions apply only to evaluation of the row functions. Ranks and access tests always use the full prescribed raw arrays before scalar clipping.

Here is one explicit set of margins used below. Let $\vartheta > 0$ be a fixed lower support threshold of a main row function, and put $\rho = \text{rk}_{k_{\text{row}}}(v)$ for its first raw row v . Define

$$\begin{aligned} h_{\vartheta}(v) &= \chi_{\vartheta/2, 3\vartheta/4}(v), \\ R_{\vartheta}(\rho) &= 1 - \chi_{\vartheta/1000, \vartheta/500}(\rho), \\ l_{\vartheta}(v, \rho) &= \chi_{\vartheta/8, \vartheta/4}(v)(1 - \chi_{\vartheta/100, \vartheta/50}(\rho)). \end{aligned} \tag{7.5}$$

We take $\vartheta = \theta_1$ for corrections, $\vartheta = 1/2$ for dense bins, and $\vartheta = \delta_0$ for the detector score. For that detector we make the allowed hierarchy choices

$$\eta_- = \delta_0/1000, \quad \eta_+ = \delta_0/500, \quad \rho(t) = 1 - \chi_{\delta_0/1000, \delta_0/500}(t) = R_{\delta_0}(t).$$

Here ρ denotes the cutoff in the hierarchy, and the same switch χ is used in both occurrences. These choices satisfy $0 < \eta_- < \eta_+ < \delta_0/3$. Thus the detector row gate here is exactly R_i in Equation (5.7). The detector does not require h_{ϑ} . The hierarchy's choices may and do satisfy $\delta_1 < \theta_1/100$ and $0 < \delta_0 < \delta_1$. Thus the correction auxiliary score l_{θ_1} can be positive only when $\mathbf{G}_{xy} > \delta_1 f$.

7.2 Separate entries by size and normalize each part

A single sampling rate is not well suited to entries of very different sizes. We split an entry into overlapping contributions at geometric scales. The smooth weights sum to one, so this introduces no approximation. At each scale, sampling and a row gate leave bounded row support; the preparatory conditional mean corrects for the gate's deletion.

Put $q_* = 2^h$, and let $\chi = \chi_{1/2, 1}$. For $w \geq 0$ define

$$\begin{aligned} \omega_0(w) &= \chi(w), \\ \omega_k(w) &= \chi(w/b_k) - \chi(w/b_{k-1}) \quad (k \geq 1). \end{aligned} \tag{7.6}$$

The top bin has no preceding switch. The weights are nonnegative and sum to one for $w > 0$, since their partial sums telescope to $\chi(w/b_k) \rightarrow 1$. Hence

$$w = \sum_{k \geq 0} w \omega_k(w). \quad (7.7)$$

At scale $a = b_k$, the bin in atom units is

$$k_a(d) = d \omega_k(ad). \quad (7.8)$$

For $k \geq 1$ its support is contained in $1/2 < d < q_*$. Its two switch transition intervals are disjoint, and

$$\chi(d) - \chi(d/q_*) = \chi(d)(1 - \chi(d/q_*)),$$

which proves directly that its st h root is Lipschitz. For $k = 0$, fix $b_{\text{top}} > \max\{k_{\mathbf{G}}, 1\}$ and evaluate the defining expression at $\text{clip}_{[0, b_{\text{top}}]}(d)$. This makes the top-bin function bounded with Lipschitz root. It preserves every true value, since $a = b_0 = 1$ and $\mathbf{G}_{xy} \leq k_{\mathbf{G}} f \leq k_{\mathbf{G}}$. For $k \geq 1$, retain the zero extension outside the support $(1/2, q_*)$; no upper clipping changes a zero bin value. Bins with $a > 2k_{\mathbf{G}} f$ vanish on the true input, so a fixed $k_f \geq 2k_{\mathbf{G}}$ suffices for the permitted scale range.

For the routed residual bin use instead

$$k_a^{(j)}(d) = k_a(d) s_j(ad/f). \quad (7.9)$$

The routing fractions have bounded Lipschitz roots. Since $a/f \leq k_f$ and the bin is zero above a fixed argument range (or is top-bin clipped), these functions have uniformly bounded Lipschitz roots as well. The same zero-extension and clipping conventions apply to routed bins. Summing Equation (7.9) over scales gives the routed residual contribution exactly on every true input.

Having split the magnitude scales, we now apply the normalization principle to one bin. Choose $R_{1/2}$ and $h = h_{1/2}$ from Equation (7.5). For the true row set

$$\rho_x = \text{rk}_{k_{\text{row}}} \{\mathbf{G}_{xz}/a : Y_a(z) = 1\}, \quad R = R_{1/2}(\rho_x).$$

Write

$$\mathbf{q}_y = \mathbb{E}[h(\mathbf{d}_y)R \mid X_u(x) = Y_t(y) = 1], \quad c(q) = \min\{1, \max\{1/2, q\}\}. \quad (7.10)$$

The number $\mathbf{q}_y$ is the deterministic conditional mean at the indicated pair. The second raw signal carries the Y_t membership mask. In addition, $\mathbf{q}_y = 0$ if $\mathbf{d}_y \leq 1/4$. The true preparatory table on the broad terminal domain Y_a is the unnormalized value $h(\mathbf{d}_y)R$. It uses only the first raw signal. Its conditional mean on the narrower slice is exactly the $\mathbf{q}_y$ in Equation (7.10). The usage rate t is part of the fixed task tuple: changing the slice on which this mean is taken would in general change the denominator. The broad preparatory domain and the narrower normalization slice must therefore be retained as separate parts of the interface.

Lemma 7.2 (Dense atom identity). *For sufficiently large fixed k_{row} , the table*

$$\mathbf{B}_{a,xy} = \mathbf{1}_{\{Y_t(y)=1\}} \frac{k_a(\mathbf{d}_y)R}{c(\mathbf{q}_y)} \quad (7.11)$$

on retained starts has bounded values and row support. On the slice $X_u(x) = Y_t(y) = 1$ its conditional expectation times a equals $\mathbf{G}_{xy} \omega_k(\mathbf{G}_{xy})$. The same assertion holds for $\mathbf{B}_a^{(j)}$ with the routing fraction included. Apart from its start membership, its true value uses only the terminal channel.

Proof. First show that the gate is usually fully open, even after conditioning on the queried destination. The row gate can fail to be full only if at least k_{row} sampled columns have $\mathbf{G}_{xz}/a > r$, where $r = (1/2)/1000$ is fixed. There are at most $m_{\mathbf{G}}/(ar)$ such deterministic columns. Conditional on $Y_t(y) = 1$, the specified column contributes at most one hit and each other column has Y_a -hit probability at most $c_{\text{bd}}a$, by Lemma 4.1. Thus the expected count is at most $1 + c_{\text{bd}}m_{\mathbf{G}}/r$. Conditioning also on the independent source channel does not alter this bound. Markov's inequality makes the failure probability at most $1/8$ by a sufficiently large k_{row} .

Now the normalization is exact on every position that contributes to the bin. Where $k_a(\mathbf{d}_y) > 0$, we have $h(\mathbf{d}_y) = 1$. Therefore $\mathbf{q}_y = \mathbb{E}[R \mid Y_t(y) = 1] \in [7/8, 1]$ there, and the clipping does nothing. The conditional mean of Equation (7.11) is consequently $k_a(\mathbf{d}_y)$. Outside that support both sides vanish. If a row output is positive, its first entry exceeds $1/2$ and its k_{row} th first rank is below $1/1000$. Hence there are at most $k_{\text{row}} - 1$ positive entries. Boundedness follows from the clipped denominator. The routed fraction is deterministic at truth and obeys the same argument. $\square$

For products of a correction with a deterministic bin contribution, we will use the following bounds.

Lemma 7.3 (Bounds for a fine product contribution). *Let $\mathbf{C}$ be the transition matrix at the current stage, and let $\mathbf{E}$ be any finite pilot or actual correction there. If a matrix $\mathbf{A}_b \geq 0$ on the same vertex set satisfies $\mathbf{A}_b \mathbf{1} \leq \mathbf{1}$ and $\mathbf{A}_{b,xy} \leq c_b b$ for some $b > 0$, then*

$$(\mathbf{E}\mathbf{A}_b)\mathbf{1} \leq k_* \mathbf{1}, \quad (\mathbf{E}\mathbf{A}_b)_{xy} \leq k_* c_b b.$$

If also $\mathbf{A}_b \leq \mathbf{C}$, then $\mathbf{E}\mathbf{A}_b \leq \mathbf{E}\mathbf{C}$. These estimates have constants independent of the bin scale b .

Proof. Use $\mathbf{E}\mathbf{1} \leq k_* \mathbf{1}$ from Lemma 5.3: $(\mathbf{E}\mathbf{A}_b)\mathbf{1} \leq \mathbf{E}\mathbf{1}$, and $(\mathbf{E}\mathbf{A}_b)_{xy} \leq c_b b \sum_z \mathbf{E}_{xz}$. The last assertion follows by multiplying $\mathbf{A}_b \leq \mathbf{C}$ on the left by the nonnegative matrix $\mathbf{E}$. $\square$

7.3 Corrections with an incoming gate

A correction is used as the left factor of a product. The algorithm moves along its edge, evaluates the right factor at the new vertex, and later returns. Both the implementation and its error analysis need a bounded number of incoming as well as outgoing accesses. A row gate alone provides no incoming bound.

We therefore add a column gate. The deterministic column-support bound proved for the hierarchy ensures that this additional gate is usually fully open on a true contributing entry. A union bound will show that both gates are open simultaneously with probability at least $3/4$; their independence is not required.

Consider $\mathbf{E} = f\mathbf{Q}(\mathbf{G}/f)\mathbf{g}_{0,y}$ and put $a = f$, $\vartheta = \theta_1$. Fix $b_{\mathbf{Q}} > \max\{k_{\mathbf{G}}, \theta_2\}$ and define the atom cutoff

$$\mathbf{Q}_{\text{at}}(d) = \mathbf{Q}(\text{clip}_{[0, b_{\mathbf{Q}}]}(d)).$$

It is bounded, has a Lipschitz sth root on the real line, and agrees with $\mathbf{Q}(d)$ for every true argument $d = \mathbf{G}_{xy}/f$. Its extension to negative arguments is zero. The exact hierarchy continues to use the uncapped $\mathbf{Q}$. Let $R = R_{\theta_1}(\rho_x)$, $h = h_{\theta_1}$, and $\mathbf{l} = \mathbf{l}_{\theta_1}$ be the functions in Equation (7.5), where the true row rank is $\rho_x = \text{rk}_{k_{\text{row}}} \{\mathbf{G}_{xz}/f : Y_f(z) = 1\}$. On a true or computed raw table, form $\mathbf{I}_{xy}$ from that table's own row entry and own rank. With $\chi_{1/4, 1/2}$ as above, define

$$\mathbf{t}_y = 1 - \chi_{1/4, 1/2}(\text{rk}_{k_c} \{\mathbf{I}_{xy} : X_u(x) = 1\}). \quad (7.12)$$

Thus $\mathbf{t}_y = 1$ through rank $1/4$ and $\mathbf{t}_y = 0$ from rank $1/2$. This rank is taken over the fixed retained source set before any incoming port filtering; there is no gate-dependent change of its index set. The true correction preparation is $h(\mathbf{d}_y)R\mathbf{t}_y$ on Y_a . Define its conditional mean by

$$\mathbf{q}_y = \mathbb{E}[h(\mathbf{d}_y)R\mathbf{t}_y \mid X_u(x) = Y_t(y) = 1]. \quad (7.13)$$

The number $\mathbf{q}_y$ is deterministic; as before, its second raw signal carries the Y_t membership mask. Define the normalized true correction using the clipped expression everywhere:

$$\mathbf{F}_{xy} = \mathbf{1}_{\{Y_t(y)=1\}} \frac{Q_{\text{at}}(\mathbf{d}_y)R\mathbf{t}_y}{c(\mathbf{q}_y)} \mathbf{g}_{0,y}. \quad (7.14)$$

Here $\mathbf{q}_y = 0$ whenever $\mathbf{d}_y \leq \theta_1/2$. Thus both kinds of second signal are supported above a fixed positive first-signal threshold. Choose $\zeta \leq 10^{-6} \min\{\theta_1, \delta_0, 1/2\}$, decreasing it later if a fixed compression margin requires this. The second signals are then uniformly tiny on the first-signal threshold scale.

Lemma 7.4 (Correction atom identity, including zero multipliers). *For sufficiently large fixed k_{row} and then k_c , the conditional mean of $f\mathbf{F}_{xy}$ on $X_u(x) = Y_t(y) = 1$ is exactly $\mathbf{E}_{xy}$. On a true position with $Q(\mathbf{d}_y) > 0$ and $\mathbf{g}_{0,y} > 0$,*

$$\mathbf{q}_y = \alpha(x, y) := \mathbb{E}[R\mathbf{t}_y \mid X_u(x) = Y_t(y) = 1] \geq 3/4. \quad (7.15)$$

No lower bound on $\mathbf{q}_y$ or $\alpha(x, y)$ is required when $\mathbf{g}_{0,y} = 0$. The true gates and normalization use only the two role channels X, Y ; $\mathbf{g}_0$ is deterministic.

Proof. We bound failure of each gate separately. The row count argument in Lemma 7.2, with the correction thresholds, gives $\Pr(R \neq 1 \mid \text{the two hits}) \leq 1/8$. If $\mathbf{g}_{0,y} > 0$, every source with $\mathbf{I}_{x'y} > 0$ belongs to the deterministic set in Equation (7.2). Conditional on $X_u(x) = 1$, the number of sampled members of that set has expectation at most

$$1 + c_{\text{bd}}ur_*/f \leq 1 + c_{\text{bd}}r_*.$$

The bound remains valid after conditioning on the independent terminal hit; it ignores the auxiliary row gate and therefore needs no additional independence between rows. If fewer than k_c such sources are sampled, the k_c th auxiliary rank is zero and $\mathbf{t}_y = 1$. Choose $k_c \geq 8(1 + c_{\text{bd}}r_*)$ to make $\Pr(\mathbf{t}_y \neq 1 \mid \text{the two hits}) \leq 1/8$. The union bound, without independence of the gates, gives $\mathbb{E}[R\mathbf{t}_y \mid \text{the two hits}] \geq 3/4$.

On a contributing entry, divide by the mean of the product of the gates. Since $h(\mathbf{d}_y) = 1$ on $\mathbf{Q}$'s support, Equation (7.15) holds and $c(\mathbf{q}_y) = \mathbf{q}_y$ there whenever $\mathbf{g}_{0,y} > 0$. The claimed conditional identity follows by division by this exact conditional mean. If $Q(\mathbf{d}_y) = 0$ or $\mathbf{g}_{0,y} = 0$, both sides of the identity are zero. In the latter case the pre-incoming quantity $Q_{\text{at}}(\mathbf{d}_y)R/c(\mathbf{q}_y)$ is still a well-defined bounded comparison target, even if the unclipped mean vanishes. $\square$

7.4 Detector scores

Detector scores already take values in a bounded interval, and their defining expectation does not need the preceding normalization. We only need to expose the row-bounded table whose column order statistic is the detector trial. The unnormalized detector table is

$$\mathbf{S}_{i,xy} = \phi(\mathbf{G}_{i,xy}/f) \mathbf{R}_{\delta_0}(\text{rk}_{k_{\text{row}}} \{\mathbf{G}_{i,xw}/f : Y_f(w) = 1\}) \quad \text{on } X_f(x) = Y_f(y) = 1. \quad (7.16)$$

Its column trial is the j th largest such score over retained starts. By the identical cutoff choice, its conditional mean given $Y_f(y) = 1$ is exactly $\mathbf{u}_{i,y}$ from Equation (5.8). The trial is an inverse rank access to a row-bounded table and does not require an incoming degree bound.

7.5 Accessing entries when the input estimates are inaccurate

So far we have proved identities for true inputs. The algorithm must also have well-defined bounded access when its inputs are inaccurate, since the final deterministic enumeration visits inaccurate environments as well. The next bounds are therefore pointwise statements about arbitrary computed arrays, with no accuracy hypothesis.

The conditional-mean identities now give the true contributions. A product estimate will query a correction at (x, z) and then a bin at (z, y) . Bounded outgoing degree limits the possible midpoints; bounded incoming degree is needed when errors at the new source z are summed over old sources x . We therefore specify access relations as well as nonzero table supports. The access relations may contain entries of weight zero.

Passing through one copy projection is partitioned by its fixed source and destination symbols. Within each part the inherited hash argument appends that symbol before the inherited suffix. Lemma 4.1 then gives the same distribution as the unpadded channel. The task tuple records these fixed symbols and suffixes. By definition, neither a right factor's task parameters nor a column vector's task parameters retain an earlier source address. The environment supplied to the table is common to all its entries. The vertices x, y are arguments of that table, rather than additional parameters. Thus two paths that discover the same entry in that environment query the same value; the path used to discover it is not part of its mathematical definition.

We will apply these row functions to arrays represented by a finite list of exceptional values and one common value at every omitted endpoint. Call that common value the *default*. Adjoin arbitrarily many dummy coordinates with the same default, and include them in every rank; they are never matrix entries or ports. For the true row arrays, dummy values are zero. Section 10 constructs the finite-list representations. The elementary rank convention below lets us prove support and access bounds before choosing that construction.

Remark 7.5 (Ranks of arrays with a common default). Suppose a row array equals a common default outside a finite list, and arbitrarily many dummy coordinates have that same default. Its k th rank can be computed from the listed coordinates and k copies of the default, with the stipulated zero padding. Indeed, every omitted coordinate merely supplies another copy of a value already available with multiplicity at least k . Additional listed real coordinates equal to the default therefore do not change the rank. The same observation applies to a rank of entry sizes when each omitted entry has the same default vector.

This is an equality of ranks for one prescribed full array. The dummy coordinates are never matrix entries or ports. A common componentwise clamp of the entries and their default preserves this representation: an omitted position still equals the clamped default. No assertion that different calculations select identical lists or ports is needed.

An access relation may safely contain an entry whose weight is zero. What matters is that it contains every nonzero entry, has bounded degree, and can be selected by repeatable finite-accuracy tests. The next lemma constructs such a relation by putting the access threshold below the nonzero threshold, with a gap large enough to absorb numerical error.

Lemma 7.6 (Support margins and exposed row classes). *Suppose a bounded nonnegative main function $k(v)$ vanishes for $v \leq \vartheta$. Its product $k(v)R_\vartheta(\rho)$ and the preparatory product $h_\vartheta(v)R_\vartheta(\rho)$ have at most $k_{\text{row}} - 1$ nonzero real entries in a row. They admit a common exposed class $\mathcal{A}$, of at most $k_{\text{row}} - 1$ real entries, containing their supports and satisfying $l_\vartheta = 1$ throughout $\mathcal{A}$. These assertions hold for arbitrary real raw estimates, without a sign or accuracy hypothesis.*

The auxiliary score l_ϑ also has at most $k_{\text{row}} - 1$ nonzero real entries and has an exposed row class with a bounded-support smooth witness equal to one on that class.

Proof. The basic counting observation is that an entry threshold above the row's k_{row} th-largest value can be met by fewer than k_{row} entries. For the main and preparatory products a nonzero entry has $v > \vartheta/2$ and $\rho < \vartheta/500$. There cannot be k_{row} such entries, since then the k_{row} th rank would exceed $\vartheta/2$.

We next leave sufficient room to perform the access test with finite precision. For an explicit class choose consistently computed approximations $\tilde{v}, \tilde{\rho}$ with absolute errors at most $\eta = \vartheta/10000$, and set

$$\mathcal{A} = \{y : Y_a(y) = 1, \quad \tilde{v}_y > 3\vartheta/8, \quad \tilde{\rho} < \vartheta/200\}. \quad (7.17)$$

Every positive main or preparatory entry belongs to $\mathcal{A}$. An entry in $\mathcal{A}$ satisfies $\mathbf{v}_y > (3/8 - 1/10000)\vartheta > \vartheta/4$ and $\rho < (1/200 + 1/10000)\vartheta < \vartheta/100$, so its witness l_ϑ equals one. Its entry threshold exceeds its rank threshold, proving $|\mathcal{A}| \leq k_{\text{row}} - 1$.

The support of l_ϑ requires $v > \vartheta/8$ and $\rho < \vartheta/50$, again giving the same row bound. To access this score itself, the class

$$\mathcal{A}_l = \{y : Y_a(y) = 1, \quad \tilde{v}_y > \vartheta/10, \quad \tilde{\rho} < \vartheta/40\}$$

contains its support and has at most $k_{\text{row}} - 1$ entries. A smooth unit witness on $\mathcal{A}_l$ is

$$J_\vartheta(v, \rho) = \chi_{82\vartheta/1000, 90\vartheta/1000}(v)(1 - \chi_{26\vartheta/1000, 27\vartheta/1000}(\rho)).$$

The approximation margins imply $\mathcal{A}_l \subseteq \{J_\vartheta = 1\}$. Moreover $82/1000 > 3(27/1000)$, so this witness also has bounded row support and the required entry versus rank separation. The witness is an analytical output; its support need not be broadened recursively.

These arguments use the rank of the prescribed entire raw array. In particular, if at least k_{row} dummy positions have a common virtual default β , then $\rho \geq \beta$. A real position still equal to that default cannot meet either exposed class: the required lower bound on $v = \beta$ exceeds the required upper bound on ρ . Thus omitted default positions do not create unlisted ports. $\square$

The fixed-accuracy approximations in this lemma are part of the access interface, not tests for equality of exact reals. The implementation must use the same approximation and tie conventions whenever it queries the same table. The strict margins prove correctness for every such choice. Auxiliary-only access classes require a row bound but no incoming bound. In particular, different computed versions may expose different zero-weight entries. The assertions are applied to each version's own final raw array and repeatable approximation convention; they do not require those classes to coincide. Accuracy of the raw array is used later in the error estimates, not in this support argument.

We can now specify the assembly order for an estimated table. First complete its raw row arrays. Apply the bounded row functions to those *final* arrays, and only then form their incoming scores and gates. No unfinished intermediate row is used as though it were the final table. In a normalized task its second input is a computed value approximating $\zeta \mathbf{1}_{\{Y_i(y)=1\}} \mathbf{q}_y$: the calculator divides that input by ζ and applies c . It does not launch another averaging operation after selecting an endpoint. Its auxiliary incoming scores are formed from its own final first-signal rows, and its incoming gate is Equation (7.12) for those scores. A preparatory task is an earlier, separate task; its raw arrays, scores, and gates are its own, and its target is the unnormalized preparatory table specified above. Corrections additionally multiply by a clipped vector estimate of the prescribed **u**-cutoff $\mathbf{g}_{0,y}$, queried with a common broad Y_f -domain tuple. The unnormalized preparations hR and $hR\mathbf{t}$ remain on their broad Y_a domain; their usage parameter t specifies the conditional slice on which they are averaged. The resulting second raw signal and the normalized main output carry the Y_t mask. Dense bins and their preparations and detector tables have no additional incoming multiplier. Because c takes values in $[1/2, 1]$, every such pre-incoming row function has a globally Lipschitz sth root as a function of its saturated first entry, first rank, and scaled second entry. The constants may depend on the fixed ζ , but not on the stage, rates, or accuracy level.

For clarity, Table 1 records the thresholds needed by the compression argument. For a pre-incoming row function h , its entry factor vanishes through e_h , and its row gate vanishes from rank $\mathbf{g}_h$ onward. Every listed pair satisfies $3\mathbf{g}_h < e_h$, including the broadest witness, for which $82\vartheta/1000 - 3(27\vartheta/1000) = \vartheta/1000 > 0$. All gates are full in a neighborhood of rank zero. The correction preparation in this table is its pre-incoming factor hR ; its incoming gate is assembled afterward.

Pre-incoming row function	e_h	$\mathbf{g}_h$
Dense bin, including routed bins	$1/2$	$1/1000$
Dense preparation hR	$1/4$	$1/1000$
Correction clipped-ratio function	θ_1	$\theta_1/500$
Correction preparation hR	$\theta_1/2$	$\theta_1/500$
Detector score	δ_0	$\delta_0/500$
Auxiliary score $\mathbf{l}_\vartheta$	$\vartheta/8$	$\vartheta/50$
Witness $\mathbf{J}_\vartheta$	$82\vartheta/1000$	$27\vartheta/1000$

Table 1: Entry and row-gate thresholds. The entry factor vanishes through e_h , and the row gate vanishes from rank $\mathbf{g}_h$ onward. Here $\vartheta \in \{1/2, \theta_1, \delta_0\}$ as appropriate.

The bounded first-entry ranges and gate transition intervals already specified give fixed sensitivity ranges for these functions. If w is a computed second raw entry, its only use is through $c(w/\zeta)$; clipping w to $[\zeta/2, \zeta]$ leaves this quantity unchanged. Its denominator stays in $[1/2, 1]$ even for negative or inaccurate inputs. Usage masks only remove entries and do not change any of these support or continuity bounds.

The last access issue is the number of exposed incoming entries. Merely keeping the first k_c would depend on an arbitrary selection and could change nonzero values. Instead, the construction discards every precursor access at an overfull column. The incoming gate has already made all those values zero, so this deletion preserves the table.

Lemma 7.7 (Port contract). *All main and preparatory row tables, auxiliary scores, and the smooth witnesses just constructed have bounded values and bounded row support. Every row access class used for a main or preparatory output, or for access to its auxiliary score, has bounded size and a smooth witness equal to one on every exposed entry, including exposed entries of weight zero. A witness itself is needed as an analytical row function, without an additional requirement to expose its support through a further witness. For a correction or its preparatory table let $\mathcal{A}$ be its pre-incoming class from Equation (7.17). Retain an exposed entry $(x, y) \in \mathcal{A}$ only when*

$$\#\{x' : X_u(x') = 1, (x', y) \in \mathcal{A}\} \leq k_c. \quad (7.18)$$

This deletion preserves all table values and gives at most $k_{\text{row}} - 1$ outgoing and at most k_c incoming exposed entries.

These assertions hold separately for every computed version, provided that its values, ranks, scores, and access classes use the same final raw arrays and common task tuple. Access ports denote distinct entries: the implementation must expose each pair at most once and pair its incoming and outgoing ports consistently. Access to auxiliary scores alone has only the asserted row bound.

Proof. The row and witness assertions are Lemma 7.6; clipping and multiplication by bounded factors do not enlarge supports. If a column has more than k_c entries in $\mathcal{A}$, its auxiliary score is one at all those entries, so its k_c th auxiliary rank is one. Equation (7.12) makes $\mathbf{t}_y = 0$ there. Consequently every value deleted by Equation (7.18) was already zero. The surviving relation has the stated two degree bounds even when zero weights are retained. The argument applies to arbitrary raw arrays, so it applies equally to the estimated and true versions.

The cap removes every precursor port at an overfull column; it does not choose k_c of them. It is applied even when the particular entry has weight zero. Every surviving port still lies in $\mathcal{A}$, so its pre-incoming witness $\mathbf{l}$ equals one regardless of a zero usage mask, incoming gate, or column multiplier. The broader auxiliary class $\mathcal{A}_\mathbf{l}$ is used to enumerate scores for an incoming rank. It is not the capped correction relation used as a paired left factor in a product.

The classes are relations on row and column IDs, rather than multisets of paths. Choosing one port per relation element gives the stated combinatorial interface. Establishing its single-address enumeration and port pairing within the space bound is an implementation obligation; the degree argument does not itself grant random access to both IDs. $\square$

The access requirements can now be summarized without conflating them with accuracy. Every query uses fixed table parameters and one supplied environment. Row ranks include the prescribed dummy coordinates. An exposed port denotes one distinct real entry and lies where its smooth witness is one. Correction and preparatory accesses obey the incoming cap. Values, scores, incoming ranks, and multipliers are all assembled from that same table. In particular, the path used to discover an entry does not change the table at which it is evaluated. Only completed earlier output types, or stated fixed local functions and aggregations of those types, may be used as raw inputs of a later task. No prefix still waiting for assembly is such an output. These requirements distinguish finite-support analytical bounds from the access properties used in the subsequent implementation. Completing a raw array here means defining its entries from the allowed earlier outputs, rather than storing the whole array at once. Its auxiliary scores and precursor classes are defined before its incoming gate and cap, so their definitions do not call that same gate or cap.

These contracts apply directly to each computed array after compression and clamping. The later comparison in Lemma 10.7 preserves relevant row-function values under its stated common-increment hypothesis. It need not preserve the finite-accuracy access descriptors of two calculations. Each calculation uses its own consistent classes, and Lemma 7.7 gives their support and degree bounds separately.

7.6 How errors propagate through table assembly

It remains to control how errors in the row functions, incoming scores, and detector vectors affect an assembled table. The incoming degree bound lets us sum these errors by columns without multiplying by the number of possible source vertices.

We measure error by an eighth moment averaged over environments and source vertices. The source-retention weight compensates for sampling that source. The destination sum stays explicit because bounded support and incoming degree, rather than a second normalization by the universe size, will control it. Root discrepancies are convenient for products of nonnegative gates: the smooth switches were chosen precisely so that their eighth roots vary continuously at zero.

For a real row array $\mathbf{Z}$ define

$$\|\mathbf{Z}\|_{u,a;s}^s = \frac{1}{n_v} \mathbb{E}_\sigma \sum_x \frac{X_u(x)}{u} \sum_{y:Y_a(y)=1} |\mathbf{Z}_{xy}|^s, \quad \mathfrak{d}_{u,a}(\mathbf{H}, \mathbf{H}') = \|\mathbf{H}^{1/s} - (\mathbf{H}')^{1/s}\|_{u,a;s}. \quad (7.19)$$

Here σ has the product law of the rank channels and independent auxiliary bits specified in Section 4. All table values are extended by zero outside their membership domain. The denominator u is the nominal rate, and actual-rate comparability in Lemma 4.1 absorbs the resulting absolute constants. For a vertex array on a channel with rate u , write

$$\|\mathbf{v}\|_{u;s}^s = \frac{1}{n_v} \mathbb{E}_\sigma \sum_x \frac{X_u(x)}{u} |\mathbf{v}_x|^s; \quad (7.20)$$

the indicated role channel replaces X when necessary. These are also the comparison norms for two sample-dependent versions. All required pre-incoming functions and smooth support witnesses are included among the row outputs whose discrepancies are to be bounded.

Bounded row support and range imply a uniform bound on these row-output norms, since $|V_l| \leq n_v$ and $\mathbb{E}X_u/u = O(1)$. Bounded vertex arrays have the analogous uniform bound. Ordinary error is controlled by root error on bounded nonnegative outputs, since

$$|z - w| \leq s c_{\text{bd}}^{(s-1)/s} |z^{1/s} - w^{1/s}| \quad (0 \leq z, w \leq c_{\text{bd}}). \quad (7.21)$$

Lemma 7.8 (Assembly comparisons). *Compare two versions with the same environment, rates, and task tuple. In the following assembly formulas, products of row outputs and column gates are pointwise: at an entry (x, y) , multiply the pre-incoming value at (x, y) by the gate and multiplier at y . Powers and roots of arrays are also entrywise. For a sparse normalized output write $\mathbf{H} = \mathbf{P}\mathbf{t}\mathbf{g}$ and $\mathbf{H}' = \mathbf{P}'\mathbf{t}'\mathbf{g}'$, where $\mathbf{P}, \mathbf{P}'$ are the pre-incoming clipped-ratio row functions (including their usage masks), $\mathbf{I}, \mathbf{I}'$ are their auxiliary score tables, and $\mathbf{g}, \mathbf{g}'$ are their column multipliers. If $\mathbf{g}, \mathbf{g}'$ are the prescribed smooth cutoffs of bounded vertex arrays $\mathbf{u}_i, \mathbf{u}'_i$, then*

$$\mathfrak{d}_{u,f}(\mathbf{H}, \mathbf{H}')^s \leq c_{\text{bd}} \mathfrak{d}_{u,f}(\mathbf{P}, \mathbf{P}')^s + c_{\text{bd}} \mathfrak{d}_{u,f}(\mathbf{I}, \mathbf{I}')^s + c_{\text{bd}} \frac{f}{u} \sum_i \|\mathbf{u}_i - \mathbf{u}'_i\|_{f;s}^s. \quad (7.22)$$

For a sparse preparation omit the last term and use its unnormalized pre-incoming row function. The corresponding dense and detector row functions have no incoming or column-multiplier assembly cost.

If $\mathbf{v}_y, \mathbf{v}'_y$ are the detector trials formed from $\mathbf{S}_i, \mathbf{S}'_i$ by their column order statistic, then

$$\|\mathbf{v} - \mathbf{v}'\|_{f;s}^s \leq c_{\text{bd}} \mathfrak{d}_{f,f}(\mathbf{S}_i, \mathbf{S}'_i)^s. \quad (7.23)$$

Proof. Separate the change in each of the three factors. Then sum only over entries where at least one of the two final outputs is nonzero. This is where the bounded incoming degree can be used. All factors are bounded. On the union E of the two final nonzero supports, the triangle inequality for the products of sth roots gives

$$|\mathbf{H}^{1/s} - (\mathbf{H}')^{1/s}| \leq c_{\text{bd}} |\mathbf{P}^{1/s} - (\mathbf{P}')^{1/s}| + c_{\text{bd}} |\mathbf{t}^{1/s} - (\mathbf{t}')^{1/s}| + c_{\text{bd}} |\mathbf{g}^{1/s} - (\mathbf{g}')^{1/s}|.$$

Outside E the left side is zero. The pre-incoming discrepancy may be charged on its entire row domain; it needs no incoming cap. Each column has at most $2k_c$ edges in E , by Lemma 7.7. By Equation (7.4), the root-Lipschitz incoming cutoff, and Equation (7.21),

$$|\mathbf{t}_y^{1/s} - (\mathbf{t}'_y)^{1/s}|^s \leq c_{\text{bd}} \sup_{x: X_u(x)=1} |\mathbf{I}_{xy}^{1/s} - (\mathbf{I}'_{xy})^{1/s}|^s.$$

Consequently its integrated contribution is at most

$$\frac{c_{\text{bd}}}{n_v u} \mathbb{E} \sum_{y: Y_f(y)=1} \sum_{x: X_u(x)=1} |\mathbf{I}_{xy}^{1/s} - (\mathbf{I}'_{xy})^{1/s}|^s,$$

which is $c_{\text{bd}} \mathfrak{d}_{u,f}(\mathbf{I}, \mathbf{I}')^s$.

For the multiplier term, the same error at a destination may be encountered from several sources. The incoming cap bounds that multiplicity, and the ratio of the two domain weights is exactly f/u . The maximum over a fixed number of $\mathbf{u}_i$ values is Lipschitz, and the multiplier cutoff has Lipschitz root. Thus the last discrepancy is bounded in sth power by $c_{\text{bd}} \sum_i |\mathbf{u}_{i,y} - \mathbf{u}'_{i,y}|^s$. Summing over E costs at most $2k_c/u$ per retained column. The vertex metric has weight $1/f$ instead, giving precisely the factor f/u in Equation (7.22). All columns in this comparison are in the common Y_f domain, including those whose final values carry a narrower usage mask. The common-tuple condition ensures that the vertex discrepancy at y is the same quantity for every old start. Explicitly, if $m_E(y)$ counts the edges of E entering y , then for each i its contribution obeys

$$\frac{1}{n_v u} \mathbb{E} \sum_{y: Y_f(y)=1} m_E(y) |\mathbf{u}_{i,y} - \mathbf{u}'_{i,y}|^s \leq 2k_c \frac{f}{u} \|\mathbf{u}_i - \mathbf{u}'_i\|_{f;s}^s.$$

This is the column-counting step; it uses no independence between the two row tables and the vector estimates.

For a detector trial, Equation (7.4) bounds the trial discrepancy by the supremum of the input score discrepancies over retained starts. Its sth power is at most the sum of their sth powers. The factor $Y_f(y)/f$ in the vertex metric is exactly the factor obtained by reversing the two sums in the row metric with $u = a = f$. Equation (7.21) gives Equation (7.23). $\square$

The required task types are therefore the dense bins $\mathbf{B}_b$ of $\mathbf{C}_l$, the routed bins $\mathbf{B}_b^{(j)}$ of $\mathbf{D}_l$, detector scores $\mathbf{S}_i$, and the pilot and actual correction tables $\mathbf{F}$, together with one earlier unnormalized preparation for each normalized type. Their true conditional-mean identities are Lemmas 7.2 and 7.4. Their computed versions use the same bounded row functions and the assembly contract above. The next sections bound their discrepancies through the additive schedule and implement their accesses without storing two unrestricted vertex addresses.

8 Conditional averaging in the shared environment

The random tables have the right conditional means. The estimator cannot average over every possible environment at every recursive call, so it needs a shorter averaging operation. Moreover,

ordinary averaging is not enough: the queried source and destination must stay in their prescribed retained sets.

Lemma 8.2 is the guarantee used by the estimator. A walk of length d reduces the discrepancy from the appropriate conditional mean exponentially in d , with constants independent of the field size and retention rates. Lemma 8.4 gives a second, computational guarantee: a path through the averaging walk can be described without saving the vertex identifiers whose retention is being conditioned on. The tests involving an identifier can be postponed until the algorithm visits that vertex.

We first prove the mixing guarantee from a uniform spectral gap, and then explain the identifier-independent path description.

The identifiers are fixed when the conditional space is defined. We condition on one retained identifier per conditioned channel; additional independent bit coordinates may be varied or frozen as specified below.

The *full environment law* already includes the frame restriction on each channel. A conditional slice additionally imposes the indicated retention events and fixes any designated frozen bits. Its averaging operator and L_s norms use the normalized probability law on that slice.

The group-theoretic input ensures that a fixed list of transformations mixes uniformly over an entire family of finite spaces. In the representation-theoretic formulation, a vector that changes little under every generator must be close to a vector invariant under the whole group. On a finite transitive action the invariant functions are constants, so this gives a quantitative gap between constants and all mean-zero functions.

The only external group-theoretic inputs are property (T) of $\mathrm{SL}_h(\mathbb{Z})$ for $h \geq 3$ and relative property (T) of

$$(\mathrm{SL}_2(\mathbb{Z}) \ltimes \mathbb{Z}^2, \mathbb{Z}^2).$$

We use Shalom's quantitative statements: [37, Theorem 2.6, p. 156] for the linear group and [37, Theorem 2.1, p. 152] for the relative property. We derive the affine finite-action gap and the conditional sampler from these inputs below.

Lemma 8.1 (A uniform affine spectral gap). *Let $\mathcal{G}_h = \mathrm{SL}_h(\mathbb{Z}) \ltimes \mathbb{Z}^h$, with fixed $h \geq 3$. There is a fixed finite symmetric generating multiset and a lazy rational random walk on it whose action on every finite transitive $\mathcal{G}_h$ -set has a spectral gap bounded below by a positive constant depending only on h .*

Proof. We first control approximate invariance under translations, then under the linear group, and finally convert this control into a mixing bound for finite actions. Let $\mathcal{S}$ consist of the elementary matrices with a single ± 1 off the diagonal, acting linearly, and the translations by $\pm \mathbf{e}_i$, where $\mathbf{e}_i$ are the standard basis vectors of $\mathbb{Z}^h$. This is a fixed finite symmetric generating set of $\mathcal{G}_h$. For a unitary representation ρ and a vector $\boldsymbol{\xi}$, put

$$\delta = \max_{g \in \mathcal{S}} \|\rho(g)\boldsymbol{\xi} - \boldsymbol{\xi}\|.$$

We first bound the distance from $\boldsymbol{\xi}$ to the invariant subspace.

For each coordinate pair $\{i, j\}$, embed the two-dimensional affine group on those coordinates, and let $\mathcal{P}_{ij}$ be the orthogonal projection onto the subspace fixed by its translations. This subspace and its orthogonal complement are invariant under that pair group, because its translation subgroup is normal. The eight generators in [37, Theorem 2.1, p. 152] belong to $\mathcal{S}$. Applying their relative Kazhdan constant $1/10$ on the orthogonal complement gives

$$\|\boldsymbol{\xi} - \mathcal{P}_{ij}\boldsymbol{\xi}\| \leq 10\delta.$$

This is the projection argument in the proof of [37, Corollary 2.3, p. 154].

To combine the two-coordinate estimates, project successively onto their translation-invariant subspaces. The projections $\mathcal{P}_{ij}$ commute: they are limits of averages of commuting coordinate translations. Their product $\mathcal{P}_{\text{tr}}$ projects onto the subspace fixed by all translations. With $q = \binom{h}{2}$, telescoping the product gives

$$\|\xi - \mathcal{P}_{\text{tr}}\xi\| \leq \sum_{i < j} \|\xi - \mathcal{P}_{ij}\xi\| \leq 10q\delta.$$

Since the full translation subgroup is normal, $\mathcal{P}_{\text{tr}}$ commutes with every $\rho(g)$. Its range carries a representation of $\mathcal{G}_h/\mathbb{Z}^h = \text{SL}_h(\mathbb{Z})$. Let $\mathcal{P}_{\text{inv}}$ project onto the full $\mathcal{G}_h$ -invariant subspace, and let $\kappa_h > 0$ be the Kazhdan constant for the elementary generators from [37, Theorem 2.6, p. 156]; adjoining inverse generators preserves the bound. Applying that constant on the range of $\mathcal{P}_{\text{tr}}$ gives

$$\|\mathcal{P}_{\text{tr}}\xi - \mathcal{P}_{\text{inv}}\xi\| \leq \kappa_h^{-1}\delta, \quad \|\xi - \mathcal{P}_{\text{inv}}\xi\| \leq c_{\text{bd},h}\delta, \quad c_{\text{bd},h} = 10q + \kappa_h^{-1}.$$

Thus the same generating set controls distance to invariants in every unitary representation, with a constant depending only on h .

We now apply the representation estimate to the finite space on which the averaging algorithm will run. Let ρ be the permutation representation of a finite transitive action, with the uniform probability measure, and put $m = |\mathcal{S}|$. The rational Markov operator

$$\mathcal{K}_h = \frac{1}{2}\mathcal{I} + \frac{1}{2m} \sum_{g \in \mathcal{S}} \rho(g)$$

is self-adjoint, and its spectrum lies in $[0, 1]$. The invariant functions are exactly the constants. For $\mathbf{f}$ orthogonal to constants, the preceding inequality and the Dirichlet identity give

$$\langle \mathbf{f}, (\mathcal{I} - \mathcal{K}_h)\mathbf{f} \rangle = \frac{1}{4m} \sum_{g \in \mathcal{S}} \|\rho(g)\mathbf{f} - \mathbf{f}\|_2^2 \geq \frac{1}{4mc_{\text{bd},h}^2} \|\mathbf{f}\|_2^2.$$

This proves the uniform spectral gap. □

The conditioning event has a particularly useful coordinate description. Once the nonconstant coefficients and the value at the conditioned identifier are specified, the constant coefficient is determined. Thus the conditional space separates into a full-rank coefficient array and a uniform point in a box. The affine action can mix these coordinates even though their moduli differ.

Lemma 8.2 (Conditional averaging). *Consider the fixed product of rank channels, conditioned on retention of one specified identifier per conditioned channel. Include any designated set of independent uniform bits, with length padded to a multiple of h , and freeze any remaining independent bits. There is a rational Markov averaging operator $\mathcal{K}$ on this conditional space such that its stationary distribution is the indicated uniform conditional law and, for each fixed $2 \leq s < \infty$,*

$$\|\mathcal{K}^d - \Pi\|_{L_s \rightarrow L_s} \leq c_{\text{bd},s} 2^{-c_s d} \quad (d \geq 0).$$

Here Π is conditional expectation over the variables being averaged. The constants are independent of the field size, retention side lengths, bit lengths, indicated identifiers, and frozen bits.

Proof. On a channel conditioned at y with side l_{box} , use coordinates $(\mathbf{F}, \mathbf{w})$, where $\mathbf{F}$ is its field frame and $\mathbf{w} = \mathbf{A}\mathbf{v}(y) \in \{0, \dots, l_{\text{box}} - 1\}^h$. The constant column is uniquely determined by $\mathbf{F}, \mathbf{w}, y$.

As observed above, $\mathbf{F}$ is uniform over frames and $\mathbf{w}$ is independent uniform in its box. Identify this box with $(\mathbb{Z}/l_{\text{box}}\mathbb{Z})^h$. For $(\mathbf{M}, \mathbf{c}) \in \mathcal{G}_h$ act by

$$(\mathbf{F}, \mathbf{w}) \longmapsto (\mathbf{MF}, \mathbf{Mw} + \mathbf{c}),$$

using reduction modulo p_{fld} on the first coordinate and modulo l_{box} on the second.

This action is transitive. Reduction of $\text{SL}_h(\mathbb{Z})$ contains the elementary generators of $\text{SL}_h(\mathbb{F}_{p_{\text{fld}}})$. That finite group acts transitively on ordered frames of length $r < h$: extend either frame to a basis and adjust a complementary column to make the change-of-basis determinant one. Once the frame has been moved, a translation adjusts $\mathbf{w}$ to any specified value. The two moduli need not agree. On an unconditioned channel use a public identifier and side p_{fld} , giving the same description.

If the free bit length is hq , identify its strings with $(\mathbb{Z}/2^q\mathbb{Z})^h$ and use the affine action on that coordinate alone. It is transitive. All currently free bit blocks can be concatenated into this one component; frozen bits are not transformed. There are a bounded number of components. Apply the lazy walk from Lemma 8.1 to a uniformly selected component. If there are j_{comp} components and the single-component gap is at least γ_h , the product walk has gap at least γ_h/j_{comp} . Here j_{comp} is bounded independently of the field and bit lengths. The walk preserves the desired product measure.

Writing its gap as $\gamma > 0$, we have $\|\mathcal{K}^d - \Pi\|_{2 \rightarrow 2} \leq (1 - \gamma)^d$ and $\|\mathcal{K}^d - \Pi\|_{\infty \rightarrow \infty} \leq 2$. The diagonal case of the Riesz–Thorin interpolation theorem, already covered by Riesz [33, Theorems II and II', p. 472], gives

$$\|\mathcal{K}^d - \Pi\|_{s \rightarrow s} \leq 2^{1-2/s} (1 - \gamma)^{2d/s},$$

which has the required form. For Thorin's complex extension, see [39, Introduction, §0.1; Russian translation, pp. 43–45]. $\square$

8.1 Describe a walk before the queried vertex is available

The previous proof describes a walk in coordinates that depend on the conditioned identifier. Saving that identifier across recursive calls would be too expensive. Instead, enumerate a bounded number of possible wrap corrections for each affine step. Each guessed correction defines an invertible transformation of the full environment without using the identifier. When the algorithm reaches the identifier, it checks which guess is valid. Exactly one survives.

This construction is useful twice: it provides short computational descriptions of averaging paths, and its invertibility permits changes of variables in the error analysis.

Algorithm 8.3 Describe a conditional walk without storing its conditioning vertices

Input. A generator word and the common channel parameters.

1. For each affine generator $(\mathbf{M}, \mathbf{c})$ on a box of side l_{box} , enumerate the fixed finite set of possible wrap vectors $\mathbf{z}$. Record the corresponding full-environment map $\mathbf{A} \mapsto \mathbf{MA} + (\mathbf{c} + l_{\text{box}}\mathbf{z})\mathbf{e}_1^\top \pmod{p_{\text{fld}}}$.
2. When the conditioned identifier is visited, check its original membership and whether its transformed integer coordinates, with the guessed wrap, lie in the same box.
3. Retain exactly the variants passing all tests. Give each surviving term the probability of its original generator word; do not divide by the number of wrap guesses.

Output. The sum over valid variants, which equals the specified conditional walk.

Lemma 8.4 (Validated walk variants). *A length- d word in the walk of Lemma 8.2 can be expanded into a bounded-alphabet family of variants described by $O(d)$ bits such that:*

1. *each fixed variant acts bijectively and measure-preservingly on the full environment before any endpoint tests;*
2. *its transformation is independent of the indicated identifiers;*
3. *for each walk word and each environment in the required conditional slice, exactly one variant passes the slice and wrap tests at the indicated identifiers, and this variant gives the conditional walk transformation;*
4. *the sum of valid variants, each with the probability of its original walk word, is exactly the conditional average.*

The tests can be performed when their indicated vertices are visited. In the admissible implementation range, field elements and identifiers have $O(b_{\text{mem}})$ bits, the total auxiliary bit string has $O(b_{\text{mem}})$ bits, and $d = O(b_{\text{mem}})$. In that range the transformations and tests use $O(b_{\text{mem}})$ shared scratch space, and the retained description of a variant uses $O(d)$ bits.

Proof. The only missing information is how the affine image wraps around the retention box. Since every generator has fixed coefficients, only a bounded number of wraps can occur in one step. Consider a fixed affine generator $(\mathbf{M}, \mathbf{c})$ on a channel with retention side l_{box} . Enumerate the transformations

$$\mathbf{A}' = \mathbf{M}\mathbf{A} + (\mathbf{c} + l_{\text{box}}\mathbf{z})\mathbf{e}_1^T \pmod{p_{\text{fld}}}, \quad (8.1)$$

where $\mathbf{e}_1$ selects the constant column and $\mathbf{z}$ ranges over a fixed finite box of integer vectors. That box contains all possible wraps: for $\mathbf{w} \in \{0, \dots, l_{\text{box}} - 1\}^h$, every coordinate of $\mathbf{M}\mathbf{w} + \mathbf{c}$ is bounded in absolute value by a fixed multiple of l_{box} plus a fixed constant. Since $l_{\text{box}} \geq 1$ and the generators are fixed, one bound for $\mathbf{z}$ works for every side length.

Each transformation in Equation (8.1) is invertible. Left multiplication preserves the frame condition, and the added term changes only the constant column. It therefore preserves uniform measure on the full channel. It does not refer to the indicated identifier.

At that identifier, let $\mathbf{w} = \mathbf{A}\mathbf{v}(y)$ in its integer representatives. Check first that $\mathbf{w}$ is in the retention box, and then that $\mathbf{M}\mathbf{w} + \mathbf{c} + l_{\text{box}}\mathbf{z}$ lies in the same box coordinatewise, using integer arithmetic before reduction modulo p_{fld} . The unique valid wrap has coordinates $\mathbf{z}_i = -\lfloor (\mathbf{M}\mathbf{w} + \mathbf{c})_i / l_{\text{box}} \rfloor$. For this vector the evaluation of $\mathbf{A}'$ is the representative of the intended affine action modulo l_{box} . Applying these checks at every step gives the unique valid wrap sequence for a whole word.

The construction applies separately to the bounded number of channels. On a channel of side p_{fld} the ordinary field reduction suffices; alternatively, the same wrap validation may be used at its public point. The free-bit torus has ordinary invertible affine operations and requires no unknown endpoint. Frozen bits remain fixed. Thus a word has only constantly many generator and wrap choices per step, giving the $O(d)$ -bit description.

The weights are important. These guesses are alternative descriptions of one walk step, not extra random choices that should be averaged uniformly. For each word and initial environment in the slice, exactly one variant survives. Its weight must be the weight of that word. In particular, the sum is not divided by the number of guessed wraps. This proves the asserted exact identity. All matrices have fixed dimensions and $O(b_{\text{mem}})$ -bit entries; evaluating a bounded-degree polynomial or a word's individual steps needs $O(b_{\text{mem}})$ arithmetic scratch. A test depending on a vertex can be delayed until that vertex is current, with the required matrix state regenerated from the recorded transformations. Theorem 13.1 accounts for reversible restoration and the simultaneous space used when these operations are nested. $\square$

Remark 8.5 (Effective environment coordinates). Conditional averaging is expressed in the channel and bit coordinates of the environment supplied to that invocation. A fixed variant from Lemma 8.4 is a measure-preserving change of variables on the full environment; its retention and wrap tests remain explicit masks. Thus free and frozen bits always refer to the supplied coordinates, which may differ from those before the change of variables.

Remark 8.6. Two distinct uses of the variants should be kept separate. For centered-noise estimates, one first recognizes the sum of valid variants as the conditional walk and applies Lemma 8.2. For crude bounds and changes of variables, each fixed variant is a measure-preserving bijection, and one may then pay for their $2^{O(d)}$ number. Counting variants before a spectral estimate would needlessly weaken the useful contraction.

9 Estimating by successive accuracy improvements

We now supply the averaging schedule described in Section 6. The exact sparse-table identities tell us what to average. The harder question is how accurate each recursive input must be, and how much averaging to apply to it.

The plan is to use estimates at successive child accuracy levels and average their differences. Before giving the schedule, we calculate the cost of changing a bin’s scale or moving the source of a query. Those costs determine the precision assignment. We then prove separate bounds for the error in the conditional mean and the noise left by a finite averaging walk.

The prefixes analyzed here are intermediate raw arrays. The full row estimator applies compression after each group and the sparse-table functions after the final group. Its complete evaluation order was given in Section 6; the destination-grouping rule is specified in Section 11. Here we first analyze the averaging part with exact destination grouping.

9.1 Tasks and dependency order

A row task specifies its input matrix, stage, iteration label, start rate u , end scale a , optional usage rate $t \leq a$, channel roles, and fixed padding suffixes. These parameters are common to its whole table. A vector task specifies the analogous vertex domain. Its argument is a vertex, rather than an additional parameter retained by another task. Each task has a nonnegative integer accuracy level m ; r will denote a child accuracy level. These accuracy levels are distinct from the numerical digit places introduced in Section 12. The ordinary and root discrepancy norms are defined in Equations (7.19) and (7.20). We always take $s = 8$.

The source weight $X_u(x)/u$ has bounded expectation, and the factor $1/n_v$ normalizes the sum over source vertices. Endpoint sums remain explicit: bounded row supports or incoming degrees, as appropriate, will control them. A bound at one specified start consequently costs a factor n_v in moment power; Lemma 14.3 makes and pays this conversion at the end of the proof.

Here is a dependency order, with only a fixed number of task types at each stage. First come the bin tasks for $\mathbf{C}_l$. At each pilot step come its input-score task $\mathbf{S}_i$, the detector vector $\mathbf{u}_i$, and then the pilot correction task. Next come the actual correction tasks in iteration order, the routed bin tasks for $\mathbf{D}_l$, and the success contribution update. Place the unnormalized preparation immediately before each normalized task of the same input. The raw first channels of both these tasks refer only to earlier inputs in this order; the normalized task has in addition its preparation as the source of its second channel.

For a correction task or its preparation, first complete its raw arrays and pre-incoming scores,

then form its incoming gate from those scores. This assembly uses the current task's completed arrays; every recursive input to their raw calculation is an earlier task in the order just specified.

9.2 Which expressions are averaged

There are only a few sample formulas to check: a direct entry, a correction followed by a bin, a copy projection, a preparatory mean, a detector trial, and a success contribution update. The retention-probability ratios below perform changes of conditioning. They are needed even though the nominal rates are comparable to the actual probabilities; an approximate ratio would lose the exact mean identity.

For a fixed row and endpoint let

$$\mathcal{C}_{xy}^{u,a} = \{X_u(x) = 1, Y_a(y) = 1\}.$$

We use conditional means and conditional walks on this slice. Write $\mathbf{B}_b$ for a normalized bin table and $\mathbf{B}_b^{(j)}$ for a bin with destination routing fraction already included; $\mathbf{F}$ denotes a normalized correction table. Their values are in atom units and are zero off their specified membership domains. In all formulas below put $\tau = \min(a, b)$.

The direct dense and correction-dense product terms, before division by the current first-channel scale a , are respectively

$$\mathbf{T}_{b,xy} = b \frac{\pi_a}{\pi_\tau} \mathbf{B}_{b,xy}, \quad (9.1)$$

$$\mathbf{T}_{b,xy}^{\text{prod}} = b \frac{\pi_a}{\pi_\tau} \frac{f}{\pi_f} \sum_{z:Z_f(z)=1} \mathbf{F}_{xz} \mathbf{B}_{b,zy}. \quad (9.2)$$

In the product, the left task has roles X, Z and rates $(u; f, f)$, listed as start, end, and usage. The right task has roles Z, Y and rates $(f; b, \tau)$. In the direct term the start rate stays u . The term $-\mathbf{E}$ in $\mathbf{D}$ is the negative of Equation (9.1) with $b = f$ and $\mathbf{F}$ in place of $\mathbf{B}_b$. Summing the direct and product terms gives the raw inputs $\mathbf{G}_i, \mathbf{G}'_i$; adding the negative term gives $\mathbf{D}$.

For a copy projection x_i, y_j from stage $l + 1$ to stage l , the raw term is

$$\mathbf{r}_{i,x} b \frac{\pi_a}{\pi_\tau} \mathbf{B}_{b,xy}^{(j)}. \quad (9.3)$$

Partition by the fixed symbols i, j . Append those symbols to the native hash arguments before the inherited suffixes on their respective channels. The parent-stage source multiplier is obtained from its detector vectors on the broad parent-stage vertex domain. No child tuple contains the old source or a particular right endpoint as an extra parameter.

For the second raw channel, use ζ times the preparatory table, take its conditional average on $\mathcal{C}_{xy}^{u,t}$, and extend the result by zero off $Y_t(y)$. For a detector vector use the trial

$$\text{rank}_{j_{\text{det}}} \{\mathbf{S}_{i,xy} : X_f(x) = 1\} \quad \text{on the slice } Y_f(y) = 1. \quad (9.4)$$

Ranks include arbitrary zero padding. For success contributions, writing $f = b_l$, the two terms for $\mathbf{w}_{l+1,x_i}$ on its inherited source slice are

$$\mathbf{r}_{i,x} \mathbf{w}_{l,x}, \quad \mathbf{r}_{i,x} \frac{f}{\pi_f} \sum_{z:Z_f(z)=1} \mathbf{F}_{xz} \mathbf{w}_{l,z}. \quad (9.5)$$

The first $\mathbf{w}_l$ call keeps the inherited source domain; the second restarts on the midpoint domain. Vector answers are finally clipped into $[0, 1]$.

Lemma 9.1 (True sample means). *Using true stochastic tables in Equations (9.1)–(9.5) gives, after the indicated conditional averages, the prescribed raw inputs and vector values on their conditioning slices, with the stated zero extensions. In particular, the first row channel has target $\mathbf{d}_y = \mathbf{G}_{xy}/a$ and the second has target $\zeta \mathbf{q}_y \mathbf{1}_{Y_t(y)}$.*

Proof. First consider how the destination’s retention condition changes with scale. If $b \geq a$, the usage hit is already the conditioning hit. The conditional normalization of a bin therefore turns $b\mathbf{B}_b$ into its deterministic bin contribution. If $b < a$, nested origin boxes give

$$\mathbb{E} \left[\frac{\pi_a}{\pi_b} \mathbf{1}_{Y_b(y)} V \mid \mathcal{C}_{xy}^{u,a} \right] = \mathbb{E}[V \mid \mathcal{C}_{xy}^{u,b}]$$

for every integrable V on the narrower slice. Thus the importance factor performs exactly the required change of conditioning.

For a product, we must justify separately why the two true normalizations can be used. For a fixed midpoint in Equation (9.2), first make this terminal change when necessary and then condition on $Z_f(z) = 1$. Apart from membership indicators, the true right bin gate and denominator use only channel Y . The true left correction uses only channels X, Z . These channels are independent, and the remaining true multipliers are deterministic. The two normalizations can therefore be applied separately. In particular, after the midpoint hit is fixed, the shared role Z does not occur in the right bin’s remaining random value. This is an identity for the true atoms; the estimates will instead be compared by their joint error norms. The factor f/π_f cancels midpoint sampling and converts its correction atom to the correction entry. Summing midpoints proves the product formula. The projection multiplies by deterministic $\mathbf{r}_{i,x}$ and includes the routing fraction in the bin, so the same argument applies on each fixed-symbol part. Fixed-suffix invariance in Lemma 4.1 justifies all inherited padding conventions.

The second channel is its defining preparatory conditional mean, on its own narrower slice; it has no importance factor. The detector mean is the definition of $\mathbf{u}_i$. True success contribution factors are deterministic, so the same midpoint argument proves Equation (9.5). $\square$

To form an accuracy level- r sample term, replace all earlier inputs in the corresponding formula by accuracy level- r estimates, with their domains and common task parameters unchanged. This includes the detector inputs to $\mathbf{r}_i$; the primary factor $\mathbf{r}_0 = 1$ remains constant. Products, source functions, and ranks are evaluated before averaging. We first compare this sample’s conditional mean with the same formula using true atoms. The scale and source-domain costs of that comparison will determine how much accuracy an accuracy level must provide.

9.3 Mean comparisons and the cost of releasing a source

Lemma 9.1 identifies the exact conditional means. For estimated factors, the comparison has two distinct costs. The factor b/a converts a bin contribution to the current entry units. A right factor at a midpoint also replaces the old source domain of rate u by the stage domain of rate f . The following estimates account for both changes without assuming independence of the errors.

The following scalar loss packages the comparison in a form that behaves well under averaging. Its denominator discounts an error in proportion to the magnitude of the values being compared. It is equivalent to the eighth-root discrepancy already used for the sparse tables, but its homogeneity makes the scale calculation more transparent.

For nonnegative h, h' define

$$\mathcal{L}(h, h') = \begin{cases} |h - h'|^s / (h + h')^{s-1}, & h + h' > 0, \\ 0, & h = h' = 0. \end{cases} \quad (9.6)$$

Lemma 9.2 (Homogeneous loss). *The loss in Equation (9.6) is comparable, by constants depending only on s , to $|h^{1/s} - (h')^{1/s}|^s$. It contracts under sums and nonnegative mixtures. If $h \leq c_{\text{bd},0}g$ for a nonnegative g , then for every $a > 0$ and $h' \geq 0$,*

$$a \min \left\{ 1, \frac{|h - h'|}{a + g} \right\}^s \leq c_{\text{bd}} \mathcal{L}(h, h'), \quad (9.7)$$

where c_{bd} depends only on $s, c_{\text{bd},0}$.

Proof. Put $v = \max(h^{1/s}, (h')^{1/s})$ and $w = \min(h^{1/s}, (h')^{1/s})$. Factoring $v^s - w^s$ bounds it between $v^{s-1}(v - w)$ and $sv^{s-1}(v - w)$; the denominator is between $v^{s(s-1)}$ and $2^{s-1}v^{s(s-1)}$. This proves comparability. For a mixture, apply Hölder to $|h - h'| = \mathcal{L}(h, h')^{1/s}(h + h')^{(s-1)/s}$ and then integrate; the triangle inequality first bounds the absolute difference of the integrals. This proves contraction, including finite sums.

If $h' \leq 2h + a$, then $h + h' \leq 3c_{\text{bd},0}g + a \leq c_{\text{bd}}'(a + g)$, which proves Equation (9.7) directly. If $h' > 2h + a$, then $|h - h'| > h'/2$, $h + h' < 3h'/2$, and $h' > a$. Thus $\mathcal{L}(h, h') \geq ch' \geq ca$, while the left side is at most a . $\square$

Every unsigned true term of the signed formula for $\mathbf{D}$ is bounded by a constant times $\mathbf{D}$, by Lemma 5.5; the corresponding assertion for the other raw inputs follows from their nonnegative sums. Consequently the preceding lemma can be applied separately to all true contributions. Triangle inequalities combine their clipped relative errors even when a contribution has a negative sign.

Lemma 9.3 (Scale and domain comparison). *Compare an accuracy level- r sample term with the same term using true stochastic inputs. For a first-channel mean contribution use its clipped relative error against $1 + \mathbf{d}_y$. In the bound for its s th-power norm, the scale factor multiplying a child root-discrepancy power is, up to fixed constants,*

$$\begin{cases} b/a, & b \geq a, \\ (b/a)^{s-1}, & b < a. \end{cases}$$

A child that releases the old source and restarts on a stage-rate f domain costs in addition at most f/u in norm power. Direct children retaining that source have no such factor. The same-scale vector and second-channel comparisons have constant scale cost, with f/u only where a source is released. No independence of estimated factors is required.

Proof. We treat coarse and fine bins separately because changing the conditioning contributes a different power of the scale ratio in the two cases. We then check vectors and preparatory means.

First suppose $b \geq a$. Apply Equation (9.7) after conditional expectation, and then the mixture contraction of Lemma 9.2 before it. Homogeneity supplies a factor b/a after division by a . The factor f/π_f in a product is bounded above and below by fixed constants. For bounded nonnegative atoms a product-root difference splits into the factor-root differences, multiplied by fixed constants times support indicators of the unaffected factor in one of the two versions.

The source-domain factor is a counting issue, not an independence claim. An error at a midpoint can be reached from only boundedly many old sources through a correction access. To determine this factor, write $\Delta_s \mathbf{F}$ and $\Delta_s \mathbf{B}_b$ for the root discrepancies. After integration on the source/end slice, the relevant sums, besides $O(b/a)/n_v$, have the forms

$$\begin{aligned} \mathbb{E} \sum_{x,z,y} \frac{X_u(x)}{u} Z_f(z) Y_a(y) |\Delta_s \mathbf{F}_{xz}|^s \mathbf{1}_{\mathbf{B}_{b,zy} \neq 0}, \\ \mathbb{E} \sum_{x,z,y} \frac{X_u(x)}{u} Z_f(z) Y_a(y) \mathbf{1}_{\mathbf{F}_{xz} \neq 0} |\Delta_s \mathbf{B}_{b,zy}|^s. \end{aligned}$$

Either support indicator can come from either compared version. In the first sum the right row cap eliminates y at constant cost. In the second sum the left incoming cap eliminates x at cost $O(1/u)$. The right child's row metric has weight $1/f$, leaving exactly $O(f/u)$. Its tuple is the same for every old source, so this elimination compares the same right error throughout. More explicitly, for either left support relation $\mathcal{P}$ with incoming degree at most k_c and a common right discrepancy matrix $\mathcal{E}$, the relevant inequality is

$$\begin{aligned} & \frac{1}{n_v} \mathbb{E} \sum_x \frac{X_u(x)}{u} \sum_{z:(x,z) \in \mathcal{P}} Z_f(z) \sum_{y:Y_b(y)=1} |\mathcal{E}_{zy}|^s \\ & \leq \frac{k_c}{n_v u} \mathbb{E} \sum_z Z_f(z) \sum_{y:Y_b(y)=1} |\mathcal{E}_{zy}|^s \\ & = k_c \frac{f}{u} \|\mathcal{E}\|_{f,b;s}^s, \end{aligned}$$

where the source channel in the last norm is Z . The incoming cap applies pointwise, so the discrepancy can depend on the same environment as $\mathcal{P}$. Using $Y_a \leq Y_b$ enlarges the remaining endpoint domain to the child's own domain. A direct term simply retains its source and endpoint sums. In a projected term, the source-multiplier error eliminates y by a row cap and changes the vertex weight from $1/u$ to $1/f_{\text{par}}$; this costs $O(f_{\text{par}}/u)$. Partitioning the fixed copy symbols keeps every tuple common during these sums.

For fine bins, using the coarse-bin estimate would miss the useful small factor. For $b < a$, change the terminal conditioning to Y_b *before* taking a power. The normalized coefficient inside is then b/a . There are only constantly many nonzero left paths in the union of the two rows. Ordinary bounded-factor difference inequalities and the finite-sum power inequality therefore apply with constant cost. The integrated change from Y_a to Y_b leaves π_a/π_b outside the power. Its total factor is

$$(b/a)^s \frac{\pi_a}{\pi_b} = O((b/a)^{s-1}).$$

The same row and incoming sums as above then apply, now with Y_b . Ordinary differences of bounded nonnegative values are controlled by their root differences. Absolute mean error bounds clipped relative mean error, so this proves the fine-bin assertion.

For the second raw channel, Jensen is applied directly on $\mathcal{C}_{xy}^{u,t}$. Its integrated endpoint mask is Y_t , already contained in the preparatory norm's Y_a domain. There is no importance factor and hence no π_a/π_t penalty. The fixed scaling ζ does not increase the cost.

A rank discrepancy is at most the supremum of its input discrepancies. For the detector trial, summing its power over Y_f/f is bounded by the score error sum over X_f/f and Y_f ; the rates match exactly. For success contributions, the finite left-path count gives an ordinary product comparison, and the same incoming cap eliminates old starts for a right-vector error. Source functions have bounded Lipschitz roots in their clipped detector inputs, so their costs are the broad vertex costs already computed. Every estimate here is pointwise or follows from Jensen and summation; none uses independence of its errors. $\square$

The preceding comparison measures errors in a retained sample term. Omitting a fine bin instead discards its whole true contribution; the next bound measures that loss.

Lemma 9.4 (Fine omission). *If one fine bin is omitted, the norm of its normalized contribution is at most $c_{\text{bd}}(b/a)^{(s-1)/s}$. This applies to direct terms, routed terms, and correction-bin products.*

Proof. The deterministic contribution $\mathbf{H}$ has entries at most $c_{\text{bd}}b$ and bounded row mass. For a product these bounds follow from Lemma 7.3. Therefore

$$\sum_y \pi_a |\mathbf{H}_{xy}/a|^s \leq \frac{\pi_a}{a^s} (c_{\text{bd}}b)^{s-1} \sum_y \mathbf{H}_{xy} \leq c_{\text{bd}}' (b/a)^{s-1}.$$

Integrating source hits gives $\pi_u/u = O(1)$, and summing at most n_v rows cancels the normalization $1/n_v$. $\square$

9.4 Choosing precision to pay for scale and domain changes

The preceding bounds explain the rate weights in the accuracy parameter. Write $h_b = \log_2(b/a)$ for the offset of a bin scale. In norm, the coarse-bin cost is $(b/a)^{1/s}$ and the fine-bin cost is $(b/a)^{(s-1)/s}$. Since $s = 8$, a weight $1/2$ between these two exponents leaves a geometric margin in both directions. The same weight pays for the source-release cost $(f/u)^{1/s}$. We now make this accuracy convention precise; the additive schedule will then trade lower child accuracy levels for longer averaging walks.

Assign positive integer ranks i_* in the task order fixed above, so every dependency decreases rank by at least one. The finite number of pilot steps and of preparations is an absolute constant, so $i_* = O(l+1)$. A task at integer accuracy level $m \geq 0$ has implicit precision

$$p_{\text{acc}} = \frac{m}{k_{\text{prec}}} - \frac{1}{2} \log_2 \frac{1}{a} - \frac{1}{2} \log_2 \frac{1}{u} - a_* i_*. \quad (9.8)$$

For a stage- l vector use $a = f = b_l$; detector vectors also have $u = f$, whereas success contributions retain their inherited start rate. The constants k_{prec} and a_* will be chosen below. If $p_{\text{acc}} \leq 0$, return the zero table or vector and empty access classes. Otherwise the task is called active. The intended induction bound is $c_{\text{ind}} 2^{-p_{\text{acc}}}$ for every required ordinary vector norm and every required root discrepancy norm, including the pre-incoming scores and witnesses. A fixed c_{ind} makes this assertion trivial for inactive tasks: values and row supports are uniformly bounded, and $\mathbb{E}X_u/u = \pi_u/u = O(1)$.

At a normalized correction task, the column multiplier is assembled from earlier detector-vector estimates at accuracy level $m - c_{\text{col}}$, for a fixed positive integer c_{col} . This is a separate final assembly call. All other calls used in one additive sample term have the common accuracy level denoted by r .

Suppose a child at accuracy level r has parameters a', u', i'_* . Subtracting Equation (9.8) for the parent gives the exact identity

$$p_{\text{acc}}' - p_{\text{acc}} = -\frac{m-r}{k_{\text{prec}}} + \frac{1}{2} \log_2 \frac{a'}{a} + \frac{1}{2} \log_2 \frac{u'}{u} + a_*(i_* - i'_*). \quad (9.9)$$

In a bin comparison $a' \geq cb$ for a fixed $c > 0$: a bin child has $a' = b$, while a correction has $a' = f$ and $b \leq k_f f$. The start is either retained or reset to the indicated broad stage rate. Fixed-stage changes in vector calls only change fixed constants. Thus, after taking s -th roots of the costs in Lemma 9.3, the scale contribution times child accuracy is bounded by

$$c_{\text{bd}} c_{\text{ind}} 2^{-p_{\text{acc}} - a_* + (m-r)/k_{\text{prec}} - 3|h_b|/8}. \quad (9.10)$$

Indeed for $h_b \geq 0$ the exponent is $h_b/s - h_b/2 = -3h_b/8$; for $h_b < 0$ it is $(s-1)h_b/s - h_b/2 = 3h_b/8$. For a child that restarts at rate f , the source-domain comparison costs $(f/u)^{1/s}$ in norm,

while its precision gains $\frac{1}{2} \log_2(f/u)$. Multiplying the cost by the resulting accuracy factor gives $(f/u)^{1/s-1/2} = (f/u)^{-3/8} \leq c_{bd}$, using the allowed start-rate condition $u \leq f$.

For the separate final column-vector call, $a = f$, its child uses $a' = u' = f$, and $m - r = c_{col}$. Thus its norm cost times child accuracy is at most

$$c_{bd}c_{ind}2^{-p_{acc}}2^{c_{col}/k_{prec}-a*}(f/u)^{-3/8}.$$

The fixed rank separation pays this fixed accuracy level loss. This calculation also explains why the source-domain ratio is charged in the error analysis rather than inserted as a numerical coefficient in a sample.

9.5 Average differences and group them by accuracy level decrease

We can now choose the schedule using the precision comparison just proved. A direct high-level child call would save little accuracy level for the caller's own work. Instead, express a high-level sample as a sum of successive improvements. Low-level improvements can tolerate long averaging walks; later improvements are smaller because their two child estimates are more accurate.

Every level-zero task is inactive under the preceding precision rule. Hence an level-zero sample term is zero: it contains at least one estimated table or vector factor. Exact base terms $\mathbf{S}_{xy}/a$ and $\mathbf{w}_{0,x}$ are inserted separately.

For every first-channel term labelled by scale b , set $c_b = \lceil c_0(1 + |h_b|) \rceil$, where c_0 is a fixed positive constant. This rule includes the sparse direct term $-\mathbf{E}$ with $b = f$, as well as direct bins, products, and projection terms. Every child factor in a term, including the detector-vector inputs to a projected source multiplier, uses that term's common accuracy level r and cutoff c_b . Use zero scale offset only for second-channel preparatory means, detector-vector trials, and success contribution terms with fixed stage-rate changes. The separate final column assembly retains its fixed drop c_{col} . Denote the corresponding normalized sample term at accuracy level r by $\mathbf{T}_{b,r}$, including division by a for the first row channel. For $1 \leq r \leq m - c_b$ form

$$\mathbf{T}_{b,r} - \mathbf{T}_{b,r-1}. \quad (9.11)$$

Apply to this difference the conditional walk of length $\ell(m - r)$, where ℓ is a sufficiently large fixed integer. Use the same generator word and the same guessed wraps for its two parts.

We call $m - r$ the slack of a child call: it is the amount by which its accuracy level has decreased. The same decrease will later pay for the description of the call. Group increments by dyadic slack $d \leq m - r < 2d$. Add groups in decreasing powers of two $d \leq m$ down to 1, including empty groups. Put exact base contributions in the first group. Let $\mathbf{s}_j$ be the uncompressed prefix through the group with slack d_j . The conditional mean prefix is obtained by replacing each walk by uniform averaging on its slice; noise is the difference from that mean. For an available bin the last accuracy level reached by this mean prefix is

$$r_j(b) = \max\{0, m - \max(d_j, c_b)\}. \quad (9.12)$$

This exact cutoff will matter in the error bound. Bins with $c_b \geq m$ have no layers and are omitted.

The walk is applied to an array indexed by actual endpoints. Thus, for each term and pair, its input is the single function

$$\mathbf{F}_{xy}(\sigma) = \mathbf{T}_{b,r,xy}(\sigma) - \mathbf{T}_{b,r-1,xy}(\sigma),$$

with all contributions at that endpoint already summed. A path description is only a way to enumerate those contributions. In particular, the two accuracy levels must be grouped at the same actual endpoint before their difference is measured. The following lemma establishes this identity for the path enumeration and also records the description length needed later by the controller.

Lemma 9.5 (Immediate paths). *At a row group of slack d , immediate candidates have descriptions of $O(d)$ bits. With exact endpoint grouping their sums equal the specified conditional-walk increments. A product visits its midpoint by a paired correction move before using any factor at that midpoint. Its right-hand task tuple does not depend on the discarded old source. Every child estimator in such a candidate has accuracy level at most $m - d$.*

Proof. We must check both what a path describes and why summing those paths gives the intended average. A descriptor records the finitely many formula and copy-part choices, its scale and accuracy level offsets, its sign in Equation (9.11), the generator word and guessed wraps, and a bounded number of constituent ports. An included scale obeys $|h_b| = O(d)$, and the walk has length $O(d)$. Lemma 8.4 therefore bounds all these records by $O(d)$ bits. Port alphabets are fixed by Lemma 7.7. There are only $O((1 + d)^2)$ choices of accuracy levels and scales before the exponential word/variant choices.

Check each constituent port, the outer source and endpoint memberships, and the guessed wraps at the vertices where their coordinates can be evaluated. For a fixed endpoint and original generator word, unique wrap validation leaves exactly its genuine conditional transformation. Give the surviving term its original walk-word probability, multiplied by the formula coefficients and sign. Guessed wraps do not change that probability. Constituent ports enumerate entries once. Hence summing over true equal endpoints is precisely the walk of the difference, even though its endpoint was not known when the variant was specified.

The first factor of every product is a correction, whose retained ports have bounded degree in both directions. Its paired move supplies the midpoint and a bounded return token. The right factor can then be queried using the common Z, Y tuple prescribed above; that tuple contains no old source. A direct unbalanced terminal motion has no further factor query at its endpoint. These are combinatorial access requirements; the controller section implements the pairing and terminal operations. Finally, the slack definition gives $r \leq m - d$, also for the $r - 1$ term. $\square$

All reachable rate logarithms remain $O(b_{\text{mem}})$ when $m_{\text{top}} = O(b_{\text{mem}})$: a scale extension at slack d changes a reciprocal-rate logarithm by $O(d)$ and reduces accuracy level by at least d . Stage resets have only their fixed stage rates, usage rates are minima of available rates, and inherited padding has at most one symbol per stage above the task. Summing along a dependency chain bounds all extensions by $O(m_{\text{top}})$. The field in the shared environment can therefore be chosen large enough for every used scale, including level-zero endpoints of available differences. Omitted finer deterministic contributions need no hash tables.

We can now estimate each mean prefix by telescoping the conditional means to accuracy level $r_j(b)$. Its deterministic omission error and its child-estimation error are treated separately. The remaining noise will be bounded by applying the conditional spectral estimate to the endpoint functions $\mathbf{F}_{xy}$ above.

Lemma 9.6 (Mean prefixes). *Assume all earlier table and vector bounds with coefficient c_{ind} . For an active task there is a fixed b_{bias} , independent of the later choice of a_* , such that every mean prefix has norm at most $c_* 2^{-p_{\text{acc}} + b_{\text{bias}} d_j}$ against its target. For the first channel the norm uses clipped relative discrepancy; for the second and for vectors it uses ordinary absolute discrepancy. The coefficient c_* can be made arbitrarily small by increasing a_* .*

Proof. There are two sources of mean error. At a scale included in the schedule, telescoping stops at a finite child accuracy level. At an omitted scale, the entire deterministic contribution is missing. We bound them separately.

Available scales. For an available scale, conditional expectations of Equation (9.11) telescope to its term at $r = r_j(b)$ in Equation (9.12). If this accuracy level is zero, the term is zero and

the inactive-child bound still applies. More generally, the child bound remains valid whenever its precision is nonpositive, by the bounded-output estimate for inactive tasks. In every case

$$m - r \leq d_j + c_0(1 + |h_b|) + 1.$$

Choose k_{prec} so that $c_0/k_{\text{prec}} < 1/8$. Equation (9.10) is then bounded by

$$c_{\text{bd}}' c_{\text{ind}} 2^{-p_{\text{acc}} - a_* + d_j/k_{\text{prec}}} 2^{-|h_b|/4}.$$

The rate grid is geometric, so summing this bound over available scales costs a fixed constant. There are only a fixed number of formula and factor types. Unbinned calls use the same calculation without the scale sum. Minkowski and subadditivity of clipped relative discrepancies combine all these terms. Exact base contributions are already present in every prefix and have no error.

Wholly omitted scales. A bin with $c_b \geq m$ contributes no layer. We bound its deterministic contribution directly by Lemma 9.4. First we show that every such bin is fine. Put $h_a = \log_2(1/a) \geq 0$. A coarse bin has $0 \leq h_b \leq h_a$ because all grid rates are at most one. Activity gives $m > k_{\text{prec}}(h_a/2 + a_* i_*)$, even after discarding the nonnegative start-rate cost. With $k_{\text{prec}}/2 > c_0$ and a_* sufficiently large, this implies $m > \lceil c_0(1 + h_a) \rceil$, so a wholly omitted bin is fine. For the sparse direct term $b = f$, the same argument gives availability when $f \geq a$. When $f < a$, the admissibility condition $a \leq k_f f$ gives $|h_f| \leq \log_2 k_f$, and increasing a_* ensures $m > c_f$. Thus this term is never wholly omitted in an active task. The inequality $c_b \geq m$ implies

$$|h_b| \geq (m - 1)/c_0 - 1.$$

Lemma 9.4 and a geometric sum consequently bound all such omissions by $c_{\text{bd}} 2^{-(s-1)m/(sc_0)}$. Since $p_{\text{acc}} \leq m/k_{\text{prec}}$ and $k_{\text{prec}} > sc_0/(s-1)$, this is at most $c_{\text{bd}} 2^{-p_{\text{acc}} - \eta m}$ for some fixed $\eta > 0$, after absorbing the fixed rounding offset. Finally $m > k_{\text{prec}} a_* i_* \geq k_{\text{prec}} a_*$ makes its coefficient arbitrarily small as a_* increases. Taking, for example, any fixed $b_{\text{bias}} > 1/k_{\text{prec}}$ proves the claim, including all losses from the last included level and from omitted levels. $\square$

9.6 Absolute difference bounds and averaging noise

The conditional-mean errors are now bounded. To control the noise left by finite averaging, we need an absolute norm bound on each paired difference before the walk is applied. This coarser bound will also be used for the fingerprint comparison.

Lemma 9.7 (Absolute differences). *At a slack- d group, the ordinary raw norm of an unaveraged paired difference is at most*

$$c_{\text{bd}} c_{\text{ind}} 2^{-p_{\text{acc}} - a_* + c_{\text{bd},0} d}, \tag{9.13}$$

where $c_{\text{bd},0}$ is fixed independently of the walk length coefficient. For two coupled versions of an accuracy level- r sample term, if the corresponding child discrepancies are bounded by $2^{-p_{\text{acc}}'} \lambda_r$, their normalized raw difference is bounded by $2^{-p_{\text{acc}} + c_{\text{bd},0}' d} \lambda_r$, summed over the finitely many child types. For a paired layer at accuracy levels r and $r-1$, apply this estimate to both parts. Its coupled discrepancy is consequently bounded by $2^{-p_{\text{acc}} + c_{\text{bd},0}' d} (\lambda_r + \lambda_{r-1})$, after enlarging the fixed constant, or by the corresponding maximum over all children in the group. Every child in either part has accuracy level at most $m - d$. For a fixed guessed measure-preserving variant and its valid endpoint restriction the same bounds hold. Counting all variants separately may increase $c_{\text{bd},0}'$ by a constant depending on the chosen walk length.

Proof. The useful cancellation comes from comparing the two child accuracy levels with one common true sample, not by bounding their magnitudes separately. Subtract the *same stochastic true-atom term* from the accuracy level- r and accuracy level- $(r - 1)$ terms, and use the triangle inequality. A product has only constantly many left paths in the union of the compared rows. Ordinary bounded-factor differences and the row and incoming caps therefore give the sums used in Lemma 9.3, without any averaging. For a coarse bin the absolute coefficient costs b/a in the norm rather than its s -th root. For a fine bin the importance factor can stay inside the power. Both are bounded by $2^{O(|h_b|)}$. Equation (9.9) pays the only separate possibly large domain ratio, f/u , as before. All other scale and accuracy changes are $2^{O(d)}$ because $|h_b| = O(d)$ and $m - r = O(d)$ in this group. Ranks and bounded source functions obey the same pointwise comparisons. This proves Equation (9.13). Replacing comparison to truth by a comparison of the two coupled child versions gives the second assertion by the identical inequalities; independence is still unnecessary.

For a fixed guessed variant, valid paths require the transformed memberships. Change variables by its full-environment measure-preserving bijection and then discard surplus validity guards. This leaves exactly the child-domain sums just bounded. When combining the paired terms, combine their contributions at the same actual endpoint before measuring the difference. The number of distinct variant descriptors is $2^{O(d)}$, so treating them separately merely enlarges the crude exponent. The estimate before such separate counting is a statement about the single-seed sample function and has no dependence on the eventual walk length coefficient. $\square$

Lemma 9.8 (Conditional prefix bounds). *Assume exact local grouping, the earlier task error bounds, and that the sample functions depend only on variables varied by the conditional walk. Their uniform slice means are then deterministic at each indicated pair or vertex. Write $\mathbf{s}_j^{\text{mean}}$ for a mean prefix and $\mathbf{s}_j^{\text{noise}} = \mathbf{s}_j - \mathbf{s}_j^{\text{mean}}$ for its noise. There are fixed b_{bias} and arbitrarily large prescribed k_n such that*

$$\left\| \min \left(1, \frac{|\mathbf{s}_j^{1,\text{mean}} - \mathbf{d}|}{1 + \mathbf{d}} \right) \right\|_{u,a;s} \leq c_* 2^{-p_{\text{acc}} + b_{\text{bias}} d_j}, \quad (9.14)$$

$$\|\mathbf{s}_j^{\text{noise}}\|_{u,a;s} \leq c_* 2^{-p_{\text{acc}} - k_n d_j}. \quad (9.15)$$

The second-channel mean uses its ordinary absolute norm, extended by zero off its usage domain, and satisfies the first bound. Vector means and noises satisfy the corresponding vertex bounds. The coefficient c_ can be chosen as small as needed by task separation and walk constants.*

Proof. The mean assertion is Lemma 9.6. For noise, apply the conditional mixing estimate to each already-grouped destination function, then sum the resulting moments. This order avoids paying separately for every guessed walk variant before obtaining the mixing gain. For a difference layer, write $\mathbf{F}_{xy}$ for its sample function on $\mathcal{C}_{xy}^{u,a}$, $\mathcal{A}_{xy}$ for the conditional walk operator of length $\ell(m - r)$, and Π_{xy} for uniform averaging on that slice. Put $w_{xy} = \Pr(\mathcal{C}_{xy}^{u,a})/(n_v u)$ and $r_{m-r} = c_{\text{bd}} 2^{-c\ell(m-r)}$. Lemma 8.2 gives a uniform operator bound, so

$$\sum_{x,y} w_{xy} \|(\mathcal{A}_{xy} - \Pi_{xy})\mathbf{F}_{xy}\|_{L_s}^s \leq r_{m-r}^s \sum_{x,y} w_{xy} \|\mathbf{F}_{xy}\|_{L_s}^s.$$

The slice weights make the sum on the right precisely the integrated raw norm to the s th power, bounded by Lemma 9.7. Thus the uniform operator estimate applies directly to the integrated input bound, even when different slices have different input norms. For the second channel replace Y_a by Y_t in the slice and its weight; the mask stays Y_t throughout. Vectors use the same calculation on their one-vertex slices. No extra conditioning factor appears.

There are $O((1+d)^2)$ accuracy level/scale terms in a group. Choose ℓ so that the contraction dominates $2^{c_{\text{bd},0}d}$ and this polynomial, leaving at least $2^{-(k_n+1)d}$. Summing groups with slack at least d_j gives Equation (9.15). Exact base terms have no noise. This contraction applies to the paired difference as one function of the same seed. The sum of valid variants is the conditional walk, so the gap applies to that sum before any separate variant counting. Increasing a_* supplies an arbitrarily small coefficient in both this estimate and the mean bound. $\square$

The determinism hypothesis in Lemma 9.8 applies at each fixed pair or vertex, in the task’s effective environment as in Remark 8.5. Uniform averaging must integrate every coordinate on which its sample function depends. In Section 11, the clean calculation removes dependence on the current block’s salt, while the parent walk varies all earlier salts used by lower-block calculations; Lemma 11.4 verifies this hypothesis. The resulting first-channel mean is a deterministic number, and the second-channel mean has only its prescribed Y_t membership mask. These are the means used in the conditional heavy-count argument of Lemma 10.5.

10 Representing a row by a default and exceptions

The averaging schedule describes a value at every retained destination, but we cannot keep a growing list of all destinations encountered. After each group we therefore replace the row by a common default value and a controlled number of exceptions. The list remains implicit: its entries will be located by recomputation.

We do not need a uniformly accurate raw value at every position. The algorithm subsequently applies bounded row functions, scores, and support witnesses to the row. These functions ignore sufficiently small entries, become constant beyond certain ranges, and shut off when a row is too crowded. The compression analysis uses precisely these properties.

The proof distinguishes the first group at which the allowed number of exceptions forces the compression threshold above its prescribed floor. Before that event, compression changes only very small values. At that event, either the true row has unusually many sampled large entries, or earlier errors have made small entries appear large. The first case is rare even after requiring a queried destination to be retained; the second is charged to those earlier errors. A comparison with the true row then controls subsequent updates.

This distinction is also why the mean error and averaging noise were kept separate. The mean discrepancy at a fixed destination is deterministic and can be multiplied by the probability of a rare sampling event. The noise can depend on that event and needs its own unconditional bound.

We first fix the increments supplied by exact endpoint grouping and prove the comparison without range clamps. We then show that clamping preserves the relevant row functions and assemble the exact-grouping error reserve. Section 11 compares the actual and clean calculations with those clamps in place.

10.1 The row properties that compression uses

Fix a row task, its start rate u , and its first-channel endpoint rate a . Write $\mathbf{d}_y \geq 0$ for its true first raw input, in units of a . Its second raw input, when present, is $\mathbf{v}_y = \zeta \mathbf{1}_{\{Y_t(y)=1\}} \mathbf{q}_y$. Here $\mathbf{q}_y$ is the deterministic conditional mean, $0 \leq \mathbf{q}_y \leq 1$, and $\mathbf{q}_y = 0$ whenever $\mathbf{d}_y$ is below the positive-start threshold of the preparatory function h . Thus $\mathbf{d}_y$ and $\mathbf{q}_y$ are deterministic at each specified endpoint; $\mathbf{v}_y$ carries the prescribed narrower membership mask. There is a fixed constant $c_{\text{bd},\text{mass}}$ such that

$$a \sum_y \mathbf{d}_y \leq c_{\text{bd},\text{mass}}. \quad (10.1)$$

The endpoint sets in this section are the retained sets, with the stipulated common padding on their hash arguments.

We apply a fixed finite family of pre-incoming row functions to a raw array. Denote this family by $\mathcal{H}$. It includes the main pre-functions, auxiliary scores, and witnesses required by the preceding construction. Each function uses the first entry, possibly the second entry, and the k_{row} th largest first entry. For each member $H \in \mathcal{H}$ there are fixed positive thresholds g_H, e_H , with $3g_H < e_H$, such that its row gate is zero when that rank is at least g_H , and its entry factor is zero when the first entry is at most e_H . The gate is constant and full in a neighborhood of zero. The s th roots of the output functions are bounded and Lipschitz in their relevant entry and rank arguments, where $s = 8$. All these arguments can be clipped to fixed sensitivity ranges without changing the functions. These are exactly the threshold buffers and saturations imposed when constructing the atoms; the thresholds for the required functions are listed in Table 1. In particular, for every real raw array, including arrays with negative entries,

$$\sum_{H \in \mathcal{H}} \#\{y : H(y) \neq 0\} \leq c_{\text{bd}, \text{supp}}, \quad (10.2)$$

where $c_{\text{bd}, \text{supp}}$ is fixed. Indeed a nonzero gate implies that fewer than k_{row} entries exceed its close threshold, and every positive output entry exceeds the larger entry threshold. This observation applies also to the broader witnesses.

Choose $\eta > 0$ below all positive entry thresholds, all positive-start thresholds for rank sensitivity, and the threshold for h , with enough fixed separation that perturbations of size several times η remain inside these flat regions. Choose $\zeta \leq \eta/100$. These constants are fixed throughout the proof.

Let $d_1 > d_2 > \dots > d_{m_*} = 1$ be the decreasing powers of two indexing slack groups. Empty groups are included. Let $\mathbf{s}_i = (\mathbf{s}_i^1, \mathbf{s}_i^2)$ be the uncompressed prefix through group i , and let $\delta_i = \mathbf{s}_i - \mathbf{s}_{i-1}$, with $\mathbf{s}_0 = 0$. Define the clipped endpoint errors and their row supremum by

$$\begin{aligned} \mathbf{e}_{i,y} &= \max \left\{ \min \left(1, \frac{|\mathbf{s}_{i,y}^1 - \mathbf{d}_y|}{1 + \mathbf{d}_y} \right), \min(1, |\mathbf{s}_{i,y}^2 - \mathbf{v}_y|) \right\}, \\ E_i &= \sup_y \mathbf{e}_{i,y}. \end{aligned} \quad (10.3)$$

omitting channel two when absent. Write $\beta_{i,y}$ for the maximum of the mean discrepancies, with the same clipping and first-channel normalization as in Equation (10.3), and $\gamma_{i,y}$ for the maximum absolute prefix noise. Thus $\mathbf{e}_{i,y} \leq \beta_{i,y} + \gamma_{i,y}$. At a specified row and endpoint, the first-channel mean discrepancy is deterministic. The second-channel mean discrepancy is a deterministic magnitude times its narrower membership mask. Prefix noise may depend on the shared environment. The deterministic assertion is a hypothesis of the compression argument: the slice average defining a mean must vary every environment coordinate on which its sample function depends. Lemma 9.8 states this requirement, and Lemma 11.4 verifies it for the clean calculations to which we will apply the result. We will use conditional rarity to control these fixed mean-error magnitudes; the noise will be controlled by its unconditional norm bound.

For row integration use

$$\int F \, d\mu = \frac{1}{n_v} \mathbb{E} \sum_x \frac{X_u(x)}{u} F(x).$$

Endpoint integration means inserting the sum over retained endpoints in this expression. The prefix estimates used as hypotheses below are

$$\left(\int \sum_y \beta_{i,y}^s \, d\mu \right)^{1/s} \leq c_* 2^{-p_{\text{acc}} + b_{\text{bias}} d_i}, \quad \left(\int \sum_y \gamma_{i,y}^s \, d\mu \right)^{1/s} \leq c_* 2^{-p_{\text{acc}} - k_n d_i}. \quad (10.4)$$

Lemma 9.8 supplies these estimates for the additive schedule. Here and below endpoint sums use the relevant masks. In particular these bounds control the $L_s(\mu)$ norms of the corresponding row suprema, which we denote by B_i and G_i .

10.2 The compression update

The update needs to preserve large entries, suppress small ones, and remain stable when its inputs change slightly. Hard deletion at a threshold would fail the last requirement. We instead clip the displacement from a new default, with a clipping radius that grows continuously from zero.

Three quantities determine the threshold: the previous default, a prescribed positive floor, and an order statistic that enforces the list capacity. The additional k_{row} th-rank adjustment makes the default compatible with the row gates. The formulas below specify this adjustment; Lemma 10.2 isolates its elementary properties before the error analysis begins.

Adjoin arbitrarily many dummy positions, all with target and increments zero. A compressed array has default $(z, 0)$, shared also by the dummies, and a finite list of exceptions. Initially $z = 0$ and the list is empty. Order statistics of an array with this default mean the order statistics of its finite exceptional values with as many default values as needed. For a rank of order r , the finite list together with at least r default copies gives exactly the prescribed rank. Additional real list positions whose values equal the default do not change this rank. Dummy positions serve only this rank computation; endpoint sums in the error bounds are over real retained IDs. The margin statement of Lemma 7.6 ensures that a default position cannot expose a row port.

Algorithm 10.1 Update a default-and-exceptions row

Input. The preceding default and exceptions, the next increment, and its slack d_j .
At group j add δ_j by actual endpoint equality, obtaining $\mathbf{w}_y$, and set

$$\begin{aligned} \rho_j &= 2^{-c_1 d_j}, & n_j &= 2^{c_2 d_j}, & \mathbf{m}_y &= \max_{\nu} |\mathbf{w}_y^{\nu}|, \\ u_0 &= \max\{z, \rho_j, \text{rank}_{n_j}(\mathbf{m})\}, \\ q_0 &= \max\{0, \text{rank}_{k_{\text{row}}}(\mathbf{w}^1)\}, \\ z' &= \max\{u_0, \min(q_0, 3u_0)\}. \end{aligned} \tag{10.5}$$

Here c_1, c_2 are fixed sufficiently large positive integers, and c_2 is chosen also so that $n_j \geq k_{\text{row}}$ in every group. With $\mathbf{b} = (z', 0)$, the updated array is

$$\mathbf{t}_{j,y}^{\nu} = \mathbf{b}^{\nu} + \text{clip}(\mathbf{w}_y^{\nu} - \mathbf{b}^{\nu}, \pm 7(\mathbf{m}_y - 2u_0)_+). \tag{10.6}$$

The notation $\text{clip}(r, \pm a) = \max(-a, \min(r, a))$ is used.

Output. The new default and the entries that differ from it. The formula is continuous in the input values; its finite access representation is supplied by the shortlist rules below.

Lemma 10.2 (Local compression properties). *The update has the following properties. It replaces every position of size at most $2u_0$ by the new default, is identity at every position of size at least $3u_0$, and lies coordinatewise between the input and the new default. The default is nondecreasing. The map from the input array and old default to the updated array and new default is uniformly Lipschitz in the sup norm. A list containing all exceptions can be chosen with at most n_j positions, using only the previous list and the new candidates.*

Proof. First inspect the two ranges where the update is especially simple. At size at most $2u_0$, its clipping radius is zero, so the entry becomes the default. Since $z' \leq 3u_0$, if $\mathbf{m}_y \geq 3u_0$ then

$$|\mathbf{w}_y^1 - z'| \leq \mathbf{m}_y + 3u_0 \leq 7(\mathbf{m}_y - 2u_0), \quad |\mathbf{w}_y^2| \leq 7(\mathbf{m}_y - 2u_0),$$

which proves identity there. Clipping a displacement toward zero proves the coordinatewise assertion. Also $z' \geq u_0 \geq z$.

If two input arrays and their defaults differ by at most δ , their size arrays, size ranks, u_0 's, and q_0 's differ by at most δ . Their new defaults differ by at most 3δ , and their clipping radii by at most 21δ . The elementary bound

$$|\text{clip}(r, \pm a) - \text{clip}(r', \pm a')| \leq |r - r'| + |a - a'|$$

therefore gives, for example, a Lipschitz constant 28 for the full update. This constant is independent of the number and magnitudes of entries.

Fewer than n_j positions have size strictly larger than u_0 . A coarse test supported above u_0 and full above $2u_0$ retains all possible exceptions within the stated cap. A position outside the old list and new candidate set has input $(z, 0)$, has size at most u_0 , and hence becomes the new default. No other position need be inspected. The selected list may retain some real positions that become exactly default after compression. Such positions still represent the same full array. No padding dummy is selected, since its input size is at most u_0 . $\square$

In particular an absolute perturbation introduced at group i propagates through the remaining updates and the final root functions with factor at most

$$c_{\text{bd}} c_{\text{bd},1}^{m_* - i + 1} \leq c_{\text{bd}} (d_i + 1)^{\log_2 c_{\text{bd},1}}, \quad (10.7)$$

for a fixed $c_{\text{bd},1} > 1$, enlarged below as needed. This statement concerns full arrays and defaults; it does not require their exceptional lists to agree.

10.3 Compare with a row whose relevant outputs stay fixed

Before integrating over randomness, we establish deterministic stability statements for one row. The most useful reference is the true row together with the small changes already caused by early compression. From the first overflow onward, run compression on this reference without adding further increments. We will show that its relevant row functions stay fixed. The actual calculation can then be compared with this reference by the prefix-error bounds.

Call group j an overflow if its u_0 exceeds ρ_j . Before the first overflow every compression change has magnitude at most $6\rho_i$ and occurs only at input size below $3\rho_i$. Because successive slacks halve, choosing c_1 large gives, uniformly in j ,

$$6 \sum_{i < j} \rho_i \leq \rho_j / 100, \quad 6 \sum_i \rho_i \leq \eta / 100. \quad (10.8)$$

The same estimates cover dummy defaults. For example, before the first overflow the preceding default is at most $3\rho_{j-1} \ll \rho_j$.

Lemma 10.3 (Preloss and stationary reference). *Let $\mathbf{l}_{j-1} = \mathbf{t}_{j-1} - \mathbf{s}_{j-1}$ before the first overflow at j . Then $\|\mathbf{l}_{j-1}\|_\infty \leq 6 \sum_{i < j} \rho_i$, and at every endpoint with $\mathbf{d}_y \geq \eta$,*

$$\max_{\nu} |\mathbf{l}_{j-1,y}^\nu| \leq c_{\text{bd}} \sum_{i < j} \rho_i \mathbf{e}_{i,y}. \quad (10.9)$$

Start a reference calculation at the input of group j with the true array $(\mathbf{d}, \mathbf{v}) + \mathbf{l}_{j-1}$, the same preceding default, and no further increments. Every subsequent compression preserves its relevant row outputs. Their root discrepancy from the true outputs is bounded by $c_{\text{bd}} \sum_{i < j} \rho_i E_i$.

Proof. First account for the changes before the first overflow. Their uniform bound is small, but at a truly substantial entry we need a stronger bound that can be charged to earlier estimation error. The uniform preloss bound follows by adding the changes described above. If a change occurs at group i at a position with $\mathbf{d}_y \geq \eta$, then $|\mathbf{w}_y^1| \leq 3\rho_i$ and the preceding preloss is much smaller than ρ_i . Thus

$$\frac{|\mathbf{s}_{i,y}^1 - \mathbf{d}_y|}{1 + \mathbf{d}_y} \geq \frac{\mathbf{d}_y - 3\rho_i - 6 \sum_{k < i} \rho_k}{1 + \mathbf{d}_y} \geq c_\eta > 0.$$

The change of size at most $6\rho_i$ is consequently bounded by $c_{\text{bd}}\rho_i\mathbf{e}_{i,y}$. Adding such changes proves Equation (10.9).

Next show that repeated compression without new increments leaves the reference outputs unchanged. We track its k_{row} th-largest first entry, because that is the statistic used by the row gates. In the reference array negative first magnitudes and all second magnitudes are at most η , by the choices of ζ and the floors. Let q be its initial k_{row} th first-coordinate rank. The default is nonnegative, so $q \geq 0$. If $q > \eta$, then $n_i \geq k_{\text{row}}$ implies that the n_i th size rank is at most q . The floor and preceding default are also at most q , so $u_0 \leq q$ and $z' \leq q$. Entries at most q stay at most q . Entries at least q are unchanged if $q \geq 3u_0$; otherwise $z' = q$, and they stay at least q . The k_{row} th rank is therefore unchanged. Negative first magnitudes and second magnitudes do not grow, so this argument repeats at every subsequent group. If instead $q \leq \eta$, the size rank, u_0 , and new default are at most η , and the next k_{row} th rank is again at most η .

Consider an individual row function. In the first case its rank stays fixed. If its gate is not already zero, $q < g_{\text{H}}$, so every entry in its support has first coordinate greater than $e_{\text{H}} > 3q \geq 3u_0$ and both channels there are untouched. A position below entry support cannot cross into support because the default is at most $q < e_{\text{H}}$. In the second case the rank stays in the common full-gate region; the same identity argument applies to every position in entry support. Thus each reference output is preserved.

Finally, positions with $\mathbf{d}_y < \eta$ stay below every entry threshold and cannot change a rank within its sensitivity region. Perturbations at all other positions satisfy Equation (10.9). Clip first entries and ranks to the fixed sensitivity ranges and use root-Lipschitz continuity of the row functions. This proves the last assertion. $\square$

If no overflow ever occurs, the same proof, together with the final prefix error, gives the bound

$$\sup_{\mathbf{H}, y} |\mathbf{H}(\mathbf{t}_{m_*}; y)^{1/s} - \mathbf{H}(\mathbf{d}, \mathbf{v}; y)^{1/s}| \leq c_{\text{bd}} \left(E_{m_*} + \sum_i \rho_i E_i \right). \quad (10.10)$$

For completeness, if the expression on the right is not small, bounded output ranges give the assertion after enlarging c_{bd} . Otherwise low-target positions remain below sensitivity. At other positions clipped first entry discrepancies are bounded by a constant times their relative error, and

second entry discrepancies by their absolute error. Adding the preloss estimate and applying the root-Lipschitz bounds proves the claim.

A raw first-channel error is measured relative to $1 + \mathbf{d}_y$, so it can be large in absolute value when the true entry is large. The next lemma must therefore use saturation and the row gates, rather than merely multiply an absolute sup-norm error by a Lipschitz constant.

Lemma 10.4 (Stability after the first overflow). *At a first overflow in group j , the discrepancy of final row-root outputs from the stationary reference of Lemma 10.3 is at most*

$$c_{\text{bd}} \sum_{i \geq j} c_{\text{bd},1}^{m_* - i + 1} E_i. \quad (10.11)$$

The constant does not depend on the largest true first entry.

Proof. Write E for the sum in Equation (10.11) without the initial constant. If E exceeds a sufficiently small fixed constant, bounded output ranges suffice. Otherwise all clipped errors involved are strictly below 1. Hence

$$|\mathbf{s}_{i,y}^1 - \mathbf{d}_y| \leq (1 + \mathbf{d}_y)E_i, \quad |\mathbf{s}_{i,y}^2 - \mathbf{v}_y| \leq E_i,$$

and later increments have first magnitude at most $(1 + \mathbf{d}_y)(E_i + E_{i-1})$ and second magnitude at most $E_i + E_{i-1}$. The input difference at group j is precisely $\mathbf{s}_j - (\mathbf{d}, \mathbf{v})$; both calculations include the same previous preloss. The factors $1 + \mathbf{d}_y$ prevent a direct absolute-error bound with a uniform constant when a true entry is large. The following two cases use the row gate and saturation to remove that dependence.

Choose g larger than every gate close threshold. A baseline at least g closes all gates, since the dummy entries force every first-coordinate rank to be at least the baseline. Closure is permanent under the unclamped update. Choose l_0 much larger than g , all sensitivity thresholds, and 1. If at least k_{row} true first entries exceed l_0 , the reference gates are closed. In the perturbed calculation either some $u_0 \geq g$, which forces permanent closure, or every $u_0 < g$. In the latter case those k_{row} coordinates stay larger than half their true value: their initial error and the total later additive variation are at most $c_{\text{bd}}(1 + \mathbf{d}_y) \sum_{i \geq j} E_i$, and their first value is then larger than $3g$. Inductively each such position lies in the compression identity region. These k_{row} positive entries close all gates as well.

It remains to consider the case of fewer than k_{row} true entries above l_0 . Lemma 10.3 bounds every reference rank and threshold by $2l_0$. Fix l_1 much larger than l_0 and all upper saturation thresholds. At the at most $k_{\text{row}} - 1$ positions where $\mathbf{d}_y > l_1$, replace the first input at group j , in both calculations, by one common positive sentinel larger than l_1 , and give that coordinate zero later first-channel increments. Do not modify second channels. The reference ranks, thresholds, and outputs are unchanged: these coordinates stay above the relevant ranks and in all upper saturation regions. The sentinels are comparison devices in this proof. They need not be identified or stored by the compression algorithm. Fewer than k_{row} such coordinates occur, so replacing them by common large values removes their possibly large absolute differences while preserving every rank that lies in a sensitivity range.

All remaining first-coordinate input errors and increments now have absolute bounds by $c_{\text{bd}}(1 + l_1)E_i$ and $c_{\text{bd}}(1 + l_1)(E_i + E_{i-1})$, respectively. Lemma 10.2, iterated from j to the last group, therefore bounds every array and threshold discrepancy by $c_{\text{bd}}E$, on taking $c_{\text{bd},1}$ larger than the fixed update constant. If E is sufficiently small, all these thresholds remain below $2l_0 + 1$. The final root-output discrepancy is at most $c_{\text{bd}}E$.

To remove the modification in the perturbed calculation, argue inductively. Every original coordinate with $\mathbf{d}_y > l_1$ remains above $\mathbf{d}_y/2$, because its total raw additive variation is at most $c_{\text{bd}}(1 + \mathbf{d}_y) \sum E_i$. It is thus above the bounded thresholds and in the identity and saturation regions. Replacing fewer than k_{row} such positive coordinates by sentinels cannot change either a k_{row} th first rank or an n_i th size rank whose sentinel-computed value is at most $2l_0 + 1$: both the original and modified values are above that bound, and $n_i \geq k_{\text{row}}$. The defaults and thresholds therefore agree, the compression is identity in both channels at these positions, and all other coordinates agree. The relevant output behavior is the same. This completes the induction and the proof. $\square$

10.4 Distinguish truly crowded rows from estimation errors

The deterministic comparisons leave two cases to integrate: no overflow, and the first overflow. To handle the latter, we separate a large retained set of true heavy endpoints from overflow caused by errors at light endpoints. For this subsection let $\mathcal{O}_j$ denote the event that group j is the first overflow. These events are pairwise disjoint. The heavy-count event defined next depends only on the true row and its retained endpoints; it does not itself assert that any overflow occurs.

At first overflow in group j , the previous default is smaller than ρ_j . Thus at least n_j retained real positions have input size strictly greater than ρ_j . Define the true heavy-count event

$$\Theta_j = \{\#\{y \text{ retained} : \mathbf{d}_y > \rho_j/4\} \geq n_j/2\}.$$

The dummy positions do not contribute to this count, since their input is the preceding default. On $\mathcal{O}_j \cap \Theta_j^c$, at least $n_j/2$ of the overflowing positions have $\mathbf{d}_y \leq \rho_j/4$. Their second targets vanish, because the floors lie below the h -threshold. Equation (10.8) then implies $\mathbf{e}_{j,y} \geq c\rho_j$ at all these positions. Consequently the indicator of this false-type first overflow is bounded by

$$\frac{c_{\text{bd}}}{n_j \rho_j^s} \sum_y \mathbf{e}_{j,y}^s. \quad (10.12)$$

A bounded row-output error may be charged to this indicator. On $\mathcal{O}_j \cap \Theta_j$, we instead use the stability estimate and the conditional rarity of Θ_j at each endpoint.

Lemma 10.5 (Conditional rarity of a true heavy count). *At every specified endpoint,*

$$\mathbb{P}(\Theta_j \mid X_u(x) = Y_a(y) = 1) \leq \frac{c_{\text{bd}}}{n_j \rho_j}. \quad (10.13)$$

The same bound holds with the narrower endpoint condition $Y_t(y) = 1$. Moreover, for every prefix mean row supremum,

$$\|\mathbf{1}_{\Theta_j} B_i\|_{L_s(\mu)} \leq c_{\text{bd}} c_* 2^{-p_{\text{acc}} + b_{\text{bias}} d_i} (n_j \rho_j)^{-1/s}. \quad (10.14)$$

Proof. Bound the number of genuinely large entries using row mass, then bound how many can be retained after one destination is forced to survive. By Equation (10.1), the deterministic heavy set

$$\mathcal{A}_j(x) = \{y : \mathbf{d}_y > \rho_j/4\} \quad \text{satisfies} \quad |\mathcal{A}_j(x)| \leq \frac{4c_{\text{bd},\text{mass}}}{a\rho_j}.$$

Conditional on one specified endpoint hit, each other endpoint in this set is retained with probability at most $c_{\text{bd}}a$, by Lemma 4.1. This remains true if the specified hit uses the narrower rate. The forced endpoint contributes at most 1, so the expected retained heavy count is at most $1 + c_{\text{bd}}a|\mathcal{A}_j(x)| \leq c_{\text{bd}}'/\rho_j$. The independent start condition has no effect. Markov's inequality at

the threshold $n_j/2$ proves Equation (10.13). This calculation uses the conditional pair bound and linearity of expectation; it permits dependence among the heavy-endpoint hits.

To exploit rarity in the mean-error estimate, condition separately at each destination before summing. This works because the magnitude of its mean error is fixed under that conditioning. For the first-channel mean error at a fixed row and endpoint, write its deterministic magnitude as $b_i(y)$. Then

$$\mathbf{1}_{\Theta_j} \sup_{y: Y_a(y)=1} b_i(y)^s \leq \sum_y \mathbf{1}_{\Theta_j} Y_a(y) b_i(y)^s.$$

Condition on the start and that endpoint before taking expectations. Equation (10.13) bounds the resulting expectation by $c_{\text{bd}}/(n_j \rho_j)$ times the endpoint-integrated mean-error power. Indeed the factor $b_i(y)^s$ is fixed under this conditioning, so

$$\mathbb{E}[\mathbf{1}_{\Theta_j} X_u(x) Y_a(y) b_i(y)^s] \leq \frac{c_{\text{bd}}}{n_j \rho_j} \mathbb{E}[X_u(x) Y_a(y) b_i(y)^s].$$

The second-channel mean error has a deterministic magnitude multiplied by $Y_t(y)$; repeat the same argument using the narrower conditional bound. Taking roots and using Equation (10.4) proves Equation (10.14). The conditioning step uses the deterministic mean-error magnitude at the specified endpoint. Prefix noise can depend on the heavy-count event, so no analogous conditional estimate is asserted for it. $\square$

Proposition 10.6 (Integrated compression bound). *For fixed sufficiently large c_1 , then c_2 , and sufficiently rapid prefix-noise decay k_n , compression preserves the prescribed row error bound, with an arbitrarily small fixed coefficient when c_* is chosen small. More precisely,*

$$\left(\int \sum_{\mathbf{H} \in \mathcal{H}} \sum_y |\mathbf{H}(\mathbf{t}_{m_*}; y)^{1/s} - \mathbf{H}(\mathbf{d}, \mathbf{v}; y)^{1/s}|^s d\mu \right)^{1/s} \leq c_{\text{bd}, \text{buf}} c_* 2^{-p_{\text{acc}}}, \quad (10.15)$$

where $c_{\text{bd}, \text{buf}}$ is fixed, independent of the number of groups, row width, and requested precision.

Proof. We add four contributions: the small changes preceding overflow, the final prefix error when no overflow occurs, overflow caused by erroneous large entries, and overflow in a truly crowded row. In the last case, mean error and noise are treated separately.

Equation (10.2) converts every row sup bound above into a bound for its summed output power with a fixed factor. In fact the support of a discrepancy lies in the union of the true and computed supports, whose total size over $\mathcal{H}$ is at most $2c_{\text{bd}, \text{supp}}$. Thus this step introduces no factor from the ambient number of endpoints. The preloss contributions have norm at most

$$c_{\text{bd}} c_* 2^{-p_{\text{acc}}} \sum_i \rho_i (2^{b_{\text{bias}} d_i} + 2^{-k_n d_i}).$$

On the event of no overflow, Equation (10.10) adds the final-prefix cost, at most $c_{\text{bd}} c_* 2^{-p_{\text{acc}}} (2^{b_{\text{bias}}} + 2^{-k_n})$. By Equation (10.12) and the disjointness of the possible first-overflow events, the total false-type error power is at most

$$c_{\text{bd}} c_*^s 2^{-s p_{\text{acc}}} \sum_j (n_j \rho_j^s)^{-1} (2^{s b_{\text{bias}} d_j} + 2^{-s k_n d_j}). \quad (10.16)$$

For a true-type first overflow at j , use Lemma 10.4 and split $E_i \leq B_i + G_i$. For the mean part, enlarge the actual first-overflow event to Θ_j , apply Equation (10.14), and use Minkowski's inequality. Its total norm is bounded by

$$c_{\text{bd}} c_* 2^{-p_{\text{acc}}} \sum_j (n_j \rho_j)^{-1/s} \sum_{i \geq j} c_{\text{bd},1}^{m_*-i+1} 2^{b_{\text{bias}} d_i}. \quad (10.17)$$

For the noise part, use the uniqueness of the first-overflow group on each row. Its tail noise sum is bounded pointwise by the single full sum $\sum_i c_{\text{bd},1}^{m_*-i+1} G_i$, regardless of dependence between noise and overflow. More explicitly,

$$\sum_j \mathbf{1}_{\mathcal{O}_j} \sum_{i \geq j} c_{\text{bd},1}^{m_*-i+1} G_i \leq \sum_i c_{\text{bd},1}^{m_*-i+1} G_i,$$

because at most one indicator on the left is nonzero. The absolute prefix-noise bound therefore gives norm at most

$$c_{\text{bd}} c_* 2^{-p_{\text{acc}}} \sum_i c_{\text{bd},1}^{m_*-i+1} 2^{-k_n d_i}, \quad (10.18)$$

without an extra sum over j .

Choose $c_1 > b_{\text{bias}}$ large enough for the floor conditions. Choose $c_2 > s c_1 + s b_{\text{bias}}$ with a fixed positive margin and also large enough that $n_j \geq k_{\text{row}}$. Equation (10.16) then converges uniformly. For its mean term the exponential factor is exactly $2^{-(c_2 - s c_1 - s b_{\text{bias}}) d_j}$. For Equation (10.17), its inner sum is at most $c_{\text{bd}} (d_j + 1)^a 2^{b_{\text{bias}} d_j}$ for a fixed a , by Equation (10.7) and dyadic grouping; the stronger choice of c_2 just made ensures convergence of the outer sum, whose exponential factor is $2^{-((c_2 - c_1)/s - b_{\text{bias}}) d_j}$. Taking k_n sufficiently large handles Equation (10.18). The preloss sum converges by $c_1 > b_{\text{bias}}$. All constants are fixed before choosing the arbitrarily small prefix coefficient c_* , giving Equation (10.15) and the asserted spare coefficient. $\square$

Proposition 10.6 bounds the compressed pre-incoming functions, including scores and witnesses. Before assembling the final atom estimates, we impose the numerical range bounds needed to evaluate these arrays.

10.5 Bound numerical ranges without changing the needed outputs

The algorithm must also avoid arbitrarily large numerical values. A clamp can be harmless if its threshold lies beyond every sensitivity range and beyond all the remaining additive variation: a value cut off there cannot later return to a range where its exact magnitude matters. This is the tail-margin argument below.

In exact endpoint grouping, the additive increments are independent of which exceptions are stored. The path description and bounded child ranges give a fixed d for which $\|\delta_i\|_\infty \leq 2^{d d_i}$, after increasing d to include immediate term counts. This also bounds a group when all its same-key increments are summed into one coordinate, with a possibly larger fixed d . Fix a common gate closure level $g \geq \max_{\mathbf{H}} g_{\mathbf{H}}$, and then choose $l_{\text{cut}} > 10g$ above every sensitivity threshold. Let $r_i = 2^{b_{\text{range}} d_i}$ and choose b_{range} sufficiently large that

$$r_i \geq 10 l_{\text{cut}} + 10 \sum_{k > i} \|\delta_k\|_\infty \quad \text{for every group } i. \quad (10.19)$$

Dyadic grouping permits this uniform choice because $d_{i+1} = d_i/2$. After each compression, clip every coordinate and the baseline componentwise to $[-r_i, r_i]$. An off-list coordinate is clipped exactly as the baseline is, so it remains a default coordinate. These clamps therefore preserve the finite-list representation.

Lemma 10.7 (Range clamping). *For fixed common increments satisfying Equation (10.19), these clamps do not change the final relevant row functions. Each clamp is sup-norm nonexpansive, so in comparisons of two calculations it preserves the uniform Lipschitz propagation bound. The analogous assertion holds for vector sums clipped beyond their final sensitivity range and all remaining additive variation.*

Proof. Compare the clamped and unclamped calculations with the same increments. For the fixed level $g < l_{\text{cut}}/10$, once both baselines are at least g , all gates are zero because of the dummies, and remain so: compression cannot lower the baseline and every clamp bound exceeds g . It remains to compare the calculations before this happens.

Maintain the following invariant. Their baselines agree while below g . Their coordinates either agree, or have matching signs and magnitudes well beyond l_{cut} ; any such differing component has enough remaining margin that the total future increments cannot bring it into $[-l_{\text{cut}}, l_{\text{cut}}]$. At the first clamp of a component this follows from Equation (10.19). A subsequent clamp restores the same margin using the current r_i . Between clamps, common additions consume at most the available tail margin.

After an addition, entries whose components differ therefore have size larger than l_{cut} in both calculations. Their size-rank statistics either are equal below g or are both at least g : differing values lie above this comparison range and all other size values agree. In the latter case both new baselines close the gates. In the former case the two values of u_0 agree and are below g . Only the first-coordinate rank clipped at $3u_0$ is needed to compute z' . Differing positive first components are above this range; differing negative first components are below zero and do not affect its nonnegative clipped value. Consequently the two new baselines agree as well. If this baseline is at least g , both calculations close. Otherwise every differing position lies in the compression identity region because its size exceeds $l_{\text{cut}} > 3g$; equal coordinates at other positions undergo the same update. Thus the invariant continues.

Before closure, all rank arguments and all entry arguments within their sensitivity ranges agree. Differing large components have matching signs and identical saturated behavior. The final row functions therefore agree. A scalar clipping map is 1-Lipschitz, which gives the comparison assertion. For a vector sum, the same tail-margin argument ensures that a clipped value stays on the same saturated side until the final bounded clipping; no rank argument is needed. $\square$

The equality assertion of Lemma 10.7 is used for the clean error bound: exact endpoint grouping supplies the same increment arrays regardless of the chosen list presentation. It preserves the values of every relevant $\mathbf{H} \in \mathcal{H}$, including scores and smooth witnesses. It makes no assertion that finite-accuracy access tests or selected descriptors agree with those of the unclamped calculation. In the adaptive comparison of Section 11, both the actual and clean calculations are clamped from their definition. Their possibly different bucket increments are compared directly, and the nonexpansiveness part of Lemma 10.7 preserves the local perturbation bound. This use allows different real representatives, including retained positions whose current value equals the default.

10.6 The accuracy available before fingerprints

The prefix estimates and compression bound now give the current task's error estimate under exact endpoint grouping. The hypotheses still require each conditional average to vary every random coordinate used by its sample function. We combine the compressed pre-incoming bounds with the atom assembly comparison; the next section will verify this averaging hypothesis for the block-clean calculation.

We now combine the averaging bounds, compression stability, and table assembly. The coefficient is chosen to use only half the available error bound. The remaining half will pay for replacing exact destination equality by short fingerprints. This is an allocation of constants within the same estimator, not an additional approximation assumption.

Proposition 10.8 (Error reserve with exact endpoint grouping). *Suppose the earlier row and vector tasks have error bound $c_{\text{ind}}2^{-P_{\text{acc}}'}$ and the support and common-parameter properties of the atom section. Use exact comparisons locally, and vary every variable on which the sample terms depend. After compression and the range clamps above, all active row outputs, pre-incoming scores, and witnesses have error at most $(c_{\text{ind}}/2)2^{-P_{\text{acc}}}$. The same holds for the final clipped vectors and the assembled correction outputs, after choosing the fixed constants in the order specified below.*

Proof. To apply Proposition 10.6, use the deterministic-mean and absolute-noise bounds of Lemma 9.8. The first target is nonnegative with bounded deterministic row mass before division by a , and its second target is bounded by a fixed multiple of ζ and vanishes below the preparatory support threshold. The buffered row functions and their witnesses have fixed root Lipschitz constants and fixed support bounds. These verify its remaining hypotheses. The compression proposition shows that all their errors are a fixed multiple of $c_*2^{-P_{\text{acc}}}$ when its convergent series of floors, capacities, and noise weights are chosen. This includes the pre-incoming quantities required by the incoming comparison. Exact endpoint grouping gives common increment arrays for the clamped and unclamped calculations. Lemma 10.7 therefore preserves these row-function values, including scores and witnesses, so the same estimates hold after clamping.

Lemma 7.8 then charges the incoming gates by those same pre-output norms. Its separate column-vector calls have a fixed positive accuracy level slack and decrease task rank. Equation (9.9) pays their broad-domain factor, and increasing a_* makes their remaining coefficient as small as required. A final vector needs no compression; its clipping is nonexpansive and the bounds at $d_j = 1$ apply directly.

Choose the fixed atom range/support bound c_{ind} first. Choose c_0 and k_{prec} to retain the margin in Equation (9.10) and fix b_{bias} . The compression proof then chooses its floor exponent c_1 , capacity exponent c_2 , and required noise decay k_n . Choose task separation and walk length sufficiently large to supply the needed small c_* . The total coefficient can thus be made at most $c_{\text{ind}}/2$. Inactive tasks retain their separate trivial c_{ind} bound. The remaining half of the active-task error bound is reserved for the coupled fingerprint error. $\square$

The reserve applies to a local exact-grouping calculation with the stated averaging property. The actual algorithm will also use short endpoint keys. In Section 11 we define an exact-grouping comparison within each accuracy level block, verify that it has this averaging property, and bound its difference from the algorithm.

11 Comparing vertices by short fingerprints

Two descriptions may reach the same destination. Their contributions must be combined, but comparing full identifiers can require too much information to remain stored during recursion. We therefore use short hash values, or fingerprints, when the accuracy level decrease is small. The vertex universe has size $2^{O(b_{\text{mem}})}$, whereas the fingerprint has only b_{mem} possible values.

A collision can change the subsequent computation, including the destinations queried later. It would be incorrect to apply a fixed-pair collision bound directly to these adaptively chosen pairs. The proof instead compares the actual computation with one using exact equality throughout one block of accuracy levels. This comparison supplies a set of candidate vertices that does not depend

on the current block's hash seed. We can bound collisions on that set and charge departures from it to errors in lower calls.

There is a second reason to define the comparison carefully. Its conditional averages must be deterministic at the queried pair, as required by the compression proof. Freezing a random coordinate on which the sample still depended would invalidate that requirement. The rules below specify both which hash seeds a task may inspect and which ones its averaging walk varies.

We first construct the fingerprints and explain how to compare them without retaining an index at every call. We then give the exact update rule and prove the adaptive error bound.

We use the root error metric $\mathfrak{d}_{u,a}$ for bounded nonnegative row outputs, its ordinary version $\|\cdot\|_{u,a;s}$ for raw arrays, and the ordinary vertex metric for vectors. For a row quantity $Z(x, \sigma)$ define also

$$\|Z\|_{\text{row},u;s}^s = \frac{1}{n_v} \mathbb{E}_\sigma \sum_x \frac{X_u(x)}{u} |Z(x, \sigma)|^s. \quad (11.1)$$

The value of s remains 8. Supremums over endpoints below are over actual ID positions, with the appropriate retained-domain masks; virtual default positions are never counted with the multiplicity of the ambient universe.

11.1 Choose one short hash seed for each accuracy level block

The fixed-pair estimate below combines polynomial evaluation, as in classical algebraic fingerprinting [36], with affine universal hashing [12]. Its proof is included to specify the fair-bit distribution and space bound. The adaptive use of these fingerprints requires the separate block comparison that follows.

Lemma 11.1 (Short fingerprints). *Suppose every endpoint has an injective, fixed-length binary encoding of length $l_{\text{ID}} \leq c_{\text{ID}} b_{\text{mem}}$. There is a uniform construction from $O(\log b_{\text{mem}})$ independent fair bits of a map $k : \{\text{endpoint IDs}\} \rightarrow \{1, \dots, b_{\text{mem}}\}$ such that, for distinct fixed IDs v, w ,*

$$\Pr[k(v) = k(w)] \leq c_{\text{bd}}/b_{\text{mem}} \leq c_{\text{bd}} b_{\text{mem}}^{-1/2}. \quad (11.2)$$

Evaluating the map at a presented endpoint needs $O(\log b_{\text{mem}})$ arithmetic scratch in addition to access to its encoding.

Proof. Pad all encodings to the same length and put $\mathbf{p}_v(t) = \sum_{i=0}^{l_{\text{ID}}-1} v_i t^i$. Choose a prime $b_{\text{mem}}^3 \leq q \leq 2b_{\text{mem}}^3$, increasing the fixed lower bound on b_{mem} if necessary. Thus $l_{\text{ID}} < q$. Let $\ell = \lceil \log_2 q \rceil$. Split a string of 3ℓ fair bits into independent integers R, A, C in $\{0, \dots, 2^\ell - 1\}$, and reduce them modulo q to obtain R_0, A_0, C_0 . Each residue has probability at most $2/q$. Let $\tau : \mathbb{F}_q \rightarrow \{1, \dots, b_{\text{mem}}\}$ partition the standard residues into consecutive sets of sizes differing by at most one, and set

$$k(v) = \tau(A_0 \mathbf{p}_v(R_0) + C_0).$$

All computations are in $\mathbb{F}_q$ before applying τ .

The nonzero polynomial $\mathbf{p}_v - \mathbf{p}_w$ has degree at most $l_{\text{ID}} - 1$, so it has at most $l_{\text{ID}} - 1$ roots. This root bound follows by repeatedly factoring $t - r_0$ at a root; it applies here because the coefficient difference 1 or -1 is nonzero in $\mathbb{F}_q$. Consequently

$$\Pr[\mathbf{p}_v(R_0) = \mathbf{p}_w(R_0)] \leq 2(l_{\text{ID}} - 1)/q.$$

For distinct evaluations z, z' , the map $(A_0, C_0) \mapsto (A_0 z + C_0, A_0 z' + C_0)$ is a bijection of $\mathbb{F}_q^2$. Under uniform parameters its two outputs are uniform and independent. The probability that their τ -values

agree is

$$q^{-2} \sum_{j=1}^{b_{\text{mem}}} |\tau^{-1}(j)|^2 \leq \frac{\lceil q/b_{\text{mem}} \rceil}{q} \leq \frac{1}{b_{\text{mem}}} + \frac{1}{q}.$$

The actual joint law of (A_0, C_0) is pointwise at most four times the uniform law and is independent of R_0 . Adding the two cases gives $2(l_{\text{ID}} - 1)/q + 4/b_{\text{mem}} + 4/q = O(1/b_{\text{mem}})$. A polynomial evaluation by Horner's rule and the residue arithmetic use $O(\log q) = O(\log b_{\text{mem}})$ bits. The prime can be chosen uniformly by searching and trial division in that space. The existence of a prime in the indicated interval is the same elementary prime-interval fact used in choosing the environment field. $\square$

A new independent hash seed for every recursive call would consume too much randomness. Instead, divide the accuracy levels into blocks and share one seed, called a salt, within each block. A block has width proportional to $\log b_{\text{mem}}$, and a salt has $O(\log b_{\text{mem}})$ bits, so the total over $O(b_{\text{mem}}/\log b_{\text{mem}})$ blocks is $O(b_{\text{mem}})$ bits.

Fix a small positive rational ϵ , to be chosen last. This coefficient controls the fingerprint block width. Write

$$h_* = \lfloor \epsilon \log_2 b_{\text{mem}} \rfloor, \quad m_b = bh_*, \quad \mathcal{I}_b = \{m_b, \dots, m_b + h_* - 1\}. \quad (11.3)$$

Enlarge the fixed minimum b_{mem} so $h_* \geq 1$ and $h_* \geq (\epsilon/2) \log_2 b_{\text{mem}}$. Give each block meeting $\{0, \dots, m_{\text{top}}\}$ an independent salt from Lemma 11.1. Pad each salt to a multiple of the fixed integer h by unused independent bits. Since $m_{\text{top}} = O(b_{\text{mem}})$, the total number of bits is

$$O((1 + b_{\text{mem}}/h_*)(\log b_{\text{mem}} + h)) = O(b_{\text{mem}}). \quad (11.4)$$

Here and throughout, constants may depend on the fixed ϵ and h . We enumerate the original fair-bit strings when enumerating environments; reducing chunks modulo q does not change that sampling convention.

The restriction on which salts a task reads is essential to the comparison argument: lower-level blocks must not inspect the current block's salt. The following access rule is part of the algorithm. A task whose accuracy level lies in $\mathcal{I}_b$ may inspect salts with indices at most b . Its conditional walks vary all earlier salts as free bits and freeze the salt of block b and every later salt. The restriction applies to structural choices, port orders, validity tests, and predecessor tests as well as to numerical values. The controller construction below implements this rule for every access operation, including inverse queries. It is not enough to impose it only on the final output values.

This rule is stated in the coordinates of the task's *effective environment*. Write σ for that environment when analyzing a task. A fixed ancestor variant may mix its free salt coordinates before supplying σ to a descendant; the descendant's table depends on the resulting environment, not on the word that produced it. Thus the rule does not forbid reading a later-indexed bit of the original master environment while replaying that word. It requires the descendant's values, port orders, and access operations to be functions only of its permitted effective coordinates. Lemma 13.11 implements this distinction. Each fixed variant is measure preserving by Lemma 8.4, so the integrated estimates may change variables to its effective environment. We do not assume that an adaptively selected variant leaves a uniform conditional law. At a block- b task, its own variants act only on effective salts earlier than b and freeze its effective block- b salt and all later ones.

11.2 Compare keys while restoring a shared bit vector

A short key is not yet enough: keeping a $\log b_{\text{mem}}$ -bit index at each of many recursive levels could still exceed the space bound. Instead, use one shared b_{mem} -bit vector as temporary workspace. A comparison may alter this vector during its execution, but it must restore exactly the contents it received. This is called catalytic use of memory; no free or uncounted memory is being assumed.

The idea is to flip the bit indexed by the first key and read the bit indexed by the second key both before and after undoing that flip. The two reads differ exactly when the keys agree. Each key is recomputed only when its vertex is visited, so no key index must survive a recursive excursion. The cancellation of arbitrary initial contents also appears in transparent programs for catalytic computation [10, Section 3]. Here the shared bit vector is included in the ordinary work-space bound.

Algorithm 11.2 Compare two fingerprints using shared memory

Input. Restoring visits to endpoints v, w , a shared bit vector $\mathbf{v}$, and a local result bit.

1. Visit v and flip $\mathbf{v}_{k(v)}$.
2. Visit w and xor $\mathbf{v}_{k(w)}$ into the result bit.
3. Visit v again and undo its flip.
4. Visit w again and xor its bit into the result.

Output. The equality bit is xored into the result, and the shared vector is restored. Only the phase and result bit persist between visits; keys are recomputed locally.

Lemma 11.3 (Restoring equality test). *Let $\mathbf{v} \in \{0, 1\}^{b_{\text{mem}}}$ be a shared vector with arbitrary initial contents. Suppose an excursion to either of two deterministically specified endpoints returns the cursor, and its path and result are independent of the incoming contents of $\mathbf{v}$. Such excursions may use nested procedures which restore $\mathbf{v}$. There is a procedure which xors the bit $\mathbf{1}_{k(v)=k(w)}$ into a local result bit, restores $\mathbf{v}$, and needs only constant local data across four endpoint excursions.*

Proof. Apply Algorithm 11.2 with initial result bit r_0 . The key is evaluated only at the vertex currently visited. Across the excursions, retain only the phase and result bit. If $\mathbf{v}^0$ is the incoming vector, the first read equals $\mathbf{v}_{k(w)}^0 \oplus \mathbf{1}_{k(v)=k(w)}$ and the second equals $\mathbf{v}_{k(w)}^0$. Thus the final result is $r_0 \oplus \mathbf{1}_{k(v)=k(w)}$ and both flips restore $\mathbf{v}^0$.

For nesting, use induction on the well-founded procedure order. A nested call sees some arbitrary incoming vector, restores precisely that vector, and returns a result independent of it. Hence the four outer endpoint paths and their keys are the same on repeated visits, even between the two outer flips. The terminal evaluation, read, or flip is a local operation with no further structural call. Only a phase and a result bit must survive across the excursions; a key index is recomputed at each visit in shared scratch. The result bit being accumulated is not used to choose either endpoint path. These facts give the induction invariant and the claimed local storage bound. $\square$

For $d > 2h_*$, exact bit-by-bit ID comparison is retained. A bit position then takes $O(\log b_{\text{mem}}) = O(d)$ suspended bits, with constants allowed to depend on the already fixed ϵ . For small slack, Lemma 11.3 avoids that per-frame index. The shared vector itself occupies b_{mem} bits and, by the lemma, may be reused by all nested comparisons. This algebra establishes restoration; the controller and traversal section supplies the deterministic excursion and call-order interfaces needed to apply it.

The endpoint action evaluates the key with the invoking comparison’s effective salt, even when the excursion visits a lower-level table. It makes no further access query and does not choose or alter the excursion. Thus the lower path remains independent of later salts. Lemmas 13.9 and 13.11 implement this separation for all restoring accesses, including inverse queries.

11.3 The actual update and the exact-equality comparison

We now specify the destination-grouping step in the estimator description. The construction is the same row update already outlined in Section 6; here the treatment of collisions is made exact. Inactive tasks still return zero outputs and empty access classes. For an active row task, the additive schedule of Section 9 supplies candidate increments from smaller-level outputs. Process the groups in decreasing slack. In each group, distribute increments by the rule below, then use the compression, range clamps, and shortlist rule of Section 10. After the last group, form the pre-incoming row functions and precursor classes from the completed arrays, then assemble any incoming gates and column multipliers as in Section 7. The separate column-vector calls have smaller accuracy level. Vector tasks use the same schedule to sum their scalar terms, with their prescribed range clamps and final clipping; they require no endpoint grouping or row-list compression.

The important choice is to group only the new increments. Combining old stored values when their keys collide could introduce an error as large as those values, even if the new increments were tiny. Keeping the old values at their existing vertices allows the collision cost to be measured using the small successive-level differences.

The row-group distribution rule is as follows. For a dyadic group with slack $d > 2h_*$, compare endpoint IDs exactly. For a group with $d \leq 2h_*$, use the key from the task’s block as follows. First discard invalid candidates, including invalid guessed wraps, individually. Sum all new candidate increments having the same key. Add a bucket’s sum to every old representative with that key. If the bucket has no old representative, introduce exactly its first valid new candidate, with its previous value equal to the old default, and add the bucket sum there. Do not combine or discard old representatives in this key step. Apply the same full-array compression, clamping, and shortlist rule as in the preceding section.

Distinct actual representatives remain distinct. Indeed two introduced representatives have different keys, and an introduced representative’s key differs from every old key. Increments at one and the same actual endpoint are always combined, even when other endpoints collide with it. Stored values are never summed by key. The two channels use the same keys and the same representative selection for the entire group.

To analyze this adaptive rule, remove the current block’s dependence on its own salt. For an analytical comparison at an accuracy level $m \in \mathcal{I}_b$, define the *clean calculation for block b* by replacing key comparisons by exact ID comparisons throughout that block, recursively in every child whose accuracy level still belongs to $\mathcal{I}_b$. Children below m_b are the actual algorithms, with exactly the same environments in the two calculations. This replacement applies to the entire numerical and structural calculation: candidate validity, representative selection, compression, exposed classes, and their port orders. Descriptor enumeration uses fixed index orders. Each finite-accuracy test for a shortlist or exposed class retains its prescribed accuracy level and tie convention, independently of any requested output accuracy. All these operations use the corresponding clean operands. Thus the clean calculation specifies its access choices as well as its represented arrays.

Denote the actual and clean versions by superscripts *ac* and *cl*. Both still perform compression and clamping. The clean calculation is a finite analytical recursion; its exact comparisons are not asserted to satisfy the eventual space bound.

Clamping is part of both definitions. The clean error reserve uses Lemma 10.7 for exact-ID

increments, which are independent of the stored exception list. That lemma preserves the relevant row functions and witnesses; it need not preserve the list or port presentation of an unclamped calculation. The coupling below compares the two clamped calculations directly. Its propagation bound uses the Lipschitz compression map and the nonexpansiveness of each common clamp, without introducing an unclamped actual calculation.

Lemma 11.4 (Independence of the clean block). *Every clean value and structural choice for block b is independent of that block's effective salt and of later effective salts. In a task of accuracy level $m \in \mathcal{I}_b$, the actual and clean row states immediately before the first group with $d \leq 2h_*$ are identical and independent of that salt. Their stored lists at this boundary have size $2^{O(h_*)}$. The deterministic slice-mean assertion required by Lemma 9.8 holds for the clean calculation.*

Proof. We must prove independence for the entire computation, including which ports it exposes; independence of the final numerical answer alone would not be enough. Induct on the accuracy level and, within a task, on its prescribed sequence of operations. Lower-block algorithms ignore the salt of block b and later salts by the access rule; same-block clean children ignore them by induction. Candidate tests use these children, exact endpoint grouping precedes compression, and the compressed row determines its exposed classes and pre-incoming scores. Incoming ranks use those completed scores. Separate column-vector queries likewise use a smaller-level clean child in this block or an actual child below it. All statements here concern the current effective environment: each same-block transformation preserves its current and later salts, and each lower-block transformation acts on a subset of its earlier salts. Each numerical recipe uses the same fixed operand order, and each structural test uses its fixed accuracy level and tie rule. Its result therefore depends only on earlier clean operands, including their finite numerical approximations. The fixed index orders then give salt-independent representatives and ports.

In particular, exact full-array addition, compression, and the prescribed row functions are salt independent. Reordering paths or choosing a different representative of the same endpoint would not change an exact sum; retaining an extra default-valued position would not change the represented array. The fixed conventions above also remove these presentation choices from the structural calculation.

For a group with $d > 2h_*$, every child accuracy level is at most $m - d < m_b$. Both runs use the same lower-block children and exact comparisons there. All their list operations, including their deterministic tie conventions, ignore the present salt. Thus their boundary states agree. The last larger dyadic slack, if there is one, is at most $4h_*$, so its shortlist has at most $2^{O(h_*)}$ positions. If there is no larger group, the boundary list is empty.

Finally, the clean estimand depends only on rank channels and earlier salts. These are precisely the variables averaged by its conditional walk, subject to its stated vertex conditions. Same-block salts are irrelevant, and lower-block salts are free variables of this walk. Projection onto the uniform measure of the slice therefore gives a deterministic quantity at its indicated pair or vertex, with the prescribed retained-domain masks. In particular, no random dependence on a frozen salt remains in this mean. For the second channel, its deterministic mean magnitude is extended by its prescribed narrower endpoint mask, as required in Section 10. The salt rule is required of all access operations, including inverse queries and port orders; Lemma 13.11 supplies its implementation. Lemmas 8.2 and 8.4 therefore apply to this function. This verifies the deterministic-mean hypothesis of Proposition 10.8; it does not assert independence between errors of different children. $\square$

11.4 Find a candidate set independent of the current hash seed

The exact-equality comparison determines a candidate set independent of the current salt. We enlarge it slightly using the smooth support witnesses, so that if an actual access leaves the set, some child witness must have changed from zero to one. This turns a discrepancy in access choices, even at a zero-weight entry, into a numerical error already covered by the induction.

We recall the relevant interfaces from Lemma 7.7. Every exposed actual path port has a broader smooth witness equal to one at that port. Each witness has bounded row support. The actual correction ports, including exposed ports of weight zero, have incoming multiplicity at most k_c . Tables contain one entry per pair of actual IDs. After fixing a copy part, the child tuple is common over the whole table; in particular, a right child reached from a midpoint does not depend on the discarded old start. All these statements hold for arbitrary computed raw arrays, not just for accurate ones.

The incoming cap applies to the entire exposed correction relation, including its zero-weight entries. The broader auxiliary score class needs only a row bound: it is queried for scores and witnesses, whereas the left factor of a two-factor path is a capped correction table. This distinction permits the source sum below to be removed even when an offending candidate has zero numerical weight.

Let $\mathcal{G}_b$ be the sigma-field generated by all coordinates of the task's effective environment except the salt of block b . Fix a row x and condition on $\mathcal{G}_b$ with $X_u(x) = 1$. Take its boundary list from Lemma 11.4, including every stored exact-default slot. For every small group and every immediate descriptor variant, evaluate the clean factors and their witnesses in the environment obtained by that fixed variant. This transformation leaves the present and later salts frozen. Include the endpoints in each direct witness support and, for a two-factor path, the endpoints in the right witness supports at all midpoints in the left witness support. Include local base endpoints when applicable. Let $\mathcal{E}_x$ be the union of these endpoints and the boundary list. Set $\mathcal{E}_x = \emptyset$ when $X_u(x) = 0$. Unused or invalid clean descriptors may enlarge this set. Actual candidates must still satisfy their wrap tests and all transformed source, midpoint, and terminal memberships; every error estimate below retains the memberships needed for its child metrics.

Lemma 11.5 (Size and independence of the envelope). *The set $\mathcal{E}_x$ is independent of the present salt and has size at most $2^{c_{\text{bd}, E} h_*}$ for a fixed $c_{\text{bd}, E}$. All clean nondefault representatives in the small groups belong to this set. Consequently, conditional on the other data,*

$$\Pr[\text{two distinct IDs in } \mathcal{E}_x \text{ have the same key}] \leq c_{\text{bd}} 2^{2c_{\text{bd}, E} h_*} b_{\text{mem}}^{-1/2}. \quad (11.5)$$

Proof. First fix every coordinate except the current salt. This also fixes the comparison computation and all of its witness supports. Each fixed variant is independent of the present salt and leaves it unchanged. Lemma 11.4 therefore makes its clean witness supports $\mathcal{G}_b$ -measurable. We use these supports as sets of actual IDs, not the order of a port enumeration. At a group of slack d there are $2^{O(d)}$ immediate descriptors, including variants, difference parts, and copy choices. The number of endpoints contributed by a direct witness is bounded. A two-factor path contributes a bounded number of left witness midpoints and, for each, a bounded number of right witness endpoints. Summing over the dyadic groups with $d \leq 2h_*$ and adding the boundary list proves the size bound. No incoming bound on a clean witness is used here.

Clean exposed ports lie in their clean witness supports. Apart from the boundary representatives, exact updates can therefore create nondefault values only at the included endpoints. Compression maps all untouched default positions to the new default. Any extra exact-default slots in a shortlist may be omitted when describing its full array. This proves the representative claim. Equation (11.5) is now the pair union bound from Lemma 11.1, conditional on the fixed envelope. $\square$

Let $\mathcal{F}_x$ be the event that some valid actual candidate in the small groups has endpoint outside $\mathcal{E}_x$, whether or not its increment is zero. As representatives are only carried forward or chosen among new candidates, all actual representatives stay in the envelope outside this event. This includes default-valued representatives: they start in the full boundary list or inherit a candidate endpoint. Dummy copies used to compute ranks are not new real representatives. Thus omitting an exact-default clean slot when describing its full array does not omit an actual address from the collision analysis.

Lemma 11.6 (Charging an escaped candidate). *Suppose actual-clean child discrepancies at accuracy level r have their prescribed metric bounds $2^{-p_{\text{acc},r}} \lambda_r$. The contribution of $\mathcal{F}_x$ to any bounded final pre-incoming row output has norm at most*

$$2^{-p_{\text{acc}}} \sum_{\substack{d \leq 2h_* \\ r \in \mathcal{I}_b, r \leq m-d}} 2^{c_{\text{bd}} d} \lambda_r, \quad (11.6)$$

where the sum includes the finitely many dependency types and their descriptor multiplicities in its constant c_{bd} . Equivalently, one may use the maximum of λ_r over the children of each slack group and increase c_{bd} .

Proof. Locate the first factor of an actual path that leaves the corresponding clean witness support. At that factor, the actual witness is one and the clean witness is zero. Thus an access disagreement produces a unit discrepancy, whether or not the path's numerical contribution vanishes.

An actual endpoint outside the envelope has an offending path factor. For a direct path its actual port is outside the clean witness support. For a two-factor path, either its actual left port is outside its clean witness support, or its midpoint lies in that support and its actual right port is outside the corresponding clean right witness support. At the offending port the actual witness is one and the clean witness is zero. Thus the indicator of this event is at most the s -th power of their root discrepancy there. Union bounding over candidate paths gives a sum of such nonnegative powers. The unit-witness property holds also for exposed zero-weight ports; no lower bound on the candidate's increment is being assumed.

Once such a path occurs, we charge the whole bounded row-output error to its occurrence. Charging only the path's increment would be insufficient: even a zero increment can affect the selection of representatives. Each output under consideration is bounded and has bounded row support. The union of the two output supports is still bounded, by Equation (10.2). Hence its entire row error power on $\mathcal{F}_x$ is at most a fixed multiple of $\mathbf{1}_{\mathcal{F}_x}$. We charge this row-failure indicator, rather than the size of the offending increment. There is no endpoint sum left on the output side to pay for.

The right-factor charge is the case where the old start must be summed out. Fix a guessed variant and a two-factor path $x \rightarrow z \rightarrow y$. Change variables by its measure-preserving transformation, as in Lemma 8.4; in the following display all memberships, ports, and witness values are evaluated in that transformed environment. Write $e_R(z, y)$ for the root discrepancy of the two right witnesses and $\mathcal{P}_{l_{\text{max}}}^{\text{ac}}$ for the actual left exposed-port relation. The candidate's validity retains $X_u(x)$, $Z_f(z)$, and the right child's endpoint membership $Y_{a'}(y)$. Dropping only surplus guards, the right-port contribution is at most a constant times

$$\begin{aligned} & \frac{1}{n_v} \mathbb{E} \sum_x \frac{X_u(x)}{u} \sum_{z: (x,z) \in \mathcal{P}_{l_{\text{max}}}^{\text{ac}}} Z_f(z) \sum_{y: Y_{a'}(y)=1} |e_R(z, y)|^s \\ & \leq \frac{k_c}{n_v u} \mathbb{E} \sum_z Z_f(z) \sum_{y: Y_{a'}(y)=1} |e_R(z, y)|^s = k_c \frac{f}{u} \mathfrak{d}_{f,a'}(\mathbf{H}_R^{\text{ac}}, \mathbf{H}_R^{\text{cl}})^s. \end{aligned} \quad (11.7)$$

The middle inequality uses the incoming cap of the *actual* left port relation. It does not use a clean incoming cap. The right task's tuple is independent of x , so the same discrepancy is being summed when the old start is removed. This condition is necessary for the displayed equality with the right metric. For the fixed variant and copy part, the right tuple has only the prescribed task parameters and transformed environment. The midpoint z is its vertex argument, and the old source x is not a parameter. Its final gate and column queries use that same effective tuple, rather than the ancestral descriptor that happened to discover the entry.

For a direct or left offending port, sum the child's witness discrepancy over its own row and endpoint domain; the bounded number of possible right ports incurs only a fixed factor. Projected paths are partitioned by the fixed copy symbols, with constant multiplicity. Normalization and preparation dependencies make the same direct witness comparison. If a candidate arose from a narrower terminal slice, its validity mask may be dropped in this row-event estimate. Thus no factor π_a/π_t is incurred. In contrast, the transformed source hit cannot be omitted before the change of variables and the incoming-cap sum; it has been retained in Equation (11.7).

At slack d , scale ratios, variants, and immediate path counts cost at most $2^{O(d)}$, as in Lemma 9.7. The possible reset factor in Equation (11.7) is absorbed by the precision gain. Admissible row tasks have $u \leq f$, as specified in Section 7. Equation (9.9) gives, after the scale-dependent $O(d)$ terms,

$$p_{\text{acc},r} \geq p_{\text{acc}} - O(d) + \frac{1}{2} \log_2(f/u),$$

$$(f/u)^{1/s} 2^{-p_{\text{acc},r}} \leq 2^{-p_{\text{acc}}+O(d)} (f/u)^{1/8-1/2} \leq 2^{-p_{\text{acc}}+O(d)}.$$

Take s -th roots and use Minkowski to sum the charges. A child below m_b is identical in the two versions and contributes zero. All remaining children satisfy $r \leq m - d$ and belong to $\mathcal{I}_b$, which proves Equation (11.6). $\square$

11.5 Bound the additional error from collisions

The escape estimate leaves rows whose candidates stay in $\mathcal{E}_x$. On these rows we separate child discrepancies from collisions among the clean endpoints. We compare full arrays with their defaults, so different shortlist presentations require no extra error term. The following observation explains the key update rule.

Lemma 11.7 (Only increments are aliased). *Consider one small group on $\mathcal{F}_x^c$. Let $\delta_{\text{ac},y}, \delta_{\text{cl},y}$ be its unaliased increments, namely its true-ID sums using actual and clean children, respectively. Let $k_d \leq 2^{c_{\text{bd}}d}$ bound its number of immediate descriptors, and let $\mathcal{C}_x$ denote a collision on $\mathcal{E}_x$. The discrepancy between the actual bucket-distributed increment and the clean true-ID increment is at most*

$$(k_d + 1) \left(\sup_y |\delta_{\text{ac},y} - \delta_{\text{cl},y}| + \mathbf{1}_{\mathcal{C}_x} \sup_y |\delta_{\text{cl},y}| \right) \quad (11.8)$$

in sup norm, simultaneously for both channels. The old full-array discrepancy is added to this bound with coefficient one.

Proof. We compare the arrays before compression, separating a change in the children from a collision among the clean destinations. When all envelope keys are distinct, the rule gives exactly the actual true-ID increment: each old occurrence is updated at its own ID, and each genuinely new ID has its own bucket. Thus the discrepancy is simply $\delta_{\text{ac}} - \delta_{\text{cl}}$.

When there is a collision, each updated representative receives a sum of increments at at most k_d distinct actual addresses. Its magnitude is at most $k_d \sup_y |\delta_{\text{ac},y}|$. At positions not updated it is

zero. At either type of position subtracting the clean increment costs at most a further $\sup_y |\delta_{\text{cl},y}|$. Apply

$$\sup_y |\delta_{\text{ac},y}| \leq \sup_y |\delta_{\text{ac},y} - \delta_{\text{cl},y}| + \sup_y |\delta_{\text{cl},y}|$$

to obtain Equation (11.8). This also covers an increment duplicated at several old representatives and a missing increment at an unchosen new representative.

At no point are old values combined or moved to a different ID. An unlisted old position has exactly the old default, and a newly selected representative starts with that value. Therefore the full-array comparison before compression is the old discrepancy plus the increment discrepancy just bounded. There is no term involving the height of an old value multiplied by a collision indicator. $\square$

In applying this lemma to a group, combine contributions at equal actual endpoints *within each paired difference* before measuring its clean height. At a fixed endpoint and original generator word, the outer slice and wrap tests are the same for accuracy levels r and $r - 1$. Their constituent port sets may differ, but each set contains all nonzero entries and enumerates each actual ID pair once; an absent entry contributes zero. Their true-ID sums are therefore exactly the two terms of the paired difference. Subtracting the same stochastic true-atom term from both sides, as in Lemma 9.7, preserves its cancellation. One can then sum the resulting sup bounds over difference layers. This remains legitimate if a first-bucket representative is chosen across several layers: the proof of Lemma 11.7 applies to their total increment, and the triangle inequality bounds the total by the sum of the paired bounds. A descriptor with zero increment contributes only to k_d in this bound. It may affect representative selection, which is already covered by Lemma 11.7 and the escape event.

The remaining probabilistic step must be applied to quantities independent of the current salt. The clean increment heights and the enlarged candidate set have this property. The event that the actual computation stays inside that set need not have it, so we do not condition on that event when applying the collision bound.

Lemma 11.8 (Direct cost of collisions). *The direct collision contribution to any final pre-incoming output has norm at most*

$$2^{-p_{\text{acc}}} 2^{c_{\text{bd}} h_*} b_{\text{mem}}^{-1/(2s)}, \quad (11.9)$$

for a fixed c_{bd} independent of d_1, d_2, ϵ introduced below. All remaining contributions on $\mathcal{F}_x^c$ are bounded by actual-clean child discrepancy norms with an amplification at most $2^{c_{\text{bd}} d}$ at slack d .

Proof. For each small dyadic group, let $\Delta_{d,xy}^{\text{cl},\nu}$ be its complete clean true-ID increment in channel $\nu \in \{1, 2\}$, including any exact base term assigned to that group. Take an absent second channel to be zero, and choose a deterministic descriptor bound $k_d \leq 2^{c_{\text{bd}} d}$. By Lemma 9.7, the clean paired increments at a group of slack d obey their crude absolute estimates in the retained endpoint metric. These estimates hold for each fixed valid variant after its measure-preserving change of variables. Counting variants and summing difference layers by the triangle inequality costs $2^{O(d)}$, without using walk contraction. Within each paired layer the true-ID cancellation is performed before this inequality is applied. If a local unpaired base term occurs in a small group, its first-group slack satisfies $m = O(d)$, so the same crude bound holds in its normalized raw units. Since $\sup_y |Z_y|^s \leq \sum_y |Z_y|^s$ over actual ID positions, the row norm of $\max_\nu \sup_y |\Delta_{d,xy}^{\text{cl},\nu}|$ is at most $2^{-p_{\text{acc}} + O(d)}$.

Lemmas 10.2 and 10.7, together with Equation (10.7), propagate an input discrepancy at slack d to the final array with a factor polynomial in $d + 1$. At each step the discrepancy is measured between the clamped full arrays, including their defaults. The next bucket increment is compared by

Lemma 11.7; compression then applies its uniform Lipschitz bound and the common clamp cannot enlarge the discrepancy. This recurrence does not require their later bucket distributions or stored lists to agree. Root-Lipschitz row functions and bounded final support then give the same type of bound for the output metric. Choose fixed $c_{\text{bd},p}, a_p$ so that $c_{\text{bd},p}(d+1)^{a_p}$ bounds this propagation. For a retained row define

$$H_x = c_{\text{bd},p} \sum_{\substack{d \leq 2h_* \\ d \text{ a group slack}}} (d+1)^{a_p} (k_d+1) \max_{\nu} \sup_y |\Delta_{d,xy}^{\text{cl},\nu}|.$$

Set $H_x = 0$ when $X_u(x) = 0$. This quantity bounds the propagated clean-height terms in Equation (11.8). The preceding estimates give

$$\|H\|_{\text{row},u;s} \leq 2^{-p_{\text{acc}} + c_{\text{bd},0}h_*}.$$

All increments defining H_x are clean, and its other factors are deterministic. Thus H_x and $\mathcal{E}_x$ are $\mathcal{G}_b$ -measurable. Conditioning on this sigma-field gives

$$\mathbb{E}[\mathbf{1}_{C_x} H_x^s \mid \mathcal{G}_b] = H_x^s \Pr(C_x \mid \mathcal{G}_b) \leq c_{\text{bd}} 2^{2c_{\text{bd},E}h_*} b_{\text{mem}}^{-1/2} H_x^s.$$

The row membership $X_u(x)$ is also $\mathcal{G}_b$ -measurable. Integrating the last inequality in the row measure therefore yields

$$\|\mathbf{1}_C H\|_{\text{row},u;s}^s \leq c_{\text{bd}} 2^{2c_{\text{bd},E}h_*} b_{\text{mem}}^{-1/2} \|H\|_{\text{row},u;s}^s.$$

Taking roots proves Equation (11.9) after enlarging c_{bd} . We have not conditioned on $\mathcal{F}_x^c$ in this independence calculation: the actual error there is bounded by $\mathbf{1}_{C_x} H_x$, which is then integrated on the whole space. Conditioning on the salt-dependent stay-inside event is unnecessary and could destroy the stated independence.

For the first term of Equation (11.8), split products using bounded-factor differences, actual or clean row caps, and the actual or clean incoming cap of the represented two-way factor, exactly as in Lemma 9.7. These are comparisons of unaliased sums using the respective children; they make no assumption about key success or independence of child errors. Fixed valid variants, scale costs, and counts are all at most $2^{O(d)}$, with the possible source-release factor absorbed by the same precision gain as in Equation (11.7). Compression propagation is again absorbed in $2^{c_{\text{bd}}d}$. This proves the remaining assertion. $\square$

11.6 Close the accuracy and comparison bounds together

There are three assertions to prove simultaneously: accuracy of the actual estimator, accuracy of the exact-equality comparison, and closeness of the two. Lower-block calls coincide in the comparison. Within the current block, the induction permits a comparison error that grows with the accuracy level, leaving a geometric gain whenever a child has lower accuracy level. That gain absorbs the cost of enumerating paths and propagating local errors.

Proposition 11.9 (Uniform block coupling). *The fixed constants can be chosen so that, for every task of accuracy level $m \in \mathcal{I}_b$ and precision p_{acc} , each prescribed actual-clean output discrepancy, including every pre-incoming score and witness, is at most*

$$2^{-p_{\text{acc}}} \lambda_m, \quad \lambda_m = 2^{-d_1 h_* + d_2(m-m_b)}. \quad (11.10)$$

The same assertion holds for the ordinary vector metrics. Adding these discrepancies to the clean estimates proves the target error bounds for the actual tables and vectors.

Proof. Perform strong induction on the accuracy level, simultaneously for the actual target estimates, the block-clean target estimates needed at that accuracy level, and Equation (11.10). An inactive task returns the same zero outputs and empty paths in both versions, so its coupling error is zero. Lower-block children are identical in a block coupling and already have their actual target bounds. Same-block clean children have the clean target bounds at smaller accuracy levels. Therefore Lemma 11.4, the prefix estimate, and the compression analysis give the clean active target bounds with the spare constants in Proposition 10.8.

For the coupling, Lemmas 11.6–11.8 bound the pre-incoming quantities. Incoming gates are then compared by their rank and root-Lipschitz bounds. In their multiplier effect, restrict to the union of the final supports of the two tables; these have bounded incoming multiplicity. Thus Lemma 7.8 costs only a fixed factor on pre-incoming errors. Its separately queried column vectors have a positive constant accuracy level drop and their f/u metric factor is covered by their precision gain. Vector raw sums have no endpoint grouping, but their same-block children may differ. The same crude individual-variant comparisons and nonexpansive clamps bound those differences. Detector rank comparisons are already part of the assembly estimate. No simultaneous key-success event over an incoming search is required.

Collect the finite number of output types and absorb their fixed assembly factors into one fixed exponent c_{bd} . For a child accuracy level $r \in \mathcal{I}_b$ used at group slack d , we have

$$\lambda_r = \lambda_m 2^{-d_2(m-r)} \leq \lambda_m 2^{-d_2 d}. \quad (11.11)$$

For a separate post-assembly query with constant positive accuracy level drop, use $d = 1$ in this inequality. Replacing the dyadic group sum by the larger sum over positive integers, all coupled child charges are at most

$$2^{-p_{acc}} \lambda_m \sum_{d=1}^{2h_*} 2^{-(d_2 - c_{bd})d}. \quad (11.12)$$

Counts of accuracy level layers and dependency types are included in $2^{c_{bd}d}$. Groups with larger slack have identical children and contribute zero. Together with Equation (11.9), this is the complete coupling recurrence.

We can now choose constants in the same order as the two error sources. First make lower-call discrepancies contract across an accuracy level decrease; then make direct collisions rarer than the smallest allowed comparison error in a block. Choose $d_2 > c_{bd}$ so large that the infinite geometric sum in Equation (11.12) is at most $1/4$. Then choose $d_1 > d_2 + 1$. Finally choose rational $\epsilon > 0$ so small that

$$\frac{1}{2s\epsilon} > c_{bd} + d_1 + 2. \quad (11.13)$$

Since $h_* \leq \epsilon \log_2 b_{\text{mem}}$, the direct relative charge satisfies

$$2^{c_{bd}h_*} b_{\text{mem}}^{-1/(2s)} \leq 2^{-(1/(2s\epsilon) - c_{bd})h_*} \leq \frac{1}{4} 2^{-d_1 h_*} \leq \frac{1}{4} \lambda_m,$$

after increasing the fixed minimum b_{mem} to absorb any fixed coefficient. The child and direct charges fit within Equation (11.10), even with room for fixed norm conventions already included in c_{bd} .

Moreover $0 \leq m - m_b < h_*$ implies

$$\lambda_m \leq 2^{-(d_1 - d_2)h_*}.$$

This is uniformly as small as required after the same minimum-size padding. The triangle inequality therefore adds the coupling bound to the spare clean active bound, for example $c_{\text{ind}} 2^{-p_{acc}}/2$, without

exceeding $c_{\text{ind}}2^{-p_{\text{acc}}}$. Inactive target bounds are the previously fixed bounded-range bounds. This completes all three inductions.

The order of choices is consequential. Fix the analytical hierarchy, table thresholds, and trivial-case bound first. Choose the precision and cutoff constants for the mean comparisons; then the compression floors, capacities, and walk lengths so the clean sums converge with their spare coefficients. Fix clamping ranges and the crude exponents in the fixed-variant comparisons next. These choices determine c_{bd} . Only after them choose d_2, d_1, ϵ as above. Thus reducing ϵ to obtain the direct collision gain does not alter an earlier clean-coefficient requirement.

Later choices of the common stage-size bound, top-accuracy level coefficient, and environment-field exponent do not change this crude exponent c_{bd} . Immediate descriptor counts depend on relative slack, fixed support bounds and fixed walk length. The sampler gap is uniform in both field size and the length of its single free-bit component. Likewise, an enlargement of the fixed identifier-length coefficient only changes the polynomial-evaluation collision term in Lemma 11.1 to at most $2c_{\text{ID}}/b_{\text{mem}}^2$; increasing the fixed minimum $b_{\text{mem}} \geq c_{\text{ID}}$ bounds it by $2/b_{\text{mem}}$. These enlargements can therefore be absorbed after d_2, d_1, ϵ have been fixed. $\square$

Theorem 11.10 (Error bounds for the actual estimators). *For every allowed task, with precision p_{acc} defined in Equation (9.8), the actual row outputs, including the pre-incoming scores and support witnesses, satisfy*

$$\mathfrak{d}_{u,a}(\bar{h}, h) \leq c_{\text{ind}}2^{-p_{\text{acc}}}.$$

The clipped vectors have the corresponding ordinary error bound. In particular, every allowed success contribution query satisfies

$$\frac{1}{n_v} \mathbb{E}_\sigma \sum_{x \in \mathbf{v}_l} \frac{X_u(x)}{u} |\bar{\mathbf{w}}_{l,x} - \mathbf{w}_{l,x}|^s \leq c_{\text{ind}}^s 2^{-sp_{\text{acc}}}. \quad (11.14)$$

These estimates are uniform over the allowed task parameters and stage, and the constants are independent of the input and of b_{mem} .

Proof. Proposition 11.9 adds the actual-clean discrepancy to Proposition 10.8 with its spare constant. The inactive-task bounds follow from the fixed bounded ranges and supports. The induction in Proposition 11.9 includes both kinds of tasks and every prescribed output. Applying its ordinary vector statement to $\mathbf{w}_l$ and taking the s -th power gives Equation (11.14). $\square$

12 Computing numerical values one digit at a time

We have defined the estimator as a mathematical function of a finite random environment. Its values involve arithmetic, conditional averages, clipping, and order statistics. This section explains how finite numerical queries evaluate those values without keeping a long approximation in every caller.

There are two notions of accuracy. The accuracy level determines which estimator value is being computed. The numerical digit place determines how accurately that fixed value is read. Increasing the latter does not improve the former, and it must not change the list or port choices that define the estimator. We therefore give each fixed task value an exact signed-digit representation and implement a request for any one digit by a terminating finite calculation.

Why return one digit at a time? If a multiplication first computed and saved a whole approximation to its left operand, that word would remain live while the right operand made recursive calls. A digit interface lets the arithmetic circuit request only the bounded pieces it needs and recompute them when necessary.

A query returns one bounded digit. During its evaluation, the space needed by a suspended caller must be bounded by the size of its structural descriptors and a constant times the decrease from its output clock to the requested operand clock. An operand clock may exceed the output clock by an amount proportional to descriptor size; the structural term also pays for this overshoot. These differences will telescope when arithmetic queries are nested inside graph procedures. Thus the relevant question is which data remain live at each operand request.

12.1 Exact values from bounded signed digits

We use integer clocks $t, j \geq 0$. Known binary scales are separated from numerical values, so a normalized value belongs to $[-1, 1]$. The fixed digit alphabet is

$$\mathcal{A} = \{-6, -5, \dots, 5, 6\}.$$

Every digit is encoded by a fixed number of bits. An operand stream $(x_j)_{j \geq 0}$ denotes the exact real number

$$x = \sum_{j \geq 0} x_j 2^{-j}, \quad x_j \in \mathcal{A}. \quad (12.1)$$

In particular, for its truncation $x^{[k_*]} = \sum_{j=0}^{k_*} x_j 2^{-j}$,

$$|x - x^{[k_*]}| \leq 6 \cdot 2^{-k_*}. \quad (12.2)$$

The stream need not be a usual binary expansion, and no uniqueness of the expansion is required. Redundant signed-digit representations have a long history in arithmetic [5]. The redundancy lets consecutive finite approximations define an exact stream even when they use different truncations.

An arithmetic root has public parameters, its clock, and a description of its operand interfaces. We require these parameters to be regenerable from the outer controller records without operand or graph queries. They are fixed inputs to each numerical computation; their full binary encodings are not copied at every gate. An operand label specifies its interface and descriptor; it never specifies an unknown vertex address from a previous cursor position.

A finite approximation at place t need not extend the approximation obtained at place $t - 1$. Rather than enforce matching binary prefixes, take their scaled difference as the next signed digit. The resulting partial sums telescope to the exact value. The error bound keeps each difference in a fixed finite alphabet, so only a fixed number of residue bits is needed to recover it.

Lemma 12.1 (Exact streams from approximation residues). *Suppose $|z| \leq 1$ and, for each $t \geq 0$, a deterministic integer $n_t(z)$ satisfies*

$$|n_t(z) - 2^t z| < 2. \quad (12.3)$$

Set

$$z_0 = n_0(z), \quad z_t = n_t(z) - 2n_{t-1}(z) \quad (t \geq 1). \quad (12.4)$$

Then $z_t \in \mathcal{A}$ and these digits represent z exactly. For any fixed integer $q \geq 4$, each output digit is determined by the required n_t residues modulo 2^q .

Proof. There are three things to check: the digits remain bounded, their weighted sum has the correct limit, and their residues determine them uniquely. Writing $e_t = n_t - 2^t z$ gives $z_t = e_t - 2e_{t-1}$ for $t > 0$, hence $|z_t| < 6$; the weaker alphabet bound $|z_t| \leq 6$ is convenient throughout. At place zero, $|n_0| < 3$, so $|n_0| \leq 2$. The partial sum through place t_* is $2^{-t_*} n_{t_*}$, which tends to z by Equation (12.3). Finally, reduction modulo 2^q is injective on the possible signed digits because

$2^q > 12$. Thus the signed difference can be recovered from the two residues without forming the full integers n_t . Concretely, for $t > 0$ compute $(n_t \bmod 2^q) - 2(n_{t-1} \bmod 2^q)$ modulo 2^q , then use the fixed finite table which sends each occurring residue to its unique representative in $\mathcal{A}$. At $t = 0$, decode the residue of n_0 in the same way. This final decoding has constant storage and introduces no comparison of unknown exact real numbers. $\square$

The integer n_t is always defined before taking its residue. Different places may use different truncations or finite circuits. Only determinism and Equation (12.3) are required; consistency between the finite approximations is supplied by Equation (12.4). In particular, when a digit at t uses n_{t-1} , the latter is evaluated by its own place- $(t - 1)$ definition.

We compute the required residues by finite Boolean circuits. To bound what survives an operand query, we evaluate those circuits from paths of child choices and regenerate their absolute addresses as needed. Here $d \geq 1$ measures the size of the immediate structural descriptors; in the application it is the current group's slack.

The local record of a suspended numerical query contains every local bit that must be retained across an operand query: its binary gate path, return modes, pending bounded results, stored descriptor choices, and any explicitly specified retained word. The input and master choice, the common vertex cursor, and one reusable scratch area are shared facilities. Temporary gate addresses, digit places and scan counters may occupy scratch while they are being computed. Any such data needed after a recursive graph query must either be in the charged local record or be reconstructible from the root parameters and that record. An operand-dependent word retained across the query is charged even if it was originally formed in scratch.

In particular, reconstructing an absolute digit place may use a binary counter in scratch, while the suspended record stores only the choices needed to recover that place. The next lemma makes this distinction precise for the finite circuits used below.

A second source of excess memory is saving an absolute circuit-gate address at every level. Instead, retain the finite choices along the route from the output gate. The absolute address of the current gate can be regenerated in shared scratch from that route. This trades recomputation for a smaller retained record.

Lemma 12.2 (Evaluation from regenerated gate addresses). *Consider a finite Boolean circuit of bounded fan-in, with a specified output gate. Conditional gates may evaluate their selector first and then only their selected branch. Suppose the type and immediate predecessors of a gate, or its operand label and digit place if it is an input gate, can be computed from the root parameters and a path of child choices from the output gate. Suppose these computations use $O(b_{\text{mem}} + t + d)$ scratch bits and make no operand or graph queries. Then depth-first evaluation retains $O(1)$ bits per gate on its current path, and requires no saved absolute gate address per gate. Its suspended record above an operand input has length at most a constant times the length of that path, plus any explicitly retained root-local record.*

Proof. At a gate, save its next-child or return mode and the already computed Boolean input, if any. At a conditional gate also save the selected branch bit. The bounded fan-in makes this a constant-size record. To locate the next child, scan the saved path from its root and regenerate successive gate descriptions in scratch. The final description determines the next child choice. On return, its Boolean result is combined with the saved bit and the local record is popped. All temporary addresses and scan counters are discarded before an operand query; a later scan recreates them from the same root and saved choices. Circuit acyclicity gives termination. This is an evaluation of the circuit unfolded into a formula; repeated gates are recomputed, so no array of previously computed values is stored. $\square$

The next theorem states exactly what the graph computation will need from arithmetic. The elementary operations, multiplication, reciprocals, and indexed reductions are proved in the appendix. Their proofs track what remains stored at a recursive operand request; a bound on temporary arithmetic workspace alone would not be enough.

12.2 The numerical interface

We state the composition result with its syntactic hypotheses. They concern finite arithmetic recipes and do not assert any property of the underlying graph operations.

Application to graph procedures requires a separate certificate. The semantic descriptor domains and structural validity conventions are fixed independently of the requested numerical clock; selected operands already have defined digit interfaces; reciprocal arguments lie in the stipulated positive intervals; and root parameters remain recoverable after lower calls. Clock-dependent truncations and comparison circuits may change n_t , but not its target recipe value. Lemmas 13.10 and 13.11 verify these conditions for the estimator procedures. An incoming rank over the whole vertex universe has no prescribed $2^{O(d)}$ descriptor bound and uses the separate construction of Lemma 13.7.

Theorem 12.3 (Uniform numerical interface). *Fix the digit alphabet, positive reciprocal intervals, and constants bounding the following recipe class. A recipe at slack $d \geq 1$ has a fixed number of stages built from bounded arithmetic operations, fixed piecewise formulas using maxima, minima and clipping, the reciprocals of Lemma A.3, indexed ranks as in Lemma A.4, and sums and known scales as in Lemma A.5. The stage and descriptor descriptions are generated from the root parameters by index computations using $O(b_{\text{mem}} + d + t)$ scratch and no operand queries, and the total non-reduction dependency depth is bounded independently of d . Each indexed domain has size $2^{O(d)}$ and descriptors of length $O(d)$. Discrete domain and validity decisions are structural leaves or buffered tests with their own fixed precision conventions. Public coefficient leaves have exact normalized digit procedures using $O(b_{\text{mem}} + d + t)$ shared scratch and making no operand or recursive graph queries.*

Then each normalized recipe value has a deterministic exact A-stream. For its digit at clock t there are uniform constants a, b, c such that every requested operand digit j satisfies

$$j \leq t + cd, \quad \text{total suspended local binary record length} \leq ad + b(t - j). \quad (12.5)$$

An output code bit satisfies the same bounds. A structural leaf is requested at clock zero and has suspended local length at most $ad + bt$. The maximum intrinsic local record length is $O(d + t)$.

All gate, digit and descriptor addresses can be regenerated from root parameters and the saved local records using $O(b_{\text{mem}} + d + t)$ shared scratch and no recursive graph calls. When $d, t = O(b_{\text{mem}})$ this is $O(b_{\text{mem}})$ scratch. If a public rational coefficient has $O(b_{\text{mem}})$ -bit numerator and denominator that can be generated in $O(b_{\text{mem}})$ scratch, its queried digits in this clock range also require only that shared $O(b_{\text{mem}})$ scratch.

Proof. The primitive constructions in the appendix define deterministic integers n_t . Lemma 12.1 turns them into exact streams. When composing primitives, their operands are these exact streams; numerical approximation errors are not accumulated along the recipe as if they were statistical errors. Instead, each primitive's deterministic n_t is an approximation to its own exact mathematical value. The construction uses only residues of n_t and n_{t-1} and a constant number of output code bits, which adds fixed overshoot and fixed local overhead.

For the record bound, choose one common positive clock coefficient b for the finite primitive and stage types. In any bound $l_* \leq a_0d + b_0(t - j)$ with $j - t \leq c_0d$, replacing b_0 by $b \geq b_0$ is justified by replacing a_0 by $a_0 + (b - b_0)c_0$. Thus the same b can be used everywhere. There are a bounded

number of stages, each with overhead $O(d)$; within a balanced reduction the elementary overheads have total $O(d)$ by Lemma A.5. Along any path the intermediate clock differences telescope, giving Equation (12.5) and total overshoot $O(d)$. The same sum, terminated at an internal arithmetic operation rather than an operand call, gives the intrinsic bound explicitly as follows. If that operation has current clock u , its own intrinsic record has length at most $c_{bd}(d + u)$, while the outer saved portion has length at most $a_*d + b(t - u)$. Choose the common b also to satisfy $b \geq c_{bd}$. Their sum is then at most $(a_* + c_{bd})d + bt$, because $u \geq 0$. Increasing the stage constants as above makes this same choice of b valid in all local bounds. The early-word case is already included in Lemma A.3: those retained bits are part of the local record and are paid for at the actual original-input clock by Equation (A.12).

Structural validity branches are finite branches in this circuit organization, with length at most its full intrinsic bound. They ask the structural operand at clock zero. Buffered tests are separate recipes with structural starting clock zero and precision fixed by Lemma A.6; they are not altered when a surrounding value is requested more accurately. Thus no discontinuous comparison of exact reals at unbounded precision is part of the interface.

All circuit constructions in the appendix use padded domains and explicit index rules. A saved gate choice selects an operand, a subtree, a bit of a known-width word, or one of the finite primitive modes. Scanning these choices from the root recovers the current descriptor and digit places by integer arithmetic. The reciprocal's extra records additionally specify its early-prefix formation state. None of these computations refers to an ancestor cursor address or requires an operand value to discover the next gate. Their integers have $O(b_{\text{mem}} + d + t)$ bits; repeated scans reuse one scratch area. Lemma 12.2 therefore applies at every Boolean portion of the recipe.

Finally, for a public coefficient p/q , $q > 0$, producing a place- r approximation uses integer division of $2^r p$ by q , with signed floors when needed. The inputs and intermediate remainders have $O(b_{\text{mem}} + r)$ bits. Standard finite binary division and multiplication use that much space, and extracting the low residue or difference digit requires no larger word. All operand bits in this computation are public, so it makes no recursive graph call and can use shared scratch throughout. For $r = O(b_{\text{mem}})$ its scratch bound is $O(b_{\text{mem}})$. In contrast, any word whose bits depend on recursive operands and must survive a new operand request has already been charged to the local record. $\square$

Remark 12.4 (Use of clocks). Theorem 12.3 is a finite-request statement for every t . Its exact-stream conclusion follows by considering these finite requests separately, and does not require a machine to choose structural domains or port orders at infinite precision. For the actual top-level computation all reached clocks will be $O(b_{\text{mem}})$ by the allowance calculation in the next section. Known rate and field coefficients have $O(b_{\text{mem}})$ -bit descriptions, and a fixed rational generator probability raised to a walk length $O(d)$ has numerator and denominator of $O(d)$ bits; these are instances of the public-coefficient provision above.

13 Evaluate the estimator with logarithmic total memory

The error theorem tells us that most environments give accurate estimates. It does not yet give an algorithm in the claimed space. We now implement the table and vector queries and count every bit retained by every active caller. This count must hold for inaccurate environments too, because the final deterministic algorithm enumerates all environments.

There are two ways a naive implementation could lose the bound. It could save the old vertex whenever it visits another one, or it could save a full numerical value while evaluating another operand. The first problem is addressed by one shared current vertex and restoring visits. The

second is addressed by the digit-query bounds of Theorem 12.3. The remaining task is to prove that these mechanisms compose across the entire recursion.

We keep one vertex identifier, called the cursor. Consider a product term $\mathbf{F}_{xz}\mathbf{B}_{b,zy}$ from Equation (9.2). For a selected valid product descriptor, a query begins with the cursor at x . The correction port moves it to z and supplies a bounded inverse port. The right factor can then be queried at z , using its own common table tuple; that tuple contains no copy of x . Returning along the inverse port restores x . Arithmetic requests the two factor digits separately, so it need not retain a complete left-factor value while computing a right-factor digit. If the operation instead asks to reach y , the right factor's terminal motion is the last recursive delegation; only local label completion follows it. These are the paths that Lemma 9.5 requires us to realize.

Incoming access reverses a different object: a finite program whose canonical initial states describe the forward row ports. We make the program deterministic with halting success states. The component of a specified success state is then a tree directed toward that state; its canonical initial states are precisely the valid row ports ending there. A contour of that tree enumerates them and returns to the success state without saving a second copy of its vertex address. This gives both capped incoming ports and incoming score ranks. To use the contour we must define program states and their predecessor queries carefully: working states of a completed lower query are not themselves part of the calling program graph.

The storage bound follows the same distinction. A caller retains its own descriptor choices and numerical path while a lower procedure runs. A slack- d layer pays $O(d)$ structural bits; a query from digit place t to place j pays for its numerical records through the bound $O(d) + c_{\text{num}}(t - j)$ of Theorem 12.3, for a fixed coefficient c_{num} . We assign a decreasing allowance to the finite operation order within a prefix, the move to its preceding prefix of slack $2d$, and the move to a child of accuracy level at most $m - d$. The resulting differences telescope over the physical stack. The proof below establishes these record bounds on the actual recipes before imposing guards on artificial program states.

Here is the order of the argument. First, define exactly what a local program state contains and prove how to query its predecessor states. Second, use a traversal of that finite program graph to implement incoming access. Third, verify the required local storage bounds for each estimator operation. Finally, assign a decreasing allowance so that these bounds telescope over the entire active stack. The distinction between a local program state and the working memory of a subroutine called by it is essential at every step.

13.1 Shared memory, call records, and query types

Fix the input and a master environment. The latter includes the rank matrices and all the fingerprint salts. A *table tuple* consists of the estimator type and stage, its accuracy level, scale and usage rates, channel roles and suffixes, and its effective environment. The tuple specifies an entire table, rather than an individual row or entry. Usage rates belong to this tuple; the separate integer clock t specifies a requested numerical digit. A generator and wrap word is a call-record descriptor used to derive a child's effective environment. It is not an additional semantic parameter of the child table: after this derivation, the child's values, forward port orders and guards depend on the resulting effective tuple, not on the ancestral word which produced it. Distinct walk descriptors are still separate summands with their prescribed weights, even if they induce the same environment transformation. The variants of Lemma 8.4 can be generated without their conditioning vertices. A vector tuple is defined in the same way, omitting the unneeded terminal parameters. All admissible absolute tuples have $O(b_{\text{mem}})$ bits.

The mutable cursor records a vertex and its stage interpretation. Other shared facilities are the

input, the top root parameters, the master environment, a catalytic bit vector of length b_{mem} , and one reusable scratch area. Its size is $O(b_{\text{mem}})$ in the actual computation, where the top numerical clock is $O(b_{\text{mem}})$; for a general top clock t it may use $O(b_{\text{mem}} + m_{\text{top}} + t + 1)$ bits. Scratch may be used to perform arbitrary local integer computations in this space. It has no content which must survive a recursive graph operation. Every such operation starts its scratch calculations canonically. Before every input access, its requested position is regenerated in shared scratch. The physical input head stays between the real endmarkers; off-data simulated positions use the base symbol routine from Lemma 3.1. These rules also apply during inaccurate calculations and guarded artificial predecessor probes. The requested coordinate and scan counters coexist only within this pure routine and are discarded before a recursive graph operation. It is harmless to copy an address temporarily inside a scratch calculation, but such a copy is discarded before a recursive graph operation begins.

We distinguish a visit that returns from a motion that does not. A paired edge supplies a short inverse token for the return. A more general visit retains a sequence of immediate choices that can be undone in reverse order; a final one-way motion has no such return obligation. The traversal construction below handles these cases separately. The finite interfaces to be constructed are as follows. A port is an index in a specified row class; the class may have holes.

1. A *restoring test* returns one bit. A *quantity query* returns one bounded signed digit of a specified value. Both restore the cursor and the catalytic vector.
2. A *terminal motion* starts at a row and a port and finishes at that entry's endpoint. No return address is provided. An *excursion* makes the same visit, performs a prescribed local action there, and returns. The action may use scratch and the catalyst but makes no recursive graph query.
3. A *paired motion* maps a valid row port to its endpoint and a bounded inverse port. Its inverse has the same table tuple. On invalid ports the restoring existence test returns false; an exported motion may instead return failure without moving the cursor.

At the final row interface all port alphabets are bounded by constants. At a prefix of slack d , indices and immediate descriptors have $O(d)$ bits. These two statements must be distinguished: an entire terminal path represented by one prefix port need not have a description of length $O(d)$.

A port code may be in range and still name a hole in its class. Its existence test returns false, its quantity query returns the zero digit, and its motion or excursion fails with cursor and catalyst unchanged. Rank operands retain their separately prescribed baseline or padding values: an absent real port is not a virtual default coordinate.

Theorem 13.1 (Realization of the estimator interfaces). *Fix the construction constants and an admissible top accuracy level $m_{\text{top}} = O(b_{\text{mem}})$. For every actual task tuple reachable from it, the row, vector and prefix interfaces defined in the preceding sections admit deterministic uniform implementations by restoring queries and cursor motions. The following assertions hold for every master environment, without an accuracy promise on its estimated values.*

1. *Every test or numerical digit query terminates, returns its defined value, and restores the cursor and catalyst. Its result is independent of incoming catalyst and scratch contents. Structural port and order choices are independent of a higher numerical request.*
2. *Terminal motions give the specified endpoints. Their excursions restore the cursor. Capped correction motions are partial paired maps with bounded inverse ports and safe behavior on invalid ports. Incoming score ranks compute the stipulated zero-padded ranks.*
3. *For a root numerical request at clock t , all records retained by the call and its descendants have total length $O(m_{\text{top}} + t + 1)$, and every descendant clock is $O(m_{\text{top}} + t + 1)$; the shared facilities*

use $O(b_{\text{mem}} + m_{\text{top}} + t + 1)$ space. In particular an actual top request with $t = O(b_{\text{mem}})$ uses total deterministic $O(b_{\text{mem}})$ space.

These implementations use no estimator oracle and retain no second vertex address across recursive graph operations.

The proof also constructs every lower operation instance admitted by the finite parameter ranges and permitted parent-to-child recipes. This includes calls made while testing possible predecessor states, whether or not those states occur in a designated execution. The larger construction is needed to compute incoming accesses by inverse traversal.

13.2 What belongs to a local program state

A terminal program has a fixed root tuple and root clock. Its initial configuration is the cursor x , a distinguished initiation tag, and the original row port p in a separate recoverable field. Distinct ports give distinct initiation states. In a scoring program the success local state contains a quantized score; otherwise it has only a success tag. The starting row and port are not root parameters. The program universe thus includes all rows and all ports of this same table. Similarly, a scoring universe at a fixed precision includes all possible quantized scores.

The program graph is an analytical and algorithmic description of one local procedure. It must not contain a fresh copy of every lower procedure's working state, or inverse traversal could repeatedly copy a growing stack. We therefore distinguish two forms of invocation inside this program. A restoring bit query, or a lower paired motion, is an *opaque transition query*: its temporary working configurations are not vertices of the terminal program graph. They are computed by the lower procedure when the transition is probed. A terminal delegation with no bounded inverse port is instead *explicit*: it pushes the lower terminal controller into the local state. Thus the vertices of the program graph comprise the cursor and a finite chain of explicit controllers, but not the working states of opaque transition queries. This distinction will also be used when graphs are traversed inversely.

A full table tuple has $O(b_{\text{mem}})$ bits, but saving a tuple at every recursive level would be too expensive. A call instead records how its parameters differ from its parent's: an operation choice, an accuracy level or scale offset, a copy symbol, or an arithmetic path. Starting from the outer parameters, a scratch computation can replay these choices whenever an absolute tuple is needed. The following definition makes this requirement precise.

Definition 13.2 (Regenerable record). A local record is a delimited binary word with a tag from a fixed finite alphabet. It may contain immediate descriptor choices, bounded port or copy symbols, offsets, Boolean evaluation paths and pending bits, and the early numerical words permitted by Theorem 12.3. Head positions in such words are marked locally. A root tuple, a root operation and its numerical clock, together with the active records, determine every descendant tuple and clock by a scratch computation. This computation does not call a graph procedure and does not use a cursor that has been left behind. We call such records *regenerable*.

The records use delimited encodings, with marked positions rather than an independent absolute stack pointer at every level. Self-delimiting encodings enlarge a nonempty word of length r by $O(r + 1)$ bits. Constant tags also identify a root boundary, the active candidate copy during a neighbor test, and the owner of a pending endpoint action. These tags permit a scratch scan to select the appropriate active view of the records. An inactive copy is not part of a descendant's program state or call recipe.

A root boundary is a constant tag, not a stored copy of an absolute table tuple. The logical root parameters of a nested query are obtained by replay up to that boundary; the temporary absolute tuple is then discarded before another graph query. Thus the phrase “fixed root tuple” specifies which mathematical program graph is being used. It does not allocate a fresh $O(b_{\text{mem}})$ -bit parameter word at every nesting level. Lemma 13.11 supplies the replay procedure.

The local storage bound must measure what the caller retains across a child call. Memory used only temporarily before that call can be reused; memory needed afterward must either be counted or recomputed. We express the retained cost using the decrease of a structural allowance and the change in requested digit place. The term *raw* in the definition means the caller’s own records, before adding the constant number of copies used by an inverse-neighbor probe.

Definition 13.3 (Raw allowance certificate). A raw controller is the caller’s own selection, arithmetic, scoring or terminal-path record, excluding the working states of a query it calls. In a terminal program graph an explicit chain contains only its terminal-path controllers and, in a scoring program, its score suffix or completed score word. The interiors of cap traversals, inverse score ranks, numerical digit queries and paired motions are opaque queries; they are never explicitly pushed working graphs in this chain.

A raw controller has an integer allowance $w \geq 1$ and a numerical clock $t \geq 0$. Fixed raw constants $a_0, b_0, a' > 0$ give the following certificate at a call to a lower query or an explicitly pushed raw terminal controller at (w', j) :

$$w' < w, \quad j - t \leq a'(w - w'), \quad \ell \leq a_0(w - w') + b_0(t - j). \quad (13.1)$$

Here ℓ counts only the suspended caller record, including its delimiters, return modes and pending data. It does not count the callee’s physical working space. Without a pending call its own raw records have length at most $a_0w + b_0t$. A structural operation has a canonical starting clock zero; numerical queries used by its finite-margin tests are included in this certificate.

We will prove this bound for the actual recipes below, before using guards to restrict artificial program states. In particular, guard insertion will not be used as a substitute for proving that a designated execution fits its bound.

We next restrict the elementary instructions so that a predecessor can be found by finitely many local guesses. In particular, an instruction may not erase a long unknown word in one step: reconstructing its predecessor would then require guessing that whole word. The following finite instruction types perform erasure and movement locally.

1. *Local edit*. Read a bounded number of marked local symbols, change a bounded number of symbols and a finite mode, and advance the marked local heads by one position. A push opens a delimiter and writes a tag in a bounded number of steps. A pop is performed only after the child word has been erased down to that tag, also by local steps.
2. *Pure or restoring branch*. Compute one bit, either in scratch without a graph call or by a lower restoring test or digit query, and change a bounded local symbol or mode. A bounded signed digit is handled as a fixed number of bits. A longer selection result is written one bit at a time, with its index stored in the local record.
3. *Paired cursor step*. With its motion tuple and short token already selected, apply one lower partial paired map, or one elementary base or label map. Its output token and finite mode are the only simultaneous local changes. The linear-size motion tuple is regenerated from records; it is not written as an opaque payload.

4. *Explicit terminal call.* Create a lower controller with a regenerable call recipe. Ordinary instructions then execute that controller. On its return, local erasure and the prescribed bounded label completion are performed by the preceding instruction types.

An endpoint action in an excursion may be a pure computation reading or toggling the catalyst. Such actions are not individual edges of a terminal program graph: that graph uses the complete restoring, catalyst-independent predicate which contains the excursion. Its other pure branches depend only on initialized scratch, the input, cursor and permitted tuple data. A value test is never combined with a cursor-changing step. Conditional validity can be a separate restoring branch; a partial paired step may simply have no edge on an invalid domain.

Every record position erased by a local edit is marked before the edit, or is a known end of a delimited word. In particular, an instruction cannot erase an arbitrarily chosen position and then forget its index. Counters, carries and loop heads are manipulated by these local steps. Pure computations and restoring tests can be recomputed to select one output bit; their scratch states and physical input heads are not included in the states whose predecessors are enumerated.

A predecessor search examines candidate states, not only states known to occur in a valid execution. A fabricated candidate can contain a stale or false validity flag. Therefore a lower procedure must check its arguments at entry rather than trust a caller's saved flag. Predecessor probes can reach artificial local states whose stored flags were not produced by a designated execution. The lower query interfaces must therefore be safe independently of those flags. We use the following entry conventions before constructing the graph.

Total calls on invalid ports. An argument is *range-valid* when its tuple, operation type, cursor type, port code and clock lie in the permitted finite ranges. A range-valid port can still name a hole. Every exported existence predicate is a total restoring test of its earlier defining masks. Every port-dependent quantity query freshly tests presence at structural clock zero before entering its value calculation. If presence fails, the query returns the zero digit of the zero stream. A forward motion or excursion instead returns failure, with cursor and catalyst unchanged and without performing an endpoint action. A paired inverse uses its target-side incoming-count and inverse-rank validation. Excursion validity is checked before building its chain of terminal-path records, not anew at the remote endpoint or during rollback. An endpoint comparison returns false if either freshly tested descriptor is invalid. Wrong types or out-of-range codes are rejected by a pure check before any lower call.

Baseline queries and rank-padding inputs retain their separately specified values: an absent port is not a virtual default coordinate. In particular, incoming score ranks are padded by zero, whereas a compressed-array rank uses copies of its baseline. Every opaque query tag invokes the canonical entry of one of these total procedures. No permitted call recipe targets an internal continuation after its presence test. A historical validity flag in a caller is never accepted as a certificate of presence. Lemma 13.10 proves that the fresh tests use earlier operations and fit the allowance bounds.

We can now form the finite local graph. Syntactic guards reject records outside the allowed ranges before invoking a lower query. These guards make artificial predecessor probes safe. They do not prove the storage bound for the intended computation: that remains a hypothesis here and is established for the actual recipes below.

Lemma 13.4 (Guarded local graph). *Suppose a fixed terminal recipe uses the preceding instruction types, has regenerable records, and obeys (13.1) on its designated executions. Suppose all its opaque transition queries are already implemented, are total on admissible arguments, and have the stated safe invalid-port behavior. Their returned bits, success flags, cursor outputs and port tokens are independent of incoming scratch and catalytic contents, and they restore the catalyst. Then there*

is a finite graph with outdegree at most one, uniformly associated with its root tuple, with these properties.

1. Its designated executions are precisely the prescribed terminal executions. Success states have no successor, and initiation and success tags are distinct.
2. Predecessor cases, ordered incidences, and paired edge movement are computable with a bounded number of case labels and one cursor. An opaque predecessor query is made only after its local bound has been checked.
3. If a transition query at (w', j) is probed while the terminal root has parameters (w, t) , the complete live local portion of this probe, including a fixed number of edited or saved controller copies, has length

$$a(w - w') + d_{\text{num}}(t - j) \quad (13.2)$$

for fixed enlarged coefficients a, d_{num} . Shared facilities and the outside stack are not copied.

The graph and all its guards are independent of an originating row, an outer stack's data, and catalytic contents.

Independence from the outer stack is understood after fixing the logical root tuple: two invocations that regenerate the same tuple define the same graph. Their physical replay histories may differ. The hypothesis on designated executions is proved for the concrete recipes in Lemma 13.10; the guards below extend query safety to artificial states reached while enumerating predecessors.

Proof. We construct the graph, explain each kind of predecessor probe, and then count the records retained by such a probe.

Use as potential states the cursor and delimited words having the specified finite tags. Check absolute parameter ranges, descriptor ranges, stage and suffix types, instruction modes, and the allowed parent-to-child recipes. Check the intrinsic word bounds of Definition 13.3. For each explicit chain also check the local bounds along that chain. Before any opaque query, check the bound from every ancestor of its local explicit chain to that query. All absolute parameters, clocks and lengths needed for these checks are regenerated or counted in scratch. No graph query is needed for a guard. At a state with an unavailable instruction omit that edge.

The designated paths pass the guards by hypothesis: their boundary bounds add along an explicit chain and telescope. Enlarging the coefficients once covers fixed instruction tags and bounded intermediate edits. The resulting graph is finite since its local words have $O(w + t)$ bits and its cursor has $O(b_{\text{mem}})$ bits. It is deterministic since every branch bit and partial paired map is deterministic. Guards may create additional dead ends, but they cannot create a new path from a canonical initiation to a success state.

The first predecessor cases keep the cursor fixed. For a local edit, predecessor candidates are obtained by guessing the bounded overwritten symbols, preceding finite mode, and bounded head moves, then testing the instruction forward. Marked positions and incremental erasure make the number of cases bounded. For a branch, perform the same fresh restoring query at the unchanged cursor and test the guessed preceding symbols. Incorrect historical branch bits do not resume the interior of a lower procedure: the query is always started again on its regenerated arguments. The guards are checked before the query, including for a candidate predecessor.

A cursor-changing step needs more care because its predecessor vertex is not available yet. First reconstruct and check the candidate's local record using the present vertex and the recorded recipe; only then invoke the lower paired inverse. For a cursor step, first construct the possible preceding local record by a bounded inverse edit, guessing the bounded preceding token if needed. Before any motion, replay that candidate's active records from the fixed root tuple. Check the instruction and

stratum, permitted child kind, accuracy level, scale and suffix ranges, salt permission, numerical clock and every ancestor-to-query raw length bound. These allocation and recipe checks do not use the predecessor cursor. The lower motion tuple is then fixed by the candidate record and the present output token. Apply the partial inverse at the present cursor and output token. If it succeeds, the returned token and preceding local symbols specify the candidate predecessor. Static checks and token comparisons can be made there. A guessed preceding port is compared with the actual port returned by the inverse. Any remaining cursor-specific type or domain check is pure at that cursor; a domain condition requiring another restoring query belongs to a separate branch edge. Thus it does not conceal a second high-cost query inside this paired step. To restore the probe, apply the paired forward map with that same tuple and the actual returned token. No saved copy of the cursor is used. Conversely, an outgoing paired probe checks the proposed target guards after its motion; if a guard fails, or if the probe is only a restoring incidence test, its inverse with the actual output token restores the starting cursor. Both directions have been allowed the same lower-operation cost. No unrelated value query is part of this cursor step.

Label an edge by its finite instruction case and bounded erased symbols. Its forward instruction determines a canonical reverse case code. An inverse probe accepts only that code. This removes any duplicate descriptions of the same predecessor edge without comparing two retained addresses. The finite case codes give a fixed local order and a bounded incidence token; a forward edge knows its reverse token.

Finally count the storage used while probing a neighbor. The important fact is that the copied material is the local explicit chain, not the older outside stack or the working state of an opaque query. For the space assertion, first ignore saved probe copies. Adding the boundary inequalities along the current explicit chain gives the bound $a_0(w - w') + b_0(t - j)$. A current state and a candidate predecessor differ in length by a constant: this follows from the bounded edit used to *construct* every candidate, before it is known to be an actual predecessor. Consequently retaining or editing both costs only a fixed multiple of that bound, with a constant adjustment absorbed by the strict allowance decrease. More explicitly, write $\Delta w = w - w' > 0$ and $\Delta t = t - j \geq -a'\Delta w$. For a fixed copy factor k and additive bounded metadata cost h , choose

$$d_{\text{num}} \geq kb_0, \quad a \geq ka_0 + (d_{\text{num}} - kb_0)a' + h. \quad (13.3)$$

Then $k(a_0\Delta w + b_0\Delta t) + h \leq a\Delta w + d_{\text{num}}\Delta t$. This also handles a negative clock decrement. These coefficients are uniform over all finite instruction cases.

The constants a_0, b_0 describe the raw skeleton uniformly over the whole recipe family. They have not been replaced by the physical probe constants a, d_{num} in a lower skeleton's guards. Fix the maximum finite copy factor k and metadata cost h and choose a, d_{num} once. A lower compiled query contributes a new physical stack segment when it runs; its contour state and saved copies are not words in the calling graph. Thus this estimate copies raw records, not another a, d_{num} -sized physical segment.

Only the local explicit chain is saved or edited. A nested query has its own root boundary and appends its local portion below the existing stack; its program graph excludes the caller's saved copies. A tag selects the candidate chain for parameter replay, ignoring an inactive saved copy. Therefore a later neighbor probe does not copy the older outside portions again. Finally, all guards and forward case codes use just the fixed tuple and its permitted records. Imposing the same salt-visibility convention as the underlying recipe preserves this independence even at synthetic states. $\square$

13.3 Find incoming entries by traversing a program tree

A terminal program sends a valid row port to a destination. To find entries coming into a given destination, start from its canonical success state and search backward through the program's possible executions. Because this local graph is deterministic and the success state halts, its component is a tree directed toward that halt.

Walking around the tree in a fixed local edge order visits all its canonical initial states. The current state and a bounded incidence token suffice to continue the walk. Returning to the distinguished success state recovers the destination without retaining a second copy of its address. This traversal is a contour of the tree.

The rotation–edge-flip traversal below is the forest permutation of Cook and McKenzie [18, Proposition 1, p. 387], with a sentinel to include an isolated root. Traversal of configuration trees also underlies space-preserving reversible simulation [21]. We prove the contour and incoming operations needed here, including their local record bounds. In particular, the use of a contour will not replace the separate allowance and guard argument.

Lemma 13.5 (Inverse contour). *Let a finite deterministic graph of the form in Lemma 13.4 have a specified canonical success halt. Its undirected component is a tree directed toward that halt. There are mutually inverse next-mark and previous-mark operations on its contour, where the marks are the root sentinel and one distinguished occurrence of each canonical initiation in the component. These operations require the current program state and bounded incidence control, but no saved root or contour-length counter.*

Proof. First establish that the component really is a tree. This property comes from determinism and the specified halt; it is not a property assumed of the original configuration graph. Following the unique successor from any node in the undirected component reaches its sink. Indeed a directed cycle in a graph of outdegree at most one cannot be joined by a path to a sink: at the first connection the unique outgoing edge is already used toward the cycle. Equivalently, on an undirected cycle all its vertices must use their outgoing edge on that cycle, so no sink lies in its component. The same argument excludes two sinks in one component. Thus the component is a finite tree with the given unique sink.

Take the incidences of its edges, with an additional sentinel incidence at the root. Edge flip exchanges the two incidences of an edge and fixes the sentinel. At each vertex cyclically rotate its incidences in the fixed local order. Rotation followed by flip, or its inverse, gives one contour cycle on these incidences. This follows by induction on the number of edges: inserting a leaf inserts its two incidences into the old cycle. The isolated-root case consists only of the sentinel.

At a canonical initiation mark its unique successor incidence; mark the sentinel at the root. Each initiation is therefore counted once. Iterating the contour successor to the next mark always terminates, since the finite contour contains the root mark. Iterating its inverse to the preceding mark is the inverse operation. A success tag and the sentinel token identify the root in this component; its address never has to be compared with an earlier address. Constant-degree adjacency and edge flip are provided by Lemma 13.4. $\square$

For correction access, we only need to distinguish at most k_c incoming entries from more than k_c . The contour can stop counting after this fixed threshold and undo its bounded number of jumps between marked states. Hence no counter proportional to the total size of the program tree is required.

Lemma 13.6 (Capped incoming ports). *Let a fixed row class have a constant port alphabet, distinct entries within each row, and a deterministic terminal program which maps its valid row ports to their endpoints. Assume the program is of the form in Lemma 13.4. For every fixed k_c there are restoring tests and mutually inverse paired motions for precisely those class entries whose endpoint has at most k_c incoming class entries. The inverse ports belong to $\{1, \dots, k_c\}$. No additional endpoint address or unbalanced path is held across these operations.*

Proof. We first ensure that marked initial states count exactly the intended entries, then describe the two directions of the paired map. Give invalid row ports a distinct failure execution. The canonical initiations in a success-root component then correspond exactly to valid class entries at that endpoint. Class validity is tested before using a forward initiation as a contour starting point.

Start at a valid initiation mark. Go forward toward the sentinel, counting other initiation marks, stopping either at the sentinel or after k_c other marks. Store the number of mark jumps, at most $k_c + 1$, and restore the start by that many inverse jumps. Do the same in the backward direction and restore again. Reaching k_c other marks certifies excess. Otherwise, let a and b be the numbers of other initiations before the sentinel in the forward and backward directions. The total is $1 + a + b$; accept exactly when this is at most k_c . The initiation's rank after the sentinel in forward order is $r = b + 1$. A forward paired motion goes to the sentinel in $a + 1$ mark jumps and returns the inverse port r .

From a canonical endpoint halt, count forward marks until returning to the sentinel or encountering $k_c + 1$ initiations. In the latter case restore the sentinel by the bounded number of inverse jumps and reject. In the former case the total $n \leq k_c$ is known. For an inverse port $r \in \{1, \dots, n\}$, go forward by r mark jumps to its initiation and return the row port stored in that initiation state. Invalid ranks are rejected at the sentinel. A halt with no preimages has $n = 0$ and is harmless. Wrong vertex types can be rejected before creating a halt state.

Every unsuccessful inverse request therefore leaves its endpoint unchanged: an excessive incoming count is undone before failure, and an invalid rank is detected at the recovered sentinel. For a successful request the returned row port is the one in the reached initiation, not a guessed port. This is the token used by a later forward restoration in Lemma 13.4.

These are inverse maps by the common contour order. Existence tests make the corresponding visits and reverse them, so they restore the cursor. All counts, ranks and incidence tokens have a fixed alphabet. The terminal program graph has the common table tuple as its root; neither direction fixes the starting row or port as an external parameter. Consequently the two directions use exactly the same graph. Restoring lower queries restore the catalyst, and the contour itself does not otherwise change it. $\square$

The incoming scores need not have bounded support in a column, so a bounded incoming-port list does not suffice to compute their rank. We instead give the terminal program a success state containing both the endpoint and a rounded score. A contour at that halt counts the entries with that rounded score; successive contours visit the scores in decreasing order. The endpoint and score belong to the current program state. During a tour they may be replaced by an earlier state of the score computation, and the contour recovers them when it returns to its unique halt. Thus neither is retained separately while an earlier state requests a score digit. The next proof combines this observation with an assembly order that stores only a short suffix before a late digit query.

Lemma 13.7 (Incoming score rank). *Let a constant-size forward row class cover every positive entry of a bounded nonnegative score table, with distinct entries in each row. Suppose pre-incoming score digits are restoring queries, and the class has a terminal program with the preceding interfaces. A fixed incoming k th-largest value, with zero padding, has exact redundant digits computed by a restoring procedure. Its local record above a place- j score query at output approximation clock t is $O(1 + (t - j)_+)$, with a fixed overshoot $j \leq t + c$, in addition to the structural allowance costs. A complete $O(t + 1)$ -bit score word is retained only over structural calls.*

Proof. The incoming score column can be long, so the previous bounded incoming list is insufficient. We quantize scores, visit their values in decreasing order, and count matching initial states by contours. The quantized value belongs to the current program state; it is not an extra word retained throughout the traversal.

Normalize the score range to $[0, 1]$, using a fixed public factor if necessary. At approximation place t , choose a fixed extra margin c . For the digit truncation $v^{[t+c]}$ define the integer

$$q_t(v) = \text{clip}_{[0, 2^t]} \left(\left\lfloor 2^t v^{[t+c]} \right\rfloor \right). \quad (13.4)$$

The digit tail has absolute value at most a fixed constant times 2^{-t-c} . Choose c so that this is less than 2^{-t} . Clipping is nonexpansive and fixes $2^t v$, so $|q_t(v) - 2^t v| < 2$. The same margin gives $q_t(0) = 0$: a truncation of the zero value has scaled magnitude less than one, so flooring and nonnegative clipping give zero. Since the forward class contains every positive score, enumerating its entries and padding by zero therefore gives the rank of the whole quantized score column.

To keep a long completed score from coexisting with a late score-digit query, assemble digits from least to most significant. Before a late digit request, only a short suffix has been formed. Use a forward terminal program which first tests the entire class and membership domain at its canonical initiation, sending invalid inputs to failure. Its success *configuration* contains the endpoint in the cursor and $q_t(v)$ in its local state. Before making its final structural motion, assemble the score digits from least to most significant. In units 2^{-t-c} , before the place- j digit is queried the accumulated suffix and its carry have $O(1 + t + c - j)$ bits. A self-delimiting offset from $t + c$ fits the same bound. After assembly, division by 2^c , flooring, clipping and loop setup are pure local computations. The complete word is then held only through structural motion and local clearing to success. Thus all its graph-query boundaries satisfy the claimed record bound. The success word is part of the graph state, not a parameter of the graph. At this fixed t the graph includes every allowed quantized value.

Now enumerate the quantized scores without saving a separate loop value across the contour. At the requested endpoint, initialize the success state at $q = 2^t$. Tour that root component and count canonical initiations, saturating at the desired fixed rank. Continue to the sentinel even if the saturated count has already been reached. At the sentinel the same endpoint and quantized value have been recovered; no other success halt occurs in that component. If the cumulative count has not yet reached the requested rank, decrement the value field by pure local steps and tour the next success component. The value field itself is the loop variable. No separate copy of it survives the tour. The saturated count and control bits have constant size. At value zero apply the stipulated zero padding.

If the desired rank has been reached, use the value field only after the tour has returned to the sentinel. The clock t is a root parameter throughout; the loop value q is not. Consequently an inverse step which reaches an earlier partial-suffix state replaces the current score state by that suffix state, and need not retain the final q or the endpoint address alongside it. Unique-sink recovery supplies both again when the component tour ends.

This computes the integer order statistic n_t of the values $q_t(v)$. A uniform perturbation of a finite list changes any order statistic by at most the same perturbation, including after zero padding. Thus n_t has error less than 2 as an approximation to 2^t times the desired rank. Keep only the fixed number of residue bits needed to form its redundant digit, clear the other local data without a graph query, and return at the unchanged endpoint. Computing n_{t-1} uses its own quantization convention; the difference-digit construction needs only the bounded errors, not nested quantizations.

During a tour the live program state may be a partial suffix state preceding a high-place query. It then contains that short suffix, rather than a saved copy of the final score. Lemma 13.4 therefore also gives the required bound for inverse neighbor queries. All root components are finite, including isolated success states, and the decreasing finite value loop terminates. Fresh lower procedures and the contour preserve the catalyst. $\square$

13.4 Verify the storage bounds for the actual estimator operations

The preceding lemmas are conditional on local storage and dependency guarantees. We now verify those guarantees for the estimator itself. This is where the representation choices from the earlier sections enter the implementation: bounded correction ports, short descriptions at each accuracy level difference, fixed-accuracy access tests, and queries for individual numerical digits.

The checks concern which procedures call which others and what data remain stored during those calls. They do not assume that an estimate is accurate.

Lemma 13.8 (Ordered terminal paths and excursions). *Every unbalanced path required by the estimator has the following ordered form: restoring selection and validity tests; local label maps and, if necessary, a paired left-factor motion; further restoring tests at that midpoint; at most one final terminal delegation; and then only local label completion. An excursion to its endpoint and back, for a pure local action, has regenerable records. It preserves the same allowance and clock bounds as this ordered path, up to fixed factors. No unbalanced spine is retained over a new high-cost query at its endpoint.*

Proof. First inspect the actual path shapes. The only unpaired motion is a final delegation; after it returns, there is no new recursive query at the remote vertex. For a direct summand the path is a local base path, a projected child path, or a child row port. A projection stores its bounded symbol before removing it and appends the prescribed destination symbol after the child terminal motion. For a product the left factor is a correction with paired access. Its inverse token is stored at the midpoint; right-factor validity and weight queries are restoring there. The right terminal path is then the last delegation. An old prefix port delegates to that preceding prefix; after its return it requires no new graph query. These are exactly the path cases of Lemma 9.5.

This ordering also permits a returning excursion. Keep only each frame's immediate choices on the way out, perform the local action, and discard deeper records before invoking an earlier inverse on the way back. There are two compiled continuations of this same terminal syntax. A one-way continuation clears its path records after performing its prescribed local completions. A visiting continuation retains the immediate choices and inverse tokens on its explicit terminal spine. After the local endpoint action it undoes the bounded completions, undoes the last delegated spine, and then uses the paired inverse token of an earlier motion. Each deeper record is cleared before invoking an earlier inverse map. For a projection, the saved symbol and prescribed append symbol give its local inverse. The identical lower paired tuple and actual returned token give the inverse of a correction move.

This is an induction on terminal delegations, not an attempt to invert an arbitrary unbalanced macro in one step. A frame stores only its own immediate descriptor, bounded choices and phase. Its returned child spine is used for local completion and the endpoint action, which make no graph calls. When a further graph query is made during rollback, the deeper spine has already been discarded. Thus the suspended records are those at the corresponding forward boundary, enlarged by bounded phase and inverse-token data. The choices can be re-created or kept on their own frame; no concatenated description of all old prefixes is passed to a new tight query.

In a restoring use, intrinsic constituent validity is checked before the visit. A product can test its right port by crossing the paired left edge and returning; this test does not use the right terminal path. Outer slice and wrap tests then use a valid intrinsic excursion. Hence none of these validity checks recursively invokes the very bucket or shortlist membership it helps to define. The endpoint action's owner is marked in the outer frame, so its tuple can be regenerated by a scratch scan even while a descendant path is retained. The action's result does not alter the selected paths or their undo. $\square$

Lemma 13.9 (Implemented endpoint comparison). *At slack d , comparisons of two valid candidate or old-list endpoints are restoring, catalyst-independent tests with $O(d)$ suspended structural control, in addition to the costs of their lower endpoint visits. They use no saved endpoint address. Their choice between an exact comparison and the designated fingerprint comparison is independent of a surrounding numerical clock.*

Proof. The two descriptors occupy $O(d)$ bits. By Lemma 13.8, each endpoint can be visited and restored for a local bit or hash action. If $d > 2h_*$, compare the $O(b_{\text{mem}})$ identifier bits in order, retaining only the bit-position counter and bounded flags. Its counter uses $O(\log b_{\text{mem}}) = O(d)$ bits, since $h_* = \lfloor \epsilon \log_2 b_{\text{mem}} \rfloor$ with a fixed positive ϵ and sufficient minimum padding.

Otherwise use the four visits in Lemma 11.3: flip the first key's bit, xor the second key's bit into a result, and repeat. The final result is key equality for every incoming catalyst and that vector is restored. Key evaluation and the indicated vector-bit action are pure endpoint operations in scratch. No key index is saved across another visit. Nested structural procedures already restore the catalyst and give results independent of its contents, by their lower operation order. The outer descriptors and all path choices ignore the accumulated result bit. The two repeated visits therefore use identical paths. This proves the same independence and restoration for the completed comparison, supplying the induction hypothesis at the next layer. The comparison mode is determined by m, d and the public block boundaries, not by the requested numerical place. $\square$

For the block comparison in Section 11, the clean calculation replaces designated key comparisons by exact endpoint equality. Endpoint comparison is an opaque restoring bit query in the terminal program graph: its internal states are not graph vertices. The exact comparison can therefore use a longer bit-position counter without enlarging the guarded raw graph states. It still returns its bit and restores the cursor and catalyst. Only the actual algorithm must satisfy the tight physical-space bound; the clean calculation supplies comparison values and paths for the error analysis.

Lemma 13.10 (Concrete recipe certificates). *All actual estimator operations admit regenerable records and a fixed finite ordering of operation strata. At a prefix of slack d , each ordinary layer has $O(d)$ structural control, and a numerical path to an operand at clock j has suspended record*

$$c_{\text{str}}d + c_{\text{num}}(t - j), \quad j - t \leq c_{\text{dig}}d, \quad (13.5)$$

with fixed coefficients. Beyond the fixed strata of this prefix, a call is to the preceding prefix of slack $2d$, or to an estimator of accuracy level at most $m - d$. The final row and incoming wrappers add

Operation	Data retained across a lower call	Local space bound
Structural tests and list selection	Immediate descriptors, loop indices, counts and flags	$O(d)$
Numerical recipes	Gate paths, pending bits and permitted early operand words	$c_{\text{str}}d + c_{\text{num}}(t - j)$
Ordered terminal paths	One frame’s choices, phase and bounded inverse tokens	$O(d)$ per immediate frame
Incoming score ranks	A partial score suffix; the full word only over structural calls	$O(1 + (t - j)_+)$ plus structural costs
Graph transition probes	A fixed number of local raw-state copies and bounded incidence data	$a(w - w') + d_{\text{num}}(t - j)$

Table 2: Suspended records at a lower call. The bounds exclude shared facilities and the callee’s physical state. Terminal-path frames are charged separately along their explicit chain; graph probes copy only that local raw chain.

a fixed number of strata at $d = 1$. Every designated terminal execution obeys the corresponding boundary bounds before guards are imposed.

Proof. We check one prefix from its lowest dependencies upward: immediate candidate paths, destination grouping, compression and list selection, and finally the exported row and incoming operations. The order is important because a presence test cannot call the value or access operation whose presence it is defining. Table 2 summarizes the records to be charged. We verify the recipes and their dependencies in the order in which they are constructed. All constants depend only on the fixed construction constants and the base machine.

Immediate paths and their values. By Lemma 9.5, a group has at most $2^{O(d)}$ immediate descriptors: term and channel part, scale and accuracy level offsets, a variant word and wrap choices, bounded copy symbols, and a bounded number of factor ports. All factor accuracy levels are at most $m - d$. A descriptor is stored literally in $O(d)$ bits; it never stores a discovered endpoint. Index and range tests are pure computations. Constituent existence tests call only these lower factors. Products cross a paired left edge for right-factor tests or digits, retaining a bounded inverse port; they return after each such query. Factor digits can therefore be requested separately by an arithmetic evaluation path. No assembled left weight remains over a late right-factor digit. Source vector factors are queried at the current source, with the prescribed local projection if needed. Their numerical coefficients and the factor products use a fixed arithmetic recipe.

For terminal motion the right unbalanced path, or the direct child path, is the last delegation. This is the ordered path of Lemma 13.8. To avoid a validity cycle distinguish intrinsic constituent existence, intrinsic motion or excursion, and outer validity. Outer validity first checks the intrinsic constituents, then uses an intrinsic excursion for its own hit and wrap conditions. A subsequent outer-valid motion may repeat that validity check and then use the intrinsic motion. Each of these is a separate stratum; none tests the updated bucket or shortlist. Invalid descriptors return failure or a zero value without relying on a motion’s promised inverse.

All rate factors multiplying these terms have logarithmic range $O(d)$: a variable bin has $|\log_2(b/a)| = O(d)$, while f/π_f is bounded. Resetting a source rate to the public stage rate does not multiply the numerical term by f/u ; that ratio was a norm cost. It therefore causes neither a large numerical shift nor a new long choice field. The fixed number of bounded-factor products, coefficients and $O(d)$ scale shifts obey Theorem 12.3.

Choose representatives before computing their updated values. The descriptor pool is the concatenation of the preceding list positions and this group’s unmerged paths, with invalid slots masked. A new descriptor is a representative exactly when it is the first valid new descriptor with its key and no old valid position has that key. Old valid positions remain separate. To decide these flags, enumerate the $2^{O(d)}$ descriptors, keeping only the current indices, bounded flags and $O(d)$ -bit counts. Each test uses lower validity and endpoint comparison operations. The comparisons use only those unmerged paths or the preceding prefix’s already-defined positions. In particular they never inspect the new value of the bucket being defined.

An intermediate entry is the old entry at an old representative, or the previous baseline at a new representative, plus the sum of all valid matching increments in each channel. That sum has a fixed $O(d)$ -bit descriptor domain and boundedly many $2^{O(d)}$ scale factors. It is evaluated by the balanced sum construction of Theorem 12.3, with the preceding domain tests as structural leaves. No high-precision sequential accumulator is held while requesting term digits. The bucket rule preserves distinct actual representatives, including when two old representatives have the same key. Thus it gives exactly the intermediate array used in Lemma 11.7.

Compute compression thresholds and select exceptions. The intermediate domain and its values are now earlier inputs. Add enough copies of the default for the size and first-signal ranks. A finite domain of $2^{O(d)}$ indices suffices: the preceding list has size $2^{O(2d)}$ and the new group has $2^{O(d)}$ descriptors. The default is included as often as needed to represent the unlimited padding in the prescribed ranks. The rank, max, min, absolute-value and clipping operations of Lemma 10.2 are the finite arithmetic primitives. The ranges are $2^{O(d)}$ by Lemma 10.7; their known normalizing shifts have magnitude $O(d)$.

Here $u_0 \geq \rho_d = 2^{-c_1 d}$. A canonical finite approximation can test a threshold strictly between u_0 and $2u_0$: with error a sufficiently small fixed fraction of ρ_d , accept every size greater than $2u_0$ and accept only sizes greater than u_0 . The required numerical clock, after normalizing the range, is $O(d)$. Fix this clock and tie convention once for that structural test. Consequently every nondefault compressed position is included, and fewer than n_d positions are selected, as in Lemma 10.2. Numerical equalities at the boundary need not be decided. No default can expose a positive row score: its rank dummies close the gate first, by the margins of Lemma 7.6.

Remap a requested list ordinal to its selected intermediate descriptor by enumerating the pool and counting the preceding accepted flags. The loop stores $O(d)$ -bit indices and a count. Its only calls are the already-defined intermediate flags and finite-margin selection tests. Compressed entry digits then query that descriptor’s intermediate value and the common compression thresholds; baseline digits use those thresholds directly. A terminal list motion selects its descriptor before delegating to an old prefix path or a new valid path. A descriptor from many prefixes ago is therefore represented by a chain of delegations, not retained as one long word above a tight child.

Check that a port exists before using it. Ordinal lookup returns a presence bit and a selected descriptor only when its finite enumeration reaches the requested ordinal. It starts with a false presence bit and returns false if enumeration ends first; the false branch supplies no descriptor. Every compressed-entry digit query and list motion branches on this fresh result before using a descriptor. The preceding-prefix lookup has the same convention. A hole denotes no real representative. Rank operands retain their prescribed structural selection of an entry, baseline, or padding value; they do not infer that value from the zero stream exported at an absent real port.

The final precursor a and auxiliary class a_I use the bounded final list-position alphabet, allowing holes. Their presence tests use list presence, role-membership masks, and the fixed-precision raw-

entry and raw-rank tests of Lemma 7.6. Source membership is checked locally; endpoint membership uses the earlier list or intrinsic excursion with a pure membership action. It does not use the exposed-class motion whose validity is being defined. These tests precede the pre-incoming score formula and use neither its incoming gate nor its cap. Every pre-score digit invocation repeats the corresponding class test and returns zero on failure. Other uncapped row-value queries use their support-covering class in the same way. Each such class contains the entire nonzero support, so the extension agrees with the mathematical table. Existence predicates evaluate their earlier masks directly; they do not ask for their own existence. Analytical witness functions without an exported port interface retain their stated full-array values and are not restricted by this convention.

Immediate-path values and visits first use their earlier constituent and, where appropriate, outer-validity tests. Endpoint comparisons likewise test their two earlier descriptors before making visits. A capped correction-value query stops with zero if its forward paired move fails; only a successful move permits a gate or column-value query at the endpoint. The paired inverse keeps its target-side validator from Lemma 13.6. Incoming ranks and vector queries return zero outside their prescribed vertex type or membership domain. These checks are repeated at canonical entry, including when the caller is a synthetic post-initiation state with a fabricated validity flag. An excursion checks before constructing its spine; its endpoint action and rollback retain the order of Lemma 13.8.

Place these entry wrappers a fixed number of strata above their validity predicates and value or motion cores. A wrapper at clock t calls its structural test at clock zero and retains $O(d)$ descriptor and control bits; its boundary is therefore bounded by $O(d) + bt$. When it subsequently calls a numerical core at clock j , the test has finished and only bounded status and descriptor data remain, so the bound $ad + b(t - j)$ persists after enlarging a . The predicates use the fixed structural precisions already proved. They do not call the score, gate, cap, new bucket or new shortlist whose value is being checked. The finite stratum list is enlarged before choosing the allowance constants. Recipe guards permit opaque calls only to total entry wrappers, so a synthetic state cannot select a continuation that skips validation. No semantic predicate is evaluated by a guard itself.

These wrappers preserve the represented relations, values and outgoing endpoints. Their insertion may change the contour ordering of incoming ports; both directions use the same updated graph, so pairing remains consistent. The tests use the same effective tuples and fixed structural clocks, and restore the catalyst on both outcomes. They require no accuracy promise and add no retained endpoint address.

Order the local operations so their dependencies decrease. For clarity, a permissible ordering, from an exported operation down to its dependencies, is:

1. compressed entries, baseline and list-port operations;
2. ordinal remapping, compression and exception selection;
3. size and first-signal ranks of intermediate entries;
4. intermediate domain flags and entry or bucket-sum values;
5. first-representative and key-match predicates;
6. endpoint comparisons of outer-valid paths;
7. outer-valid visits, then outer path validity;
8. intrinsic visits, constituent tests, factor digits and ordered motions, subdivided according to the preceding distinction.

The wording of an interface name does not force a call at the same stratum. For example, the “visit” used to verify outer validity is the intrinsic visit, while the visit used by key comparison is an

outer-valid visit. Arrange these finite subtypes in their indicated dependency order: outer-valid visits use outer validity; outer validity uses intrinsic visits; key comparisons use outer-valid visits; and bucket predicates use those comparisons. Arithmetic on intermediate values uses only their domain flags and lower matching predicates. This refines the displayed list to a finite acyclic list. Repeated enumeration over descriptors stays in a loop of its layer, rather than creating one recursive frame per descriptor. Between these layers Theorem 12.3 gives (13.5); structural loops add $O(d)$ bits. All remaining path calls go to the preceding prefix or to the lower-accuracy level factors already described.

Final row classes and pre-incoming values. At the final slack $d = 1$, the list size, port alphabet and all threshold gaps are constants. The classes and witnesses in Lemma 7.7 are defined by canonical finite-margin entry and rank tests on the final raw arrays. They can be enumerated without testing positivity of a computed score. Their tests and pre-incoming score digits query only the final buffer and a fixed arithmetic recipe. The same raw-array tuple is used for every row. A normalized dense entry uses its second raw argument, divided by the fixed scaling constant and clipped before the reciprocal. It does not start a new preparation experiment at its unbalanced endpoint.

The broader auxiliary class used to score an incoming gate depends only on the corresponding pre-incoming row calculation. Its validity and score queries do not depend on that incoming gate or cap. The precursor a for a correction has the same property. Its canonical terminal program begins with a restoring check of the full precursor class and membership domain, and sends every invalid initiation to failure. Thus Lemma 13.6 counts exactly its valid entries, not merely entries promised valid by an external caller. The auxiliary scoring program has the same initial check. Its score digits are pre-incoming, so Lemma 13.7 invokes only lower forward operations. By Lemma 7.7, discarding a precursor with more than k_c incoming entries preserves its value.

Final incoming and vector wrappers. Place the capped-pair and inverse-score wrappers above their respective terminal programs, and those programs above their pre-class, selection and pre-score queries. They are separate terminal programs: the gate score program does not call this task's cap pairing, and the cap program does not call its gate rank. A final correction-value query first completes the paired move. At the endpoint its motion-return data consist only of the bounded inverse port while it queries a gate-rank digit or a lower-accuracy level column-vector digit, and then returns. This can be done separately for every arithmetic operand bit. Only numerical path bits and immediate descriptors coexist with a late query. The factor r_i at a projected source is handled in the same way at that source.

A detector trial is an inverse-score wrapper of its lower-accuracy level input row task. Success contribution terms query source factors locally and cross paired corrections before midpoint vector queries. Vector prefix sums use the simpler balanced arithmetic form of the same dyadic schedule. These are all the vector and incoming uses in the additive formulas. There are only a fixed number of final wrapper types. Their short counts and return ports cost a fixed allowance gap. In an inverse score program the saved numerical suffix satisfies the clock bound in Lemma 13.7; its full word is held only over structural motion. The cap wrapper's counts have fixed size.

Generate the descriptors and add the retained records. Each enumerated object above has a literal $O(d)$ -bit index or is a Boolean path in one of the stated arithmetic recipes. Uniform index-generation uses finite integer loops on these indices, including rank comparison pairs and balanced-tree nodes. Constants, field/rate coefficients and walk probabilities have $O(b_{\text{mem}})$ -bit

rational descriptions. Their requested digits use integer arithmetic in scratch and make no recursive graph call. In particular, a walk word has length $O(d)$ and a fixed rational generator probability, so its probability denominator has $O(d)$ bits. Rate and field expressions involve fixed powers of $O(b_{\text{mem}})$ -bit integers. The arithmetic theorem’s uniformity and public-coefficient hypotheses are thus satisfied.

The finite arithmetic list covers the formulas used here explicitly. A fixed polynomial switch is evaluated on an affine-rescaled clipped argument; positive parts and absolute values use maxima, and a variable clip is a minimum followed by a maximum. The normalizing reciprocals have their arguments clipped to $[1/2, 1]$. The eighth roots used in the error estimates express regularity of the row functions and are not additional numerical operations requested by the controller. Ranks inside a prefix use the finite descriptor construction above; the unbounded incoming classes use Lemma 13.7. This accounts for the arithmetic primitives without adding a real-number comparison or an incoming-rank oracle.

Structural selection words and loop counters have $O(d)$ bits. The only longer local numerical words are the early operand records of Theorem 12.3 and the just-described score suffix; they obey their explicit clock-decrement bounds. Late operand calls hold only numerical paths, pending bounded results, immediate descriptors and short inverse tokens. A fixed number of arithmetic and structural layers enlarges the constants in (13.5); its clock differences add. Ordered terminal delegations add their own immediate records, with the same estimate at each boundary, including on rollback as proved in Lemma 13.8. This verifies the record certificate on the designated executions, independently of any graph guards. $\square$

Two remaining requirements link the implementation back to the error analysis. Recomputing a table’s parameters must recover the same table after the cursor has moved. Also, every access decision in a lower accuracy level block must respect the restriction on later hash seeds, including decisions made during predecessor probes. We verify both by replaying the stored relative choices.

Lemma 13.11 (Parameter replay and salt visibility). *The records in Lemma 13.10 are regenerable in the sense of Definition 13.2. With respect to their own effective environments, actual procedures at an accuracy level in block b ignore all salt blocks later than b , including in structural orders, graph universes, guards and predecessor probes. Re-querying a table entry or using a paired port in either direction uses the same effective table tuple.*

Proof. Start with the fixed top parameters and scan the active records. An estimator edge specifies a task type, a stage change prescribed by the formula, an accuracy level offset, a bin-scale offset if present, a bounded copy choice and channel-role recipe. These determine the next absolute stage, rates and suffixes. A projection appends the stored symbol before the inherited suffix. A fresh midpoint or vector role uses the specified public stage rate and its prescribed suffix; it does not carry the old source. These are the common-tuple recipes in Lemma 7.7 and Lemma 9.5.

An averaging edge also carries its generator and wrap word. Apply these fixed transformations in order to the current environment. Their descriptors are independent of the indicated endpoint; membership and wrap consistency are tested separately on visits to that vertex. Thus no discarded conditioning address is needed to regenerate the environment. For an old-prefix edge the tuple stays that of the same row calculation. At final assembly a gate or column multiplier uses this common tuple, rather than the internal variant which happened to lead to a selected representative. An action-owner tag allows an endpoint-local calculation to replay just to that ancestor tuple.

Numerical paths specify arithmetic operands, gates, digit places and the early-prefix modes of a reciprocal. Their finite index rules are those of Theorem 12.3; an absolute place is recovered from the root clock and the path or offset, not saved afresh per frame. Descriptor selections which

used the cursor were stored explicitly as their bounded choices before movement. No later tuple computation needs to repeat such a selection at an unknown old cursor.

At every step the tuple and the temporary integers have $O(b_{\text{mem}})$ bits in the actual parameter and clock ranges. Keep a current tuple in scratch and overwrite it as each record is processed. Repeating the scan reconstructs an ancestor or a selected candidate chain. Delimiters and marked active-copy tags specify which chain is used. This costs one scratch facility, including its scanning counters, rather than one absolute pointer, input-head position or tuple per frame.

A pure endpoint action belongs to its outer invoking predicate and may read that owner's permitted salt. It is excluded from the lower program graph and cannot affect the lower path or rollback. Nested lower queries still restore their arbitrary incoming catalyst contents.

Finally, the only salt read by a new group comparison is its task's own block salt. Descendants do not increase their accuracy level. Walks alter only the already designated earlier free salt blocks and leave the present and later ones frozen. All descriptor and instruction orders are fixed index orders; all range and length checks depend only on public bounds and accessible tuple fields. Give each recipe the syntactic permission to address only those salt blocks at most its accuracy level block in its effective environment. This is a direct range check on a requested salt index, not a condition on the salt's value. A synthetic record cannot change that permission, choose a higher-accuracy level recipe, or bypass the fixed parent-to-child rules. The same replay and checks are used before predecessor queries. Consequently the entire lower-block structure is independent of later salts. The clean calculation in Lemma 11.4 uses the same fixed descriptor orders and structural clocks, with its stipulated replacements of comparison predicates. The error proof uses the salt independence of the actual lower algorithms and of the clean recipes. $\square$

Remark 13.12 (Effective coordinates and physical replay). The salt assertion in Lemma 13.11 is a statement about a task as a function of its effective environment. For fixed other task parameters, write this function schematically as $F_b(\eta)$. Its values, access orders and guarded program graph use only the rank channels and the permitted salt coordinates of η . An ancestor can invoke it at $\eta = \varphi(\sigma)$, where σ is the master environment and φ is a recorded environment transformation. Computing the permitted coordinates of $\varphi(\sigma)$ can read physical master bits whose original block indices exceed b . Those reads derive the effective argument; they do not give F_b a new semantic parameter or permission to inspect later coordinates of that argument.

For a clean calculation in the current block, the current task's own walks vary the earlier effective salt coordinates and freeze its current and later ones. Same-block clean comparisons omit the current key, and lower-block procedures obey the local permission rule just proved. This is the coordinate convention used for the slice means in Lemma 11.4. A callback owned by a higher task remains the separate pure endpoint action described above; it cannot enter the lower graph, change its selected path or affect rollback through catalytic contents.

13.5 Add the memory retained by all active calls

All local procedures and their retained-record bounds are now available. It remains to choose one decreasing allowance for every permitted kind of dependency and sum the physical memory costs. The proof has two separate totals to control: the sum of retained caller records, and the largest numerical digit place reached by any descendant. The latter determines the size of shared arithmetic scratch.

Two linear quantities organize the final accounting. For positive physical-space coefficients a, d_{num} , chosen below, define

$$\Phi(w, t) = aw + d_{\text{num}}t, \quad \Psi(w, t) = t + a'w, \quad (13.6)$$

where a' is the clock-overshoot coefficient in Definition 13.3. The decrease of Φ will pay for a suspended physical segment, including its local probe copies. The nonincrease of Ψ will bound every descendant numerical clock and hence the shared arithmetic scratch.

Proof of Theorem 13.1. We first assign an allowance that decreases on all three kinds of call: a lower local operation, a preceding prefix, and a smaller accuracy level. We then construct the procedures by induction on this allowance. Finally, we sum the local bounds and include the shared memory.

Let s_{op} exceed the finite number of operation strata in Lemma 13.10, including the subdivision of path validation and all final wrappers. Choose fixed integers $k_0 > s_{\text{op}} + 1$ and $k_1 > 2(k_0 + s_{\text{op}} + 1)$, increasing them as needed for the local constants. Assign to an operation at accuracy level m , prefix slack d and stratum $s_0 \in \{0, \dots, s_{\text{op}}\}$ the allowance

$$w = k_1(1 + m) - (k_0 + s_0)d. \quad (13.7)$$

Final wrappers use $d = 1$, with increasing s_0 down their dependencies. For a nontrivial prefix, $1 \leq d \leq m$; trivial tasks use fixed top strata and terminate locally.

Check the allowance decrease separately for the three kinds of dependency. A same-slack descent lowers w by at least d . A preceding-prefix descent to $2d$, even with a reset to an earlier stratum, lowers it by at least $(k_0 - s_{\text{op}})d > d$. A statistical descent to $r \leq m - d$ lowers it by at least $(k_1 - k_0 - s_{\text{op}})d$, since the child allowance is at most $k_1(1 + r)$ minus a nonnegative stratum term. Hence every required boundary has $w - w' \geq cd$ for a fixed $c > 0$, and all allowances remain positive and $O(1 + m)$. The estimates (13.5) and the ordered-path record certificates give Definition 13.3, with common coefficients. When a common clock coefficient is increased, compensate the allowance coefficient using the bounded clock overshoot, just as in (13.3). Add a fixed allowance gap for a traversal wrapper's bounded incidence, count and status data.

Before using induction, separate the caller's raw records from the copies used by physical predecessor probes. Enlarging constants for those copies must happen once, not recursively at every level. There are two distinct estimates in this induction. First, Lemma 13.10 certifies the raw caller words with the same a_0, b_0 at every accuracy level; a lower opaque operation contributes its call recipe, but none of its internal working states, to that certificate. Second, Lemma 13.4 converts a copied raw portion to its physical bound with one fixed pair a, d_{num} . The induction below appends a lower operation's physical portion; it never substitutes that physical portion into an upper raw graph state. The raw certificate therefore does not depend recursively on the already enlarged physical constants.

We can now construct the procedures in increasing allowance. The induction covers all recipe instances admitted by the finite range and parent-to-child checks: both designated calls and calls made when probing candidate predecessor records. It does not require such a candidate record to have occurred on a designated execution. Each opaque call still enters the lower procedure at its canonical entry, where its semantic presence tests are performed. At the bottom are local base ports, rank and salt evaluation, rational arithmetic without graph operands, and trivial zero tasks. Suppose lower operations are total, have their claimed cursor and catalyst behavior, and obey the local bounds. The finite restoring and numerical recipes of Lemma 13.10, using Theorem 12.3, then compute exactly their specified bit or digit. Their finite loops and lower calls terminate. Endpoint visits use Lemma 13.8, and completed endpoint comparisons use Lemma 13.9. The latter restores arbitrary catalyst contents and has an independent answer, preserving the induction hypothesis. The other recipes make no direct catalyst-dependent choice.

The canonical terminal programs have finite loops and only lower terminal delegations or opaque calls, so their prescribed executions terminate. Their class checks send invalid canonical initiations to failure. The certificates already proved show that all their designated boundaries fit the local

bounds. By Lemma 13.11, their parameters and static guards are uniform and independent of a discarded cursor. We may therefore apply Lemma 13.4. Guards remove no designated execution. At synthetic states they ensure that every probed query has an admissible lower tuple and available space; fresh safe lower queries do not rely on the truth of an earlier stored flag.

The finite graph contours of Lemma 13.5 now implement the higher cap and inverse-rank wrappers in Lemmas 13.6 and 13.7. They terminate on success-root components even if other graph components contain cycles or guard-created dead ends. They count only the canonical initial executions of these programs, so their paired maps and ranks are the mathematical ones specified by the tables. By the common root tuple and salt convention the same maps are obtained in either direction and on every re-query. This completes the well-founded construction. An opaque query in this induction denotes its already constructed finite procedure; recursively substituting these procedures gives an ordinary deterministic machine.

The allowance is a runtime parameter of this fixed finite collection of instruction types, rather than an index for input-dependent program text. Operation tags select the finite recipe family; loop counters, gate paths and regenerable descriptors select its current instance. Thus the induction constructs one uniform interpreter for these instructions. It requires neither a separately compiled machine for each accuracy level nor stored transition graphs for the contours.

The final accounting is over the physical stack that exists at one instant. It includes inactive caller records and saved local predecessor states, not just the currently executing routine. It remains to add all simultaneously live data. By Lemma 13.4, a traversal root at (w_i, t_i) with a pending query at (w_{i+1}, t_{i+1}) contributes at most

$$a(w_i - w_{i+1}) + d_{\text{num}}(t_i - t_{i+1}) = \Phi(w_i, t_i) - \Phi(w_{i+1}, t_{i+1}) \quad (13.8)$$

local bits, including its fixed number of neighbor-state copies. Ordinary restoring and numerical calls have the same bound after one enlargement of a, d_{num} . Inverse traversal copies only its local explicit chain, not older stack segments or the internal configurations of a transition query. Choose a, d_{num} also to dominate the bottom operation's intrinsic linear bound. If the active chain has boundaries $0, \dots, r$, with boundary zero at the top, put $\Phi_i = \Phi(w_i, t_i)$. The total local space is at most

$$\sum_{i=0}^{r-1} (\Phi_i - \Phi_{i+1}) + \Phi_r + O(1) = \Phi_0 + O(1) = aw_{\text{top}} + d_{\text{num}}t_{\text{top}} + O(1).$$

Figure 2 shows which records enter each summand.

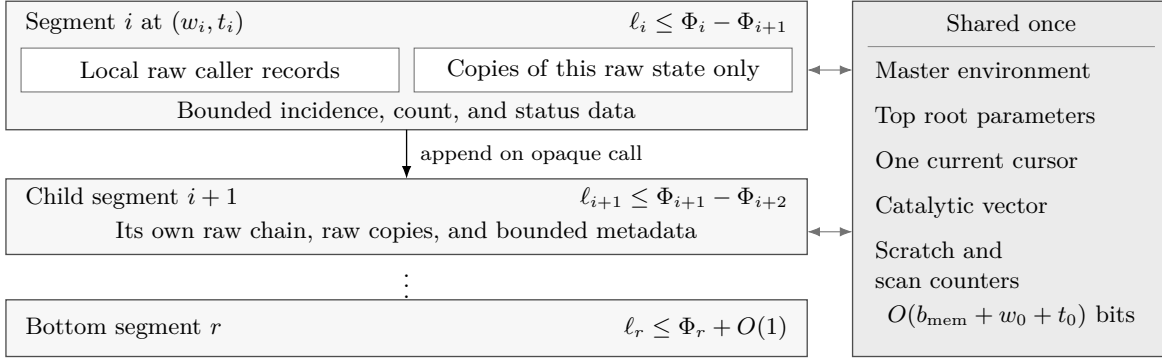
The telescoping sum bounds the call records. We still have to bound shared scratch when a child asks for a later digit than its parent. The separate clock inequality in (13.1) gives

$$\Psi(w_{i+1}, t_{i+1}) \leq \Psi(w_i, t_i). \quad (13.9)$$

Since $t_i \leq \Psi(w_i, t_i)$, all clocks are $O(t_{\text{top}} + w_{\text{top}})$. This also bounds the public scratch calculations in the arithmetic theorem. An uncharged full coefficient or cursor copy cannot persist there: any operand-dependent early word crossing a query was included in its record certificate, while pure public computations have no recursive query. The master environment, cursor, catalyst, scratch and their global scan counters are single facilities. Their total space is $O(b_{\text{mem}} + t_{\text{top}} + w_{\text{top}})$.

For exact-stream semantics the induction is valid for each finite requested clock, allowing scratch to scale with that clock. Every structural domain uses its fixed finite-margin clock, so its definition does not change in this argument. In particular the integer n_t of a score rank or arithmetic operation is always evaluated by its own place convention, even when used in a request for the next digit. The actual application has $m_{\text{top}}, t_{\text{top}} = O(b_{\text{mem}})$, and hence all these facilities and active records use $O(b_{\text{mem}})$ bits. $\square$

Physical stack segments (earlier segments omitted)



$$\sum_{i=0}^r \ell_i \leq \sum_{i=0}^{r-1} (\Phi_i - \Phi_{i+1}) + \Phi_r + O(1) = \Phi_0 + O(1)$$

Figure 2: Physical stack space, with ℓ_i the length of segment i and $\Phi_i = \Phi(w_i, t_i)$. Opaque child work is appended outside raw copies. The proved record certificates and local graph bound, with enlarged a, d_{num} , pay for copies and metadata. Shared facilities are counted separately. The displayed copies occur at traversal calls; older physical segments are not copied into the child. Ordinary restoring and numerical segments obey the same difference bound.

14 From the estimator to deterministic simulation

We now put the three parts together. The exact hierarchy supplies a success contribution close to acceptance probability. The estimator approximates that success contribution for most environments. The controller evaluates every environment in logarithmic space and always halts. These statements have different quantifiers, and the final reduction needs each in its stated form.

There are three errors to allocate: the residual transition contribution, the estimator's statistical error at the designated start, and truncation of its numerical digits. We choose the parameters so that their sum is below the requested accuracy. Enumerating the conditional environments and taking a median then produces a deterministic approximation, without a promise that the acceptance probability lies outside a decision gap.

All transducers below have read-only input and a one-way write-only output tape. Output contents are never available as work memory. For a fixed randomized machine $\mathcal{M}$, write $p_{\mathcal{M}}(x)$ for its acceptance probability on input x .

Theorem 14.1 (Fixed inverse-polynomial accuracy). *Fix a randomized machine $\mathcal{M}$ with polynomial worst-case running time and $O(\log(n+2))$ work space, and fix an integer $d \geq 1$. There is a uniform deterministic transducer that, on every input x of length n , outputs a dyadic rational $z_x \in [0, 1]$ with*

$$|z_x - p_{\mathcal{M}}(x)| \leq (n+2)^{-d}.$$

Its work space and output length are $O(\log(n+2))$, and its running time is polynomial in $n+2$. The constants may depend on $\mathcal{M}$ and d , but not on the input.

The fixed exponent in this theorem is enough to decide bounded-error computations. After proving it, we derive the class equality and its promise version. We then allow the requested accuracy to be part of the input and construct accepting computations under an inverse-polynomial probability promise.

Algorithm 14.2 Deterministic probability approximation

Input. A fixed resource-bounded randomized machine, an input, and a fixed inverse-polynomial error requirement.

1. Build the implicit time-layered model and choose the hierarchy depth so that its deterministic residual error is within its assigned tolerance.
2. Choose the top accuracy level to pay for the target statistical error at the specified primary start. Choose the numerical digit precision separately.
3. Enumerate original environment encodings satisfying the top start-retention condition, preserving their multiplicities. For each, evaluate the final success entry to the prescribed numerical precision.
4. Find a median using repeated counting passes over those encodings. Recompute values when needed; keep no list of estimates.

Output. The median approximation. The proof below allocates the errors, proves a majority of accurate estimates, and bounds every seed's computation.

14.1 Choose constants before choosing the input-dependent ranges

Several parts of the proof ask for a fixed constant to be sufficiently large or small. We record their order here to ensure that no later choice invalidates an earlier estimate. In particular, the averaging gap is uniform in the field size and number of free bits; the field can therefore be chosen after the required identifier and sampling ranges are known.

The construction has the following order of choices.

1. Fix $h, h_0, s = 8$ and the fixed number of independent channels. The finite-action gap is uniform in every later modulus and bit length.
2. Choose the correction thresholds and cutoff, then the iteration count k_* from the factorial bound and its resulting uniform magnitude constants. These estimates hold for every column multiplier in $[0, 1]$.
3. Choose d_0 for the density reduction, the routing thresholds, and the finite ladders of entry and gate thresholds. The row-gate rank k_{row} can then be chosen from the row-mass bounds alone.
4. Choose the detector rank j_{det} large enough for the incidence bound on extra copies. Choose r_* large enough for overload detection, and finally the incoming rank k_c large enough for correction normalization. The earlier row-gate choice does not depend on k_c .
5. Fix the bounded-output constant, the scale cutoff and precision coefficient, and the prefix mean exponent. Choose the buffer floor and capacity exponents and its noise coefficient so that all buffer sums converge. Choose task separation and walk length coefficients sufficiently large for the small constants in the clean estimation induction.
6. Fix the crude range and propagation constants, then the block gain coefficients d_2, d_1 and a sufficiently small rational block width coefficient ϵ . These choices close the coupled fingerprint induction of Theorem 11.10.
7. Choose the fixed operation-stratum and allowance coefficients and the guard constants in Theorem 13.1. Their finite enlargement does not alter the analytical tables or the distribution of the environment.
8. Choose the coefficient of the top accuracy level, the field-size exponent, and the minimum padding of b_{mem} .

Within each step only finitely many strict inequalities between fixed constants are needed. The proofs in the corresponding sections give these inequalities. In particular, the row rank is fixed before the incoming rank, the block constants follow the crude error constants, and the field is chosen after the reachable resource ranges. There is no dependence of an earlier sampler gap on that last choice.

The coefficient ϵ in this order is the fixed fingerprint block-width coefficient. Once it has been chosen, increasing the fixed linear top accuracy level increases the number of salt blocks but leaves the sampler's gap uniform: all earlier salt bits form one free component in Lemma 8.2. The storage estimate (11.4) therefore remains $O(b_{\text{mem}})$.

For clarity, one may bound the stage index by l_{max} , the inherited suffix length by the stages remaining above a task, and logarithms of reciprocal requested rates by

$$hl_{\text{max}} + c(m_{\text{top}} - m) \quad (14.1)$$

with a sufficiently large fixed c . A child at slack d can extend the variable scale range by only $O(d)$ and loses at least d of accuracy level. Stage-rate resets are already covered by hl_{max} . The finitely many local task operations use the stated widths and fixed subranks. Thus Equation (14.1) is closed under all estimator recipes. The controller guards allow these designated executions.

There is consequently a fixed $c_{\text{bd}, p_{\text{fld}}}$ such that a prime

$$2^{c_{\text{bd}, p_{\text{fld}}} b_{\text{mem}}} \leq p_{\text{fld}} \leq 2^{c_{\text{bd}, p_{\text{fld}}} b_{\text{mem}} + 1}$$

is larger than all identifiers, makes the radix invertible, and gives $p_{\text{fld}} 2^{-k} \geq 1$ for every rate ever requested. Bertrand's postulate provides such a prime; testing consecutive integers by trial division finds one using $O(b_{\text{mem}})$ space. The search range and running time are $2^{O(b_{\text{mem}})}$. Field arithmetic and fixed-dimensional rank tests also use $O(b_{\text{mem}})$ space.

14.2 Pass from average error to the specified start

The estimator theorem averages over source vertices. We need the value at one particular vertex, so we cannot use that theorem without paying for the loss of averaging over vertices. The next elementary calculation keeps one nonnegative summand and makes the cost explicit. Conditioning on the start being retained also identifies the exact family of environments that the deterministic algorithm will enumerate.

Lemma 14.3 (Conditioning on one start). *Let $\mathbf{w} \in [0, 1]^V$, and suppose an estimator satisfies*

$$\frac{1}{n_v} \mathbb{E} \sum_{v \in V} \frac{X_f(v)}{f} |\bar{\mathbf{w}}_v - \mathbf{w}_v|^8 \leq c_{\text{ind}}^8 2^{-8p_{\text{acc}}}.$$

For any specified $z \in V$ and every $\eta > 0$,

$$\mathbb{E}[|\bar{\mathbf{w}}_z - \mathbf{w}_z|^8 \mid X_f(z) = 1] \leq n_v \frac{f}{\pi_f} c_{\text{ind}}^8 2^{-8p_{\text{acc}}}, \quad (14.2)$$

$$\Pr[|\bar{\mathbf{w}}_z - \mathbf{w}_z| > \eta \mid X_f(z) = 1] \leq n_v \frac{f}{\pi_f} c_{\text{ind}}^8 2^{-8p_{\text{acc}}} \eta^{-8}. \quad (14.3)$$

Proof. Keep only the nonnegative summand at z . Since $\Pr[X_f(z) = 1] = \pi_f > 0$,

$$\frac{\pi_f}{f} \mathbb{E}[|\bar{\mathbf{w}}_z - \mathbf{w}_z|^8 \mid X_f(z) = 1] = \mathbb{E}\left[\frac{X_f(z)}{f} |\bar{\mathbf{w}}_z - \mathbf{w}_z|^8\right] \leq n_v c_{\text{ind}}^8 2^{-8p_{\text{acc}}}.$$

Dividing by π_f/f proves Equation (14.2). In particular, choosing one specified coordinate costs the factor n_v from the integrated norm. Markov's inequality gives Equation (14.3). $\square$

Proof of Theorem 14.1. We choose the hierarchy depth and accuracy level first, then convert each estimator value to a finite dyadic number. After proving that most such numbers are accurate, we explain how to find their median and count the total memory and time.

Apply Lemma 3.1 to $\mathcal{M}$, with no promise on its acceptance probability. Set

$$r = d \lceil \log_2(n+2) \rceil, \quad \Delta = 2^{-r}, \quad l_{\max} = \left\lceil \frac{c_{\text{bd},0} b_{\text{mem}} + r + 3}{h-1} \right\rceil,$$

where $|V_0| \leq 2^{c_{\text{bd},0} b_{\text{mem}}}$ is an explicit configuration bound. Since d is fixed, $r, l_{\max} = O(b_{\text{mem}})$. Let z be the all-primary copy of the start at stage $l_{\max}$. Proposition 5.8 gives

$$0 \leq p_{\mathcal{M}}(x) - \mathbf{w}_{l_{\max},z} \leq \Delta/8. \quad (14.4)$$

This bound holds for every acceptance probability. Choose a fixed c_{bd,n_v} large enough to bound all the resulting stage domains by $n_v = 2^{c_{\text{bd},n_v} b_{\text{mem}}}$.

For the final success contribution task take $a = u = f = b_{l_{\max}}$. Its rank satisfies $i_* = O(l_{\max} + 1)$, and Equation (9.8) reads

$$p_{\text{acc}} = m_{\text{top}}/k_{\text{prec}} - h l_{\max} - a_* i_*.$$

By Lemma 4.1, fix an integer $c_{\text{bd},\text{tail}} \geq 1$ bounding $(f/\pi_f) c_{\text{ind}}^8$ throughout the admissible range. Choose the fixed linear coefficient of $m_{\text{top}} = O(b_{\text{mem}})$ large enough that

$$p_{\text{acc}} \geq r + 3 + \frac{c_{\text{bd},n_v} b_{\text{mem}} + \lceil \log_2 c_{\text{bd},\text{tail}} \rceil + 3}{8}. \quad (14.5)$$

This is the extra accuracy needed to pay the factor n_v in the passage to a specified vertex. The top-level and field-size choices in the preceding subsection allow this enlargement; Equation (14.1) still bounds every descendant rate. Theorem 11.10 supplies the success contribution moment estimate. Lemma 14.3, with $\eta = \Delta/8$, now shows that

$$\Pr[|\bar{\mathbf{w}}_{l_{\max},z} - \mathbf{w}_{l_{\max},z}| > \Delta/8 \mid X_f(z) = 1] \leq 1/8. \quad (14.6)$$

Thus the factor n_v lost on selecting one coordinate has been paid by the statistical precision. Explicitly, the bound in Equation (14.3) is at most

$$c_{\text{bd},\text{tail}} 2^{c_{\text{bd},n_v} b_{\text{mem}} - 8p_{\text{acc}} + 8r + 24} \leq 2^{-3},$$

by Equation (14.5).

We next turn the estimator values into a finite list of dyadic numbers. For each master environment, Theorem 12.3 gives the digits $w_j \in \{-6, \dots, 6\}$ of its one fixed value $\bar{\mathbf{w}}_{l_{\max},z}$. Structural choices do not depend on the requested digit. Put $k_{\text{digit}} = r + 6$, compute

$$a_{\text{raw},\sigma} = \sum_{j=0}^{k_{\text{digit}}} w_j 2^{k_{\text{digit}}-j}, \quad a_{\sigma} = \min\{2^{k_{\text{digit}}}, \max\{0, a_{\text{raw},\sigma}\}\}, \quad v_{\sigma} = a_{\sigma} 2^{-k_{\text{digit}}}.$$

The integer $a_{\text{raw},\sigma}$ is obtained by starting at zero and applying $a_{\text{raw}} \leftarrow 2a_{\text{raw}} + w_j$ in order, so it uses $O(k_{\text{digit}})$ bits. By Equation (12.2), the truncation error is at most $6 \cdot 2^{-k_{\text{digit}}} < \Delta/8$. Clipping cannot increase error relative to $\mathbf{w}_{l_{\max},z} \in [0, 1]$. Equations (14.4) and (14.6) imply that, on more than three quarters of the environments conditioned on $X_f(z) = 1$,

$$|v_{\sigma} - p_{\mathcal{M}}(x)| < 3\Delta/8. \quad (14.7)$$

The majority statement refers to a particular conditional distribution, so the deterministic enumeration must reproduce it exactly. Enumerate each field-matrix entry in $\{0, \dots, p_{\text{fld}} - 1\}$ and each raw salt bit string, retaining precisely those encodings whose positive-degree columns form a frame and for which $X_f(z) = 1$. The retained encodings are equally weighted and give the law in Equation (14.6). In particular, enumerate the original salt strings, not their reduced fingerprint parameters. Different raw strings that induce the same fingerprint remain separate encodings; their multiplicities reproduce the original probability law. At least one encoding is retained: a frame exists, and its independent constant column can place the designated evaluation in the nonempty retention box. The matrices have fixed dimensions and $\log p_{\text{fld}} = O(b_{\text{mem}})$; together with Equation (11.4), this gives $O(b_{\text{mem}})$ bits per master environment and at most $2^{O(b_{\text{mem}})}$ encodings to enumerate.

We find the median by counting, not by storing or sorting the estimates. Let $e_{\text{count}} \geq 1$ be the number of retained encodings, found by one pass, and let $h = \lfloor e_{\text{count}}/2 \rfloor + 1$. Binary-search the integer interval $[0, 2^{k_{\text{digit}}}]$ for the least a such that at least h retained encodings satisfy $a_{\sigma} \leq a$. Each count is computed by a fresh pass, recomputing each numerator a_{σ} . This finds the upper median of the finite multiset, including all repetitions. Since more than three quarters of its members satisfy Equation (14.7), so does its median: fewer than $e_{\text{count}}/4$ values can lie below the indicated interval, and fewer than $e_{\text{count}}/4$ can lie above it. The algorithm retains only the current count and search registers; it never stores the multiset. Output the integers (a, k_{digit}) representing $z_x = a2^{-k_{\text{digit}}}$. Then

$$|z_x - p_{\mathcal{M}}(x)| < 3\Delta/8 \leq (n+2)^{-d}.$$

We account for the memory held during this computation. The master environment, current cursor, catalyst, shared scratch, and suspended controller records together use $O(b_{\text{mem}})$ bits by Theorem 13.1, because $m_{\text{top}}, k_{\text{digit}} = O(b_{\text{mem}})$. The outer computation keeps only a fixed number of additional $O(b_{\text{mem}})$ -bit registers: its top parameters and designated start, the environment enumeration index, e_{count} , h , the current count, the two binary-search endpoints and candidate, the digit index, and the signed numerator accumulator. Each count is bounded by the number of environment encodings, so it also has $O(b_{\text{mem}})$ bits. These outer registers remain live while a digit query runs; in particular, both its environment-enumeration state and its numerator accumulator are included. They are present once for the whole computation, not once per recursive level. Every query terminates on every environment, including the inaccurate environments, by the same realization theorem. The output is written only after the median is computed.

To bound time, use totality together with the complete configuration count. This is why termination on inaccurate environments was needed, rather than only on the majority of accurate ones. Each digit query is a halting deterministic computation with $O(b_{\text{mem}})$ work bits and physical input heads confined between the real endmarkers. Counting its finite control, work-tape contents, and all work and input head positions gives at most $2^{O(b_{\text{mem}})}$ complete configurations on its fixed input. A repeated complete configuration would force a deterministic loop, so totality bounds the query by $2^{O(b_{\text{mem}})}$ steps. There are $2^{O(b_{\text{mem}})}$ environment encodings, $O(k_{\text{digit}} + 1)$ binary-search passes, and $k_{\text{digit}} + 1$ digit queries per numerator. Their product is still $2^{O(b_{\text{mem}})} = (n+2)^{O(1)}$. All finite parameters are generated uniformly, and all constants are fixed for $(\mathcal{M}, d)$. $\square$

14.3 The class equality and promise separation

Proof of Theorem 1.1. Let $\mathcal{M}$ recognize a language in BPL, with acceptance probability at most $1/3$ on no inputs and at least $2/3$ on yes inputs. Apply Theorem 14.1 with $d = 3$. For every input length, including $n = 0$, its error is at most $(n + 2)^{-3} \leq 1/8$. Hence the dyadic answer is at most $1/3 + 1/8 = 11/24$ on a no input and at least $2/3 - 1/8 = 13/24$ on a yes input. Comparing this answer with $1/2$ decides the language uniformly in logarithmic space. This proves $\text{BPL} \subseteq \text{L}$ directly from probability approximation.

A deterministic machine may ignore its coins, giving $\text{L} \subseteq \text{RL}$. Two independent repetitions of an RL machine, accepting if either repetition accepts, have acceptance probability zero on no inputs and at least $3/4$ on yes inputs. They use logarithmic space and polynomial time, so $\text{RL} \subseteq \text{BPL}$. The three inclusions prove $\text{L} = \text{RL} = \text{BPL}$. $\square$

A promise problem is a pair $(\mathcal{Y}, \mathcal{N})$ of disjoint sets of inputs. Here **PromiseBPL**, also written **prBPL**, requires one polynomial-time randomized logarithmic-space machine that halts on every input, accepts each input in $\mathcal{Y}$ with probability at least $2/3$, and accepts each input in $\mathcal{N}$ with probability at most $1/3$. The class **PromiseL** consists of pairs separated by a total deterministic logarithmic-space decider; there is no correctness requirement outside $\mathcal{Y} \cup \mathcal{N}$.

Corollary 14.4. *Under these conventions, $\text{PromiseBPL} = \text{PromiseL}$.*

Proof. The same $d = 3$ approximation and threshold $1/2$ used in the preceding proof accept every input in $\mathcal{Y}$ and reject every input in $\mathcal{N}$. The resulting machine halts also outside the promise because its approximation queries are total. The reverse inclusion follows by ignoring the coins. $\square$

14.4 Handle an accuracy request supplied as input

The fixed-accuracy theorem allows constants to depend on a fixed inverse-polynomial exponent. Theorem 1.2 instead needs one algorithm whose constants do not depend on the input accuracy request.

Merely asking for more digits would not achieve this: it approximates the same estimator more accurately while leaving its statistical error unchanged. We instead represent the computation as a padded input to one fixed inverse-polynomial approximator. The padding length increases the size parameter used by that approximator, while its bits can be generated on demand in the desired space.

Proof of Theorem 1.2. Fix a padded machine $\mathcal{M}_{\text{pad}}$ as follows. On a string beginning with $1^n 0x$, where $|x| = n$, it simulates $\mathcal{M}(x)$ and ignores all remaining symbols. If the delimiter or the following n symbols are absent, it rejects. This is one fixed randomized machine with polynomial worst-case running time and logarithmic work space in its full input length. On every string with the specified prefix its acceptance probability is $p_{\mathcal{M}}(x)$. Fix once and for all the transducer of Theorem 14.1 for $\mathcal{M}_{\text{pad}}$ and $d = 1$.

On input $(x, 1^q)$, with $n = |x|$ and $q \geq 1$, set

$$m = 2^{q + \lceil \log_2(2n+4) \rceil + 4}.$$

Simulate that fixed transducer on the virtual length- m string

$$y = 1^n 0x 0^{m - (2n+1)}.$$

An input query is answered by comparing its address with the prefix boundaries and either returning the appropriate bit of x or a fixed bit. The simulated input-head address and all these comparisons use $O(\log m)$ bits; the string y is never stored. The simulated transducer returns a dyadic rational $z \in [0, 1]$ satisfying

$$|z - p_{\mathcal{M}}(x)| \leq (m + 2)^{-1} < 2^{-q-4}.$$

Its numerator and exponent have $O(\log m)$ bits and can be retained in work space while its finite output is simulated.

Compute by exact dyadic arithmetic

$$a = \left\lfloor 2^{q+2}z + \frac{1}{2} \right\rfloor.$$

Then $0 \leq a \leq 2^{q+2}$, and rounding gives

$$\left| a2^{-(q+2)} - p_{\mathcal{M}}(x) \right| \leq 2^{-q-3} + 2^{-q-4} < 2^{-q}.$$

Write the binary numerator a to the output tape. It has at most $q + 3$ bits and represents the denominator specified in the theorem. The simulator state, virtual input address, actual input counters, captured dyadic answer, and rounding operands occupy a fixed number of records of total size

$$O(\log m) = O(\log(n + 2) + q).$$

The simulation and the synthesis of each input bit take polynomial time in m , so the total time is $m^{O(1)} = (n + 2)^{O(1)} 2^{O(q)}$. The padded machine and its exponent-one approximator were fixed before reading q . Thus their constants are independent of the requested accuracy. Throughout their simulation, the original size convention is $b_{\text{mem}} = \Theta(\log(m + 2))$, exactly as required by the estimator and controller theorems. $\square$

14.5 Construct an accepting computation from continuation values

We return to fixed inverse-polynomial accuracy to construct an accepting computation when the machine's acceptance probability has an inverse-polynomial lower bound. For a transducer, the conclusion concerns outputs certified by an accepting run: its output tape must be write-only, so past emitted symbols do not influence later transitions.

Corollary 14.5 (Certified accepting computations). *Fix a randomized logarithmic-space machine $\mathcal{M}$ that halts within a fixed polynomial bound $\mathfrak{t}(n) \geq 1$ on every input and every coin sequence, and fix an integer $c \geq 0$. There is a uniform deterministic logarithmic-space, polynomial-time transducer that outputs an accepting computation of $\mathcal{M}$ whenever*

$$p_{\mathcal{M}}(x) \geq (n + 2)^{-c}.$$

On every other input it outputs either an accepting computation or a failure symbol. If every accepting run of $\mathcal{M}$ emits a valid witness for a specified relation, the transducer can instead output such a witness with the same guarantee.

Proof. The decision approximation can also guide a path. At each configuration, some successor has continuation probability at least the current weighted average. Approximate those successor probabilities closely enough that the loss incurred by choosing the largest approximation is small over the entire path.

Give halted configurations constant self-transitions through time $t(n)$. A configuration and its elapsed time use $O(\log(n+2))$ bits. There is a fixed randomized logarithmic-space continuation machine which, on a valid encoding of (x, v, t) , starts from configuration v and simulates the remaining $t(n) - t$ steps with fresh coins. It checks the fixed work-space box and time bound, rejecting malformed encodings and any attempted violation. Its acceptance probability on the reachable configurations is the continuation value $\mathbf{p}_{t,v}$, and it has a polynomial worst-case time bound in its input length.

Write $\gamma = (n+2)^{-c}$. Choose a fixed exponent d large enough that Theorem 14.1, applied to the continuation machine, approximates every queried value within $\delta \leq \gamma/(8t(n))$. Such a fixed exponent exists because t is a fixed polynomial and the continuation input contains x . The extra fields have logarithmic length. We supply them virtually from saved configuration and time registers, so the approximation and those registers together use logarithmic space.

Starting at the initial configuration, at each time estimate the continuation value of every positive-probability successor and choose one with maximum estimated value, breaking ties by a fixed port order. Since $\mathbf{p}_{t,v}$ is the weighted mean of the successor values, the chosen successor v' satisfies

$$\mathbf{p}_{t+1,v'} \geq \max_w \mathbf{p}_{t+1,w} - 2\delta \geq \mathbf{p}_{t,v} - 2\delta.$$

On an input satisfying the stated lower bound, after $t(n)$ choices the terminal continuation value is at least $\gamma - 2t(n)\delta \geq 3\gamma/4 > 0$. A terminal continuation value is zero or one, so this computation accepts.

A low-probability input need not yield a successful path, so do not emit a purported accepting computation before checking where the deterministic choices lead. First run this construction without writing output. If its terminal state rejects, output the failure symbol. Otherwise repeat the same deterministic choices and stream the configurations through the original first halting state, or the symbols emitted by the simulated transducer. The artificial self-transitions after halting are omitted from the output. This second pass follows the same accepting computation. Consequently no rejecting computation is output, even outside the probability promise.

At any time the current, candidate, and best successor configurations, the time counter, a constant number of dyadic scores, the virtual input address, and the approximation workspace occupy a constant number of $O(\log(n+2))$ -bit records. The finite successor lists have constant size, and there are only $t(n)$ steps in each pass, so the total running time is polynomial. Each approximation has a deterministic guarantee at its queried configuration; no simultaneous accuracy event for all adaptive queries is needed. $\square$

A Detailed arithmetic constructions

Section 12 introduced the signed-digit representation and stated the numerical guarantee used by the controller. This appendix proves the primitive operations from which that guarantee is assembled. For each operation, there are two separate questions: how accurately a finite computation approximates its exact value, and how much local information survives a request to one of its operands. We follow the same order for the elementary operations, multiplication, reciprocals, and indexed reductions.

A.1 Compare finite truncations before taking a digit

We first implement the elementary operations on finite truncations, and then bound their distance from the exact answer. Comparisons are between finite rational values; the algorithm never has to decide equality of two arbitrary exact reals. The truncation margin absorbs any change in which operand a maximum or minimum selects.

Lemma A.1 (Finite comparisons and bounded elementary operations). *The normalized operations of taking a pair average, a maximum or minimum, and a fixed binary shift admit deterministic integers satisfying Equation (12.3). Their residue procedures, for any fixed q , have the following properties. There are constants $c_{bd,0}, c_{bd,1}$ such that an operand place j obeys $j \leq t + c_{bd,0}$, its suspended local record has length at most $c_{bd,0} + c_{bd,1}(t - j)$, and the intrinsic local space is $O(t + 1)$. Fixed finite compositions of these normalized operations have the same bounds, with enlarged constants. Fixed Boolean operations on returned digit codes, including sign decoding and conditional selection of code bits, add only constant local overhead.*

Proof. First consider comparing two truncations through $k_* = t + h$, where h is a sufficiently large fixed integer. Their difference digits have magnitude at most 12. Scanning from place zero to k_* , update a signed prefix by $p \leftarrow 2p + \delta$. While $|p| \leq 12$, retain p exactly. Once $p > 12$ or $p < -12$, retain only the corresponding absorbing sign state. This sign is correct for every finite continuation: the remaining suffix, in the current scale, has magnitude strictly less than 12. The state space is finite, so each transition has a fixed bounded-fan-in Boolean circuit. The sign at the end determines the exact order of the two finite truncations. The path from that sign bit to an input at place j has length $O(k_* - j + 1) = O(t - j + 1)$.

Let v_t be the average, maximum, or minimum of the two truncations, and put $n_t = \lfloor 2^t v_t \rfloor$. Each operation is 1-Lipschitz for the sup norm on its operands. Consequently $|v_t - v| \leq 6 \cdot 2^{-k_*}$, where v is the corresponding exact output. Choosing h so that $6 \cdot 2^{-h} < 1/2$ gives $|n_t - 2^t v| < 3/2$.

To obtain $n_t \bmod 2^q$, only a fixed number of low places of the selected truncation are needed. More explicitly, at scale 2^{k_*} , terms of input place $j \leq t - q$ are multiples of 2^{q+h} . Modulo this modulus the remaining $q + h$ places suffice. For an average with integer input prefixes a_{k_*}, b_{k_*} at scale 2^{k_*} , use $n_t = \lfloor (a_{k_*} + b_{k_*}) / 2^{h+1} \rfloor$. Its residue needs modulus 2^{q+h+1} and one additional retained place: only places $j \leq t - q - 1$ can be omitted. The finite number of remaining digits can be added and shifted by a constant-size Boolean circuit. A maximum or minimum first uses the comparison bit to select these residue bits. Its long comparison branch has the path bound proved above, and its residue branch reaches only places within a fixed distance of t . A fixed shift changes these places by a fixed amount. Fixed normalization factors are handled in this way whenever an intermediate range exceeds $[-1, 1]$.

The topology of the comparison chain and the low-residue circuits depends only on the root clock, the fixed margins, and the gate path. Regenerating a place by replaying that path needs no saved binary place counter at each transition. Lemma 12.2 therefore gives the suspended-space bounds. Their longest paths have length $O(t + 1)$.

For composition, use one common coefficient of $(t - j)$ for the finite primitive list. Increasing such a coefficient by Δ requires increasing its constant term by at most $\Delta c_{bd,0}$, because the operand overshoot is at most $c_{bd,0}$. With these common coefficients, the clock differences telescope along the composed path. The finite number of constant terms remains a constant. $\square$

We will also use the following explicit Boolean arithmetic facts. A carry bit in binary addition is described by a generate/propagate pair (g, p) . Composition of adjacent intervals is the associative operation

$$(g_2, p_2) \circ (g_1, p_1) = (g_2 \vee (p_2 \wedge g_1), p_2 \wedge p_1).$$

Balanced trees of these pairs give every carry in $O(\log(w_* + 1))$ depth for w_* -bit words. A balanced tree adding r_* such words modulo 2^{w_*} has depth

$$O(\log(r_* + 1) \log(w_* + 1)). \quad (\text{A.1})$$

The trees can be padded by zero words to powers of two, so their gate addresses are computable by ordinary index arithmetic. For counts of r_* Boolean bits one can do better: constant-depth full adders replace three same-width rows by two rows of their sum, with carries placed in the next column. In $O(\log(r_* + 1))$ rounds only two rows remain. The word width is $O(\log(r_* + 1))$; a final carry computation gives total depth $O(\log(r_* + 1))$. These constructions use bounded fan-in throughout.

A.2 Multiply without retaining a long partial product

Expanding a product of digit streams produces many pairs of input digits. A straightforward sum can leave a long partial result stored while requesting another digit. We instead partition the pairs according to the smaller of their two indices. In a group whose smaller index is large, both digits are early enough relative to the requested output place to pay for the circuit path leading to them.

The proof has four parts: define finite band sums, bound the omitted tails and rounding error, compute only the residues needed for the output digit, and charge each circuit path to the difference between its output and input places.

Lemma A.2 (Multiplication). *For normalized exact input streams x and y , multiplication has all the conclusions of Lemma A.1. In particular, its suspended storage at an operand place j is $O(1) + c(t - j)$, despite the potentially long product expansion.*

Proof. First partition the formal product and specify which pairs are retained. Fix q and write $a_* = 6$. Partition ordered pairs (i, j) of nonnegative places into a zero band, where $\min(i, j) = 0$, and positive bands

$$\mathcal{P}_{d_*} = \{(i, j) : d_* \leq \min(i, j) < 2d_*\}, \quad d_* = 1, 2, 4, \dots$$

Set $h_0 = h_o$ for the zero band and $h_{d_*} = h_o + \lfloor d_*/4 \rfloor$ for positive bands. The fixed integer h_o will be chosen below. Retain in band d_* only pairs with $i + j \leq t + h_{d_*}$, and let

$$s_{d_*} = \sum_{\substack{(i,j) \in \mathcal{P}_{d_*} \\ i+j \leq t+h_{d_*}}} x_i y_j 2^{t+h_{d_*}-i-j}. \quad (\text{A.2})$$

For the zero band the same formula uses its defining pair set. These are mathematical integers; no bound on their full binary length is assumed in the residue computation.

The omitted tails can be bounded without forming the band sums. At any fixed diagonal $i + j = s$, a positive band contains at most $2d_*$ ordered pairs. Indeed the smaller place has d_* possibilities, and an off-diagonal pair has two orientations. The zero band has at most two pairs. Thus, in units of 2^{-t} , the total omitted product terms have absolute sum at most

$$E_{\text{omit}} \leq 2a_*^2 \sum_{d_* \in \{0,1,2,4,\dots\}} (d_* + 1) 2^{-h_{d_*}}. \quad (\text{A.3})$$

The geometric summation of the diagonal weights gives this bound also when the truncation threshold lies below the first possible diagonal. Both the series in Equation (A.3) and $\sum_{d_*} 2^{-h_{d_*}}$ converge.

Choose h_o sufficiently large that their displayed combined error bound, including coefficient $2a_*^2$, is less than $1/2$.

There are only finitely many nonempty retained bands. For a positive one, $2d_* \leq t + h_{d_*}$, whence

$$d_* \leq \frac{4}{7}(t + h_o). \quad (\text{A.4})$$

The left side of the nonemptiness condition grows monotonically faster than its right side along the dyadic bands. We may therefore stop at the last band satisfying this inequality, retaining empty bands within the finite prefix if necessary.

Starting with the most distant retained band, define integers recursively by

$$f_{d_*} = s_{d_*} + \left\lfloor 2^{h_{d_*} - h_{d_*^+}} f_{d_*^+} \right\rfloor, \quad n_t = \left\lfloor 2^{-h_o} f_0 \right\rfloor, \quad (\text{A.5})$$

where d_*^+ is the next band, and the last missing tail is zero. One internal floor, divided back to scale 2^t , changes the sum by less than $2^{-h_{d_*}}$. Induction down the bands consequently shows that the difference between $2^{-h_o} f_0$ and the sum of all retained terms in scale- t units is at most $\sum_{d_*} 2^{-h_{d_*}}$. Equation (A.3) and the choice of h_o make the error before the final floor less than $1/2$. The final floor costs less than one more unit. Therefore Equation (12.3) holds for $z = xy$.

The integers in this recurrence define the approximation; they need not be stored in full. We next implement Equation (A.5) using residues. At band d_* use modulus $2^{q+h_{d_*}}$. An integer term in Equation (A.2) with $i + j \leq t - q$ is a multiple of this modulus and can be omitted from the circuit. It remains present in the mathematical integer s_{d_*} . Only the diagonals

$$t - q < i + j \leq t + h_{d_*} \quad (\text{A.6})$$

need to be processed, giving $O((d_* + 1)(h_{d_*} + q + 1)) = O((d_* + 1)^2)$ candidate pairs.

Signed floor shifts are well defined on the required residues. For $h' \geq h$ and $a - b = k2^{q+h'}$,

$$\left\lfloor \frac{a}{2^{h'-h}} \right\rfloor - \left\lfloor \frac{b}{2^{h'-h}} \right\rfloor = k2^{q+h}. \quad (\text{A.7})$$

This identity holds for all signed integers a, b . Thus a next-band residue modulo $2^{q+h'}$ supplies the correctly floor-shifted tail modulo 2^{q+h} . Apply the same observation to the final shift by h_o to obtain $n_t \bmod 2^q$. In the finite binary circuit, this shift just discards the lowest $h' - h$ bits of the canonical $(q + h')$ -bit residue. The displayed identity proves that this is valid also when the underlying integer is negative.

For completeness, the finite gate structure can be indexed without storing an absolute operand place at each gate. For a positive band enumerate triples (r, s, o) , where $d_* \leq r < 2d_*$, $t - q < s \leq t + h_{d_*}$, and $o \in \{0, 1\}$. The triple supplies $(r, s - r)$ if $s - r \geq r$ and $o = 0$; it supplies $(s - r, r)$ if $s - r > r$ and $o = 1$; otherwise it supplies a zero term. This enumerates the relevant pairs exactly once. For the zero band use $r = 0$ with the same orientation rule. Negative candidate places are invalid and supply zero. The offsets $r - d_*$ and $s - (t - q + 1)$ have $O(\log(d_* + 2))$ bits. They and o are obtained from paths in padded binary selection trees. The full places are then computed from the root clock in scratch.

Each nonzero term is a constant-bit signed product $x_i y_j$, shifted into a word of width $w_{d_*} = q + h_{d_*} = O(d_* + 1)$ and interpreted modulo $2^{w_{d_*}}$. A bit of this word is a fixed Boolean function of the two bounded digit codes, the bit index and the known shift; negative products use their ordinary residue, equivalently sign extension before reduction. Add these words and the shifted tail using Equation (A.1). A band contributes depth $O(\log^2(d_* + 2))$. Bit selection for the shifted tail is wiring and does not require evaluating a full tail word.

It remains to connect circuit depth with the digit place of an operand. This is the reason for grouping by the smaller input index. The sum of these depths through the bands up to d_* is $O(\log^3(d_* + 2))$, hence at most $c_{\text{bd}}(d_* + 1)$ for a fixed c_{bd} . For an original input digit in a positive band, its partner place is at least d_* . Equation (A.6) gives

$$j \leq t + h_{d_*} - d_* \leq t + h_o - \frac{3}{4}d_*. \quad (\text{A.8})$$

Equivalently, $t - j \geq \frac{3}{4}d_* - h_o$, so the full path cost through these bands satisfies

$$c_{\text{bd}}(d_* + 1) \leq c_{\text{bd}} \left(1 + \frac{4}{3}h_o\right) + \frac{4c_{\text{bd}}}{3}(t - j).$$

This gives $c_{\text{bd},0} + c_{\text{bd},1}(t - j)$ with fixed constants. The zero band has constant depth and only constant overshoot. All its gate descriptions, term coordinates, bit positions and tail shifts are computable by replaying paths and doing integer index arithmetic from t and q . Lemma 12.2 converts these depth bounds to suspended storage. Finally, Equation (A.4) bounds every intrinsic path and local word width by $O(t + 1)$. $\square$

A.3 Take a reciprocal using an early prefix and a small remainder

For a reciprocal, a prefix through roughly half the requested digit place is enough to define a linear approximation whose remaining error is quadratically small. Forming that prefix requires a long word. The memory bound is still possible because the word is present only while the algorithm queries early input places, where the output-to-input digit difference is large enough to pay for it.

The crucial order is to compute a needed coefficient digit from the early prefix, release the prefix word, and only then request a late digit of the input remainder. Treating the prefix as a permanent local constant would lose the storage bound.

Lemma A.3 (Reciprocal on a positive compact interval). *Fix constants $0 < a \leq b$. On inputs $x \in [a, b]$, a reciprocal, after a fixed known normalization, has the same operand-clock, suspended-record, and intrinsic-space bounds as in Lemma A.1. Its local record may contain an $O(t + 1)$ -bit word only while querying input places $j \leq \lfloor t/2 \rfloor + O(1)$.*

Proof. The input can first be normalized by a fixed binary scale. It is enough to prove the result with a fixed digit alphabet and fixed positive interval; all constants below may depend on a, b . Choose a fixed power of two c_{bd} large enough that the desired output $z = 1/(c_{\text{bd}}x)$ lies in $[0, 1]$ and that the coefficients below have absolute value at most one. Put

$$k = \lfloor t/2 \rfloor + c, \quad x^0 = x^{[k]}, \quad e = x - x^0,$$

where c is a sufficiently large fixed integer. By Equation (12.2), $|e| \leq 6 \cdot 2^{-k}$. Choosing c first ensures $x^0 \in [a/2, 2b]$ and $|e| \leq 1$ at every t . Increasing c_{bd} if necessary gives bounded coefficients

$$\alpha = \frac{1}{c_{\text{bd}}x^0}, \quad \beta = -\frac{1}{c_{\text{bd}}(x^0)^2}.$$

The exact algebraic identity

$$\frac{1}{c_{\text{bd}}x} = \alpha + \beta e + \frac{e^2}{c_{\text{bd}}(x^0)^2x} \quad (\text{A.9})$$

shows that the linear expression $t_t = \alpha + \beta e$ differs from z by at most a fixed constant times 2^{-2k} . Choose c large enough that

$$2^t |t_t - z| \leq \frac{1}{4} \quad \text{for every } t \geq 0. \quad (\text{A.10})$$

Clipping t_t to $[-1, 1]$ cannot increase this error; denote the clipped expression by $\hat{t}_t$.

For this fixed original t , the coefficients α, β are exact rational numbers. Their digits are produced by integer arithmetic, as described below. The suffix e has the exact stream with digit zero through k and digit x_j thereafter. Thus Lemmas A.1 and A.2 provide exact streams for $\hat{t}_t$, using only a fixed number of normalized operations. Explicitly, form the product βe , then the pair average $u = (\alpha + \beta e)/2$, clip u to $[-1/2, 1/2]$ by a maximum and a minimum, and finally apply the fixed shift by 2. Every exported intermediate value is normalized, and the final value is $\hat{t}_t$. Forming this expression introduces no additional reciprocal operation.

Let $p_s(\hat{t}_t)$ be the integer approximation furnished by those operations at place $s = t + 3$, and define

$$n_t(z) = \left\lfloor p_{t+3}(\hat{t}_t)/8 \right\rfloor. \quad (\text{A.11})$$

The cutoff k and the coefficients remain those of the original t , not those of s . Equations (12.3) and (A.10) give

$$|n_t(z) - 2^t z| < 1 + \frac{2}{8} + \frac{1}{4} = \frac{3}{2} < 2.$$

To obtain this integer modulo 2^q , ask for the approximation p_{t+3} modulo 2^{q+3} and use Equation (A.7).

The approximation error is now controlled. We separately verify the lifetime of the operand-dependent words, since their rational form alone does not make them public constants. It remains to justify storage in the coefficient computations. The signed numerator of x^0 in units 2^{-k} has $O(k+1)$ bits. Form it by querying places $0, \dots, k$ and updating the integer prefix. The numerator, a place counter, and all data that must survive the next query are part of the local record. To return a coefficient digit of place r , numerator squaring and integer division require $O(k+r+1)$ bits. Here $r \leq t + O(1)$, because the enclosing expression has fixed depth and constant overshoot. After the input prefix is obtained, these integer computations make no further operand queries. Standard binary long multiplication and division, performed by bounded loops with their partial products or remainder, use only this linear number of bits. For $r > 0$, computing floors of the coefficient scaled at r and $r-1$ gives its exact difference digit; at $r = 0$ use the place-zero floor alone, as in Lemma 12.1. Clipping is unnecessary because the coefficients were normalized in advance.

Every original-input query during this formation has $j \leq k \leq t/2 + O(1)$. Hence

$$t + 1 \leq c_{\text{bd},2} + c_{\text{bd},3}(t - j) \quad (\text{A.12})$$

for fixed constants, at all such queries. The enclosing gate path also has length $O(t+1)$, so the complete suspended local record satisfies the required bound there. Equivalently, the coefficient branch is charged against the original reciprocal clock t , not against its possibly small coefficient digit place r .

On a suffix branch with $j > k$, the coefficient word is absent. The Boolean evaluation asks for a coefficient digit at a separate leaf, returns its bounded code bits, and discards the coefficient word. If the other child of a gate subsequently asks for a late suffix digit, only the bounded pending bits and gate choices are retained. The path bound from the preceding arithmetic lemmas therefore gives $O(1) + c(t-j)$ also on this branch. Repeated coefficient requests rebuild the word. The parameter k and the choice of coefficient are regenerated from the original t and the saved gate path; no earlier cursor address is required. All intrinsic records have $O(t+1)$ bits. $\square$

The coefficients in this proof are operand-dependent, even though they are rational once x^0 is known. They therefore do not qualify as public coefficient leaves. A coefficient request first forms its charged early prefix, then uses scratch for integer arithmetic, returns only the requested bounded digit, and releases that prefix before a suffix branch is evaluated. The original reciprocal clock t remains regenerable throughout. This order of operations is the lifetime rule used in the suspended-record bound. In particular, $j \leq \lfloor t/2 \rfloor + c$ gives the explicit payment $t + 1 \leq 2(t - j) + 2c + 1$ for the early word.

A.4 Order statistics, sums, and finite-accuracy access tests

The remaining operations have an explicitly indexed domain of size at most exponential in the local descriptor length. We can recompute a value whenever it is compared or added, rather than store the whole domain. Numerical order statistics may select different approximating entries at different digit places, but they must approximate one fixed exact order statistic. By contrast, a structural access test has a fixed precision chosen once for that test.

Lemma A.4 (Indexed finite ranks). *Fix $\kappa > 0$ and $d \geq 1$. Consider at most $2^{\kappa d}$ normalized values, indexed by descriptors of length at most κd , and an order statistic specified by an $O(d)$ -bit index. The semantic list length n is public; the requested index is in $\{1, \dots, n\}$, or an out-of-range request returns a prescribed normalized constant. Absent descriptors may be assigned prescribed constant dummy values by structural validity predicates. This order statistic has deterministic integer approximations n_t satisfying Equation (12.3). Their residue procedures have constants a, b, c , depending only on the descriptor bounds, such that an operand digit query at j obeys*

$$j \leq t + cd, \quad l_* \leq ad + b(t - j), \quad (\text{A.13})$$

where l_* is the total suspended local record length. A structural validity query is assigned clock zero and has suspended length $O(d + t)$. The intrinsic local length is $O(d + t)$.

Proof. For $1 \leq k \leq n$, the k th largest of the semantic n -entry list is its $(n - k + 1)$ st ascending order statistic, before any extra padding. Pad to a fixed descriptor domain whose cardinality is a power of two, adding entries of value $+1$. These entries do not change any of the first n ascending order statistics. An empty list or an out-of-range request returns its specified default immediately. Replace invalid original descriptors by the prescribed dummies. An operand access evaluates validity first and enters the actual value branch only when valid; an invalid access returns the dummy's digit. Thus a dummy does not require a query to an undefined value. For each valid numerical value use its truncation through $k_* = t + h$ for fixed large h . Order these finite rational values, breaking ties by descriptor index. Compare every pair using the signed comparison of Lemma A.1. For each descriptor count how many descriptors precede it, then compare that count to the requested rank minus one. Exactly one descriptor is selected. Select its low-residue bits using Boolean AND and OR trees and finally floor to output scale 2^t . The selected descriptor is internal to this residue computation. It may change with t ; the exact order statistic being approximated is fixed by the semantic list and the requested index.

If all coordinates of two finite lists differ by at most η , their corresponding order statistics differ by at most η . Indeed, at least the specified number of the perturbed entries are no larger than the original order statistic plus η ; interchanging the lists gives the reverse inequality. Here $\eta = 6 \cdot 2^{-k_*}$. Thus the selected truncation, followed by the final floor, satisfies Equation (12.3) for sufficiently large fixed h . Descriptor ties affect neither this estimate nor the uniqueness of the finite selection. This argument perturbs the entire padded exact list. Some truncated original entries may exceed

+1 slightly, but their error, and hence the error of the selected padded order statistic, is still at most η . The padding argument requires no separate claim that every finite truncation remains in $[-1, 1]$.

Each count has $O(d)$ bits. Carry-save counting as described above, the comparison to the requested rank, and selection among $2^{O(d)}$ descriptors add $O(d)$ Boolean depth above the pairwise comparison bits and selected residue bits. A place- j input to a pairwise comparison consequently has path length $O(d) + O(t - j)$; a residue input is within a fixed distance of t and has path length $O(d)$. The paths to validity predicates have length at most the whole circuit depth $O(d + t)$, and the predicates themselves are separate structural leaves of clock zero.

Each gate is specified by its pair of descriptor indices, its counting-tree or selection-tree position, and a comparator or bit position. The paths determine these indices in $O(d + t)$ scratch, using only the fixed domain and root parameters. No numerical comparison is needed to determine the circuit topology. Apply Lemma 12.2. The constants in Equation (A.13) can absorb the fixed digit overshoot because $d \geq 1$. $\square$

For the compression ranks, copies of a previously computed baseline are ordinary indexed operands: their digit interfaces query the baseline's exact stream one bounded digit at a time. A slot that chooses between an entry and the baseline first evaluates its structural flag and then queries the selected, already-defined stream. Treating this as a defined operand interface leaves no absent value for the rank procedure to evaluate. The required baseline copies belong to the semantic list; the extra power-of-two padding in the proof is then supplied by the constant +1. The structural choice adds only a fixed conditional branch, so the same digit-clock and suspended-record bounds apply.

A rank over the full incoming vertex universe need not have a $2^{O(d)}$ descriptor domain. That operation uses the terminal-program construction of Lemma 13.7, rather than the finite-domain result proved here.

A sum is evaluated by a balanced tree of normalized pair averages, followed by a known scale adjustment. This avoids retaining a sequential accumulator across late operand queries. The number of tree levels is paid for by the descriptor length.

Lemma A.5 (Bounded sums and known scales). *Suppose an immediate descriptor domain has at most $2^{\kappa d}$ terms, all required binary scale exponents have magnitude at most κd , and every intermediate output has a known range bound $2^{\kappa d}$. A sum over an empty domain is zero, and an average is taken over a public nonempty domain. Balanced sums or averages of these terms, followed by the required known rescalings and bounded clipping, satisfy Equation (A.13) and have intrinsic local space $O(d + t)$.*

Proof. Choose one common known power-of-two range bound for the immediate terms and divide each term by it, so every term lies in $[-1, 1]$. Pad their number to a power of two 2^r , $r = O(d)$, and use a complete binary tree of pair averages. Every internal value stays normalized. Along one path there are r elementary operations. Choose common coefficients in their bounds, as in the last paragraph of the proof of Lemma A.1. The constant terms sum to $O(r)$ and the differences of successive digit clocks telescope. Thus the local length is $O(d) + b(t - j)$ and the total overshoot is $O(d)$. Restore the common normalization scale only after the tree of pair averages. For a sum, combine this factor with the known padding count 2^r in one binary shift of magnitude $O(d)$. For an average over an original domain of size m , combine the scale restoration with multiplication of the padded average by the public factor $2^r/m$. With minimal padding the latter factor is in $[1, 2]$, so it requires only one normalized public multiplication and a fixed shift in addition to the common scale restoration.

More explicitly, for a shift by 2^v , truncate the operand at $k_* = \max(0, t + v + h)$. If $t + v + h \geq 0$, its scaled truncation error in scale- t units is at most $6 \cdot 2^{-h}$. If the scaled output is known to be

normalized, floor in the desired output scale. Otherwise, when a bounded clip is required, apply that clip to the scaled truncation before flooring. For the normalized clip to $[-1, 1]$, for example, define the approximation using $\min(1, \max(-1, 2^v x^{[k_*]}))$; its error cannot increase. This shift-and-clip procedure does not first export an unbounded intermediate as a normalized stream. The low residue requires the constant number of places near $t + v$, and a comparison needed for clipping has a constant-state chain through k_* . Comparisons to the scaled clipping endpoints are equivalently comparisons of $x^{[k_*]}$ to the known endpoints divided by 2^v ; their finite constants and shifted place data are public. When $v \geq 0$ these endpoints belong to $[-1, 1]$; when $v < 0$ the normalized operand is already in the clip interval after scaling, so no clipping comparison is needed. Its path cost is $O(1) + O(\max(0, t + v - j))$, which is at most $O(d) + O(t - j)$ on an actual operand query. If $t + v + h < 0$, zero is already a sufficiently accurate approximation after increasing h , and no operand query is needed. A fixed number of such rescalings and clips preserves the bounds by telescoping. Known non-power-of-two coefficients can instead be handled as public normalized streams and bounded multiplication. $\square$

An access decision does not need to distinguish values exactly at one sharp threshold. The allowed gap between acceptance above $2u$ and rejection at or below u makes a fixed finite precision sufficient. This is the numerical basis for the support margins used in the table and compression constructions.

Lemma A.6 (Finite buffered tests). *Let $d \geq 1$, let s and u have known magnitude bound $2^{\kappa d}$, and suppose $u \geq \rho \geq 2^{-\kappa d}$. There is a deterministic finite test that accepts every $s > 2u$ and accepts only $s > u$. It uses $O(d)$ -place digit queries after normalization, with intrinsic and suspended local lengths $O(d)$. Its result is independent of any subsequent numerical output clock.*

Proof. Compute finite rational approximations $\tilde{s}, \tilde{u}$ with individual errors at most $\rho/16$. The known range and lower bound on ρ make $O(d)$ normalized digits sufficient. Accept exactly when $\tilde{s} > \frac{3}{2}\tilde{u}$, using a fixed convention at equality. If $s > 2u$, the left side minus the right side is greater than $u/2 - 5\rho/32 > 0$. If $s \leq u$, it is at most $-u/2 + 5\rho/32 < 0$. The finite comparison and all previous arithmetic have the bounds already proved. The test chooses its precision from d and ρ , not from an enclosing numerical clock. With its structural starting clock defined as zero, its operand requests have $j = O(d)$; increasing a in Equation (A.13) pays also for the term $-bj$. $\square$

B Compile a supplied machine into a deterministic simulation

The approximation theorem begins by fixing a randomized machine. There is a separate uniformity question: can one algorithm take a machine description and its supplied resource bounds, and output a deterministic simulator together with explicit bounds for that simulator? This appendix gives that compiler.

The supplied time, space, and error bounds are hypotheses about the source. The compiler does not decide whether they are true. Instead, it uses one fixed resource-capped interpreter and one fixed deterministic approximator for that interpreter. Source-dependent padding puts every promised source computation within the interpreter's caps.

Theorem B.1 (Effective simulation with explicit bounds). *There is a terminating algorithm which, given a finite randomized machine description $\mathcal{M}$ and positive integers a, b , emits a deterministic machine $\mathcal{D}_{\mathcal{M}}$ and positive integers k_{space}, h, c with the following guarantee. If, on every input of length n and every random tape, $\mathcal{M}$ halts within $(n + 2)^a$ steps, uses at most $b \lceil \log(n + 2) \rceil$ work bits, and has acceptance probability at most $1/3$ on no inputs and at least $2/3$ on yes inputs, then*

$\mathcal{D}_{\mathcal{M}}$ decides the same language. On each input, that same execution uses at most $k_{\text{space}} \log(n+2)$ work bits and $h(n+2)^c$ elementary bit operations.

The compiler's own running time need only be finite, and its output exponent may depend on the source.

The proof has four steps. Construct a single interpreter with unconditional resource bounds on every encoded input. Apply the approximation theorem to it once. Encode any promised source run as a sufficiently padded input to that fixed approximation program. Finally, generate the virtual-input routines and derive explicit bounds from their finite descriptions.

We prove Theorem B.1 by specializing one fixed deterministic program. First we construct a randomized interpreter $\mathcal{U}$ that enforces polynomial-time and logarithmic-space bounds on every input. Theorem 14.1 supplies a deterministic machine $\mathcal{A}$ that separates the inputs on which $\mathcal{U}$ accepts with probability at least $2/3$ from those on which it accepts with probability at most $1/3$. Both $\mathcal{U}$ and $\mathcal{A}$ are fixed before the compiler receives a source description $\mathcal{M}$ and bounds a, b .

When the source satisfies its supplied resource bounds, padding places each encoded computation within $\mathcal{U}$'s caps without changing its acceptance probability. The generated machine runs $\mathcal{A}$ on that padded input, generating each requested symbol from the real input. We then compute the output bounds k_{space}, h, c from the generated finite transition table and the fixed library data. The compiler never has to select a new probability approximator or infer resource bounds from the source's behavior.

Universal deterministic estimation has a substantial antecedent in Pyne, Raz and Zhan [27, full version, Theorem 5.1]. They construct one deterministic algorithm that, on input $(1^n, \mathcal{B})$ with $n \geq 1$ and $\mathcal{B}$ an ordered branching program of length and width at most n , approximates its acceptance probability to error less than $1/n$. For every space-constructible $s(n) \geq \log n$, this single algorithm uses $O(s(n))$ space exactly when $\text{prBPL} \subseteq \text{SPACE}[O(s(n))]$. Here the preceding sections establish the separator used by one fixed library, and we compute explicit resource bounds for its source-specific specialization.

All quantities in this section are local to the compiler construction. In particular, p is a padding exponent, d_{code} is a program length, $b_n = \lceil \log_2(n+2) \rceil$, and n_{pad} is a virtual input length. They do not denote the accuracy levels, field modulus, or domain sizes used in the preceding sections. We use the usual counted-cell convention for work tapes: traversed writable cells count toward space, whether or not they contain a nonblank symbol. The finite control and the read-only input have their standard treatment.

B.1 An interpreter that enforces its own resource limits

The interpreter must be small-space even on malformed or maliciously large descriptions. It therefore measures the whole input first, checks the code length before allocating storage, and imposes explicit time and tape limits during simulation. The coding below makes those checks finite scans of listed data.

Fix a binary coding of ordinary finite transition lists. One concrete choice uses a fixed record grammar and the field encoding $w \mapsto 1^{|w|}0w$. Lists specify the finite state set, tape and alphabet declarations, initial and halting states, blanks and endmarkers, and transition records. A transition record lists its tested state, scanned symbols and coin pattern, followed by its next state, written symbols and local head movements. The grammar has fixed nesting depth; its fields are actual listed data, not recursively executable descriptions. State and symbol IDs are canonical binary integers used only as names. The first matching transition is used; a missing transition causes rejection. Wildcards for ignored coins can equivalently be expanded into a finite explicit list.

Write z for a program in this coding and $d_{\text{code}} = |z| \geq 1$. The coding permits at most d_{code} work tapes and a block of at most d_{code} fresh fair bits per simulated transition. Every ID has at most d_{code} bits. A harmless padding field can enlarge the program to meet these inequalities. All tape movements are local. Work coordinates are measured from their initial origins; the ordinary model has one input head and one head per work tape. For a source input x of length n , let $E_{d_{\text{code}}}(x)$ encode each of its symbols by its ID padded to exactly d_{code} bits.

Preserving the supplied source bounds. An ordinary source machine $\mathcal{M}$ can be converted effectively to this format by renaming its states and symbols and listing its transitions. Its tapes are represented directly, so a source transition retains its work writes and head movements and counts as one interpreted transition. The conversion does not pass through a slower single-tape simulation.

Under the optional one-way read-only random-tape convention, the source may reuse its current random bit. Store a cached bit and a validity flag in finite control, initially invalid. At each transition obtain one fresh bit. Use it if the cache is invalid; otherwise use the cached bit and discard the fresh one. Execute the source transition, then retain the used bit if the random head stayed and invalidate the cache if it advanced. This samples a newly visited random cell lazily on its next use. It preserves the joint distribution of the source run and needs no initialization transition. An initially halting state is recognized before the first transition. The same argument uses a fixed finite block for a convention allowing finitely many fresh bits per elementary transition, discarding unused bits. The finite cache changes neither counted work space nor the number of source transitions. Thus the supplied parameters a, b remain valid; only the static description length can increase.

Lemma B.2 (A total resource-capped interpreter). *There is one fixed randomized Turing machine $\mathcal{U}$ with polynomial worst-case running time and $O(\log(m+2))$ counted work bits on every binary input of length m , with the following behavior. Set*

$$\lambda_m = \lceil \log_2(m+2) \rceil, \quad r(m, d_{\text{code}}) = \left\lfloor \frac{\lambda_m}{100(d_{\text{code}} + 1)^3} \right\rfloor. \quad (\text{B.1})$$

On an input with prefix

$$1^{d_{\text{code}}} 0 \approx 1^n 0 E_{d_{\text{code}}}(x), \quad (\text{B.2})$$

it first requires $1 \leq d_{\text{code}} \leq \lambda_m$ and $r(m, d_{\text{code}}) \geq 1$. It then simulates the encoded machine for at most m transitions, allowing each counted work tape only coordinates in $[-r, r]$. It rejects malformed data, a requested move outside these bounds, or failure to halt by the end of the transition limit. Otherwise it returns the encoded machine's decision. The suffix after the complete prefix has no parsing significance.

Proof. First validate the input before allocating any description-dependent storage. Count m before interpreting the prefix. Scan the unary code-length header with a bounded counter, rejecting if its length exceeds λ_m or the required delimiter is absent. Check the two eligibility conditions before copying z or allocating simulation storage. Then verify that the following d_{code} code bits, the unary data-length header, and the required data bits are present. The parsed n is at most m , and $d_{\text{code}} n \leq \lambda_m m$, so even size calculations preceding a failed length check use only $O(\lambda_m)$ bits. List counts, ID lengths, tape declarations, and movements are checked against the actual code and the stated bounds before any loops or allocations using them. There is no enumeration of an implicit domain of symbol or state names.

The eligibility conditions now imply a simultaneous storage bound for all encoded tapes and administrative records. Store the encoded tapes sequentially on a fixed number of real work tapes

of $\mathcal{U}$. At most d_{code} strips, each with $2r + 1$ cells of d_{code} bits, require

$$d_{\text{code}}^2(2r + 1) = O(\lambda_m) \quad (\text{B.3})$$

bits. Indeed, eligibility gives $d_{\text{code}}^3 \leq \lambda_m$ and $d_{\text{code}}^2 r \leq \lambda_m$. The copied program, current state, scanned and written symbol tuples, coin block, strip coordinates, step counter, and parsing scratch fit simultaneously in $O(\lambda_m)$ bits. A single simulated input-head coordinate has magnitude $O(m)$ by the transition cap and therefore also uses $O(\lambda_m)$ bits. Read the corresponding source symbol by rescanning the encoded data, supplying the appropriate markers or off-data blanks according to the encoded input-tape convention.

All routines are bounded scans and ordinary binary arithmetic. They iterate over actual code positions, input positions, a bounded tape/head list, strip positions, and bit positions, with a fixed nesting depth. Each range is polynomially bounded in $m + 2$; binary arithmetic is implemented by bit scans, not unit-cost operations on growing words. Matching a listed transition, fetching input symbols, updating the strips, and obtaining its finite coin block therefore take polynomial time in $m + 2$, with one fixed exponent determined by these routines. The same bounds hold when a check fails. There are at most m interpreted transitions. Checking the initial halting state and the state reached by the last permitted transition gives the asserted total behavior, including $m = 0$. The resulting fixed $\mathcal{U}$ has the required polynomial worst-case time on every random tape. $\square$

No bounded-error language promise is asserted for $\mathcal{U}$ on arbitrary inputs. Its acceptance probability may lie between $1/3$ and $2/3$. This is why we use probability approximation rather than a bare language-class containment.

B.2 Fix the deterministic approximator once

The interpreter need not itself decide a bounded-error language on arbitrary encodings. This causes no difficulty because our theorem approximates arbitrary acceptance probabilities. Apply Theorem 14.1 to the fixed machine $\mathcal{U}$ and accuracy exponent 3, and compare its dyadic answer with $1/2$. The error on an input of length m is at most $(m + 2)^{-3} \leq 1/8$. Hence the resulting deterministic machine accepts whenever $\Pr[\mathcal{U} \text{ accepts}] \geq 2/3$ and rejects whenever that probability is at most $1/3$, since

$$\frac{1}{3} + \frac{1}{8} = \frac{11}{24} < \frac{1}{2} < \frac{13}{24} = \frac{2}{3} - \frac{1}{8}.$$

It halts on every input, including inputs between the two thresholds, as also stated in Corollary 14.4. Its logarithmic space bound counts all simultaneously live work in the approximation algorithm. Store the $O(\log(m + 2))$ -bit dyadic output in work space for the threshold test.

We fix this library once in the following convenient form. It is an ordinary finite-tape deterministic decision machine $\mathcal{A}$ with one input head confined to the endmarked input, initially at its left endmarker, such that every work head stays in

$$[-(\lambda_m + 1), \lambda_m + 1] \quad (\text{B.4})$$

relative to its origin. To obtain this form, first consolidate any input heads by keeping their coordinates in logarithmic work space and scanning for each read. Even an off-data coordinate under an unbounded-input convention has logarithmic length because the original library has polynomial running time. A fixed change of initial control state handles its input-head start convention.

Next choose an integer w at least a coefficient bounding each counted work head's distance from its origin by $w\lambda_m$. Such one fixed w exists from the logarithmic bound, enlarged if necessary

for the finitely many small lengths. Group consecutive work cells into blocks of width w , storing within-block offsets in finite control. The resulting finite alphabet and finite transition list realize Equation (B.4), also for two-sided tapes. All symbol widths will be charged in bits below; grouping only exposes a uniform address bound and gives no uncharged bit compression.

This choice of $\mathcal{A}$, including its one blocking width, precedes every compiler input. More formally, the remaining construction is a computable syntactic function of a finite library list and $(\mathcal{M}, a, b)$. Fixing one valid finite list yields one ordinary computable compiler on all those source inputs. It does not require an algorithm that discovers a separator or a logarithmic-space coefficient from an arbitrary machine description. The existence of the fixed list was proved by Theorem 14.1; no constant is selected as a noncomputable function of the source machine or its ordinary input.

The fixed finite data are the program grammar, the transition list of $\mathcal{A}$ with its blocked work alphabets and head conventions, and the finite callback templates described below. The construction constants, cutoffs, generator choices and finite instruction routines used by the separator are included in that list. Its blocking width is fixed at the same time. These are data specifying one compiler; subsequent source-dependent integers are computed from these data and the supplied $(\mathcal{M}, a, b)$.

B.3 Use padding to fit the promised source computation

The deterministic approximator is now fixed, independently of the source description. We enlarge its virtual input enough that the interpreter's work strips and transition limit contain every promised source run. No branch of such a run is cut off, so acceptance probability is preserved exactly.

Given the source description and its supplied positive integers a, b , construct z as above and compute

$$\begin{aligned} p &= 1000(d_{\text{code}} + 2)^4(a + b + 2), \\ b_n &= \lceil \log_2(n + 2) \rceil, \quad n_{\text{pad}} = 2^{p b_n}. \end{aligned} \tag{B.5}$$

The generated machine will run $\mathcal{A}$ on a virtual binary input y_x of length n_{pad} whose prefix is Equation (B.2) and whose remaining bits are zero. Its prefix length is exactly

$$l_{\text{pre}} = (d_{\text{code}} + 1) + d_{\text{code}} + (n + 1) + d_{\text{code}}n = (d_{\text{code}} + 1)(n + 2). \tag{B.6}$$

Since $b_n \geq 1$, $2^{b_n} \geq n + 2$, and $2^{p-1} \geq d_{\text{code}} + 1$, this length is at most n_{pad} . The interpreter's length parameter is

$$\ell_{\text{pad}} := \lambda_{n_{\text{pad}}} = p b_n + 1. \tag{B.7}$$

All eligibility checks pass. In particular,

$$\frac{p}{100(d_{\text{code}} + 1)^3} = 10 \frac{(d_{\text{code}} + 2)^4}{(d_{\text{code}} + 1)^3} (a + b + 2) \geq 30(a + b + 2),$$

so

$$r(n_{\text{pad}}, d_{\text{code}}) \geq 30(a + b + 2)b_n - 1 \geq (a + b)b_n + 4. \tag{B.8}$$

Also $p > a$, whence $(n + 2)^a \leq 2^{a b_n} < n_{\text{pad}}$. Local motion from an initial work origin to a cell entails visiting the intervening counted cells. The source bound of $b b_n$ work bits therefore places every source work cell in the allocated strip; the slack in Equation (B.8) also covers a final one-cell move. This uses the source's original cell costs, each at least one bit, and does not charge it for the larger interpreter IDs.

Consequently no resource cutoff changes a promised source run, and

$$\Pr[\mathcal{U}(y_x) \text{ accepts}] = \Pr[\mathcal{M}(x) \text{ accepts}]. \tag{B.9}$$

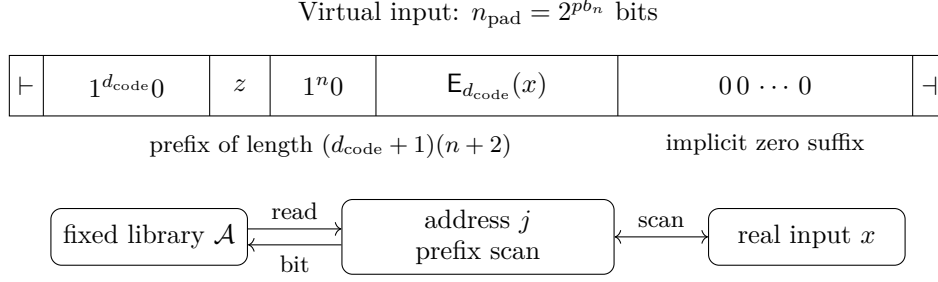


Figure 3: The fixed library reads the padded input through a callback. The callback regenerates prefix symbols from the real input and returns zero for an address in the padding interval.

There is no conditioning or additional approximation in this equality. The decision of $\mathcal{A}(y_x)$ is therefore the decision required for $l_{\mathcal{M}}$. Although n_{pad} is given exponentially, it is at most $(n + 2)^{2p}$ for the fixed source. More importantly, it need never be written as an input string.

Remark B.3 (Optional head conventions). The construction also applies when the fixed coding and $\mathcal{U}$ are chosen to allow at most d_{code} heads per tape, with the usual common initial origin. These conventions are fixed before choosing $\mathcal{A}$. There are then at most d_{code}^2 work-head coordinates and scanned symbols. Their storage is bounded by

$$O(d_{\text{code}}^2 \log(2r + 1) + d_{\text{code}}^3) = O(\lambda_m).$$

For multiple input heads, or an optional entirely blank tape whose cells are not charged, use signed r -bit coordinates and reject on overflow. At most d_{code}^2 such coordinates still cost $O(d_{\text{code}}^2 r)$ bits. These optional conventions do not change the interpreter's bounds. The padded instances just constructed never overflow those coordinates.

Under the optional head conventions, local motion bounds an input or uncharged blank-tape coordinate by the source time, up to the fixed endmarker offset. Its signed encoding fits because $r \geq ab_n + 4$.

Figure 3 shows how the generated machine supplies the virtual input requested by $\mathcal{A}$. The library's work tapes remain live while the callback scans the real input; the resource bound below counts the library and callback storage simultaneously.

B.4 Read the padded input without storing it

The padded string can be much longer than the real input. Only its current queried address needs to be stored. A finite routine answers that query by scanning the short encoded prefix and returning zero if the address lies in the padding. We now give the routine explicitly so that its memory is counted alongside the suspended approximator.

Inline the finite transition list of $\mathcal{A}$, giving it its own work tapes. The generated machine $\mathcal{D}$ suspends its library control state in finite control during an input operation. Four auxiliary binary words suffice: the length count $n + 1$, the virtual right-endmarker address $n_{\text{pad}} + 1$, the virtual input-head position j , and a prefix position i . Each has a separate tape, with origin and termination marks and digits in low-end-first order.

Initialize the length count to one and increment it once per real input symbol. Its bit length is exactly

$$\text{bitlength}(n + 1) = \lfloor \log_2(n + 1) \rfloor + 1 = \lceil \log_2(n + 2) \rceil = b_n \quad (n \geq 0). \quad (\text{B.10})$$

For every bit place in that count, write p zero bits on the bound tape, then append a one. This produces the low-end-first representation of $2^{pb_n} = n_{\text{pad}}$; increment it to obtain $n_{\text{pad}} + 1$. The repetition of p is hardwired finite control. Initially set $j = 0$, the library's left-endmarker position.

At a virtual read, first test $j = 0$ and $j = n_{\text{pad}} + 1$ and return the corresponding endmarker. Otherwise initialize $i = 1$ and regenerate the prefix in its natural order:

1. emit the fixed string $1^{d_{\text{code}}}0z$;
2. scan x , emitting one 1 per symbol, and then emit 0;
3. rewind and scan x again, emitting each fixed-width symbol code in $E_{d_{\text{code}}}(x)$.

Here “emit” means compare the current i with j , return the bit if they agree, and otherwise increment i and continue. If the scan finishes without a match, return zero. The strings z , the integer d_{code} , symbol codes and emission phases are fixed in the generated control. The real input head is rewound by scanning to its endmarker, without storing its position. In particular, $j = n_{\text{pad}} + 1$ is answered before regeneration, while a padding address such as $j = n_{\text{pad}}$ returns zero after the finite prefix scan. The largest prefix counter value is $l_{\text{pre}} + 1 \leq n_{\text{pad}} + 1$.

For explicit counter routines, an increment uses a finite carry flag. Compare two low-end-first words by scanning both, retaining only a finite comparison flag that is overwritten by each higher differing bit. After one terminator, treat its later bits as zero using a finite “ended” flag. Rewind each tape to its origin mark afterward. Zeroing and copying are likewise bounded bit scans. No growing arithmetic register beyond the named words is required. All words have at most $pb_n + 2$ bits; including delimiters and rewinds, their heads need visit only positions through $pb_n + 10$.

After the callback, perform the indicated library transition and update j in place for its input-head motion. The library's work heads and contents are untouched by the callback. Its state, the returned input symbol, and any step-local scanned tuple are fixed finite-control data. The callback does not invoke $\mathcal{A}$ recursively or maintain a stack of callbacks. The real input head stays between its real endmarkers.

Every listed routine terminates. For example, a complete regeneration uses at most l_{pre} emissions, each with an $O(pb_n + 1)$ -bit comparison and increment, together with two real-input scans. Initialization makes n increments and writes $pb_n + 1$ bound bits. These observations give direct polynomial bounds for the auxiliary operations. The whole simulation halts because $\mathcal{A}$ is total. Its more uniform explicit time bound, including every such operation, is derived next from the complete configurations of this same machine.

All the above instructions can be generated as finite transition tables, unrolling fixed repetitions and strings if necessary. Formation of the source list, the integer p , and this transition table is a finite effective process. It makes no test of the source's semantic error or resource promises. Syntactically invalid compiler inputs may be assigned a fixed default output; correctness is required on the stipulated class. The generated simulation is total and obeys the bounds below even when the supplied source promises fail; those promises are used to establish that its decision agrees with the source.

B.5 Compute explicit bounds for the generated machine

All source-dependent choices have now been made by finite operations on code and supplied integers. The remaining bounds use three concrete counts: work tapes, alphabet sizes, and control states. We first bound all live work bits, then count complete configurations of the halting machine, and finally convert finite-alphabet transitions to elementary bit operations.

Use the finite-alphabet description just constructed to compute t , its number of work tapes; $g \geq 2$, the maximum of two and its work alphabet sizes, including all blank, origin and termination

symbols; and $q_{\mathcal{D}}$, its number of states, including callback phases. Let $u \geq 1$ dominate its explicit transition-table bit length, all alphabet sizes (including the input alphabet), the number of states, and the number of tapes. Padding this integer is effective. Return the positive integer bounds

$$\begin{aligned} \rho &= 100(p+2), & s_{\text{bits}} &= 1 + (t+1)(2\rho+1)(g+2), \\ k_{\text{space}} &= 4s_{\text{bits}}, & c &= 2s_{\text{bits}} + 1, & h &= 2^{(u+100)^2}(q_{\mathcal{D}}+1). \end{aligned} \quad (\text{B.11})$$

These depend on the fixed source and the fixed library, and are computed from finite syntax. They have no dependence on n or x .

All live work bits. Every work head stays in $[-\rho b_n, \rho b_n]$. For a library tape this follows from Equations (B.4) and (B.7), since $\ell_{\text{pad}} + 1 = pb_n + 2 \leq \rho b_n$. For an auxiliary tape it follows from $pb_n + 10 \leq \rho b_n$, using $b_n \geq 1$. Thus the library's live data and all callback data satisfy the same bound at the same time. Encoding a work alphabet of size g_i in $\max\{1, \lceil \log_2 g_i \rceil\} \leq g$ bits per symbol bounds their total data by

$$t(2\rho b_n + 1)g \leq s_{\text{bits}}b_n. \quad (\text{B.12})$$

Even twice this quantity satisfies the stronger logarithmic bound: with $l_n = \log_2(n+2) \geq 1$ we have $b_n \leq 2l_n$ and

$$2s_{\text{bits}}b_n \leq 4s_{\text{bits}}l_n = k_{\text{space}}l_n.$$

All finite-alphabet transitions. For a fixed real input x , a complete configuration is specified by the state, the real input-head position, all work-head positions, and all work symbols in the stated intervals. Everything outside these intervals remains at its initial value. There are at most

$$q_{\mathcal{D}}(n+2)(2\rho b_n + 1)^t g^{t(2\rho b_n + 1)} \leq q_{\mathcal{D}}(n+2)2^{s_{\text{bits}}b_n} \leq q_{\mathcal{D}}(n+2)^c \quad (\text{B.13})$$

such configurations. For the first inequality use

$$2\rho b_n + 1 \leq (2\rho + 1)b_n, \quad \log_2 g \leq g, \quad \log_2(2\rho b_n + 1) \leq (2\rho + 1)b_n;$$

the second uses $b_n \leq 2\log_2(n+2)$ and $c = 2s_{\text{bits}} + 1$. A deterministic halting computation cannot repeat a complete configuration. Therefore Equation (B.13) bounds its total transitions, including initialization, counter arithmetic, every regenerated input bit, and every library transition. The virtual head position is already in the counted word j ; it is not an additional uncounted configuration component.

Elementary bit operations. The preceding count refers to ordinary finite-alphabet transitions. To charge them in bits, encode each work symbol by a fixed-width block, mapping blank to the all-zero block. Read and write within the current block, return to its first bit, and move to the first bit of the target block. Within-block phases and scanned finite tuples are stored in finite control. No per-block writable delimiter is needed, and these operations visit only blocks whose macro positions occur in the proved work intervals. Existing origin and termination symbols have already been included in g . Thus the block implementation obeys the bit-space estimate above; the factor two also allows the usual blank/bit tape convention. Input-symbol access has fixed width and is covered by the same finite description parameter u .

There are at most $u+1$ scanned symbols of at most u bits each. An explicit sequential comparison against at most u transition rows, followed by the fixed-width reads, writes and moves, uses at

most a fixed polynomial in $u + 1$ bit actions per transition. For example, the deliberately loose bound $100(u + 1)^8$ suffices for these scans and comparisons, with all temporary phase and symbol information in finite control. It is dominated, for $u \geq 1$, by

$$100(u + 1)^8 \leq 2^{(u+100)^2}.$$

The binary microcode may have more states than $q_{\mathcal{D}}$, but no new configuration count is needed: multiply the number of original transitions by this per-transition bit cost. Equations (B.11) and (B.13) give the simultaneous bound

$$\text{time}_{\text{bit}}(\mathcal{D}, x) \leq h(n + 2)^c. \quad (\text{B.14})$$

No growing-word arithmetic is charged as a single elementary bit action.

Completion of the proof of Theorem B.1. The compiler returns the described machine $\mathcal{D}$ and the integers k_{space}, h, c in Equation (B.11). Its finite syntactic construction terminates. On every promised source input, Equation (B.9) and the fixed separator’s threshold guarantee give the correct decision. The totality of $\mathcal{A}$ and the finite callback loops prove halting. Equations (B.12) and (B.14) give at most $k_{\text{space}} \log_2(n + 2)$ work bits and $h(n + 2)^c$ bit operations for this same machine.

All claims include $n = 0$: then $b_n = 1$, the real-input scans are empty, the count $n + 1 = 1$ has one bit, and every displayed padding, strip, space, and time inequality remains valid. The library’s grouping width already includes its finitely many small-length exceptions. All construction constants are fixed before the ordinary input is read.

The work and time internal to probability approximation—environment enumeration, numerical digits, recursive interfaces, recomputation, and precision management—belong to the one complete library machine proved in Theorem 14.1. Its tapes remain counted throughout virtualization. The compiler adds only the exact input operations analyzed here, without an evaluation oracle or an external stack. Configuration counting gives polynomial time because this entire deterministic machine uses exact logarithmic space. $\square$

C Historical context and related work

The complexity of probabilistic Turing machines was developed by Gill [19], including measures of both time and tape usage. Random walks made the logarithmic-space question concrete: Aleliunas, Karp, Lipton, Lovász and Rackoff obtained polynomial-time randomized algorithms for undirected connectivity and explicitly asked whether polynomial-time randomized logarithmic space could be simulated in deterministic logarithmic space [3]. Early work also studied probabilistic machines without a polynomial-time restriction. Borodin, Cook and Pippenger obtained a quadratic-space deterministic simulation in this broader setting [8]. The distinction between a space bound and simultaneous time and space bounds is essential throughout this history.

Pseudorandom generators attack the problem by replacing a long sequence of fresh random bits with the output generated from a short seed. Ajtai, Komlós and Szemerédi gave early logarithmic-space constructions that hit small-space tests using short random strings [2]. Babai, Nisan and Szegedy subsequently obtained pseudorandom generators through multiparty communication bounds [6]. Nisan’s generator uses $O(s \log r)$ seed bits for a space- s computation reading r random bits [23]. For logarithmic space and polynomially many bits this gives $O(\log^2 n)$ seed length; direct seed enumeration has a quasipolynomial time bound. Nisan’s later simultaneous simulation achieves polynomial time and $O(\log^2 n)$ space [24]. Although its title uses RL, that result includes two-sided bounded error. For $s(n) \geq \log n$, Nisan and Zuckerman showed that a space- s computation using

poly(s) random bits can be simulated using only $O(s)$ random bits and $O(s)$ space [26]. The restriction on the original number of random bits is substantive; for logarithmic space it permits polylogarithmically many bits.

Saks and Zhou reduced the deterministic space needed for general bounded-error logarithmic-space computation to $O(\log^{3/2} n)$ [35]. Cai, Chakaravarthy and van Melkebeek established a simultaneous tradeoff: for each fixed rational $0 \leq \alpha \leq 1/2$, the simulation uses time $n^{O(\log^{1/2-\alpha} n)}$ and space $O(\log^{3/2+\alpha} n)$ [11]. The polynomial-time endpoint has squared-logarithmic space. Hoza subsequently improved the general space bound to $O(\log^{3/2} n / \sqrt{\log \log n})$ [20, Theorem 1.6 and Corollary A.5]. These smaller space containments do not assert the same polynomial-time endpoint. Hoza’s work also develops weighted pseudorandomness, in which signed or real weights replace an ordinary probability distribution; the space-containment improvement and the new weighted-generator construction are distinct results of that paper.

Acceptance-probability estimation also connects derandomization to stochastic matrix products and inverses. Ahmadinejad, Kelner, Murtagh, Peebles, Sidford and Vadhan developed high-precision small-space algorithms for random-walk probabilities, with stronger bounds for Eulerian graphs [1]. Cohen, Doron, Sberlo and Ta-Shma later used varying precision across recursion to improve the space complexity of long stochastic matrix products [17]. The construction in these notes works with a finite inverse on a time-layered configuration graph; its correction and copy hierarchy controls the information retained while individual entries are evaluated.

Structural restrictions on a computation provide another route. For undirected connectivity, Nisan, Szemerédi and Wigderson obtained deterministic space $O(\log^{3/2} n)$ [25], subsequently improved to $O(\log^{4/3} n)$ by Armoni, Ta-Shma, Wigderson and Zhou [4]. Reingold’s logarithmic-space algorithm, whose conference version appeared at STOC 2005, settles undirected connectivity [29, 30]. Reingold, Trevisan and Vadhan studied regular directed graphs and gave logarithmic-space path search for Eulerian directed graphs [31]. Their oblivious walk constructions impose additional labeling conditions. General directed configuration graphs need not have these properties, so those results did not settle the unrestricted randomized computation problem. A read-once branching program is a layered deterministic computation that reads the input symbols once in a fixed order (bits in the binary case). Its length is the number of steps, and its width is the largest number of states in a layer. A 1/2-hitting set for a class of programs contains an accepting input for every member that accepts at least half of all input strings. The relation between one-sided and two-sided derandomization also requires care. Cheng and Hoza proved that one uniformly logarithmic-space-enumerable 1/2-hitting-set family for all width- n , length- n read-once branching programs would yield $L = BPL$ [14]. The premise is a hitting-set construction; it is not merely the language-class equality $RL = L$.

Recent results refine both general and structured models. Chen, Cohen, Doron, Khaskelberg and Ta-Shma improve error reduction for weighted generators; in the polynomial-width regime their construction has seed length $\log |\Sigma| + O(\log^2 t + \log(1/\delta))$, where t is branching-program length, Σ its alphabet and δ the error [13]. Cheng and Wu obtain simultaneous polynomial-time and polylogarithmic-space simulations for regular branching programs, with extensions to polynomial-time randomized logarithmic-space machines whose two-way read-only random tape visits each cell at most constantly often, and to such machines with read-once randomness and an auxiliary stack whose push, pop and idle schedule is independent of the random bits. Consecutive stays on a random-tape cell count as one visit [15, Theorems 5–6]. Hardness-versus-randomness approaches additionally give conditional, certified and per-instance alternatives; Pyne and Tell survey this line of work [28].

Effectivity has its own substantial antecedents. Pyne, Raz and Zhan construct a universal

deterministic estimator whose $O(s(n))$ space guarantee follows from $\text{prBPL} \subseteq \text{SPACE}[O(s(n))]$, for space-constructible $s(n) \geq \log n$ [27, full version, Theorem 5.1]. Here prBPL denotes promise problems solvable in polynomial time and randomized logarithmic space with bounded error. The compiler in Appendix B uses the fixed separator established by the preceding construction and supplies explicit source-specific numerical resource bounds. Its construction should be read in this context of universal simulation, with the source bounds supplied as part of the compiler input.

D Notation and a guide to the source symbols

The following tables collect the recurring objects. Each is also defined near its first use. Local cutoff constants, finite program tags, and temporary variables are specified in their own sections.

Symbol	Meaning
$\mathbf{S}, \mathbf{e}$	Base time-layered transition matrix and final acceptance indicator.
$\mathbf{p}_0$	Vector of exact continuation probabilities in the base model.
$\mathbf{C}_l, \mathbf{w}_l$	Stage- l residual transition matrix and accumulated success vector.
$\mathbf{E}, \mathbf{D}$	Cumulative correction and residual $\mathbf{C} - \mathbf{E} + \mathbf{E}\mathbf{C}$ before copying.
$\mathbf{G}_i, \mathbf{G}'_i$	Raw pilot and actual propagation inputs, respectively.
$\mathbf{g}, \mathbf{u}_i$	Frozen column multipliers and exact detector scores. The source writes the latter as U_i .
$\mathbf{P}_l^r, \mathbf{P}_l^c$	One-stage and cumulative weighted projections from copies to their ancestors. The superscript c means cumulative.
$\mathbf{B}_l$	Cumulative matrix taking the base indicator to $\mathbf{w}_l$; distinct from a scale-indexed sampling table.
$f, b_k = 2^{-hk}$	Current entry bound and the fixed geometric grid of nominal retention rates.
a, u, t	Entry scale, start-retention rate, and optional usage rate in a sampling task. In numerical sections t is instead a digit clock; the change is stated there.
$X_a(y), Y_a(y), Z_a(y)$	Scalar membership indicators on different sampling channels.
π_a	Actual retention probability at nominal rate a .
$\mathbf{A}, \mathbf{F}$	Polynomial coefficient matrix and its frame of nonconstant columns. A local exact correction may instead be denoted $\mathbf{E}$; the normalized correction table is $\mathbf{F}$.
$p_{\text{fld}}, l_{\text{box}}$	Prime modulus and retention-box side length.
$\mathcal{G}_h, \mathcal{K}$	Integer affine group and a conditional averaging operator.
$\mathbf{B}_b, \mathbf{F}$	Normalized sparse bin and correction tables (the source's $\mathcal{B}_b, \mathcal{F}$).
m, r	Parent accuracy level and child accuracy level. The source uses M, m .
p_{acc}	Effective statistical accuracy exponent, after scale, domain, and rank charges; the source uses p .
i_*, d	Dependency rank and a dyadic parent-child level decrease.
b_{mem}, n_v	Size parameter comparable to $\log(n+2)$ and common upper bound on one stage's vertex count; the source uses B, N .
l_{max}, h, d_0	Number of outer stages, fixed dimension/rate exponent, and copy radix; the source uses L, H, D_0 .
$k_{\text{row}}, j_{\text{det}}, k_c$	Fixed row rank, detector rank, and incoming cap; the source uses K, J, K_c .

Symbol	Meaning
k_*, k_{entry}	Fixed sweep count (also a correction row-sum bound), and the residual-entry constant; the source uses k_*, K_* .
$\mathbf{s}_i, \mathbf{t}_i$	Uncompressed and compressed raw-row prefixes in the compression section.
$h_*, \mathcal{I}_b$	Fingerprint block width and the b th block of accuracy levels.
$\mathbf{v}$	Shared bit vector used for restoring fingerprint comparisons; the source calls it V in that subsection.
w, w'	Parent and child call allowances in the memory accounting, distinct from the success vector $\mathbf{w}$.
t, j	Requested output and operand digit clocks in the arithmetic and controller sections.

An overbar marks an estimator when it is compared with its exact target; a pilot correction instead carries a hat. Neither boldface nor case encodes accuracy. In particular, a matrix stays bold uppercase after fixing its random environment. Functions such as the scalar cutoff Q and the row functions H are named functions, not scalar random variables.

The three independent measures of progress are worth keeping distinct. The outer stage reduces the exact residual. The accuracy level determines which recursive estimator is used. The numerical digit clock asks for a more precise reading of that fixed estimator. Raising the last does not improve the statistical approximation supplied by the second.

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